BIMopen user manual

BIMopen
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BIMopen user manual

Home

- BIMopen user manual 16
  - How this manual is organised 16
  - Conventions 17
  - What this manual describes 17

Getting started

- Getting started 18
  - Running BIMopen on the desktop 18
    - Start screen 18
    - Language and theme 18
  - Running BIMopen in the browser 18
  - The sample project 19
  - Your first house 19
    - 1. Start a new project 19
    - 2. Open the ground floor plan 20
    - 3. Draw four walls 20
    - 4. Add a door and a window 21
    - 5. Add a floor and a roof 21
    - 6. Look at it in 3D 22
    - 7. Put the plan on a sheet 22
    - 8. Undo, redo and save 23
  - What next 23

The interface

- The interface 24
  - File panel 24
    - Recent Files 25
    - Options 26
  - Top bar and quick access tools 27
    - About BIMopen 28
    - The Commands window 28
    - Updates 29
  - Rail and tool panel 29
    - The Model tools 32
    - The Annotations tools 33
    - The Views tools 35
    - The Project tools 35
    - The contextual tab Edit 37
  - Options strip 38
  - Element panel 39
  - Project Index 42
  - Views and components strip 43
  - Panels 44
  - Dialogs 44
  - Drawing area and view windows 44
    - Views in separate windows 46
    - Drawing in every view 46
    - View bar 46
    - Zoom 47
    - Moving around in the views 47
    - The context menu of a view 49

- Snap 49
- Command search and command line 51
- Status bar 51
- Selecting, in short 51

Interface styles

- Interface styles 53
  - Choosing the style 53
  - The Classic style 54
  - Everything of the Classic style 54
  - Touch mode 55
    - Turning it on 55
    - Phone, landscape and tablet 55
    - Gestures and the pen 60

Projects and files

- Projects and files 61
  - Creating a project 61
    - Templates 62
  - BIMopen files 62
  - Opening a project 62
  - Saving 63
    - Saving for archiving 63
  - Project settings 63
    - Project information 63
    - Project units 63
    - Imperial units 64
    - Typing lengths 67

- Sun Settings 67
- Options 67
- Graphics: graphics card and edges 68
- Copying settings from a project 69
- Exchanging files 69
- Exporting to IFC 69
- Opening IFC files 69
- Exporting to USD 70
- Project bundles 70
- Other exports 70

Importing and converting

- Importing and converting 71
- Opening an IFC file 71
- Import IFC Options 71
- Importing an IFC file into the project 72
- Conversion Status 74
- Converting an imported object 76
- The three ways 76
- Components from imported objects 77
- Identical shapes 78
- Selecting imported objects 78
- Imported objects in the Project Index 78
- Restoring the originals 79
- Coming next 79

DWG and DXF

- DWG and DXF 81
  - Exporting a view or a sheet 81
  - Importing or linking a drawing 81
  - The CAD file in the project 82
  - Manage Links 82

Selecting and modifying

- Selecting and modifying 83
  - Highlighting and selecting 83
  - Elements close together: TAB 84
  - Chains of walls and lines 84
  - Filtering a selection 85
  - Select All Instances 85
  - Dragging elements 85
  - The Modify tools 86
    - Move 86
    - Copy 86
    - Rotate 86
    - Mirror 87
    - Repeat in Series 87
    - Offset 87
    - Align 87
    - Shorten/Lengthen 87
    - Split Element and Split with Joint 88
    - Scale 88
    - Fix and Unpin 88

- Delete, undo and redo 88
- Copy, cut and paste 88
  - Edit pasted 89
- Transfer type 89
- Groups 89
- Typed shortcuts 90
- Not yet available 90

Walls, doors and windows

- Walls, doors and windows 91
  - Drawing walls 91
  - Curved walls 92
  - Elliptical walls 93
  - The alignment and flipping 93
  - Attaching walls to roofs and floors 94
  - Editing a wall's profile 94
  - Wall joins 94
    - The Wall Joins tool 94
    - Disallowing a join 95
  - Wall types and layers 95
    - Type parameters 95
    - Type layers 96
  - Wall properties 97
  - Placing doors and windows 98
  - Tags 98
  - Flipping a door or window 98
  - Moving a door or window to another wall 99
  - Door and window properties 99
  - How walls, doors and windows are drawn 100

- Curtain walls 100
  - Grid lines and mullions 101
  - Mullion and panel types 101
- Stacked walls 102
  - The sub-walls of the type 103
- Levels 103
- Work planes 103
  - Setting the work plane 103
  - Showing the work plane 104
  - Reference planes 104
- 3D lines and 3D text 104
  - 3D lines 104
  - 3D text 105
- At the command line 105
- Not available yet 105

Floors, roofs, stairs and components

- Floors, roofs, stairs and components 106
  - Sketch mode 106
    - Drawing lines 106
    - Picking and changing sketch lines 107
    - Editing the sketch of an element 107
  - Floors 107
    - Sloped floors 108
    - Floors over walls 108
    - Floor properties 108
  - Roofs 109
    - Roof properties 110
    - Extrusion roofs 110

- Joining roofs 111
- Sloped glazing 111
- Soffits 111
- Stairs 112
  - How a stair is built 113
- Ramps 115
- Railings 115
- Components 117
  - Casework 119
- Schedules and IFC 119
- Not yet in BIMopen 119

Ceilings and columns

- Ceilings and columns 120
  - Ceilings 120
    - Types 120
    - The ceiling grid 121
    - Properties 121
  - Ceiling plans 121
  - Columns and structural columns 121
    - How they are drawn 121
    - Types 122

Openings

- Openings 123
  - Openings in floors, roofs and ceilings 123
  - Shafts 123
  - Openings in walls 124

- Properties, schedules and files 124
- Not yet in BIMopen 125

Phases and demolition

- Phases and demolition 126
  - Phase statuses 126
  - Project phases 126
  - Phase filters 126
  - Graphic overrides 127
  - The phase of views 127
  - The phases of elements 127
  - Demolish 127
  - Infills 128
  - Rooms and schedules by phase 128
  - Phase legends 128
  - Exchange 129

Structure and foundations

- Structure and foundations 130
  - Structural framing types 130
  - Beam 130
    - Beam properties 130
    - How it is drawn 131
    - Joins 131
  - Brace 131
    - Plan symbols and Structural Settings 131
  - Beam layout 131
  - Truss 132

- Structural wall and structural floor 133
  - Shape editing 133
- Foundations 134
- Rebar cover 135
- Views, tags, schedules and IFC 136
- From the command line 136

Site plan and terrain

- Site plan and terrain 137
  - Terrain 137
    - Shape editing 137
    - Type, layers and contours 138
  - Editing the terrain 138
  - Building pad 138
  - Site elements and parking 138
  - Property lines 139
  - Contour labels 139
  - The site plan 139
  - Site settings 139
  - Not yet 139

Design options

- Design options 140
  - Option sets and options 140
  - Editing an option 140
  - Moving elements into options 141
  - Dedicated views 141
  - Rooms, joins and outputs 141

- Make primary, delete, accept 141

Views and graphics

- Views and graphics 142
  - 3D views 142
    - Section box, projection and orientation 142
    - Perspective views 143
  - Graphic styles 144
  - Realistic style 145
  - Pastel style 146
  - Live view and walk mode 148
    - Disable Collision 149
    - Walkthrough 149
  - Sun and shadows 150
    - Sun settings 151
    - Sun study 152
    - Graphic display options 152
    - Sketchy lines 156
  - Plans and their cut heights 158
  - Underlay 159
  - Light and dark background 159
  - Opening a view from its symbol 160
  - Scale, detail and crop region 160
  - Hiding and isolating 162
  - Visibility and Graphics 163
  - Filters 163
  - View templates 164
  - Managing views 165

Annotation and 2D details

- Annotation and 2D details 167
  - Dimensions 167
    - Aligned and linear dimensions 167
    - Radial, diameter and arc length dimensions 167
    - Changing a dimension 168
    - Temporary dimensions 169
    - Constraints 170
    - Spot dimensions 171
    - Arrows, fonts and alternate units 172
  - Text 173
    - Text format and leaders 174
  - Symbols 175
  - Tags 175
    - Loaded tags and numbering 177
    - Multi-category tags 178
    - Material tags 178
    - Tags with several leaders 178
    - Several values 178
    - Keynotes 179
  - Rooms 179
  - Areas 180
  - Colour schemes 180
  - Legend views 181
  - Detail lines and regions 183
    - The tangency padlock 185
  - Detail components, repeating details and insulation 185
  - Draw order 186

- Free 2D views 186
- At the command line 187

Sheets, schedules and printing

- Sheets, schedules and printing 188
  - Sheets 188
    - Creating a sheet 188
    - Title blocks 189
    - Views on sheets 189
      - Text, lines and dimensions on a sheet 190
    - Guide grids 190
    - Sheet list 190
    - Revisions 191
  - Schedules 191
    - Creating a schedule 191
    - Schedule Properties 191
    - Working in a schedule view 192
    - Schedules on sheets 192
    - Graphic schedule by type 192
    - Key schedules and material takeoffs 193
    - Project and shared parameters 193
    - Exporting and reusing schedules 193
  - Printing 194
    - The Print dialog 194
    - View/sheet sets 195
    - Print Setup 195
    - Print Preview 196
  - Exporting images 196

Command line and MCP

- Command line and MCP 198
  - The command line in the window 198
    - Typed shortcuts start the tools 198
    - Points and lengths 198
    - Arguments in JSON 198
    - Commands you will use often 199
    - Clipboard, types and groups 199
  - bimopen-cli 200
  - MCP: BIMopen for AI agents 201

Keyboard shortcuts

- Keyboard shortcuts 202
  - Keys 202
  - Mouse 202
  - Typed shortcuts 203
    - Modelling 203
    - Modifying 204
    - Annotating and detailing 204
    - Views and settings 205

Troubleshooting

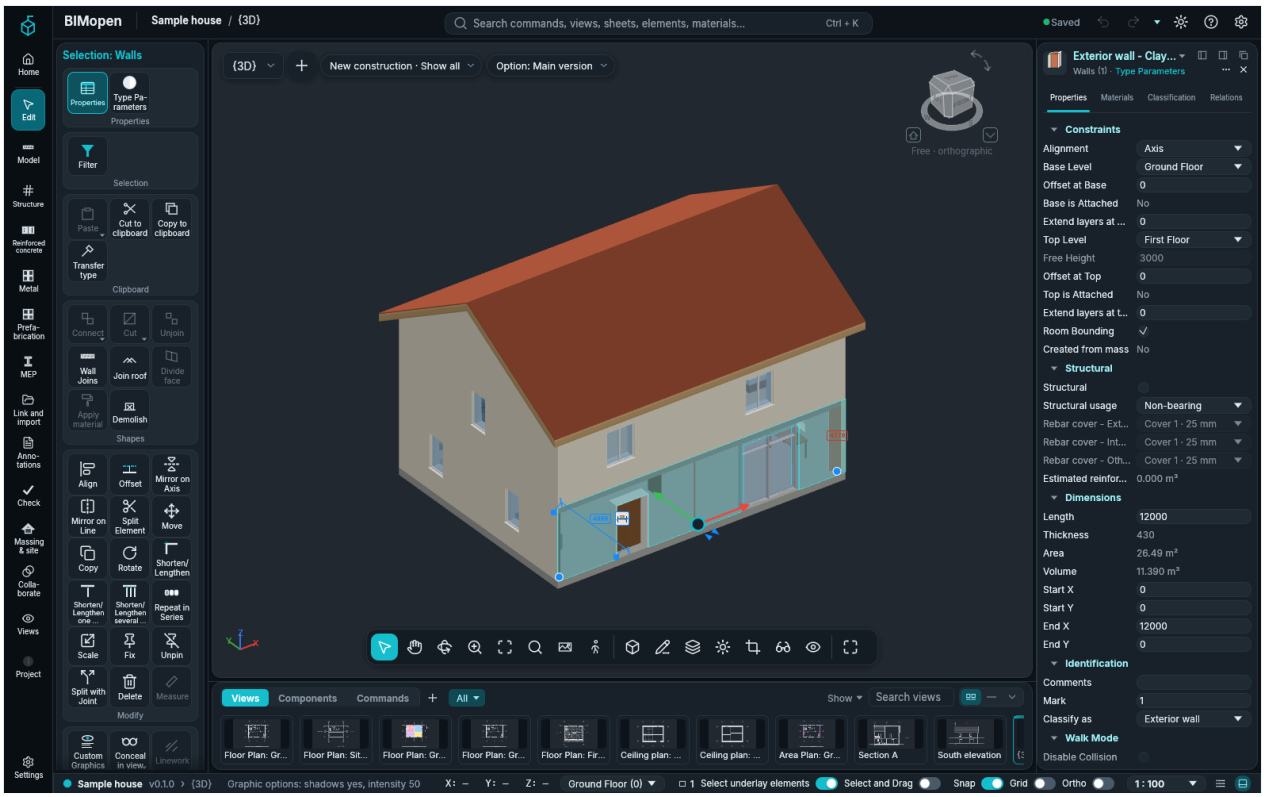
- Troubleshooting 206
  - Starting BIMopen 206
    - The web app shows «BIMopen non si è avviato» and an error 206
    - The desktop app does not build on Linux 206

- The interface 206
  - The element panel or the Project Index has disappeared 206
  - What I type ends up at the command line 206
  - A tile is greyed 206
  - DL starts a dimension instead of a detail line 206
- Modelling 206
  - «no wall at that point» 206
  - A height set at the command line is enormous 206
- Files 206
  - Unsaved work is gone after New, Open or Sample project 206
  - Open… in the browser does not show a file dialog 207
  - bimopen-cli prints its usage and stops 207

BIMopen user manual

BIMopen is an open-source application for building information modelling (BIM). You model the building with walls, doors, windows, floors, roofs, stairs and rooms, and the drawings come from the model: plans, sections, elevations, 3D views, schedules and sheets are views of the same model and change together when you edit it.

BIMopen runs on the desktop (Linux, Windows, macOS) and in a web browser, and it can be driven from a command line and by AI agents over MCP. Its interface speaks six languages: Italian, English, German, French, Spanish and Arabic, the last one written from right to left.



The BIMopen window with the sample house in the 3D view

How this manual is organised

Chapter	What it covers
Getting started	Installing and starting BIMopen, the sample project, and a first small house from an empty project to a sheet
The interface	The BIMopen style: the rail and the tool panel, the top bar, the command search, the options strip, the element panel, the Project Index, the views strip, the drawing area, the status bar and how to move around in the views
Interface styles	The Classic style, and touch mode on touch monitors, tablets and phones
Projects and files	Opening, saving, exporting and importing projects
Selecting and modifying	Selecting elements and the Modify tools
Walls, doors and windows	The building envelope and its openings
Floors, roofs, stairs and components	Floors, roofs, stairs, railings, curtain walls and placed components
Views and graphics	Plans, sections, elevations, 3D views and how they are drawn
Annotation and 2D details	Dimensions, text, tags, rooms, areas and 2D details
Sheets, schedules and printing	Sheets, schedules, printing and project settings

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Chapter	What it covers
Command line and MCP	Typing commands, `bimopen-cli` and the MCP server for agents
Keyboard shortcuts	Every key and typed shortcut
Troubleshooting	What to do when something does not work as expected

The chapters from *Projects and files* to *Sheets, schedules and printing* are being written area by area: each one is completed when the features it describes have been checked one by one through the interface. Until then they list what they will contain.

Conventions

- The manual describes and pictures the **BIMopen** interface style, the one BIMopen starts in. Interface styles shows the **Classic** style and touch mode, which run the same commands.
- Names of the interface are written in **bold**, as they appear on the screen in English: **Create** on the rail, **Wall** tile, **Properties** tab.
- A path through the interface is written with arrows: **Create** ▸ **Build** ▸ **Wall** means **Create** on the rail, then the **Build** group of the tool panel, then its **Wall** tile.
- Keys are written like `Ctrl+Z`; a shortcut typed at the keyboard, letter by letter, is written in capitals, like `WA`.
- Commands typed at the command line, and their output, are written `like this`. Points are `x,y` in metres.
- Notes, tips and warnings are set apart:

> **Note**
>
> A note adds something worth knowing.

> **Tip**
>
> A tip shows a quicker or better way.

> **Warning**
>
> A warning tells you about something that can lose work or give a wrong result.

What this manual describes

This manual describes BIMopen as it is today, and only what has been tried in the program. When a feature is not described yet, it may still exist: the roadmap *(https://bimopen.it/roadmap.html)* lists what is available and what is planned. The whole manual is also a PDF, one per language: the last entry of the navigation, **PDF**, downloads it.

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Getting started

This chapter shows how to start BIMopen on the desktop and in the browser, what the sample project contains, and how to model a first small house: four walls, a door, a window, a floor, a roof, a 3D view and a sheet with the plan on it. It takes about ten minutes.

Running BIMopen on the desktop

BIMopen is built from its source code with Rust, version 1.90 or later. On Linux the graphical interface also needs the development files of xkbcommon, Wayland and GTK 3.

```
sudo apt-get install -y libxkbcommon-dev libwayland-dev libgtk-3-dev   # Debian, Ubuntu
cargo run --release -p bimopen -- --sample                             # opens the sample house
```

The program is `bimopen`. It accepts a file to open and one option:

Command	What it opens
`bimopen`	A new, empty project
`bimopen --sample`	The sample house
`bimopen house.bpr`	The project saved in `house.bpr`
`bimopen --no-splash`	As `bimopen`, without the start screen

You can also open a project by dragging a `.bpr` file from the file manager into the window.

Start screen

While BIMopen loads, the start screen appears: a window without a frame, centred on the screen, with the name, the version, the loading stage under way with its percentage and a turquoise line growing along the bottom edge. The stages are the real ones (preferences, fonts, graphics card, families, opening the project); the screen stays at least half a second, so it does not flash, and fades out when the main window is ready. It speaks the language chosen in the app, otherwise the system's when it is one of the six, otherwise English.

To leave it out, clear **Options ▸ General ▸ Show the start screen** (BIMopen remembers it), or start `bimopen --no-splash`. On Wayland the compositor decides where the window appears (usually in the middle).

Language and theme

The first time, BIMopen speaks the language of the operating system when it is one of its six languages (Italian, English, German, French, Spanish, Arabic), and English otherwise. To change it, click the gear at the right end of the top bar (**Options**) and choose the **Language** under **User Interface**. The interface changes at once.

In **Options** you also choose the **Theme**, **Light** or **Dark** (the sun or the moon in the top bar switches it too), and whether the background of the views is inverted (**Invert background color**: dark views with light lines). Choosing the dark theme also inverts the background; you can turn it back off. At the first start BIMopen uses the dark theme, with the background inverted. It remembers the language, the theme and the background for the next time.

Running BIMopen in the browser

The web version is the same program, compiled for the browser. Open bimopen.it/app/ *(https://bimopen.it/app/)*: it starts with the sample house. Nothing needs to be installed, and the project stays in your browser: it is not sent anywhere.

The address accepts these options, which can be combined with `&`:

Address	Effect
`app/?lang=de`	Opens in German for this visit (`it`, `en`, `de`, `fr`, `es`, `ar`)
`app/?empty`	Starts with a new, empty project instead of the sample
`app/?webgl`	Draws with WebGL2 even where WebGPU is available

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The browser must support WebGPU or WebGL2. In the browser some things work differently:

- **Open…** cannot show a file dialog: drag a `.bpr` file into the page to open it;
- **Save** and the exports download the file, with the name of the project.

To build the web version yourself:

```
cargo xtask web                          # builds dist/web
python3 -m http.server 8080 -d dist/web  # then open http://localhost:8080
```

`cargo xtask web` needs the `wasm32-unknown-unknown` target of Rust and the `wasm-bindgen` command of the version written in `Cargo.lock`; with binaryen's `wasm-opt` installed, the result is smaller.

The sample project

The sample house shows most of what BIMopen can do. Open it with **File ▸ Sample project** (the File panel opens from BIMopen's mark, at the top of the rail), with `bimopen --sample`, or by typing `sample` at the command line.

It contains:

- two levels, each with its floor plan, and five grid lines;
- fourteen walls of four types, with layers, joined at the corners;
- six doors and twelve windows, tagged in the plans;
- two floors, a pitched roof, a stair with its railings and a storefront curtain wall;
- furniture and plumbing fixtures, eight rooms coloured by use, with a legend;
- a section, an elevation, an area plan, a saved 3D view and a free 2D view with a wall detail;
- dimensions, text notes and the schedules of walls, doors, windows, rooms and materials;
- its place in **Cagliari**, where its address is: the sun and the shadows are those of that city.

> Tip
>
> The sample is a good place to try a tool before using it on your own project: a new sample is always one click away, and nothing you change in it is saved unless you save it.

Your first house

The steps below start from an empty project and end with a sheet. Every step names what to click in the **BIMopen** interface style, the one BIMopen starts in; the interface chapter explains each part of the window.

1. Start a new project

1. Click BIMopen's mark at the top of the rail: the **File** panel opens.
2. Click **New…** in the **Projects** group. The **New project** dialog proposes BIMopen's default template: click **OK**.

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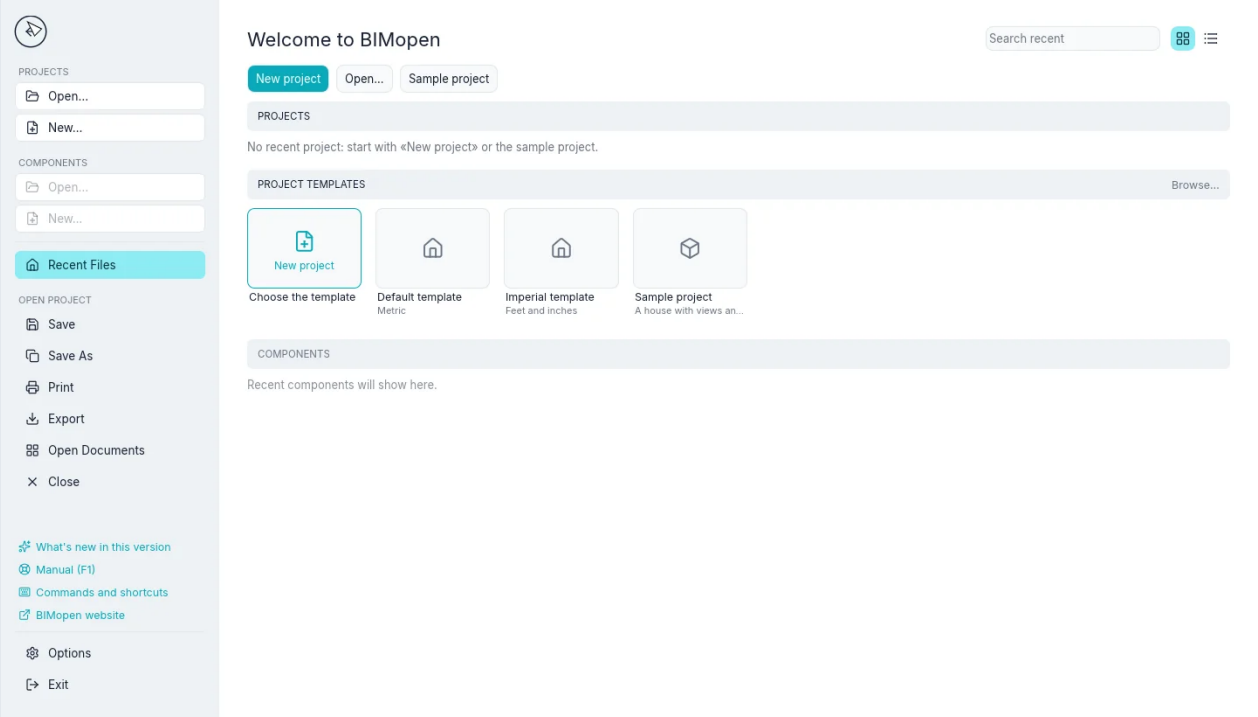

Step 1: the File panel

The new project has two levels, **Ground Floor** and **First Floor**, a floor plan for each, and the default 3D view. It has no walls yet. Its wall, door, window, floor and roof types are ready to use.

2. Open the ground floor plan

In the views strip under the drawing area, click the thumbnail **Floor Plan: Ground Floor**. The plan opens in the drawing area, and the top bar and the view selector show **Floor Plan: Ground Floor**.

> Note
>
> **Views** on the rail also shows the **Project Index**, the tree of every view: there, a single click on a view shows its properties in the element panel without opening it, and a double-click opens it.

3. Draw four walls

1. In the tools of **Create**, the rail entry BIMopen opens with, click **Wall** in the **Build** group (if the tool panel is folded, click **Create** on the rail first). The contextual card **Draw Wall** comes on top of the tool panel, the options strip over the view shows the settings of the wall, and the element panel shows the type of the next wall, **Exterior wall - Clay block 300 + ETICS**.
2. In the status bar, turn on **Ortho**: the walls stay horizontal or vertical.
3. Click in the plan where the first corner goes. The status bar asks for the next point: `WALL next point [Esc to finish]:`.

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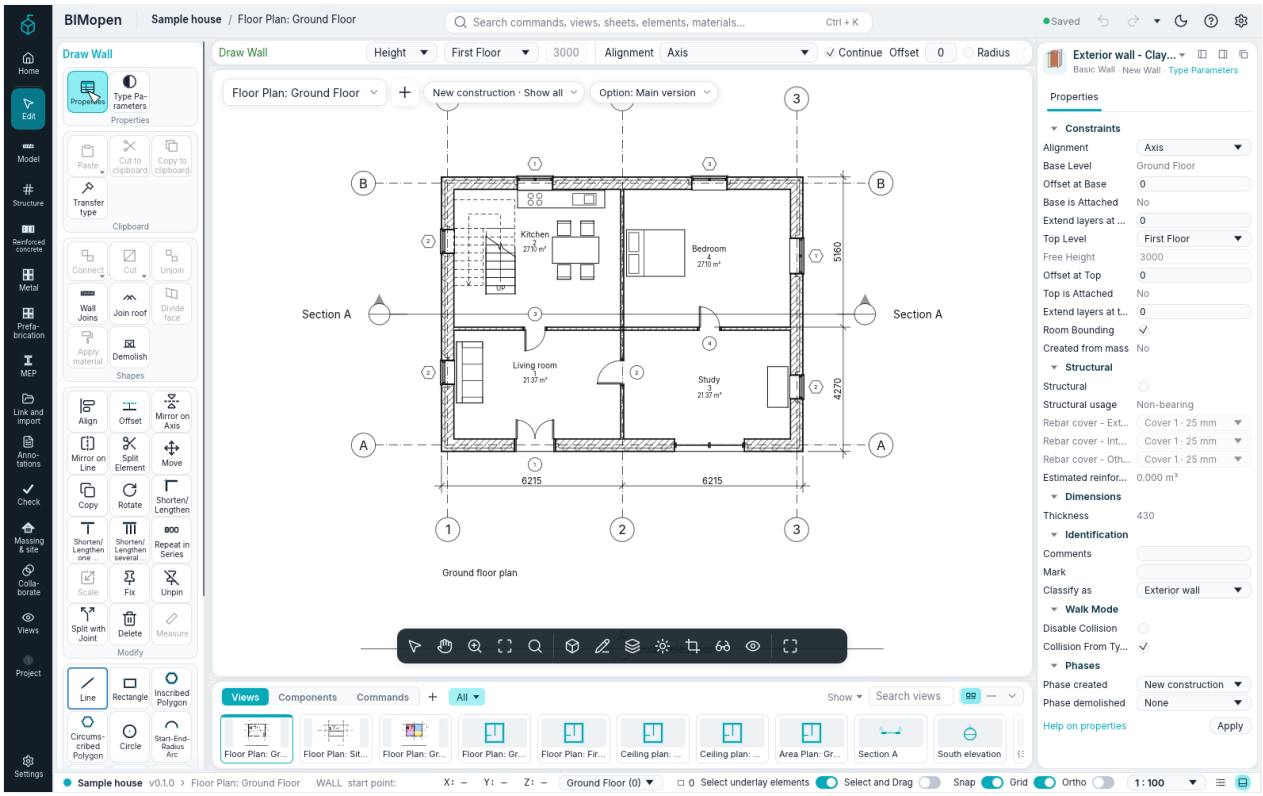

1. Click the second, third and fourth corners. While you move the mouse, the length and the angle of the wall being drawn are shown next to it.
2. Click the first corner again: the four walls are joined at the corners.
3. Press `Esc` to stop the chain, and `Esc` again to end the tool.

> **Tip**
>
> You can draw the same walls by typing their corners in metres: `wall 0,0 10,0 10,6 0,6 c` . The final `c` closes the chain. See Command line and MCP.

4. Add a door and a window

1. Click **Create ▸ Build ▸ Door**, then click on a wall where the door goes. The door is placed in the wall, which is cut around it. The command log confirms it: `Door «Single swing door 80x210» in wall …` .
2. Press `Esc`: the tool ends and the contextual card goes away.
3. Click **Create ▸ Build ▸ Window**, click on another wall, and press `Esc`.

A door or a window can only be placed in a wall: after a click away from the walls, the command line answers `no wall at that point` and nothing is placed.

5. Add a floor and a roof

1. Click **Create ▸ Build ▸ Flooring**. Sketch mode opens: the contextual card **Edit | Create Floor Boundary**, with **Pick Walls** active.
2. Move the cursor over a wall and press `Tab`: the four walls are highlighted. Click: their lines go on the inner faces of the walls.
3. Click **Finish Edit Mode**: the floor is made and the command log gives its area.
4. Click **Create ▸ Build ▸ Roof**. **Pick Walls** is active again; the options strip has **Defines Slope** (ticked) and the **Overhang** (500 mm by default).
5. Press `Tab` over a wall, click, then **Finish Edit Mode**: a hip roof sloping at 30° covers the walls, with the overhang all around.

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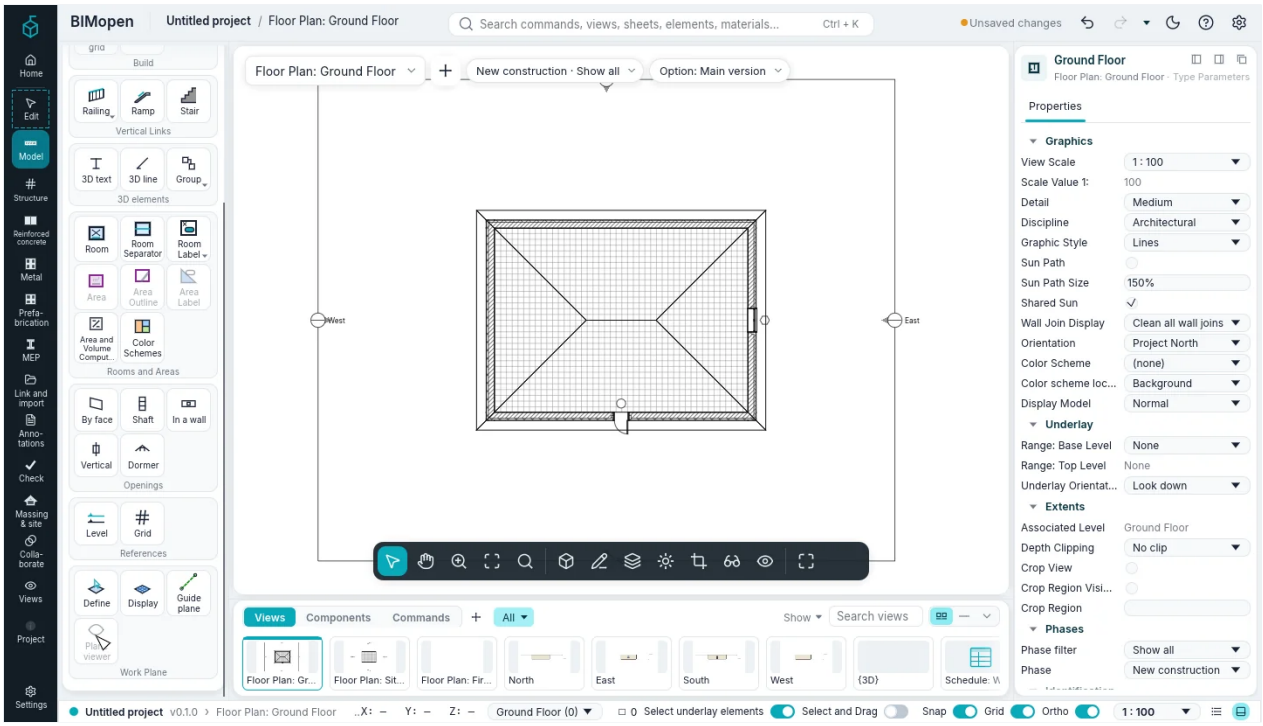
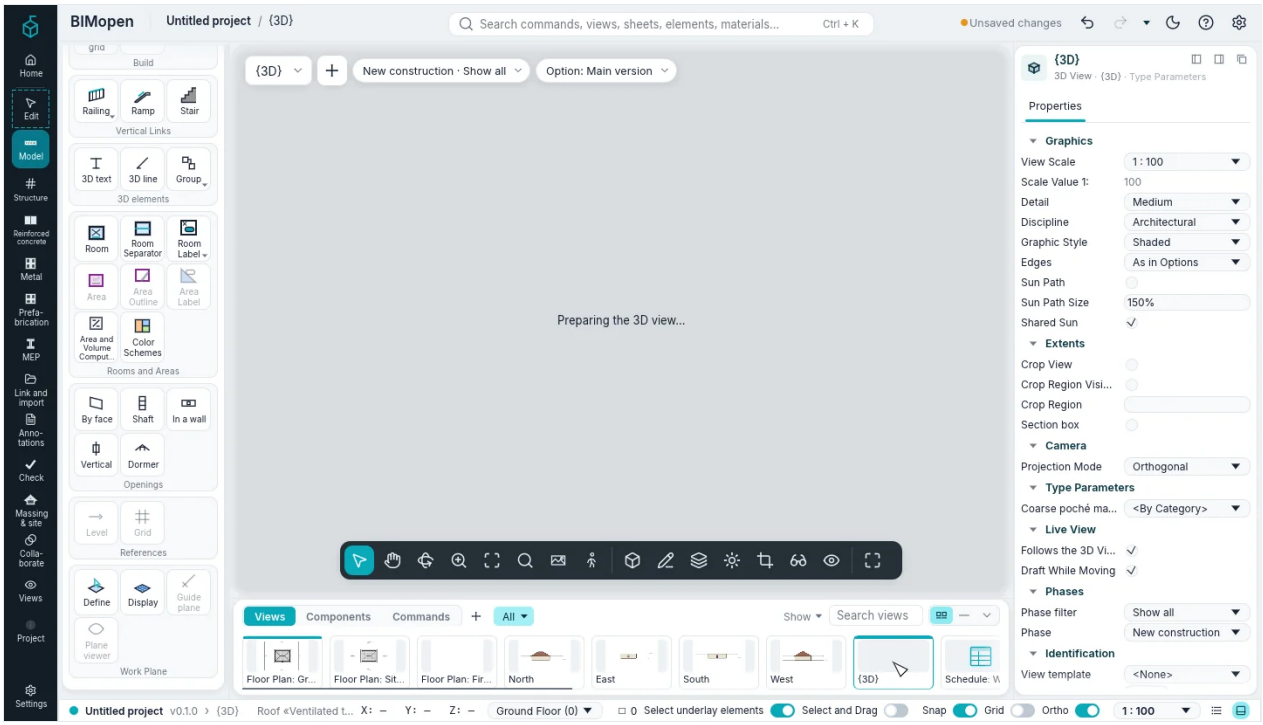


Step 5: walls, door, window, floor and roof

Floors, roofs, stairs and components tells more about sketch mode.

6. Look at it in 3D

Click the **{3D}** thumbnail in the views strip. The default 3D view, **{3D}**, opens. Drag with the right mouse button (or with `Shift` and the middle button) to orbit, drag with the middle button to pan, and turn the wheel to zoom; the left button selects, as in plans. `Home` fits the whole model in the view.



Step 6: the house in 3D

7. Put the plan on a sheet

1. Click **Document** on the rail, then **Sheet** in the **Sheets** group. The **New Sheet** dialog lists the title blocks: none, A0, A1, A2 or A3, with their size in millimetres.
2. Click **OK**. The sheet is created with the next free number (**A101** for the first one, then **A102**...) and opens.

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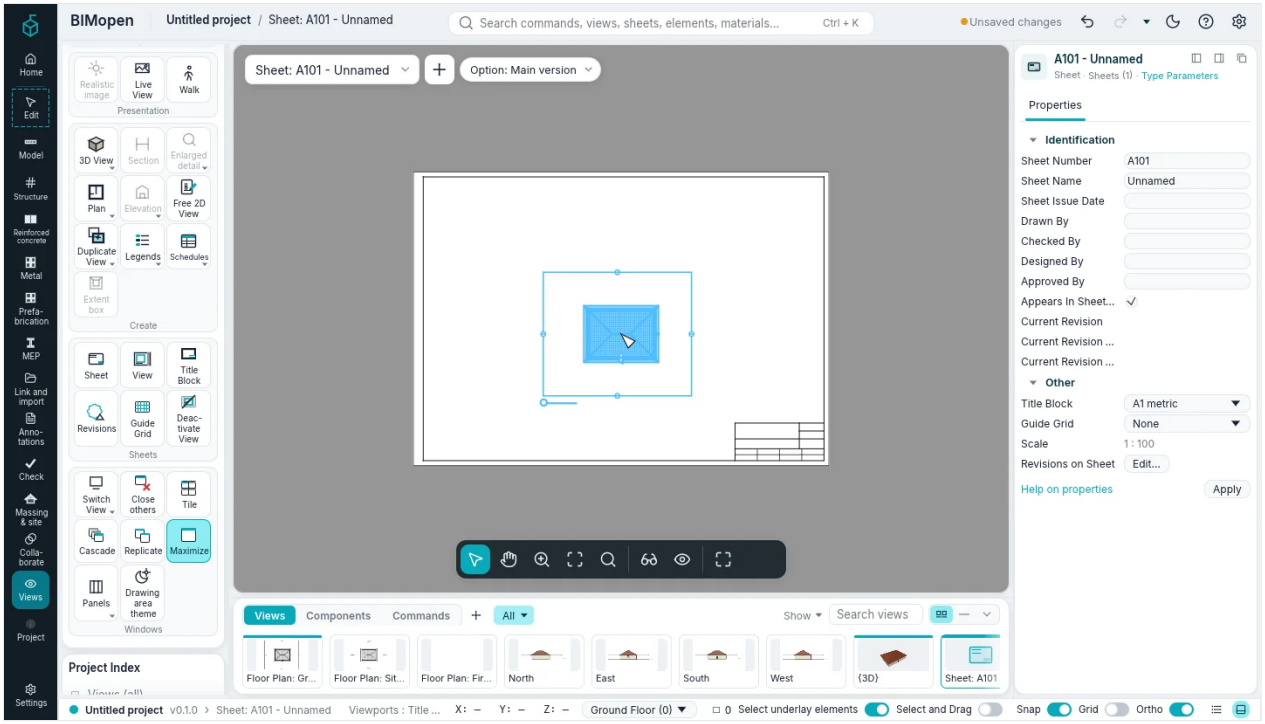
3. Click **Document ▸ Sheets ▸ View**. The **Views** dialog lists the views that are not on a sheet yet.
4. Choose **Floor Plan: Ground Floor** and click **Add View to Sheet**. A rectangle of the plan's size follows the cursor: click on the sheet where the plan goes. It is placed at its own scale, 1 : 100, as a viewport, which stays selected.



Step 7: the plan on a sheet

8. Undo, redo and save

- `Ctrl+Z` undoes the last change (here, the viewport goes away); `Ctrl+Y` redoes it. The arrows of the top bar, **Undo** and **Redo**, do the same.
- `Ctrl+S` saves the project, as a click on the save state in the top bar. A project that has never been saved asks for a file name first; the file has the extension `.bpr`. In the browser, the file is downloaded.

The top bar says **Unsaved changes** while there are changes to save, and **Saved** after saving.

What next

- The interface describes every part of the window.
- Keyboard shortcuts lists the keys and the two-letter shortcuts, such as `WA` for a wall.
- Command line and MCP shows how to type commands, run scripts and let an agent model for you.

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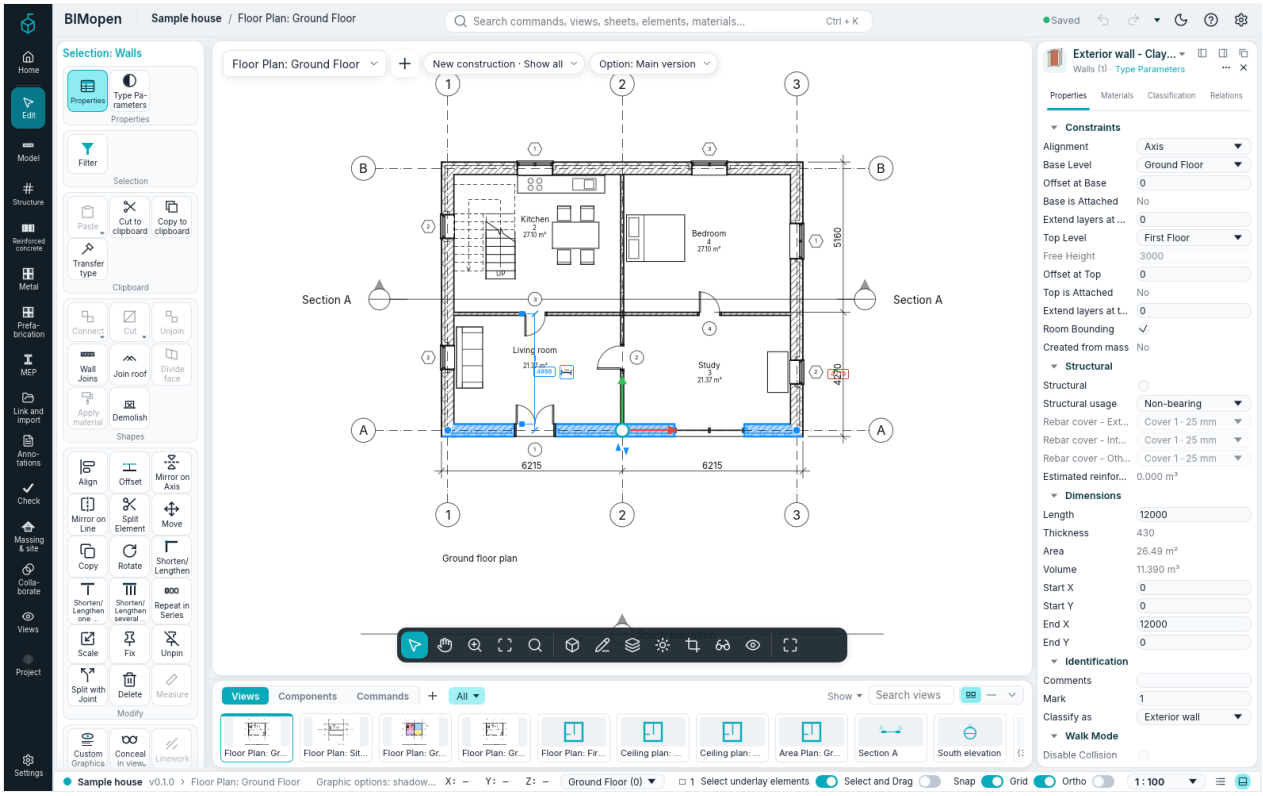
The interface

This chapter describes the **BIMopen** interface style, the one BIMopen starts in, on the desktop and in the browser. The other looks, the **Classic** style and touch mode for tablets and phones, run the same commands with the same names; Interface styles shows them and tells how to switch.



The BIMopen window with a wall selected in the ground floor plan

The window has these parts:

1. the **rail**, on the left edge: BIMopen's mark, which opens the File panel, then **Home**, the contextual tab **Edit**, the tabs **Model**, **Structure**, **Reinforced concrete**, **Metal**, **Prefabrication**, **MEP**, **Link and import**, **Annotations**, **Check**, **Massing & site**, **Collaborate**, **Views** and **Project**, and at the bottom **Settings**;
2. the **tool panel**, beside the rail: the groups of the tab chosen on the rail, as framed blocks of tiles;
3. the **top bar**: the project and the active view, the command search, which is also the command line, the save state, **Undo** and **Redo**, the quick access tools, the light or dark theme, the **?** menu and **Options**;
4. the **drawing area**, in the middle: the views, each with its view selector at the top and the view bar at the bottom; while a tool works, the **options strip** runs over it;
5. the **element panel**, on the right: the properties of the selection, or of the active view;
6. the **views and components strip**, under the drawing area, which you can put away (F9);
7. the **status bar**, along the bottom.

The panels keep their width whatever they show: a long name is cut with «...» and its tooltip shows it whole. Drag the edge of the tool panel or of the element panel to make it wider or narrower; BIMopen remembers it for the next session. A click on the tab already chosen on the rail folds the tool panel, and a second click opens it again. Without touch the interface is compact, to leave the drawing area as much room as possible; in touch mode the buttons stay big (see Interface styles).

In Arabic the whole window is mirrored: the rail and the tool panel are on the right, the element panel on the left, and the strips read from right to left.

File panel

BIMopen's mark, at the top of the rail, opens the **File** panel: a full-window page with the file commands and the recent projects. In the **Classic** style **File**, left of the tabs, opens it; in touch mode **Menu ▸ File**. **Back** (the round button at the top) or Esc goes back to the model exactly as it was.

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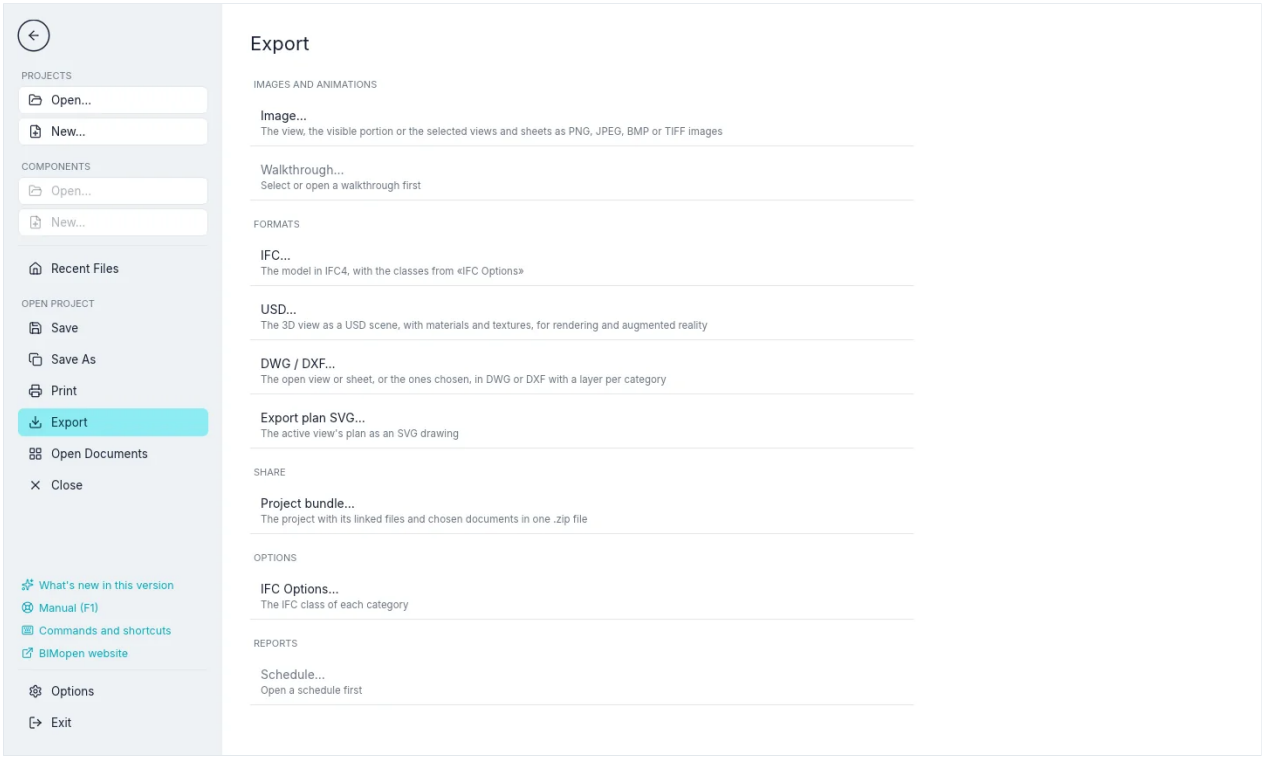
On the left:

Item	What it does
Projects ▸ Open...	A page with **Project...** (`Ctrl+O`), **IFC...** and **IFC Options...**
Projects ▸ New...	Opens **New project**, with the template file to start from
Components ▸ Open..., New...	Greyed out: components are not available yet
Recent Files	The panel's start page, described below
Save `Ctrl+S`	Saves the project in its file; a project never saved asks for a name
Save As	A page with **Project...**, **Template...** and **Options...** (the save options)
Print	A page with **Print...** (`Ctrl+P`), **Print Preview** and **Print Setup...**
Export	A page with **Image...**, **IFC...**, **Export plan SVG...**, **IFC Options...** and **Schedule...**
Open Documents	The open view windows; a click brings the chosen one to the front
Close	Closes the project and goes back to the start page
What's new in this version, **Manual (F1)**, **Commands and shortcuts**, **BIMopen website**	The links, at the bottom
Options	Opens the **Options** dialog
Exit	Closes BIMopen (desktop only)

The commands of the open project (from **Save** to **Close**) are there only while a project is open. Each page gives every item a line of explanation.



The File panel, Export page

Recent Files

The **Recent Files** page has three bands:

- **Projects**: the recent projects as square thumbnails, with the name and the date modified; a label tells which view the thumbnail comes from (**3D**, **Plan**, **Sheet**...). On hover come the pin, which keeps the project at the top of the list, the ⋯

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menu (**Open**, **Open as copy**, **Open folder**, **Copy path**, **Pin**, **Remove from list**; also with the right button) and the ×, which takes the project off the list: for a few seconds **Undo** puts it back where it was. The file on disk is never touched. A moved or deleted file shows grey, with **File not found**, **Locate…** to point to its new place and **Remove**; **Remove missing** takes them all off, **Clear list** takes off those not pinned.

- **Project templates**: **New project**, the default template, the imperial template, the templates found in the files folder, the sample project and **Browse…** for another template file.
- **Components**: the recent components (`.bco`, `.bct`), with their thumbnail and the same commands as projects. In the web version, files you open are kept in the browser (up to 1.5 MB each) and open again from here.

At the top, the search filters by name and folder, and two buttons choose between **Thumbnails** and **List**; the list also shows the folder, the date and the size, and sorts by name or by date. The choice is remembered.

Every save stores a thumbnail in the file: the default 3D view, or the view chosen in **File Save Options ▸ Preview**. A file that cannot be read shows a neutral icon.

At startup without a file the panel is the start page, without **Back**; at the first start, with the list empty, it offers **New project**, **Open…** and **Sample project**. In touch mode the thumbnails are bigger, the actions always show and a long press opens the ⋯ menu. In the browser there is no **Open folder**.

New…, **Open…**, a recent project and **Sample project** take the place of the open project; if it has unsaved changes, they first ask whether to save them.

Options

The gear at the right end of the top bar, **Settings** at the bottom of the rail, or **File ▸ Options**, opens **Options**. It has, among others, these settings:

Setting	Values	Effect
User Interface ▸ Language	Italiano, English, Deutsch, Français, Español, العربية	The language of the whole interface, at once
User Interface ▸ Theme	**Light**, **Dark**	The colours of the window. **Dark** also turns on **Invert background color**. The sun or moon in the top bar switches it too
User Interface ▸ Interface	**BIMopen**, **Classic**	The interface style, see Interface styles
User Interface ▸ Touch mode	**Automatic**, **Always**, **Never**	The layout for touch screens, see Interface styles
Graphics ▸ Invert background color	on, off	Views with a dark background and light lines
Orientation cube ▸ Orientation cube size, **Inactive opacity**	**Small**, **Medium**, **Large**; 0 % to 100 %	The size of the cube, and how much it shows while the pointer is elsewhere
Orientation cube ▸ Keep scene upright	on, off	Dragging the cube never turns the model upside down; off, the view can pass over the top
Orientation cube ▸ Fit to view	on, off	A click on the cube also fits the model in the window

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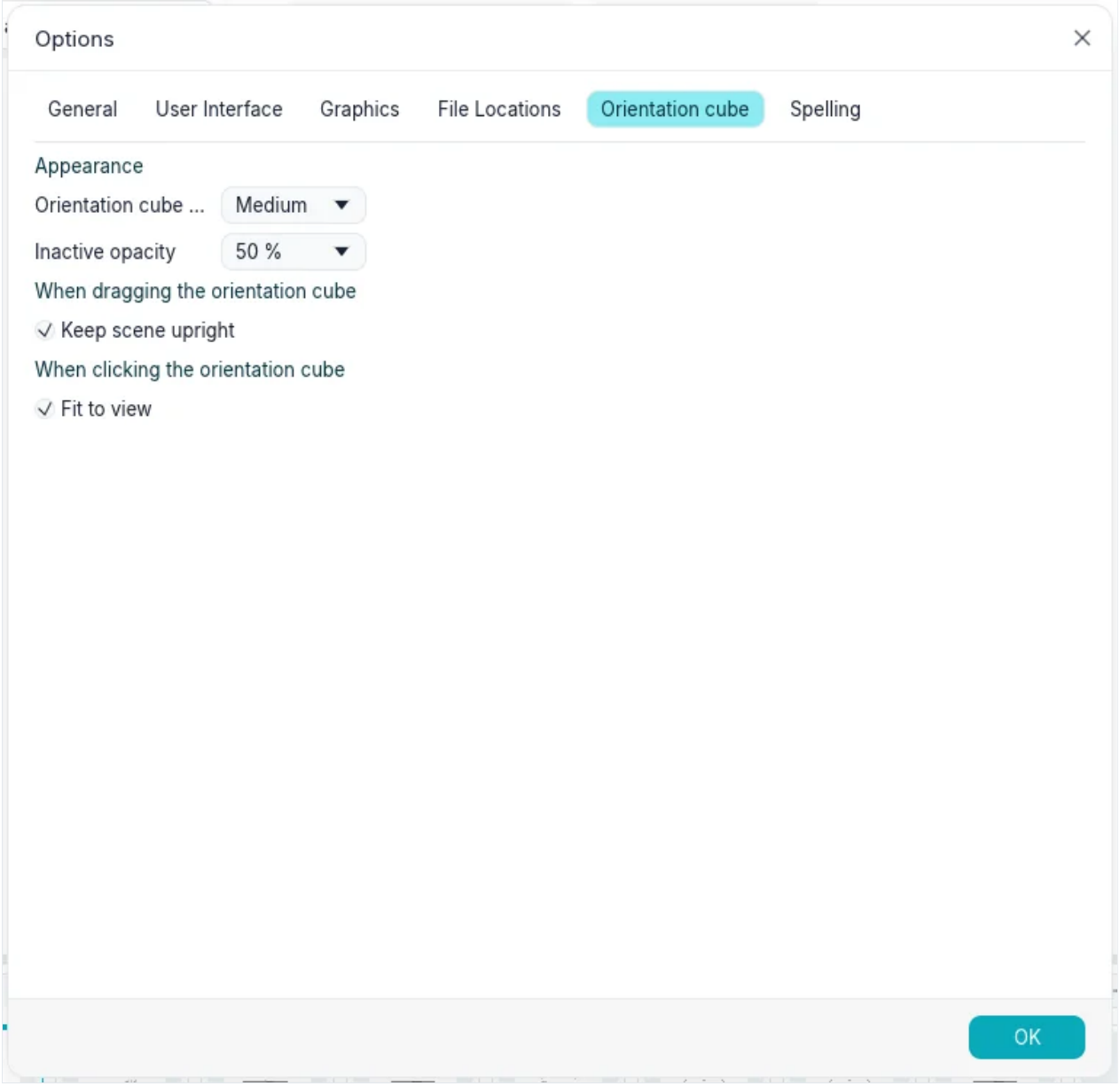

Options, Orientation cube tab

BIMopen remembers the language, the theme and the style the next time it starts.

Top bar and quick access tools

The top bar shows, from the left, **BIMopen**, the name of the project (a click opens **Project information**) and the active view; in the middle the command search; on the right:

The top bar

Control	What it does
Saved / **Unsaved changes**	The save state; a click saves, `Ctrl+S`
Undo	Undoes the last change, `Ctrl+Z`. Greyed when there is nothing to undo
Redo	Redoes the change just undone, `Ctrl+Y`. Greyed when there is nothing to redo
The quick access tools	The tools you added, then the arrow of the quick access toolbar
The sun or the moon	Switches to the light or to the dark theme
?	**Help** F1, **Commands** (every command, in a window: The Commands window) and **About** (About BIMopen)

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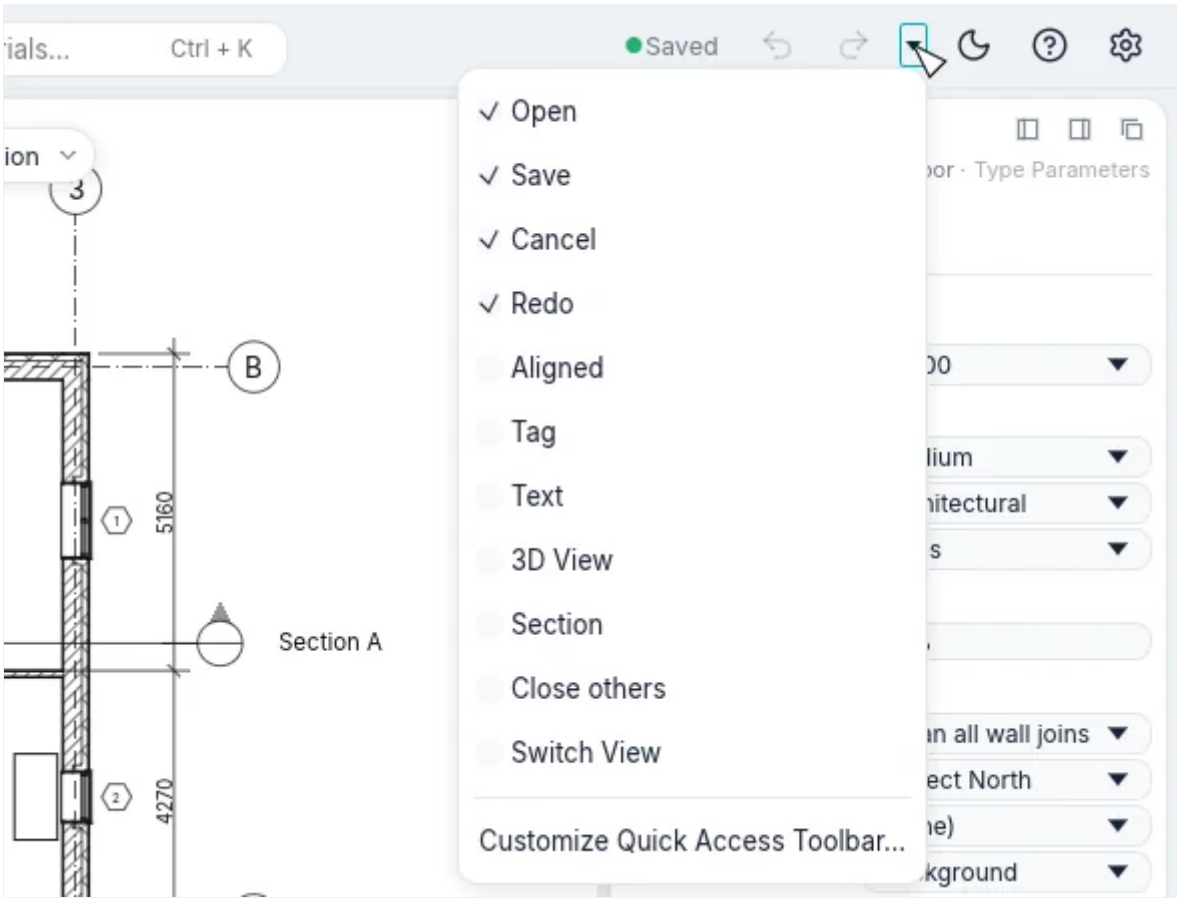
Control	What it does
The gear	**Options**

Right-click a tile of the tool panel and choose **Add to Quick Access Toolbar**: the tool shows in the top bar and starts from there whatever rail entry is open. Right-click it there for **Remove from Quick Access Toolbar** or **Add Separator** (a line to its right). The small arrow after the tools, **Customize Quick Access Toolbar**, lists the default tools (**Open…**, **Save**, **Aligned dimension**, **Tag by Category**, **Text**, **3D View**, **Section**, **Close Hidden**, **Switch View**): tick one to show it, untick it to remove it. In the same menu, **Customize Quick Access Toolbar…** opens a window with the tools in order: **Move Up** and **Move Down** move the one selected, **Add Separator** puts a line after it, **Remove** takes it (or a line) away, **OK** applies. The tools are kept for the next sessions.



Customize Quick Access Toolbar

? ▸ **Help** (F1) opens this manual, in the language of the interface, in the browser. F1 opens the page of what you are doing: the open dialog, the tile under the mouse (its tip ends with **Press F1 for more information**) or the active tool.

About BIMopen

? ▸ **About** shows the program's mark and name, the version with the date and the commit it was built from (read when the program is built: a build in development says so), and the platform (the system and whether desktop or web). The note under it recalls that this is a test version. The rows open in the browser the **Website**, the **User manual** and **What's new** (the news of every version); **Third-party licences** opens in place the list of the works the program contains (typefaces, icons, the textures of the materials, the sky, the Moon and the stars), with their licences; the full texts are in the NOTICE file. **Copy system information** puts in the clipboard the version, the system, the graphics card and the backend in use: paste it when you report a problem.

The Commands window

? ▸ **Commands**, or `help` typed at the command line, opens the list of every command. On the left the ribbon's tabs, in their order, then **Modify** and **Command line only**; in the middle, for each command, its name in the language of the interface, the name to type, its synonyms and its shortcut; on the right the description, the usage and an example of the command chosen. The search finds a command by any of these, in every tab (`help door` opens the window with the door already found). A command not available yet is greyed; **Available only** hides them. A double-click, Enter or **Start** starts the command as its button does; a command of the command line alone is written in the command line, to complete. **Copy** puts the command

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with its example in the clipboard. The arrows go through the list, Esc closes it. In touch mode the window opens from **Commands ▸ List of commands** (the command field with nothing typed), with rows the size of a finger.

Updates

At startup, at most once a day, BIMopen reads from the website the file of the latest version (nothing else is sent). When a newer version is out, the **bell** in the top bar gets an orange dot (on the right of the title bar in the Classic style, in the top bar in touch mode). A click opens its panel: the version and its date, its main news, **All the news**, and **Download and update**, **Later** (no dot until the next start) and **Skip this version** (no dot until a newer one).

Download and update downloads the program with a progress bar (**Continue in the background** hides the panel, **Cancel** stops). Before anything is replaced, BIMopen checks the file: its SHA-256 checksum must match the version's, and the version's file must carry BIMopen's digital signature. A package that is not signed, signed with another key or altered is discarded and nothing is installed. Then **Restart and update** asks whether to save the changes, puts the new program in place of the old one and starts it on the same project and view.

- **Windows**: the running program is renamed and replaced at the restart. Until the executable is signed, Windows may warn at the first start: **More info ▸ Run anyway**.
- **Android**: **Download and install** downloads the package and the system installer asks to confirm (the first time it asks for the **Install unknown apps** permission).
- **Web**: **Reload** loads the new version.

Without a connection, with an invalid signature or without enough disk space the panel says so and BIMopen keeps working as before. **Options ▸ General ▸ Check for updates** turns the check off; **Check now** checks at once.

Rail and tool panel

The rail lists the tabs of the ribbon of the **Classic** style, with the same names and in the same order. Click a tab: the tool panel shows its groups as framed blocks, each with its name at the bottom, and in each block its commands in the ribbon's order, three to a row, as tiles with an icon and a name. A long name goes on up to three lines, broken between words or at a syllable with a hyphen. A tile with a small ▾ mark in its corner has a menu: a click on the tile starts the command, a click on the mark opens the menu with its variants. Resting the mouse on a tile shows what it does and, in brackets, the shortcut you can type to start it. The commands BIMopen does not run yet are greyed, and their tip says «Not available yet»; **Level**, **Grid** and **Guide plane** are greyed in a 3D view («Not available in the 3D view»). When a tab has more commands than the panel shows at once (**Model**, **Structure**), the panel scrolls. **Settings**, at the bottom of the rail, opens **Options**.

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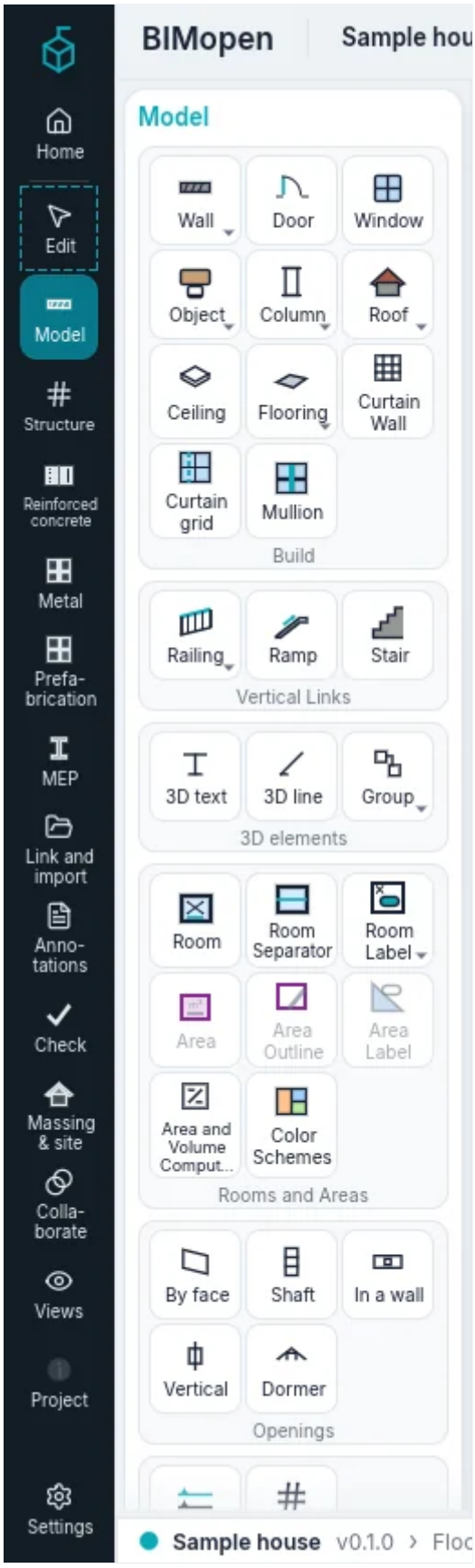

The rail and the Model tab

Tab	Group	Commands
Model	**Build**	**Wall** ▾, **Door**, **Window**, **Object** ▾, *Column* ▾, **Roof** ▾, *Ceiling*, **Flooring** ▾, **Curtain Wall**, **Curtain grid**, **Mullion**

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Tab	Group	Commands
	Vertical Links	**Railing** ▾, **Ramp**, **Stair**
	3D elements	*3D text*, *3D line*, *Group* ▾
	Rooms and Areas	**Room**, **Room Separator**, **Room Label** ▾, **Area** ▾, **Area Outline**, **Area Label** ▾
	Openings	*By face*, *Shaft*, *In a wall*, *Vertical opening*, *Dormer*
	References	**Level**, **Grid**
	Work Plane	**Define**, **Display**, **Guide plane**, *Plane viewer*
Structure	**Structural elements**	*Beam*, *Structural wall* ▾, *Structural column*, *Structural floor* ▾, *Truss*, *Brace*, *Beam layout*
	Connections	*Connection* ▾, *Automatic joints*
	Foundations	*Pad footing*, *Strip footing*, *Raft foundation* ▾
	3D elements	**Object** ▾, *3D text*, *3D line*, *Group* ▾
	Openings	*Normal to a face*, *Shaft*, *In a wall*, *Vertical opening*, *Dormer*
	References	**Level**, **Grid**
	Work Plane	**Define**, **Display**, **Guide plane**, *Plane viewer*
Reinforced concrete	**Reinforcement**	*Rebar*, *Reinforcement along a path*, *Adaptive rebar*, *Reinforcement over an area*, *Mesh over an area*, *Single mesh*, *Concrete cover*, *Coupler*
	Settings	*Cover options*, *Rebar options*
	Work Plane	**Define**, **Display**, **Guide plane**, *Plane viewer*
Metal	**Connections**	*Connection* ▾, *Automatic joints*
	Steelwork parts	*Plate*, *Bolts* ▾, *Weld*
	Fabrication	*Cope*, *Skewed cut*, *Shorten*, *Shaped cut*
	Automatic cuts	*End cut*, *Corner joint* ▾, *Intersection*, *Cut with*
	Work Plane	**Define**, **Display**, **Guide plane**, *Plane viewer*
Prefabrication	**Division**	*Divide*
	Joints	*Assembly parts*
	Reinforcement	*Rebar*, *Custom mesh*
	Production	*Workshop drawings*, *Export for CAM*
	Settings	*Prefabrication settings*, *Cold-formed profiles*
	Automatic update	*Suspend* ▾
MEP	**Air**	*Duct*, *Duct from a path*, *Duct fitting*, *Duct accessory*, *Duct insulation*, *Flexible duct*, *Air outlet*
	Prefabricated parts	*Prefabricated part*, *Multi-point route*
	System details	*Duct bracing*
	P&ID diagrams	*P&ID editor*
	Mechanical	*Mechanical equipment*, *Control device*
	Other systems	*Plumbing* ▾, *Electrical* ▾, *Modelling* ▾, **Work Plane** ▾
Link and import	**Link**	*Link project*, *Link IFC*, *Link CAD*, *Link topography*, *Decal* ▾, *Point cloud*, *Manage links*, *Link PDF*, *Link image*, *Federated model* ▾
	Import	*Import CAD*, *Import gbXML*, *Import PDF*, *Import image*
	Load from library	**Load component**, *Load as group*, **Insert from File** ▾
Annotations	**Dimension**	**Aligned**, **Linear**, **Angular**, **Radial**, **Diameter**, **Arc length**, **Spot dimension** ▾
	Detail	**Detail line**, **Filled region** ▾, **Detail symbol** ▾, **Revision Cloud**, *2D group* ▾

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Tab	Group	Commands
	Text	**Text**, *Spell checker*
	Tag	**Tag**, **Tag all**, *Material*, **Color legend**, *Beam tags*, *Multi-category tag*, *View reference*, *Tread numbers*, *Rebar tag* ▾, *Coded note* ▾, *Duct legend*, *Pipe legend*
	Symbol	**Symbol**, *Span direction*, *Rebar symbol*, *Stair path*
Check	**Energy**	*Energy analysis*
	Check	*Check clashes*, *Show loads*, *Disconnected connections*, *Electrical loads*, *Escape route*
	Analytical model	*Analytical member*, *Analytical panel*, *Analytical node*, *Supports and restraints*, *Automatic analytical model*, *Analysis results*
	Spaces and zones	*Space*, *Space separator*, *Space tag*, *Space names*, *Drawn zone*, *Zone from spaces*
Massing & site	**Massing**	*Display masses* ▾, *New mass*, *Place a mass*, *Draw in 3D*
	From mass faces	*Curtain Wall*, *Roof*, *Wall*, *Slab*, *Terrain from face*
	Site	*Terrain*, *Site element*, *Parking*, *Building pad*, *Surrounding context*
	Edit site	*Divide terrain*, *Merge terrains*, *Sub-area*, *Property line*, *Grading*, *Contour labels*
Collaborate	**Shared work**	*Worksets*, *Synchronize*, *Show history*, *Enable sharing*, *Active workset*, *Dim other worksets*, *Reload changes*, *Release elements*, *Permission requests*, *Sharing status*
	Archive	*Restore backup*, *Publishing options*
	Coordination	*Copy and monitor* ▾, *Coordination review* ▾, *Coordination options*, *Realign hosts*, *Federated model changes*
Views	**Graphics**	**View presets** ▾, **Visibility and Graphics**, **Filters**, **Thin lines**, *Show hidden lines*, *Remove hidden lines*, *Cut outline* ▾
	Presentation	*Realistic image*
	Create	**3D View** ▾, **Section**, **Enlarged detail** ▾, **Plan** ▾, **Elevation** ▾, **Free 2D View**, **Duplicate View** ▾, **Legends** ▾, **Schedules** ▾, *Extent box*
	Sheets	**Sheet**, **View**, **Title Block**, **Revisions**, **Guide Grid**, **Deactivate View**
	Windows	**Switch View** ▾, **Close others**, **Tile**, **Cascade**, **Replicate**, **Maximize**, **Panels** ▾
	Project Index	The tree of the project's views, schedules, sheets and components
Project	**Settings**	**Materials**, **Styles by Category**, **Snap**, **Project information**, **Project parameters**, **Project units**, **Copy Settings from Project**, *General parameters*
	Phases	*Phases*
	Options	*Design options*, *Main version*
	Numbering	*Renumber*
	Position	*Shared coordinates* ▾, *Move project* ▾
	Project management	*Project images*, *Opening view*
	Selections	*Save selection*, *Load selection*, *Edit selection*
	Inquiry	*Select by number*, *Selection numbers*, *Warnings*

In italics the commands BIMopen does not run yet; ▾ marks a menu.

Press and release `Alt`, or press `F10`, to show the key tips: a letter on each tab of the rail and, on the tiles of the open tab, their tool's shortcut. Type a tab's letter to open it, then a tile's letters to start its tool; `Esc` hides the tips. While they show, what you type does not go to the command line.

The Model tools

The tools you use most to model are in **Model**:

Group	Tiles
Build	**Wall** `WA` , **Door** `DR` , **Window** `WN` , **Roof** `RF` , **Flooring** `SB` , **Object** `CM` , **Curtain Grid** `CG` , **Mullion** `MU`

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Group	Tiles
Vertical Links	**Railing** `RA`, **Stair** `ST`
Rooms and Areas	**Room** `RM`, **Room Separator** `RS`, **Area** ▾ (**Area Plan**, **Area Outline**, **Area**), **Legend** `LG`, **Settings** ▾ (**Area and Volume Computations**, **Color Schemes**)
References	**Level**: drawn in an elevation or a section, two clicks at the new level's height (from a plan it opens an elevation); **Grid** `GR`

Area Outline and **Area** work in an area plan: until one is open they are greyed, and their tip says to open an area plan first.

The Annotations tools

Group	Tiles
Dimension	**Aligned** `DI`, **Linear** `DL`, **Angular** `DA`
Text	**Text** `TX`
Detail	**Detail line**, **Filled region**, **Masking region**, **Detail component** `CD`, **Repeating detail** `RD`, **Insulation** `IS`
Tag	**Tag All** `TG`

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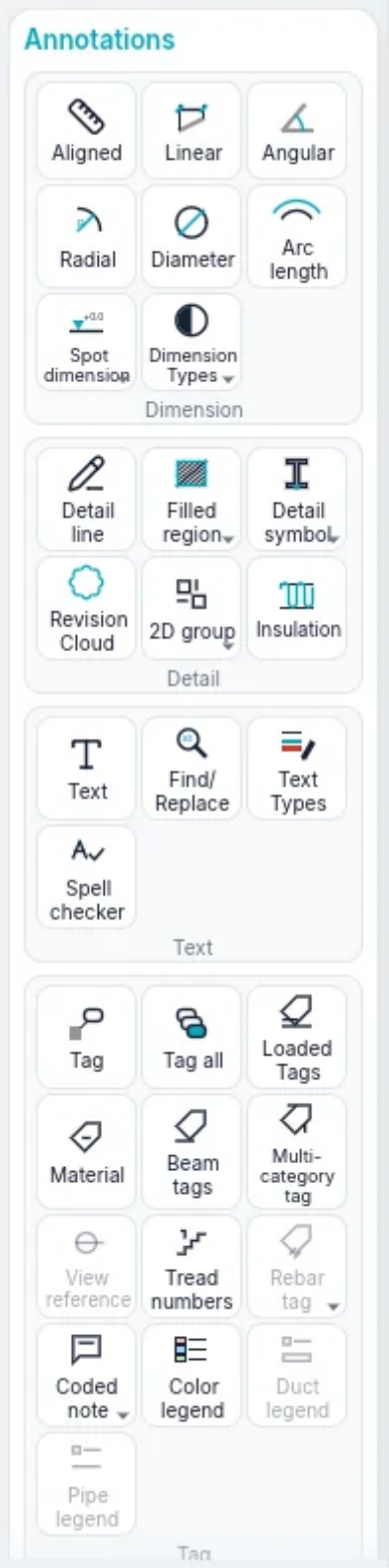

The Annotations tools

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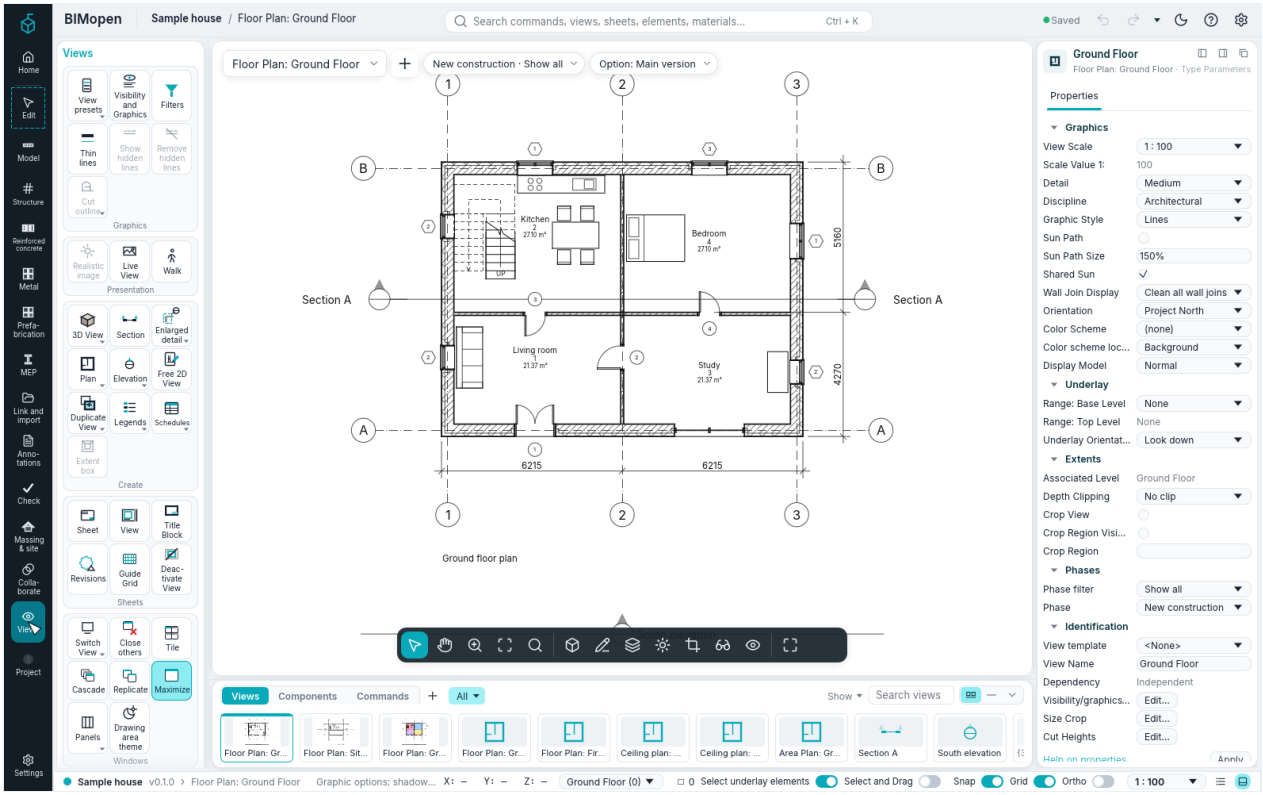
The Views tools



Views on the rail: the tools of the views and the Project Index

Group	Tiles
Create	**3D View** ▾ (**Default 3D View**, **Save View**), **Plan** ▾ (**Floor Plan**, **Area Plan**), **Duplicate View** ▾ (**Duplicate**, **Duplicate with Detailing**), **Section** `SE`, **Elevation** ▾ (**Elevation** `EL`, **Framing Elevation**), **Enlarged detail** `DT` (a callout), **Free 2D View** `VD`, **Schedules** ▾ (**Schedule/Quantities**, **Material Takeoff**, **Sheet List**, **Note Block**, **Predefined schedules**)
Sheets	**Sheet**: a new sheet with the title block you choose; **View**: places a view on the open sheet
Graphics	**Visibility and Graphics** `VG`, **Filters**, **View templates**, **Apply template to current view**, **Create template from current view**
Windows	**Switch View**, **Replicate**, **Cascade**, **Tile**, **Maximize**, **Close others**, **Panels**, **Drawing area theme**

Sheets ▸ View works only while a sheet is open.

Save View works only while a 3D view is open, and **Duplicate with Detailing** only for views that are not 3D. Under the tools, the **Project Index** card holds the whole tree of the project.

Every ▾ menu also lists, greyed, the entries not available yet; the command search finds them all.

The Project tools

Group	Tiles
Settings	**Materials**, **Project information**, **Styles by Category**, **Line styles**, **Purge unused**, **Project units** `UN`, **Snap**, and **Additional settings** (**Line styles**, **Fill patterns**)
Inquiry	**Check**: checks the openings and references of the model and reports at the command line

Snap turns snapping on and off for the whole program: it is highlighted while on, as the **Snap** switch of the status bar. **Additional settings ▸ Snaps...** opens the **Snap** window: the length and angular increments, the object snaps and the overrides (see Snap).

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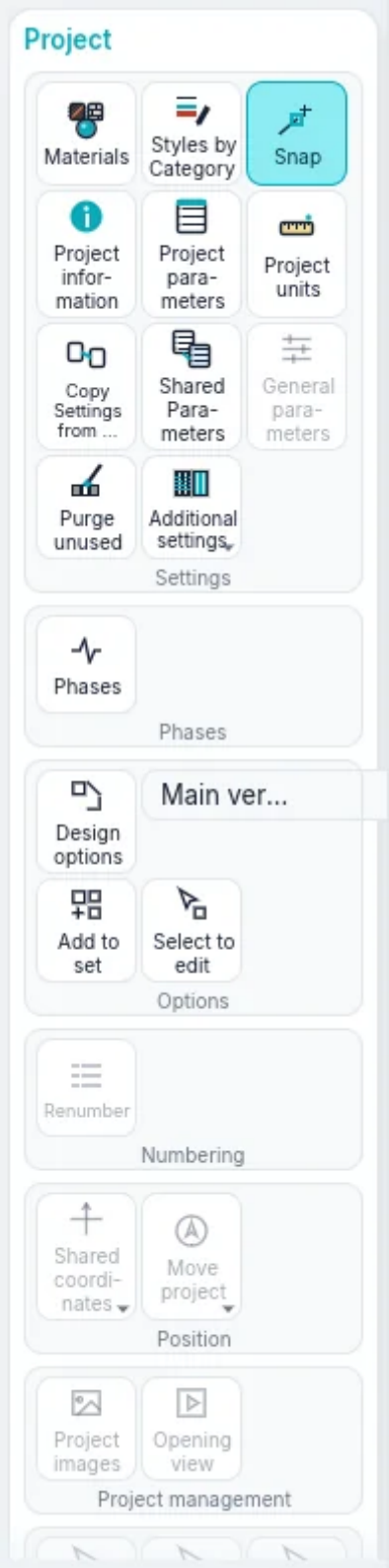

The Project tools

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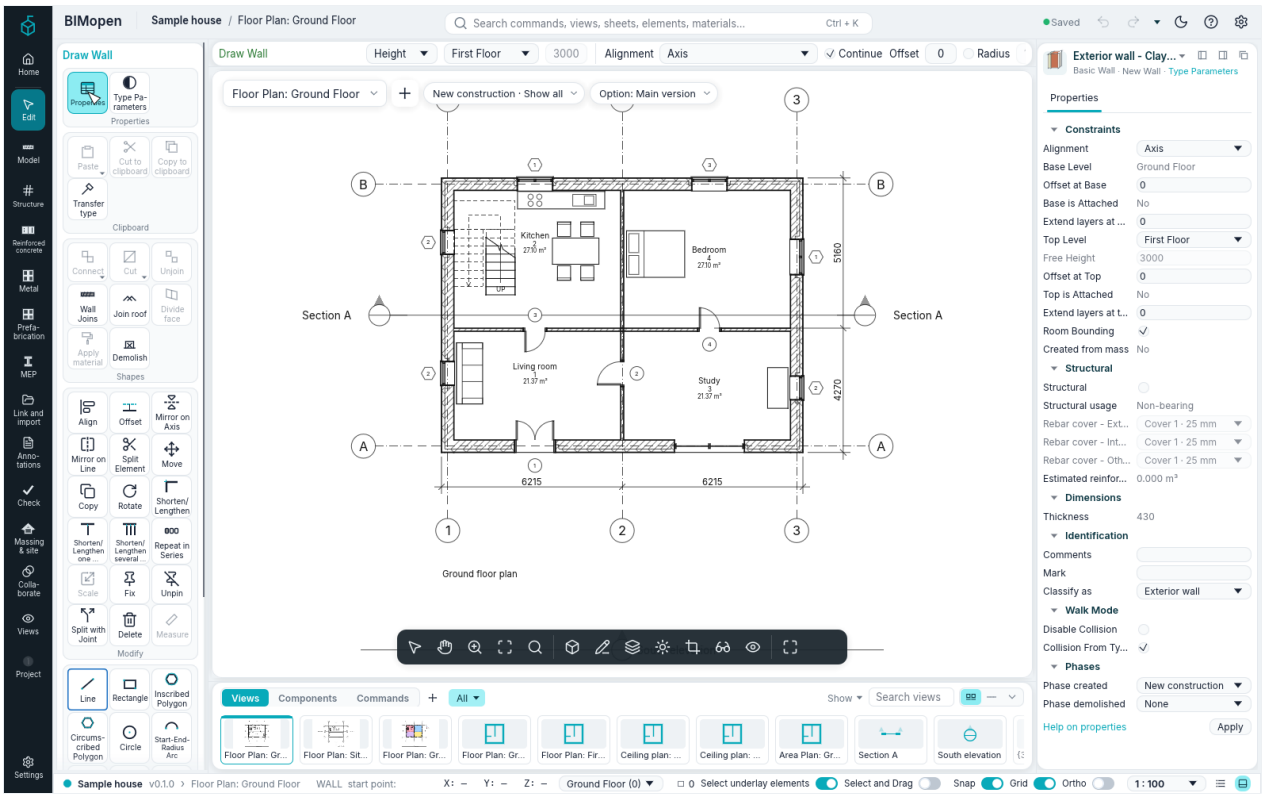
The contextual tab Edit



The wall tool at work: the contextual tab Edit in the tool panel and the options strip over the view

When you start a tool or select something, the rail chooses by itself the contextual tab **Edit**, drawn with a dashed outline, and the tool panel shows its groups under the name of what you are doing, in the accent colour: **Draw Wall** while drawing walls, **Selection: Walls** with walls selected, **Multi-Select** with elements of different categories, **Edit | ...** in the other modes. When the tool ends, or the selection is cleared, the tool panel goes back to the tab you were on; `Esc` or a click on empty space does the same. A click on **Edit** shows its groups at any time.

It always has these groups:

Group	Tiles
Properties	**Properties**: shows or hides the element panel; **Type Parameters**: the properties of the type of the selection, or of the next element the tool will place
Modify	**Align** `AL`, **Offset** `OF`, **Mirror on Axis** `MM`, **Mirror on Line** `DM`, **Split Element** `SL`, **Split with Joint** `SG`, **Move** `MV`, **Copy** `CO`, **Rotate** `RO`, **Shorten/Lengthen** `TR`, **Shorten/Lengthen one element**, **Shorten/Lengthen several elements**, **Repeat in Series** `AR`, **Scale** `RE`, **Fix** `PN`, **Unpin** `UP`, **Delete** (`Del`)

Delete works only when something is selected. The selecting and modifying chapter describes the Modify tools.

Other groups appear for what is selected or for the tool:

When	Group and tiles
Anything selected	**Selection**: **Filter** (see Selecting and modifying)
Doors or windows selected	**Opening**: **Flip facing**, **Flip hand**
Walls selected	**Modify Wall**: **Link Top/Base** (the walls rise to the roof above them), **Unlink Top/Base**
A level selected	**View**: **Open plan**
Sections or elevations selected	**Section**: **Split Segment** (sections only), **Flip**, **Go to View**
A callout selected	**Detail**: **Go to View**
Curtain walls selected	**Curtain Wall**: **Curtain Grid**, **Mullion**, **Panel**
Flooring or **Roof** tool	**Draw**: **Boundary Line**, **Pick Walls**
Stair or **Railing** tool	**Mode**: **Finish Edit Mode**; **Draw**: **Run** or **Lines**

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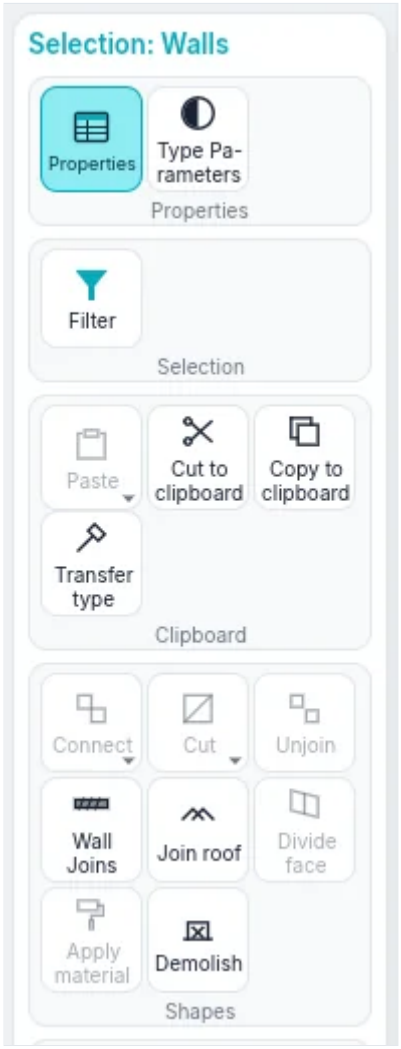
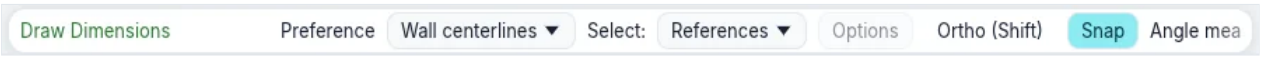
When	Group and tiles
Mullion tool	**Placement**: **Grid Line**, **Grid Line Segment**, **All Grid Lines**
A detail tool	The drawing shapes of the tool



The contextual tab of a selection

Options strip

While a tool works, a strip runs over the top of the view: the name of what you are doing, then the settings of the tool. They apply to the next elements you place.



The options strip

Tool	Settings on the options strip
Wall	**Height** or **Depth**; the level the wall reaches, or **Unconnected** with its value; **Alignment**; **Continue**; **Offset**; **Radius** (greyed: curved walls are not available yet); **Join at the ends** and **Automatic joining** (greyed); **Exterior on the left** or **Exterior on the right** (the space bar swaps them)
Door, **Window**	**Model on site** (greyed); **Tag on Placement** with **Horizontal** or **Vertical**, **Leader** and its length; **Loaded tags...** (greyed)
Object	**Model on site** (greyed); **Rotate after inserting**: the object placed stays selected and **Rotate** starts
Roof	The shape: **Gable**, **Hip** or **Flat**; **Slope**, from 5° to 60°; **Overhang**, up to 2 m
Stair	**Width**, in millimetres; **Railing Type**, or **None**
Dimensions	What the cursor picks first on a wall: **Wall centerlines** or **Wall faces**

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Tool	Settings on the options strip
Enlarged detail (callout)	The scale of the new detail view, or **Reference other view** and the detail or free 2D view the callout points at (always on in a free 2D view)
Area Outline	**Line** or **Pick lines**; with **Pick lines**, **Apply area rules** or an **Offset**

The **Ortho** and **Snap** switches of the status bar work for every tool that draws lines: **Ortho** keeps the segments horizontal or vertical, **Snap** catches endpoints, midpoints and intersections. With **Ortho** off, holding `Shift` while drawing gives the same constraint; with it on, `Shift` releases it.

Element panel

The element panel, on the right, shows and changes the parameters of what you are working on. Its header is one line: the type with its menu, and under it the family, the count of what is selected, as **Walls (1)**, and **Type Parameters**, which opens the parameters of the type. On its right, **Close** (which clears the selection) and **More actions** (⋯: **Type Parameters** and the actions of the contextual tab). Under the header, its tabs; the rows of the parameters are low, with a wide column for the values:

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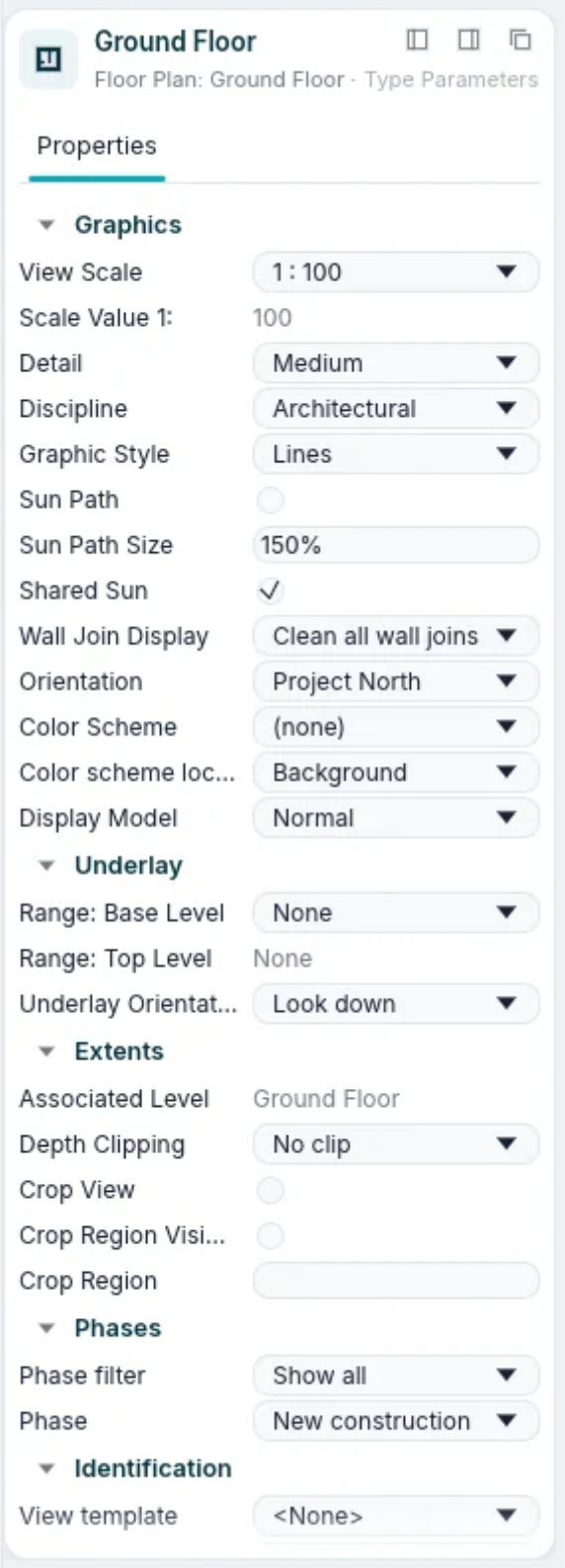

The element panel: the properties of the view

Tab	What it holds
Properties	The type selector, then the parameters, in groups (**Constraints**, **Graphics**, **Identification**...). Click the name of a group to fold it. Values in grey are read-only: they are computed from the model. **Apply**, or `Enter`, confirms a value you are typing
Materials	For walls, floors and roofs: the layers of the type with their colours and thicknesses, and a preview of the layers
Classification	The category, the type and the IFC class
Relations	The level, the host, the doors and windows it hosts, the views and the schedules it appears in

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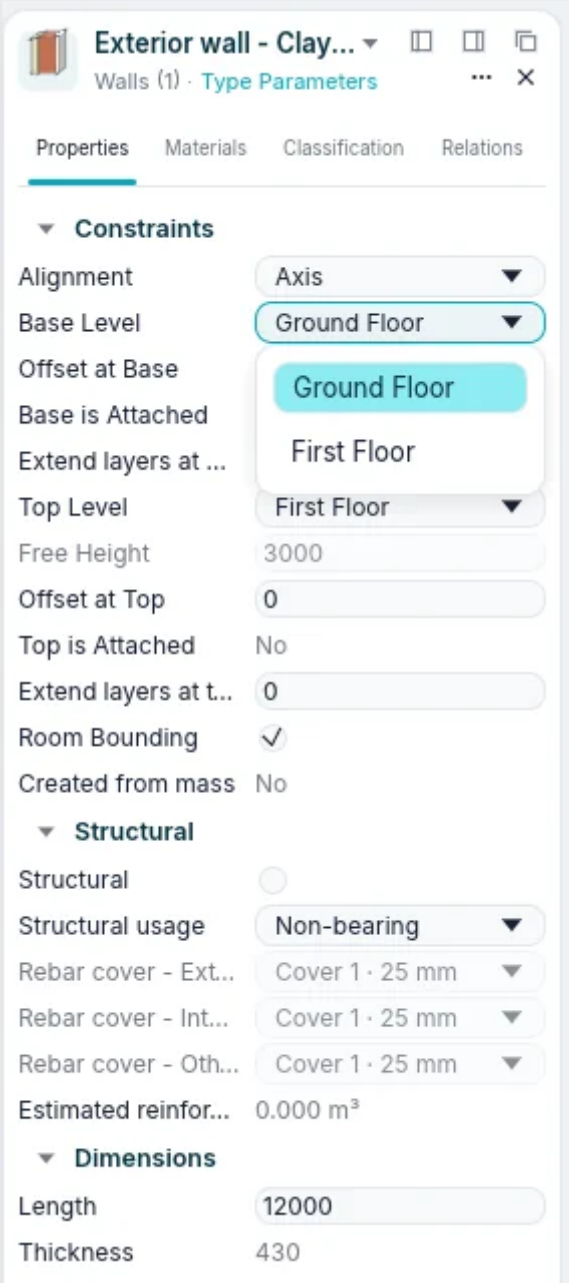
The **type selector** is the header of the panel: a small preview of the type (walls, floors and roofs as their layers, doors, windows and components as a picture), the name of the type with ▾, and the family under it. Click it to choose another type: the selected elements change type, or, while a tool is active, the next element gets that type. The list opens over the rows: search the types by name; they are grouped by family, each with its preview. The arrow keys move, `Enter` picks, `Esc` closes the list.



The type selector

What the panel shows depends on the situation:

Situation	The panel shows
Nothing selected, no tool	The properties of the active view (scale, detail, graphic style...), or of the view clicked in the Project Index
A tool active	The type of the next element, as **New Wall**, **New Door**...
One element selected	Its type and its instance parameters
Several elements of one category	Their number, as **Walls (3)**
Elements of different categories	**Multi-Select**, **Common (5)**, and how many of each category, as **3 × Walls**

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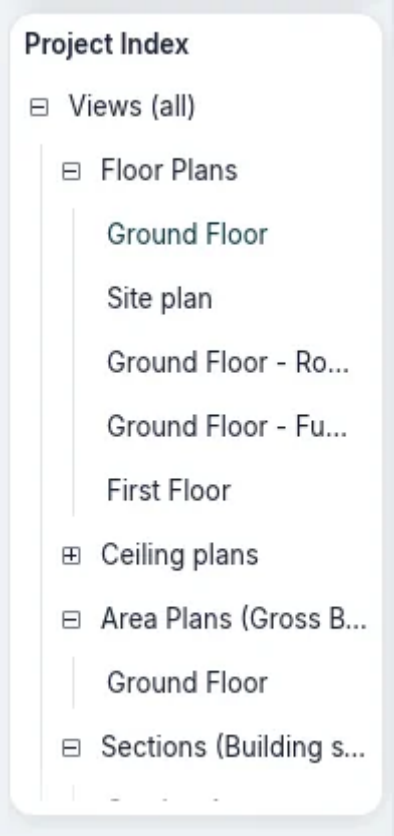
Lengths are written and read in the project's units (millimetres in a new project). You can type a value with its unit, for example `2.7m`.

Project Index

Views on the rail shows, under its tools, the **Project Index**: the views, schedules, sheets and components of the project in a tree. Click ⊟ or ⊞ to fold or unfold a branch.



The Project Index

Branch	What it contains
Views (all) ▸ Floor Plans	The floor plans of every level, from the lowest, and the callouts drawn in plans
Views (all) ▸ Area Plans (...)	The area plans, one branch per area scheme
Views (all) ▸ Sections, Elevations	The sections and the elevations
Views (all) ▸ Detail Views	The callouts drawn in sections
Views (all) ▸ 3D Views	**{3D}**, the default 3D view, and the saved 3D views
Views (all) ▸ Free 2D Views	The free 2D views
Schedules/Quantities	The project's schedules, then the predefined schedules
Sheets (all)	The sheets, as number - name
Components	The types of the project, by category: walls, doors, windows, floors, roofs, **Furniture**, **Plumbing Fixtures** and the others; under a category of components, each component, then its types

The branches of area plans, sections, elevations, detail views and free 2D views appear only when they have something in them. The active view is written in bold.

With a view:

- a **click** shows its properties in the element panel, without opening it;

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- a **double-click** opens it;
- a **right-click** opens its menu: **Open**, **Close** (closes its windows), **Duplicate View ▸** (**Duplicate**, **Duplicate with Detailing**), **Apply view template...**, **Create view template from view...**, **Delete**, **Rename...**, **Properties**.

The menu of a sheet has **Apply view template to all views...** instead of the two template items, and no **Duplicate View**. **{3D}** cannot be deleted or renamed.

With a schedule of the project, a click selects it, a double-click opens it, and the right-click menu has **Open**, **Schedule Properties...** and **Delete**. A predefined schedule opens with a click.

With a type, under **Components**, a double-click opens its type parameters; the right-click menu has **Type Parameters**, **Duplicate...**, **Create Instance** (starts the tool that places it, with that type) and **Delete**.

After a click in the Project Index, the keyboard moves in it: `Up` and `Down` go from row to row (the view's properties show in the element panel), `Right` opens a branch, `Left` closes it or goes up to the branch the row is in, `Enter` opens the view.

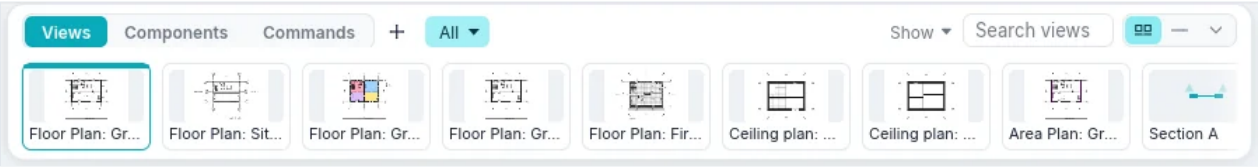
Views go on sheets from here too. Drag a plan, section, elevation, free 2D view, legend, saved 3D view or schedule onto the open sheet: its viewport follows the cursor and stays where you drop it. Or drop it on a sheet's name in the tree: the sheet opens and the viewport follows the cursor until you click. While you drag, the tree scrolls when the cursor nears its edge. The right-click menu of a sheet has **Add View...**. The views on a sheet are listed under it; the right-click menu of one has **Remove from Sheet**, and the menu of **Sheets (all)** has **New Sheet...**. In a 3D view the right-click menu also has **Zoom to 3D Extents**: the whole model, the elements hidden in the view too. A view already on a sheet is refused, with the reason on the command line; legends and schedules can go on several sheets. A saved 3D view is drawn on the sheet with the edges seen from its camera.

The right-click menu of a plan, section or free 2D view has **Find Referring Views**: the views that show its symbol and the sheets it is on; **Go to View** opens the one chosen.

Views and components strip

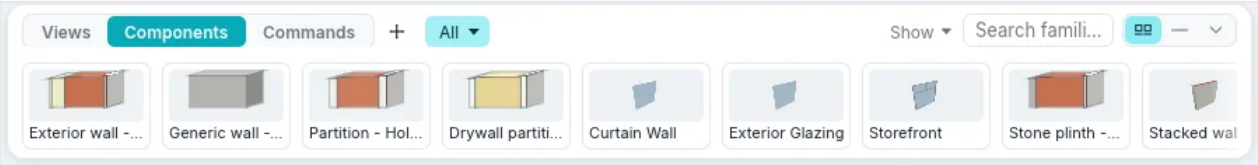
The strip under the drawing area has three modes, chosen with the buttons on its left:

- **Views**: live thumbnails of the views, grouped by **All**, **Open (n)**, **Plans**, **Sections**, **Elevations**, **3D**, **Legends**, **Schedules** and **Sheets**. The current view is framed, open views have an accent line. Click a thumbnail to open the view; right-click for the view's menu, with **Open in the active window**; drag it onto the open sheet, or onto a sheet's thumbnail, to place it there. **+** makes a new view: **Default 3D View**, **Save View**, **Duplicate**, **Sheet** or **Schedule/Quantities**.
- **Components**: one preview for each type, grouped by category. Click a preview to place that type; drag it into a plan to insert it where you let go. A door or a window dropped near a wall goes into it, along it.
- **Commands**: the history of the command line. Click a line you typed to run it again.



The strip in Views mode

On the right, **Show** filters the thumbnails by discipline and phase, the search box by name, and three buttons set the size: **Previews**, **Compact** and **Hide**.



The strip in Components mode

When the groups do not fit beside the modes, they fold into one menu, as **All ▾**. To give the room of the strip to the drawing area, put it away with **Views strip**, the last button of the status bar, or with `F9`; the same button or key brings it back.

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Panels

The element panel can move. Drag it by its header: it docks on the left or on the right of the drawing area, or floats where you let it go. The buttons at the top right of the panel dock it left, dock it right or make it float; **Views ▸ Windows ▸ Panels** shows or hides it and the other panels. Each panel stays where you put it in the next sessions.

While you drag a panel, only the places where it really docks are drawn, each as the panel it will become. The place under the pointer is drawn strongly, the others faintly. Elsewhere nothing is drawn, and the panel floats where you let it go.

Dialogs

Every dialog opens in the middle of the screen, each time it opens, even if it was moved the last time or the window has changed size. Once open, it stays where it is: drag it by its title bar to move it. A change inside it, such as the **Preview** switch, does not move it.

Every dialog has a fixed size, which does not change with its content. The exception is the dialogs built around a long list or table (materials, visibility/graphics, object styles, filters, line styles and patterns, fill patterns, line weights, schedule properties, project and shared parameters, keyboard shortcuts, purge unused, IFC export classes, view templates): dragging one of their edges makes them larger, never smaller than they first open nor larger than the screen, and the list grows with them. They open again in the middle with the last size given.

Drawing area and view windows

Every view you open is a window in the drawing area. A view opened again comes to the front; at least one window always stays open. At the top left of the view, the **view selector** shows its name: its menu lists the views and arranges the windows, and **+** beside it makes a new view. **Views ▸ Windows** arranges them too:

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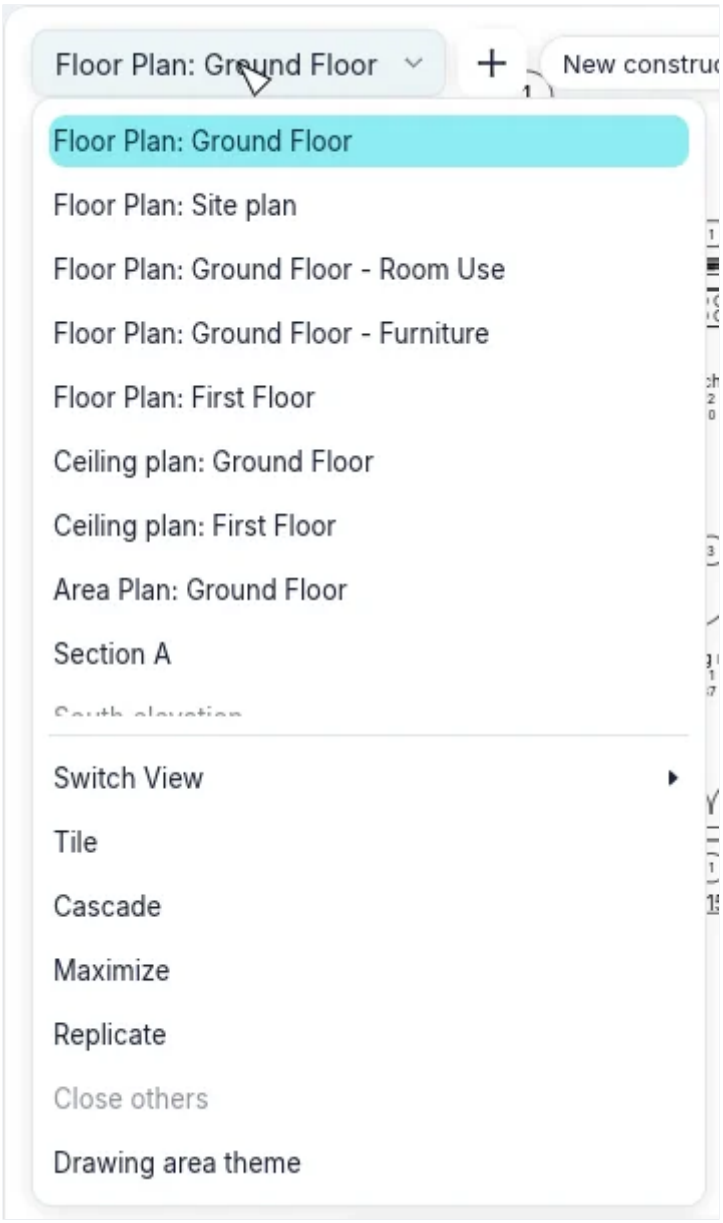

The view selector

Tile	What it does
Switch View	Lists the open windows; the active one is ticked
Replicate	Opens a second window on the active view, with its own zoom
Cascade	Overlapping windows, each with its title bar: drag a title to move a window, its edges to resize it
Tile	All the windows side by side, each with a title bar; the active one has a coloured title and frame
Maximize	The active window fills the drawing area; the others wait underneath
Close others	Closes every window except the active one
Panels	Shows or hides the **Properties**, the **Project Index**, the **Command line**, the **Navigation bar**, **Keyboard Shortcuts** and **Recent Files**
Drawing area theme	Each click switches the drawing area background between black and white; the rest of the interface does not change. It is the same choice as **Invert background color** in **Options**, and it stays for the next sessions

Click in a window to make it active: the tools work in the active window, the others only show the model. In **Tile** and **Cascade**, the **×** on a window's title bar closes it. `Ctrl+Tab` goes to the window you used before.

Tile puts the active window at the top left and the others after it, from the most recently active; **Switch View** lists them in the same order. Tile again after working in another window and that window comes first.

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Options ▸ User Interface ▸ Keyboard Shortcuts lists the commands with their shortcuts. Search a command by name, or **Filter** the list (**All**, **All Defined**, **All Not Defined**, **All Reserved**); select a command, type the keys in **Press new keys** and click **Assign**. The same keys may go to several commands: typed in the drawing area, the status bar shows the first, the arrow keys go through the others and `Enter` or `Space` runs the one shown; **Export…** writes every shortcut to an XML file and **Import…** reads such a file back as your shortcuts (a file naming no command here is refused, and your shortcuts stay); **Remove** takes away the keys you added. Your shortcuts are kept for the next sessions and start the command when typed, as the built-in ones do.

A tool never changes the active view by itself. Started in a view where it cannot work (a roof or a room in a 3D view, a level in a plan, an area outside an area plan), it opens **Go to View**: the views that suit it, with the plan of the active level chosen first. **Open View** opens the chosen view and starts the tool there; **Cancel** leaves everything as it was. **Edit Boundary** of a floor, roof, stair or railing, from a section or an elevation, asks the same way for a plan of its level, and a new section or elevation leaves the plan you drew it in active. **Move**, **Copy**, a linear **Repeat in Series** and **Mirror on Line** also work in sections and elevations: along the view; the axis is the vertical plane at right angles to the view through its first point, however it is drawn. A radial **Repeat in Series** works in plans and 3D views: in a section or an elevation it stays chosen, but its clicks make nothing and the status bar asks for **Linear**.

Views in separate windows

A view can leave the drawing area: **Separate window** in its menu (right-click it in the **Project Index**, on its thumbnail in the views strip, or on the title bar of a tiled window), or drag the title bar of a tiled or cascaded window out of the drawing area. On the computer the view opens in a system window of its own, which you can move to another monitor too, resize and maximise; on the web it floats in the page, because the browser opens no other windows. A view floating in the program also has **System window**, to move it into a system window.

You can separate several views at once. Each keeps its tools, its view bar and, in 3D, its orientation cube; the selection shows in all of them, and the active view is the last one you clicked. **Back to the main window**, at the top of the view, puts it back in the drawing area; a floating view also goes back when dragged onto the band at the top of the drawing area, which lights up. Closing the system window brings it back the same way. Where each separate view was stays with the project, and when it opens again the separate views come back to their places.

Drawing in every view

A wall is drawn the same way in a plan and in a 3D view, on the level's plane: from the first click the wall being drawn follows the cursor, in the plan as its footprint and in 3D as its see-through solid, at its height and thickness; a dashed line carries its axis on, and others hold it in line with the points of the chain already clicked. The temporary dimension, blue with its witness lines and ticks, gives the length and the angle; type a length and press `Enter` to draw the wall that long. The snaps (endpoints, midpoints, intersections, perpendicular, other walls' faces and axes) work in 3D too. The wall selected in a 3D view shows, as in a plan, its temporary dimensions, the flip arrows and the ends to drag. In sections and elevations a wall is not drawn: **Go to View** offers a plan or the 3D view.

Roofs and floors are drawn in the 3D view too, on the plane of their level: **Pick Walls** takes the wall under the cursor (its top lights up in orange, `Tab` takes the whole chain), the lines are drawn with every sketch tool and with temporary dimensions, and **Edit Footprint** of a roof or a floor opens its sketch in 3D. **Edit Boundary** of a wall works in 3D on the wall's plane, and in a section or elevation that sees it face on.

While you draw or edit a sketch you can switch views (thumbnails, **Project Index**, separate windows): the tool stays active and goes on in the new view if it can work there. A chain of walls begun in a plan goes on in 3D from its last point, and a profile sketch begun in an elevation goes on in 3D. The wall being drawn and the sketch show in every open view at once.

View bar

At the bottom centre of every view, a dark bar of buttons:

Button	What it does
Select and modify	Ends the navigation mode, back to selecting, `Esc`
Pan, **Orbit** (3D), **Zoom**	With one on, the left button drags the view; `Esc` ends it
Zoom to Fit	Fits the whole view in the window, `Home`
Zoom Options	The zoom menu, see below
The graphic style	**Wireframe**, **Lines**, **Shaded**, **Shaded with Edges**, **Flat Colors**
The detail	**Coarse**, **Medium** or **Fine**
Sun and shadows	In 3D views, the sun and its shadows

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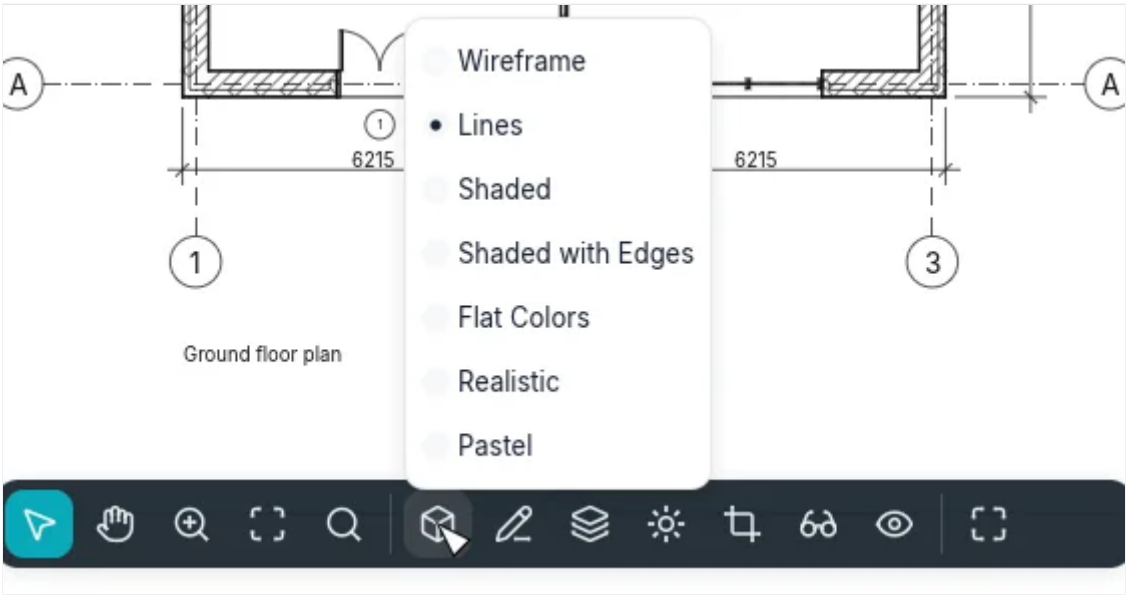
Button	What it does
Crop	**Crop View** crops the view to its crop region, or stops cropping; **Show Crop Region** shows or hides its frame
Isolate Temporarily	Hides or isolates elements or categories in this view until you reset it; framed while active
Reveal Hidden	Shows the hidden elements, to show them again
Full screen	The view alone in the window; **Exit full screen** comes back



The view bar and the graphic styles

The scale of the view is on the right of the status bar, for example **1 : 100**: it opens the list of scales; under **Custom** type another one, as `1 : 75`, and press `Enter`. A sheet has no scale of its own: its views have theirs.

Zoom

Zoom Options, on the view bar, opens the zoom menu: **Zoom In**, **Zoom Out**, **Zoom in Region** (drag a rectangle on the view; `Esc` gives up), **Zoom Out (2x)**, **Zoom to Fit**, **Zoom All to Fit** (every open window), **Zoom Sheet Size** (the view at the size it prints), and **Previous Pan/Zoom** and **Next Pan/Zoom**, which go back and forth through the views of that window. Typed: `ZR`, `ZO`, `ZF`, `ZA`, `ZS`, `ZP`.

The navigation wheel is not available.

Moving around in the views

In plans, sections, elevations and sheets	
Mouse wheel	Zooms in and out at the cursor
Drag with the middle button	Pans
Drag with the left button	A selection window, or moves the selected elements (Selecting and modifying)
`Home`	Fits the whole view in the window

In 3D views	
Drag with the right button, or `Shift` + middle button	Orbits around the model
Drag with the middle button	Pans
Drag with the left button	A selection window, or moves the selected elements
Mouse wheel	Zooms in and out
`Home`	Fits the whole model in the window

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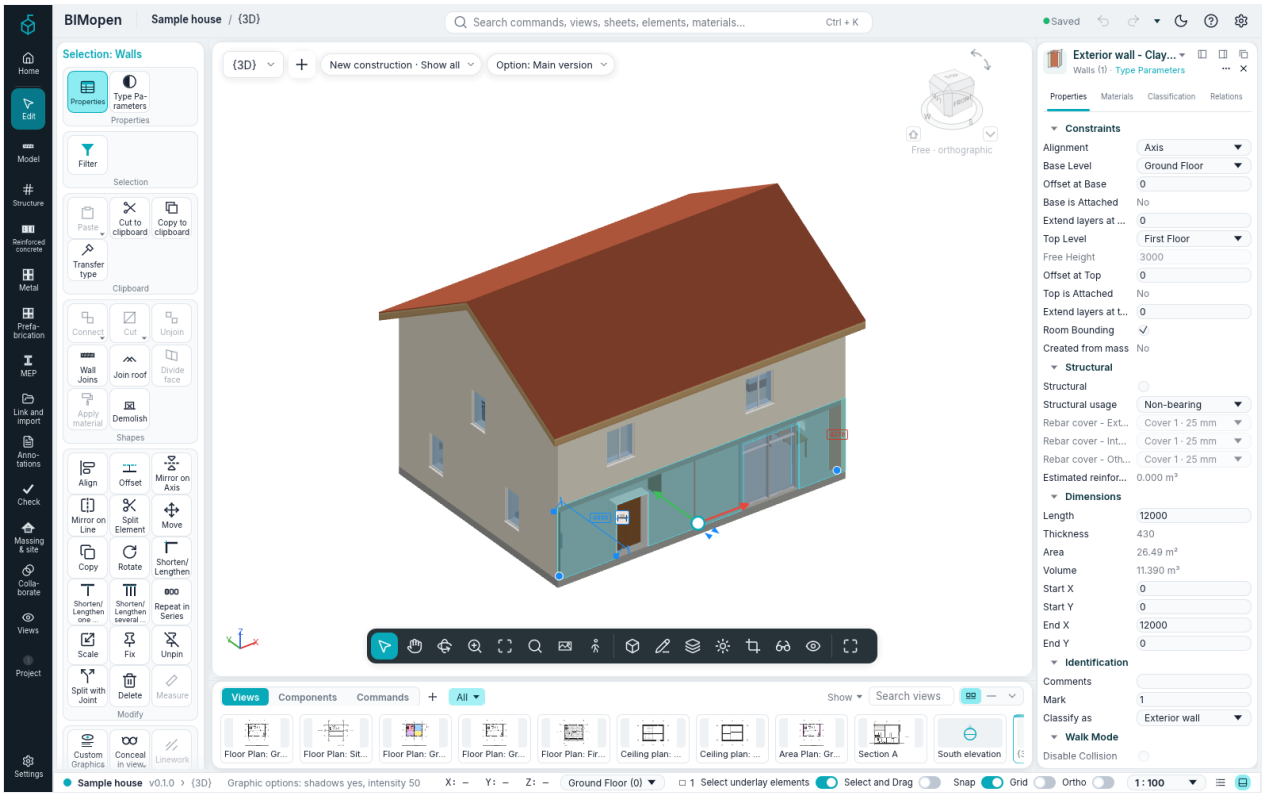

The BIMopen window with the sample house in the 3D view

A small tripod in the lower corner of the 3D view shows the directions of the axes: X red, Y green, Z blue. A saved 3D view remembers where you left it. On the selection, the move arrows let you drag it: drag an arrow to move along it, the centre to move freely in plan; the distance shows as you drag, and the release moves as **Move** does.

In 3D, the wheel zooms keeping the model point under the cursor in place, and the orbit turns about the centre of the selection, or about the point the view looks at when nothing is selected.

The **Orientation cube**, in the top corner of the 3D view, turns with the model. Its edges and corners are bevelled and every part is a target of its own: the six faces (**TOP**, **BOTTOM**, **FRONT**, **BACK**, **RIGHT**, **LEFT**, their names written on them in perspective and turning with them) look straight at that side; the twelve edge strips look from between two sides; the eight corner triangles from between three (an isometric view). The part under the pointer is tinted with the highlight colour, the one the view looks from now stays highlighted; a click turns the view there in a brief transition. Drag the cube to orbit.

The cube is light grey, its faces shaded by a soft light, in both the light and the dark theme; a compass ring at its foot carries the initials of the cardinal points (**N** towards project north).

The orientation cube

Under the cube the view's orientation is written: **Front · perspective**, **Top – Front · perspective**, **Isometric · perspective**, or **Free · perspective** when the view looks from no part of the cube.

When a face is seen straight on, four triangles round the cube (**Turn up**, **Turn down**, **Turn left**, **Turn right**) turn the view a quarter to the face beyond that side. The two curved arrows at the top right, **Roll left** and **Roll right**, turn the picture a quarter turn counterclockwise or clockwise.

The small house button at the bottom left, **Starting view**, goes back to the starting view; right-click it for **Set as starting view**, which makes the present view the starting one (each 3D view keeps its own, with the project). The small button at the bottom

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right, **Cube menu**, or a right-click on the cube, opens the menu: **Orthographic** / **Perspective** (for now the view is perspective only: **Orthographic** is greyed, «Not available yet»); **Set as starting view**; **Zoom to Fit**, which fits the model in the window; **Options...**, which opens the **Orientation cube** page of **Options**. Right to left (Arabic) the cube sits at the top left and the two small buttons change sides; the arrows stay where they point.

The context menu of a view

Right-click in a view while no tool is active. If the cursor is on an element that is not selected, the element is selected first. The menu has, with a selection, the modify commands first (**Move**, **Copy**, **Rotate**, **Align**, **Offset** and the others), then:

- **Cancel**: clears the selection;
- **Select All Instances ▸ Visible in View** or **▸ In Entire Project**: the elements of the same type; **Select Previous**: the selection made before;
- **Hide in view ▸ Elements** or **▸ Category**: hides the selected elements, or their whole categories, in this view;
- **Custom Graphics ▸ By category...**: opens **Visibility and Graphics** on the categories of the selection;
- **Visibility and Graphics...**.

While a tool is active, a right-click opens the tool's menu instead, and the step in progress stays: **Cancel** (what `Esc` does, which still cancels the step), **Repeat [command]**, **Snap Overrides ▸** (the snap of the next point, as the two-letter keys do; in every view, see Snap), **Zoom in Region**, **Zoom Out (2x)**, **Zoom to Fit**, **Previous Pan/Zoom**, **Next Pan/Zoom** and **Properties**. It is the same in plans, sections, elevations, 3D views and on sheets. With the menu open, `Esc` or a click in the view only closes it.

Snap

Additional settings ▸ Snaps... (in the BIMopen style the **Snaps...** command) opens the **Snap** window, the same in both styles:

- **Snaps Off** turns everything off (as the status bar's **Snap** switch, which stays as it was when the program opens again).
- **Length dimension snap increments** and **Angular dimension snap increments**, each with its box: values separated by semicolons (by default 1000; 100; 10 mm and 90°; 45°; 15°; 5°; 1°). The zoom chooses the step: zoomed in, finer steps.
- **Object snaps**: a box for **Endpoint**, **Midpoint**, **Nearest**, **Work Plane Grid**, **Quadrant**, **Intersection**, **Center**, **Perpendicular**, **Tangent** and **Remote Objects**, with their two letters; **Check all** and **Check none**. With **Nearest** off, the line under the cursor still offers its ends and its midpoint, even far away. **Remote Objects** (on at first) lines the point up, across or along, with the ends of lines far away, with a dashed guide.
- **Temporary overrides**: the list of the two letters that hold for one point.
- **Restore Defaults** puts everything back as it was at first.

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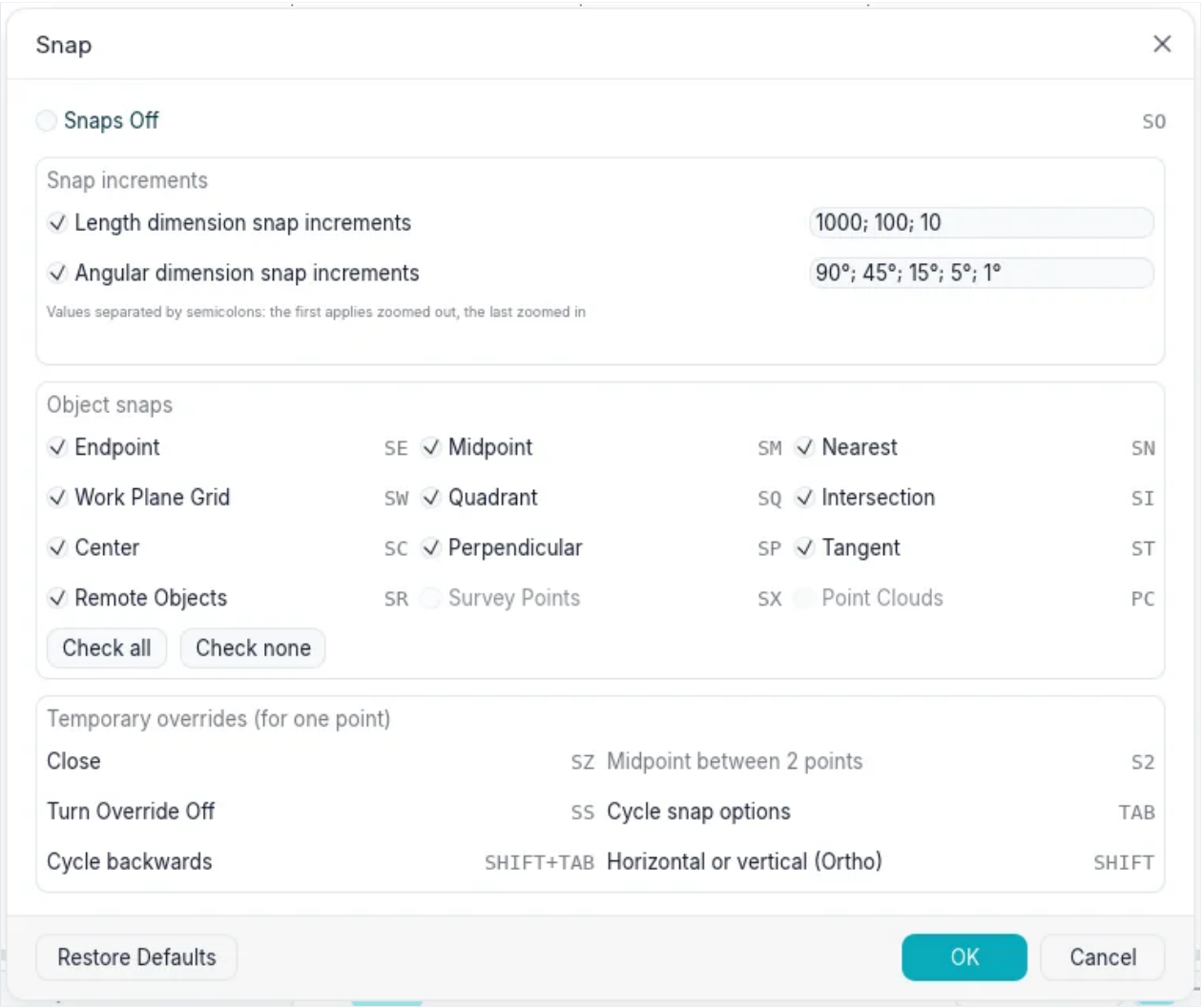

Snap

The settings belong to the program, not to the project: they are kept between sessions.

While drawing from a point (walls, lines, grids, reference planes, sketches, and **Rotate**'s second ray) the segment snaps its direction the same way in every view: horizontal or vertical, parallel or perpendicular to a line nearby, or a multiple of the angular increment the zoom gives; only when the cursor is a few pixels from that direction, otherwise the angle stays free. A dashed guide follows the direction, an arc with the value shows the angle, and the status bar says for instance *Snap: Angle 30° · step 15°*. The length is rounded to the length increment. **Angle measured from**, on the options bar, counts angles from the view's **Horizontal axis** or from the **Previous segment**. In the 3D view everything is measured on the work plane (the level's plane). At the command line `snap {"angles":[45,15],"objects":{"quadrant":false}}` changes the settings and `snap` alone shows them.

The `anglesnap 0,0 3,0.02` command (also through MCP, and `snap` with two points) returns the direction a segment snaps to, by the same rules as every view.

At the top of the menu, **Repeat [command]** starts the last tool you used again; `Enter` does the same when no tool is active and the command line is empty. The menu is the same in plans, sections and 3D views.

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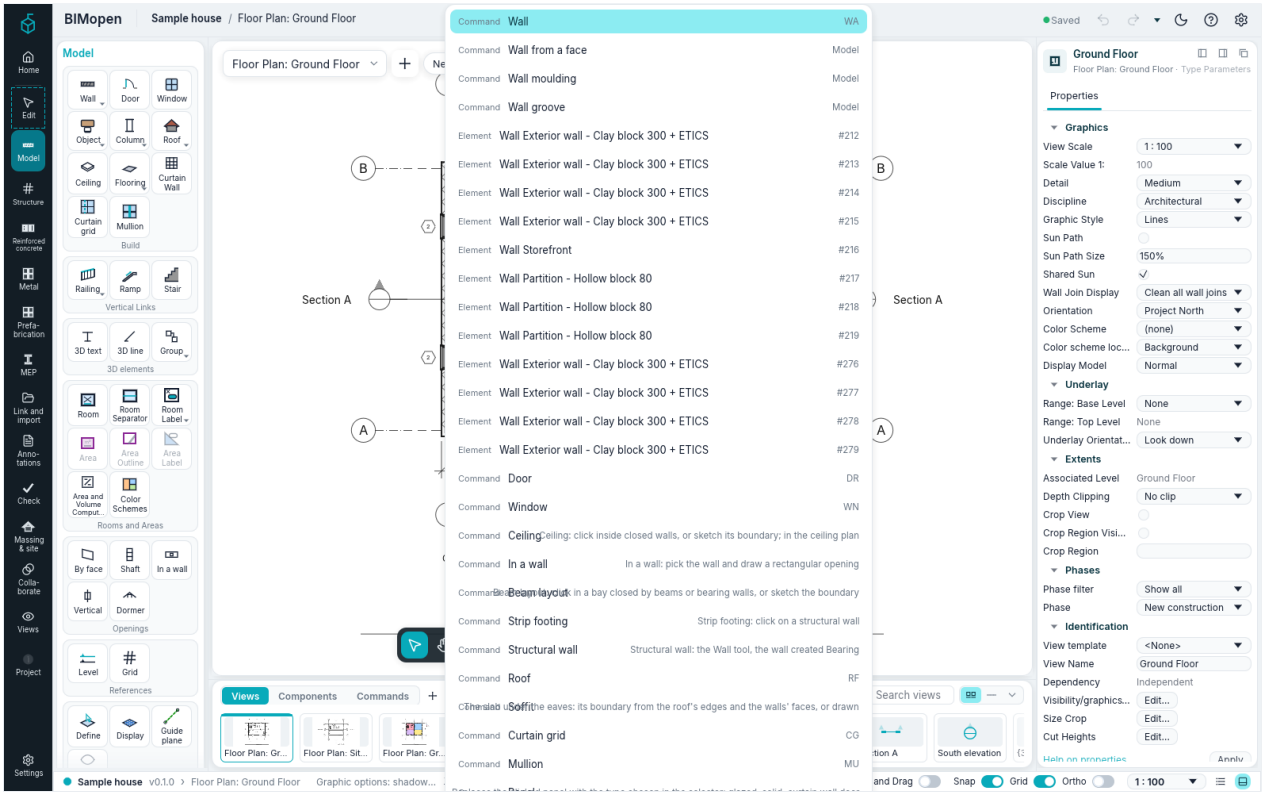
Command search and command line



The command search: commands, elements and views that match what is typed

Press `Ctrl+K`, or click the search box in the middle of the top bar, and type a few letters: the list shows the commands, views, sheets, schedules, elements and materials that match. Click a result, or choose it with the arrow keys and press `Enter`: a command starts, a view opens, an element is selected and shown, a material opens in **Materials**. The search finds every command of the tool panel, the contextual card's too (with a view selected on a sheet, «activate» finds **Activate View**), and runs it as its tile does.

The search box is also the command line. You do not need to click it: whatever you type in a view while no field is being edited goes there. `Enter` runs a keyboard shortcut (`WA`, `DR`, `ZF`…) or a command with its arguments (`wall 0,0 5,0`) as typed; `Up` and `Down` bring back the lines typed before. The log of what has been done, the commands run, their results and the errors in red, is in **Command log** at the right end of the status bar and in the **Commands** mode of the strip. The command line chapter describes the commands.

Status bar

On the left, the status bar shows the project, the version and the active view, then:

The status bar

- while a tool is active, what the tool expects, for example `WALL next point [Esc to finish]:`;
- with no tool, the category and the type of the element under the cursor, as **Walls : Exterior wall - Clay block 300 + ETICS**;
- while you drag, the length of the move, as **Move: 1500**;
- otherwise the last line of the command log.

On the right: the coordinates of the cursor (**X**, **Y**, **Z**) in metres; the active level, whose menu opens the plan of another level; the number of selected elements, which opens **Filter**; the switches **Select and Drag** (drag elements without selecting them first), **Snap**, **Grid** and **Ortho**; the scale of the view; **Command log**; and **Views strip** (`F9`), which puts the views strip away or shows it again.

Selecting, in short

The selecting and modifying chapter describes selection in full. The essentials:

- click an element to select it; click an empty point to clear the selection;

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- `Ctrl` + click adds an element to the selection, `Shift` + click takes it out; `Tab` highlights the next element under the cursor, or a chain of joined walls;
- drag from left to right to select what is entirely inside the rectangle, from right to left to select what it touches;
- drag a selected element to move it;
- `Ctrl+A` selects the elements of the open plan's level, or of the whole model in a 3D view; `Del` deletes the selection; `Esc` clears it.

On a touch screen the same is done with the fingers and the pen: see touch mode.

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Interface styles

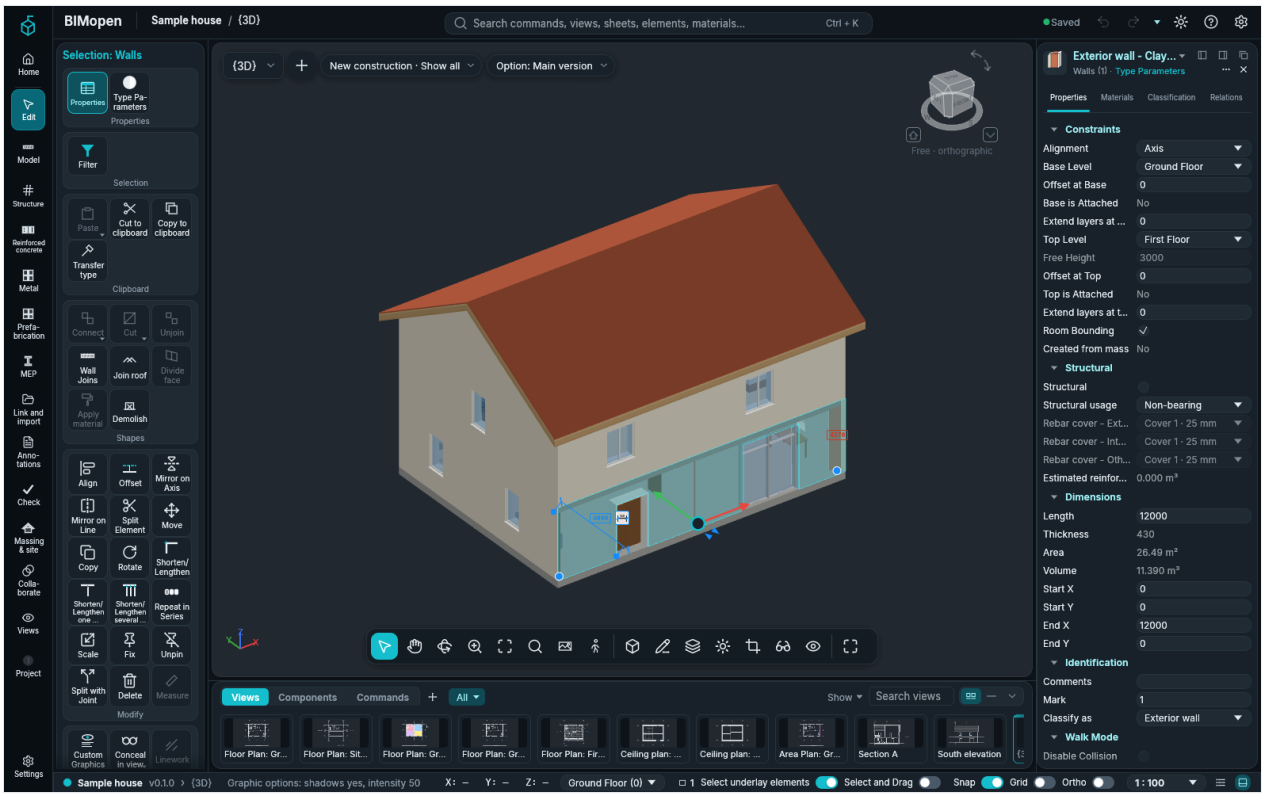
BIMopen has two interface styles and a touch mode. All of them run the same commands, with the same names, keyboard shortcuts and results; only the look and the place of the tools change.

- The **BIMopen** style, described in The interface and pictured throughout this manual: the rail, the tool panel, the command search, the element panel and the strip of views. BIMopen starts in this style.
- The **Classic** style: the ribbon with its tabs, the options bar, the **Properties** palette and the **Project Index** in panels.
- **Touch mode**: the layout for touch screens and pens, on a touch monitor, a tablet or a phone.

Choosing the style

1. Click the gear at the right end of the top bar, or open the **File** panel and choose **Options**.
2. Under **User Interface**, set **Interface** to **Classic** or **BIMopen**. BIMopen starts in the **BIMopen** style.

The window changes at once. The choice is remembered, on the desktop and in the browser, and the next session starts in the same style. **Theme** (**Light** or **Dark**) applies to both styles.



The BIMopen style in the dark theme

In Arabic the whole window is mirrored, in every style: the rail and the tool panel, or the palettes, are on the right, the element panel on the left.

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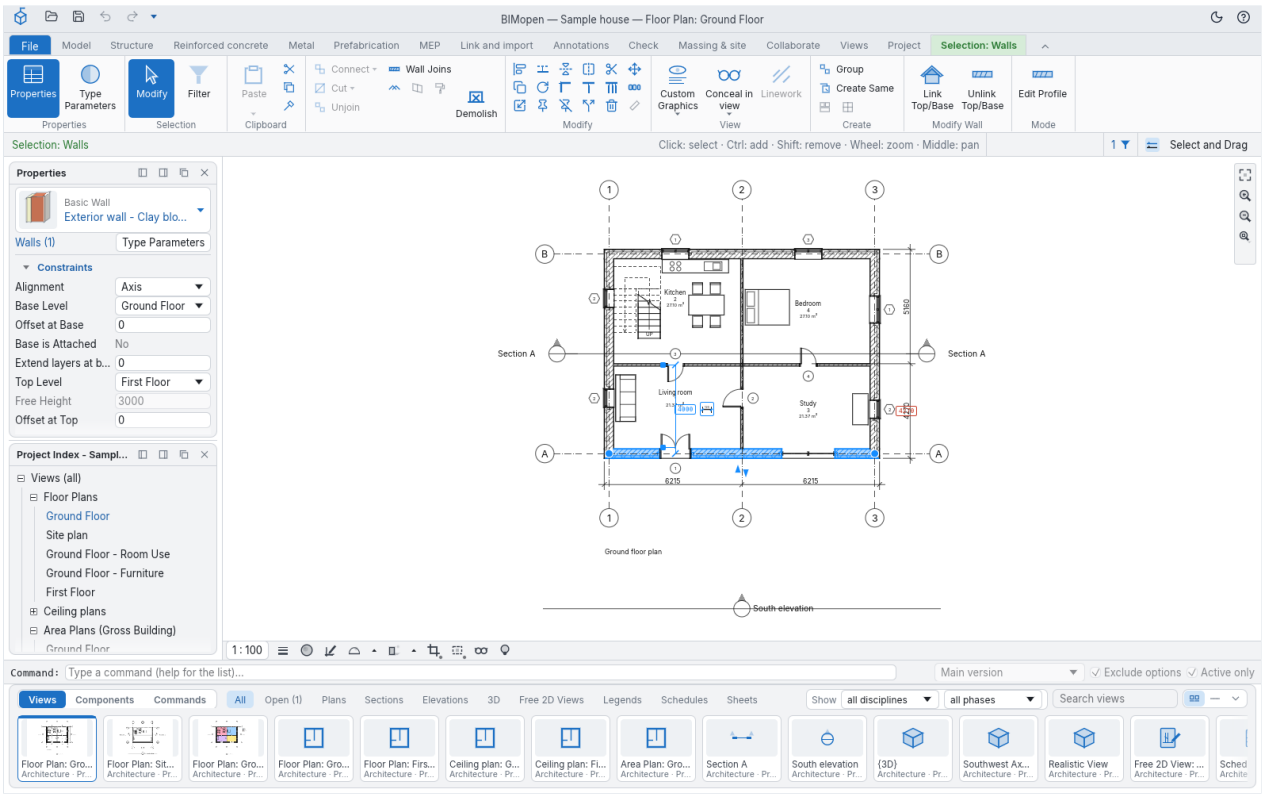
The Classic style



The Classic style

Part	Where	What it holds
Title bar	Top	The quick access toolbar (**Open...**, **Save**, **Undo**, **Redo** and the tools you add), the project and the active view, the light or dark theme and the **?** menu
Ribbon	Under the title bar	**File**, which opens the File panel, then the tabs (**Model**, **Structure**, **Reinforced concrete**, **Metal**, **Prefabrication**, **MEP**, **Link and import**, **Annotations**, **Check**, **Massing & site**, **Collaborate**, **Views**, **Project**) with their groups of buttons, and the contextual tab while a tool works or something is selected
Options bar	Under the ribbon	The options of the active tool, as the options strip of the **BIMopen** style, and at its right end what the tool expects or the element under the cursor, the selection count and **Select and Drag**; right-click it for **Dock at Bottom**
Properties and **Project Index**	Left	The element panel's **Properties** tab, and the **Project Index** of **Views**, as panels that dock on either side or float
Drawing area	Centre	The views, each with its view control bar at the bottom (scale, detail, crop, isolate) and the navigation bar on its right edge (**Zoom to Fit**, **Zoom In**, **Zoom Out**, **Zoom Options**)
Command line	Under the drawing area	The log and the prompt; what you type in a view goes there
Views and components strip	Bottom	The same strip as in the **BIMopen** style

The rail of the **BIMopen** style lists the same tabs as the ribbon, with the same names and in the same order, and the contextual tab as **Edit**; **Views** also holds the **Project Index**.

Everything of the Classic style

Every function of the **Classic** style is in the **BIMopen** style, once:

- every button of the ribbon is a tile of the tool panel, under the same tab of the rail and in the same group, and the contextual tab is **Edit** on the rail, chosen by itself while a tool works or something is selected;

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- BIMopen's mark at the top of the rail opens the File panel; the top bar has undo, redo, the quick access tools and the **?** menu;
- **Views** holds the whole **Project Index**, with its menus;
- a thumbnail of the view strip can be dragged onto the open sheet, or onto a sheet's thumbnail, and its right-click menu has the browser's actions and **Open in the active window**;
- the view selector's menu arranges the view windows (**Switch View**, **Tile**, **Cascade**, **Maximize**, **Replicate**, **Close Hidden**); `Ctrl+Tab` goes to the previous window;
- the right button on a tile offers **Add to Quick Access Toolbar**: the tool shows in the top bar, works whatever tab is open, and its right button offers **Remove from Quick Access Toolbar**;
- the key tips work as on the ribbon: tap `Alt` (or press `F10`), then a tab's letter opens that tab on the rail (its letter shows on the rail) and a tool's letters start it.

Touch mode

Touch mode lays the interface out for touch screens and pens: phones, tablets and touch monitors. It runs the same commands as the two styles, with the same names; only the layout changes, and every control is big enough for a finger.

Turning it on

Options ▸ User Interface ▸ Touch mode has three values:

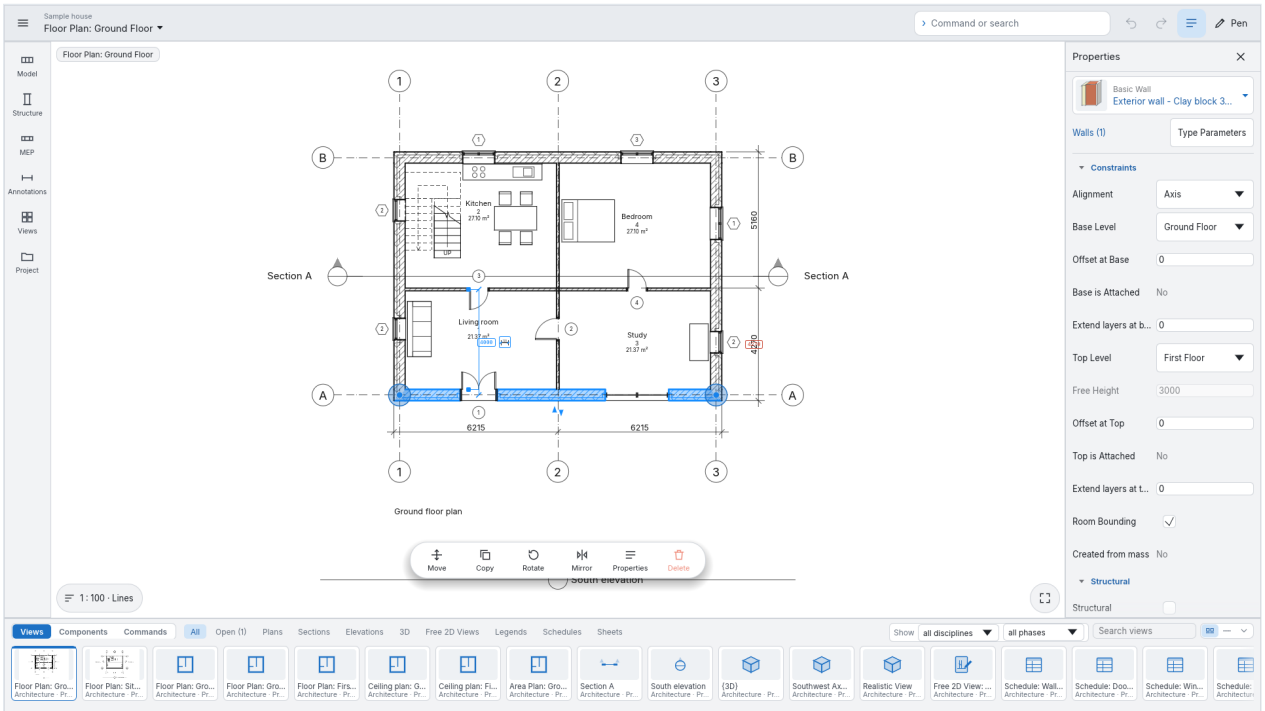
Value	Effect
Automatic	On where the device is touch first: in a browser with a finger as pointer or a window at most 900 pixels wide, and on Android. On the desktop it stays off
Always	Always on, on the desktop too (a touch monitor, a tablet PC)
Never	Never on

BIMopen remembers the choice with the other preferences. **Menu ▸ Leave touch mode** brings back the other interface until the next start.

Phone, landscape and tablet

The layout follows the room the window has.

On a touch monitor, or any wide window (here 1920 × 1080): the rail on the left opens a palette of tools for each category (**Model**, **Structure**, **MEP**, **Annotations**, **Views**, **Project**), the properties stay on the right, the views and components strip at the bottom, and the **Command or search** field in the top bar, beside **Undo**, **Redo** and **Pen**.



Touch mode on a touch monitor, 1920 × 1080

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On a tablet, upright (800 × 1280) or sideways (1280 × 800): the same layout as on a monitor. Where the window is narrow the properties are put away, to leave the room to the drawing: **Properties** in the top bar brings them back.

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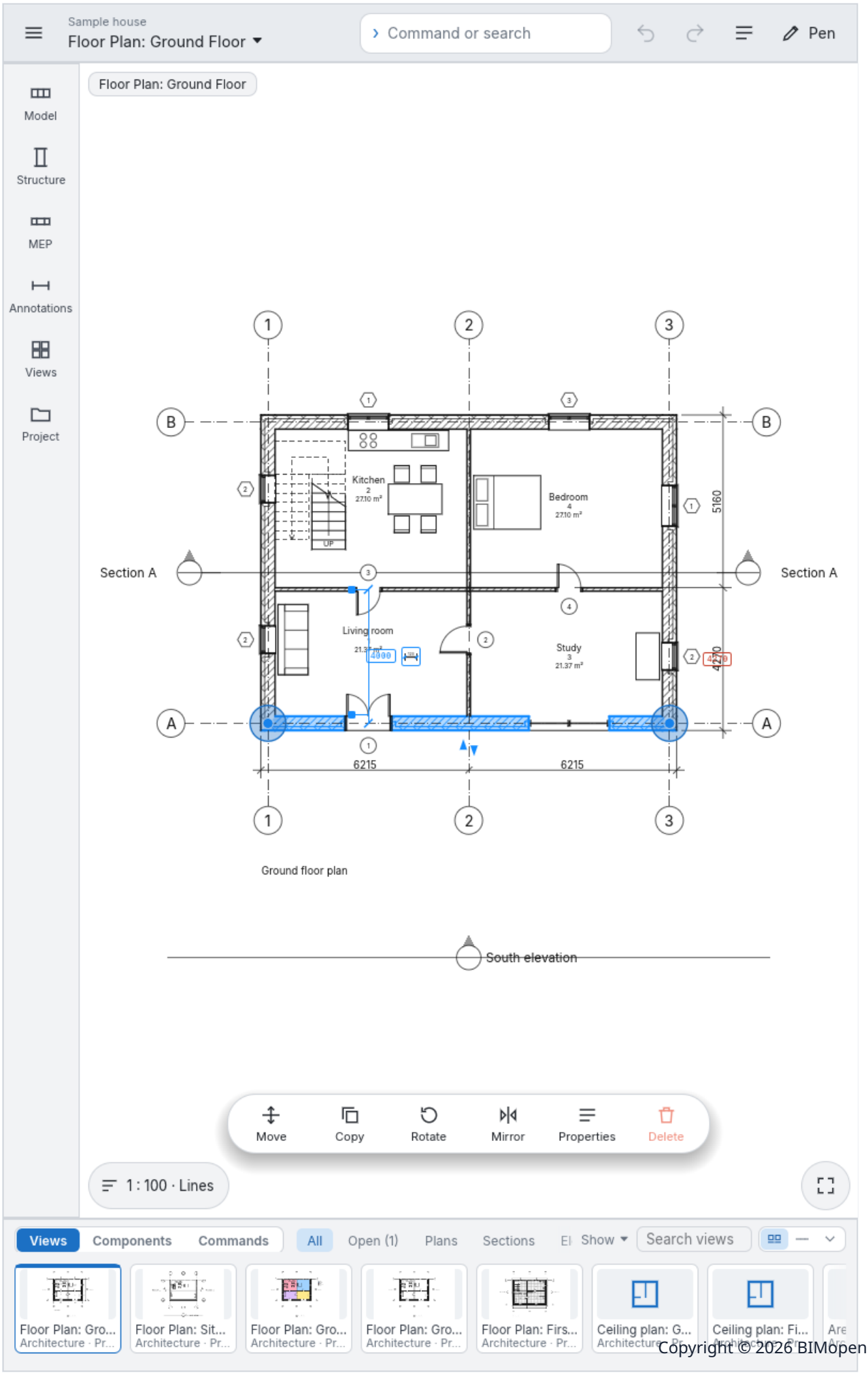

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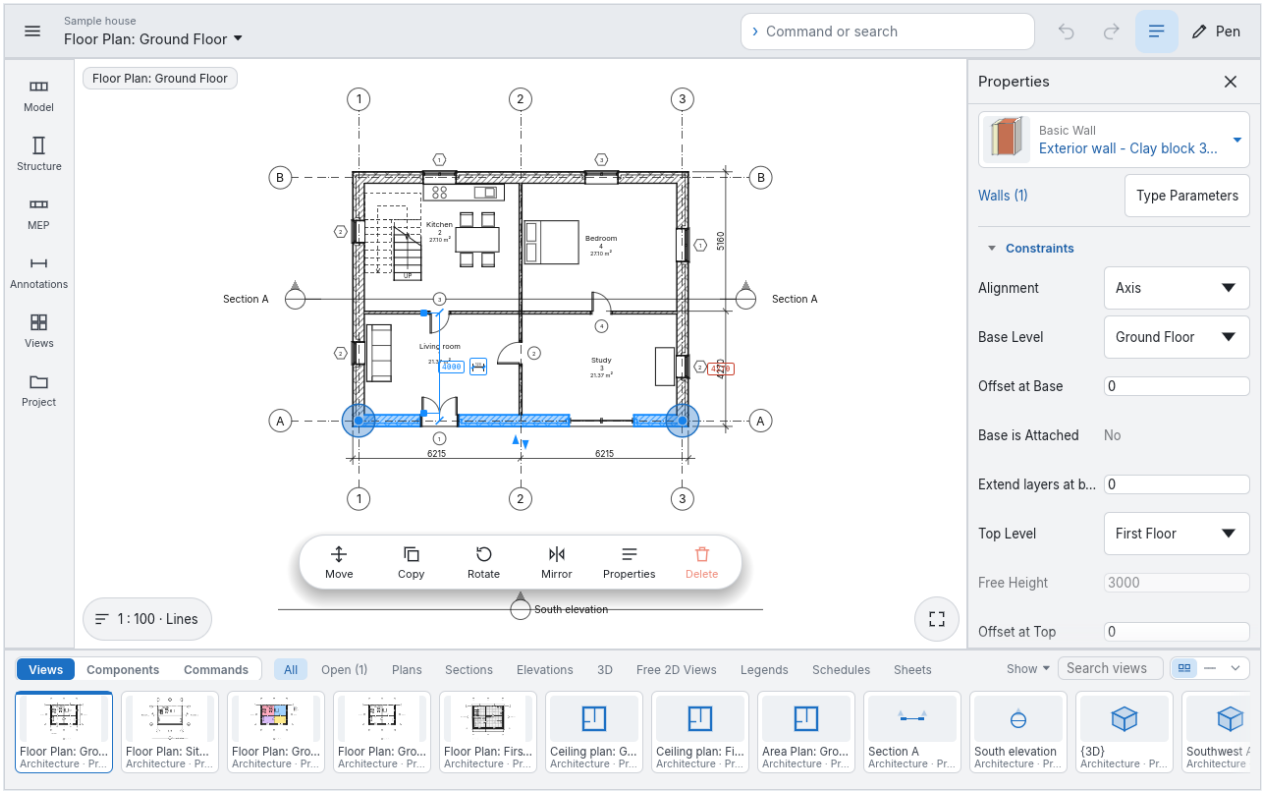
Touch mode on a tablet held upright, 800 × 1280



Touch mode on a tablet held sideways, 1280 × 800

On a phone, upright (narrow and tall): a compact top bar with the menu, the project and the view (a tap opens the views), **Undo**, **Redo**, **Command or search** and **Pen**; the drawing takes the whole width; five big buttons at the bottom, **Views**, **Components**, **Tools** (**Modify** while something is selected), **Properties** and **Commands**, each opening a sheet from the bottom. Drag the sheet's handle up to make it taller, down to make it smaller or close it; a tap outside closes it.

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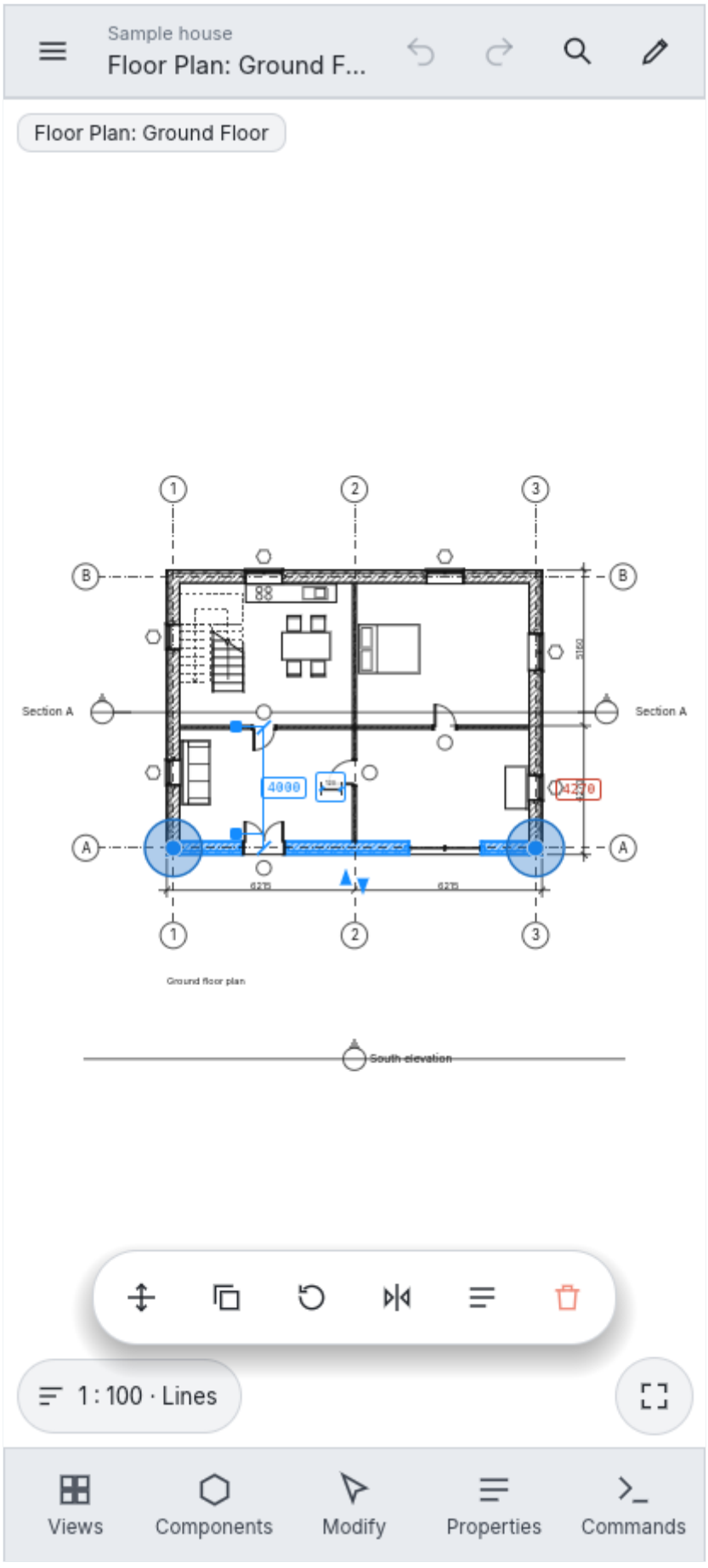

Touch mode on a phone held upright, 412 × 915

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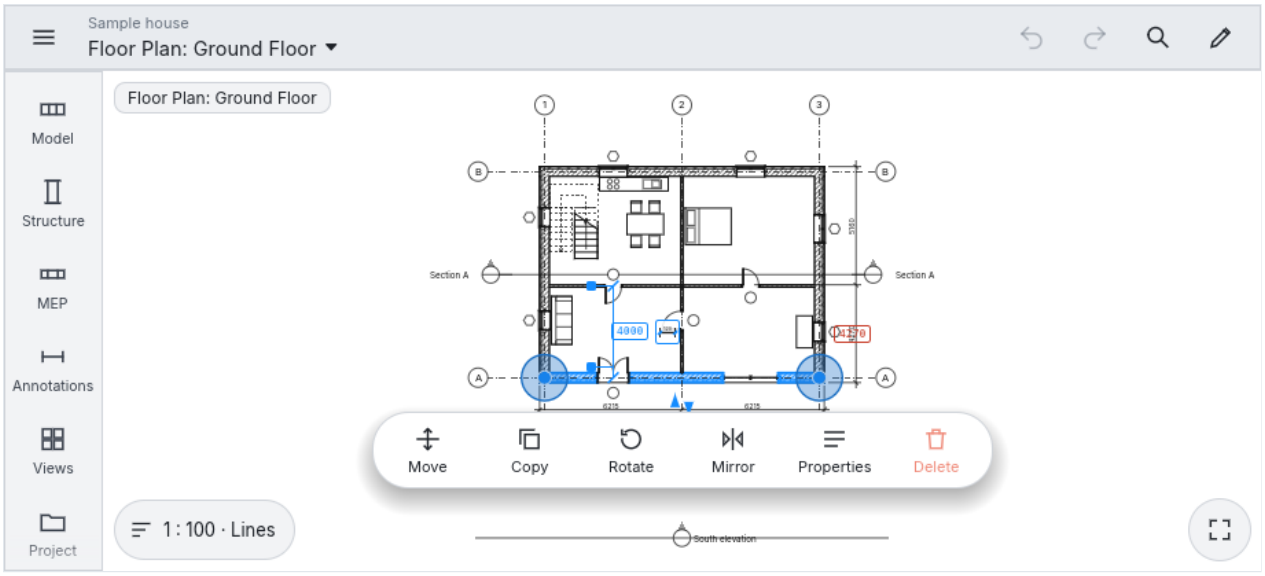
On a phone sideways (short and wide): a rail on the left with the categories **Model**, **Structure**, **MEP**, **Annotations**, **Views** and **Project**, plus the views and the properties; the sheets open on the right.



Touch mode on a phone held sideways, 915 × 412

The tools are large tiles with an icon and a name. A tool still to come is greyed, **Not available yet**; a tool that does not work in the 3D view (**Level**, **Grid**, **Reference Plane**, the annotations) is greyed there, **Not available in the 3D view**. A tap on a greyed tile says why.

While a tool works, a green strip under the top bar shows its options and **Done**, which ends it (or finishes a sketch). While you draw a wall, **Length…** opens the command line: type the length of the next wall and, if needed, its angle, as `4.5` or `4.5 90`.

A tap selects, and the selected walls show big grips at their ends. Over the drawing a floating bar offers **Move**, **Copy**, **Rotate**, **Mirror**, **Properties** and **Delete**; a long press opens the context menu. At the bottom left the view button shows the scale and the graphic style and opens **Scale**, **Detail**, **Graphic style**, **Sun and shadows**, **Crop** and **Isolate Temporarily**; at the bottom right **Zoom to Fit**.

Gestures and the pen

Gesture	What it does
tap	selects, or places the point of the active tool
drag with one finger	as a mouse drag: a selection window, a grip, an element
long press	the context menu
two fingers	pan; pinch to zoom; in the 3D view twist to turn the model
double tap	fits the view in the window

Pen in the top bar is for a pen: the pen draws and selects, fingers only move the view (one finger pans, two fingers pan, zoom and turn). A pen palette on the side of the drawing has **Select**, **Wall**, **Door**, **Window**, **Detail line**, **Aligned**, **Text** and **More tools…**. In a browser a pen is told from a finger by the browser; in the desktop app a touch that reports a pressure is taken for a pen.

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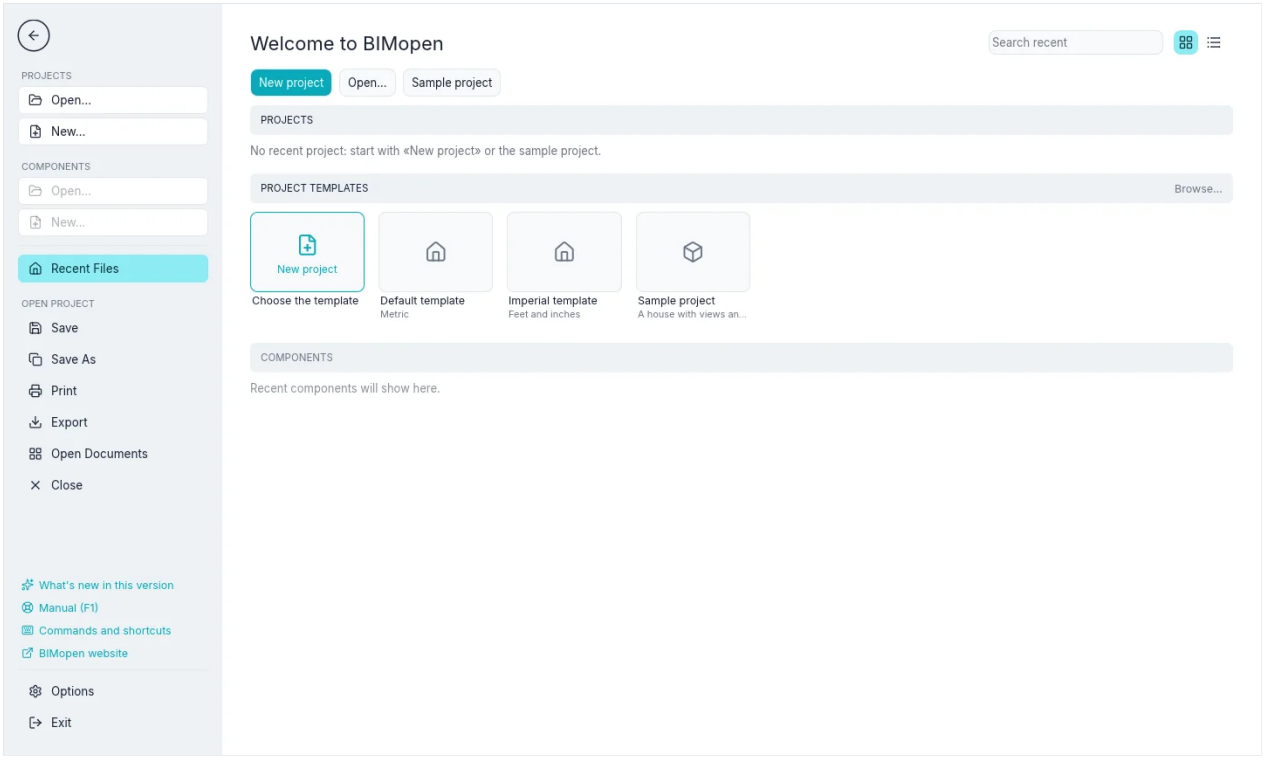
Projects and files

This chapter describes how a project is created, opened, saved and exchanged with other programs, its settings and the options of BIMopen itself.

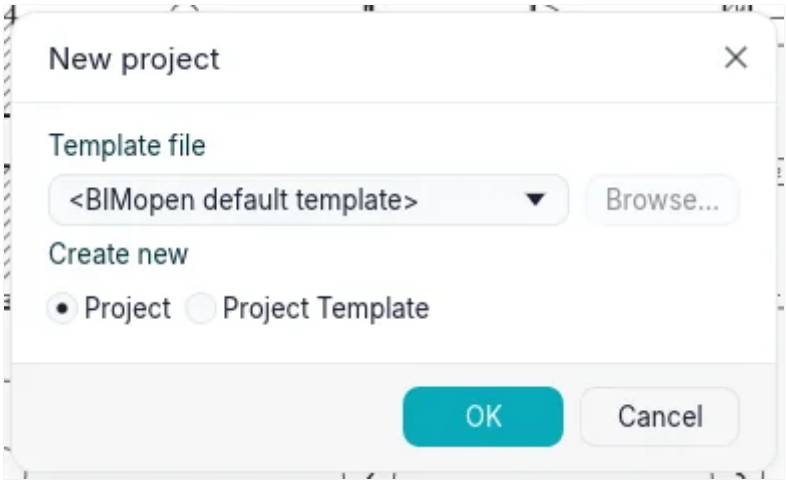


The File panel, on BIMopen's mark

Creating a project

File ➤ New... opens **New project**:

- **Template file**: BIMopen's default template, or the one set in **Options ➤ File Locations**; the **BIMopen imperial template** (see Imperial units); a template file chosen with **Browse...**; or **None**, which starts from two levels and one type of each kind and asks for the system of measurement: **Imperial** or **Metric**.
- **Create new**: **Project** or **Project Template**.



New project

A project made from a template has everything the template had: types, settings, views, sheets, project information and print settings. Saving it asks for a name, so the template is not overwritten.

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Templates

A template is a `.bpt` file. **File ➤ Save As ➤ Template...** saves the open project as a template; a project created as a **Project Template**, or a template opened, is saved as a template too.

BIMopen files

BIMopen saves in its own format, **format 2027**, with four extensions:

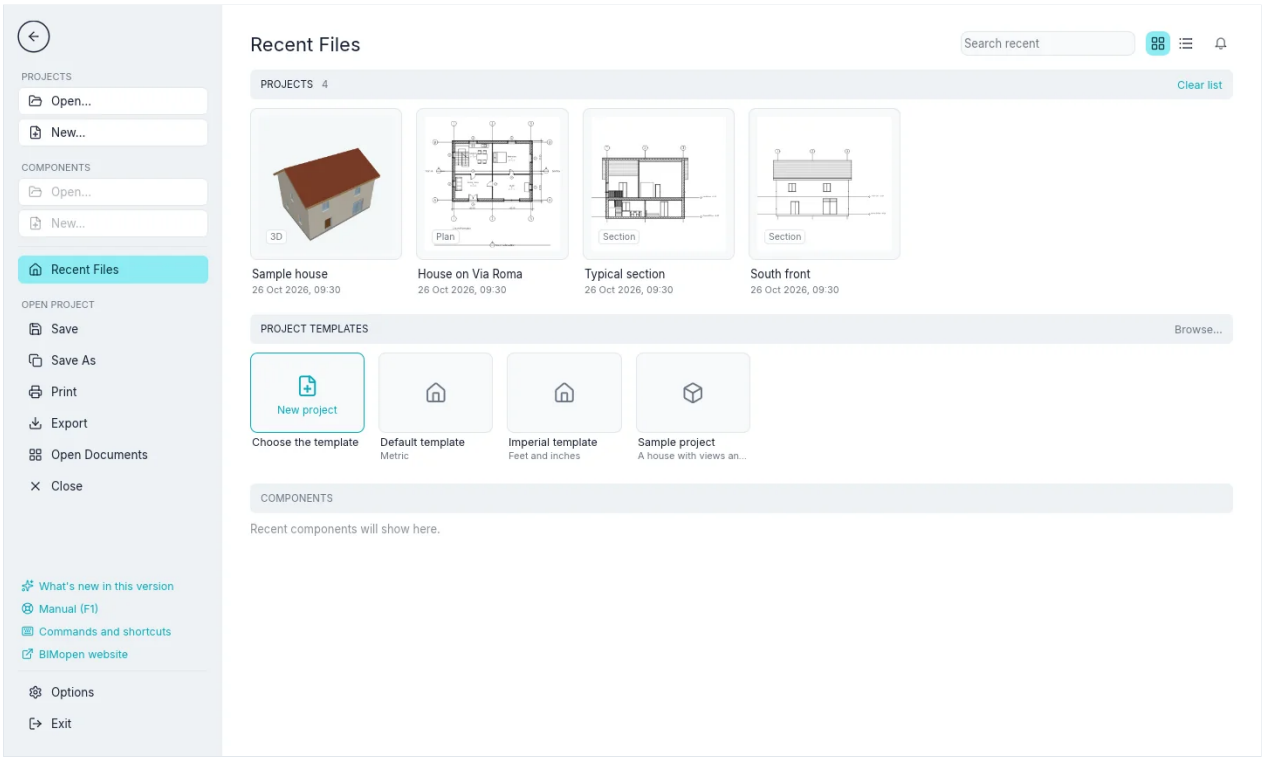
Extension	What it is
`.bpr`	A project
`.bpt`	A project template, which new projects start from
`.bco`	A component, loaded into projects
`.bct`	A component template, which new components start from

The format is made so that every file of format 2027 also opens in later versions of BIMopen. Every element has a permanent identifier, saved in the file, which IFC export uses as its GlobalId: the same element has the same GlobalId in every export, and a copy gets a new one.

A file that is not of format 2027 does not open, and the command line says why: it is not a BIMopen file, or it was saved with a newer version of BIMopen, whose number it gives. Models are exchanged with other programs in IFC: see Import and conversion.

Opening a project

`Ctrl+O` or **File ➤ Open... ➤ Project...** opens a project. **File** shows the recent projects as thumbnails, the most recent first; a pin keeps a project at the top of the list, the ✕ takes it off the list (the file stays where it is) and a moved file is found again with **Locate...** (see Recent Files). Without a file, BIMopen starts on the File panel (unless **Enable Recent Files page at startup** is unticked in the options). Dropping a project file on the window opens it too.



Recent projects in the File panel

A project already open in another BIMopen window opens read-only: **Save** asks for a new name. If BIMopen stopped without saving (a crash, a power cut), opening the project again brings the unsaved work back as unsaved changes.

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Saving

`Ctrl+S`, or a click on the save state in the top bar, saves the project; until then the file stays as it was last saved, and the top bar says **Unsaved changes**. **File ➤ Save As ➤ Project...** writes it to a new file and goes on there.

Save As ➤ Options... opens **File Save Options**: **Maximum backups** keeps that many earlier versions beside the file (`house.0001.bpr`, `house.0002.bpr`...), deleting the oldest; **Compact file** rewrites the file without its obsolete parts at the next save; **Preview** chooses the view of the thumbnail that every save stores in the file and that the File panel shows: the **Default 3D View**, the **Active view** or a chosen view.

When a changed project stays unsaved for the interval set in **Options ➤ General** (**Save reminder interval**, 30 minutes to start with), **Save Reminder** offers to save it, not to save it, or to change the interval.

One project is open at a time: the project opened takes the place of the one open, and the views of that one close with it. Opening the project already open leaves it as it is, when it has no unsaved changes; with changes, answering **No** to the question to save them drops them and opens the file again as last saved, as closing it without saving and reopening it.

Before **New**, **Open...**, a recent document, a dropped file, `open` at the command line, the sample project, or closing the window, a changed project asks **Save changes to ...?**: **Yes** saves it, **No** drops the changes, **Cancel** keeps it open.

In the browser, **Open...** uses the browser's file chooser, and **Save** downloads the file; backups, crash recovery and read-only opening need a file on disk and are desktop-only.

Saving for archiving

The project file keeps a cache of the views slow to make again (elevations, sections and thumbnails taking 300 ms or more): reopened, it shows them at once. It holds only for the same model, language and BIMopen version; otherwise it is dropped by itself, and the file opens the same without it. **Save As ➤ For archiving...** saves a compressed copy without the cache, smaller, to hand over or archive: the copy makes those views again the first time, and the open project stays on its file. **Options ➤ Use the view cache in the project file**, turned off, removes it at the ordinary save too.

Project settings

Project information

Settings ➤ Settings ➤ Project information sets the project's name, number, client, address, issue date and status: the title blocks of every sheet show them.

Project units

Settings ➤ Settings ➤ Project units lists the quantities (length, area, volume, angle, slope and the others) with a sample of each, and the **Decimal symbol/digit grouping**. **Format...** of a quantity sets its unit, its **Rounding** (a number of decimals, or **Custom** with an **Increment factor**, such as 5 mm), its **Unit symbol**, **Suppress trailing zeros**, the plus sign for positive values and digit grouping.

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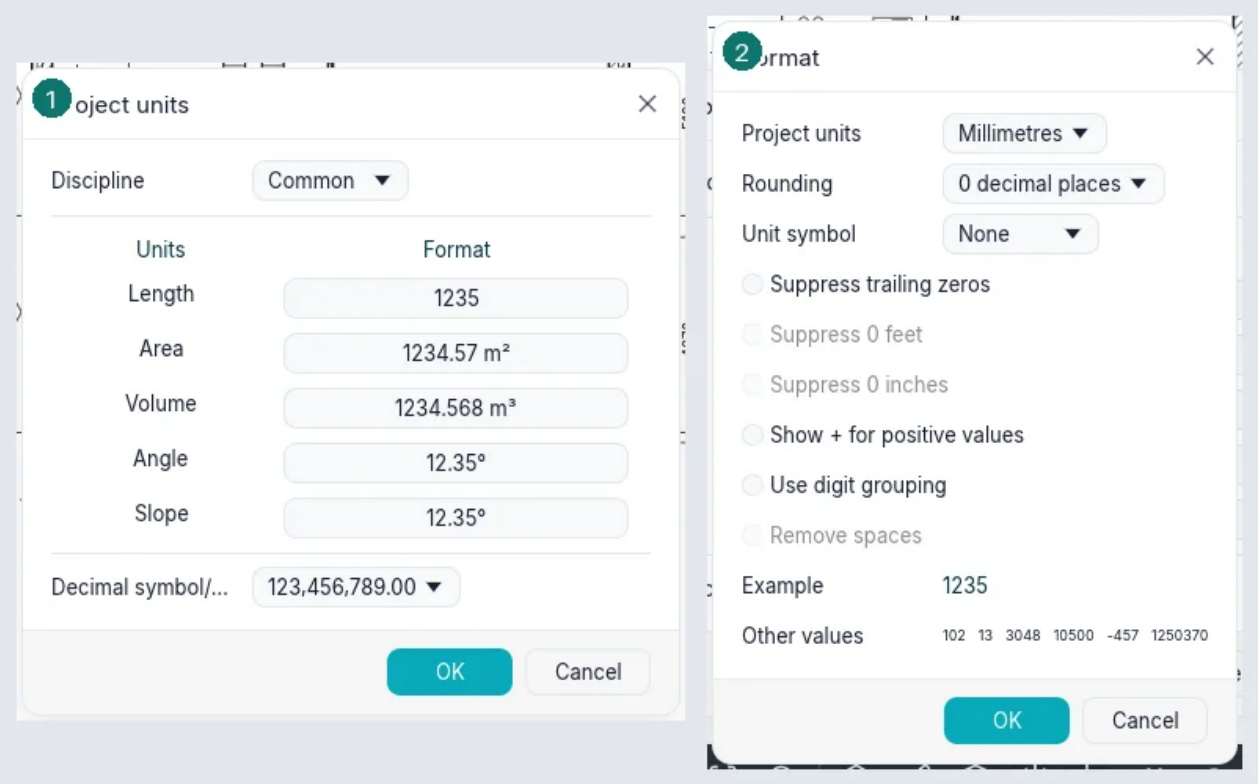

Project units, then Format... of a quantity

Imperial units

In **Format...** the units of a length are in two groups, **Metric** and **Imperial**: **Feet and fractional inches** (`5' - 6 1/2"`), **Fractional inches** (`66 1/2"`), **Decimal feet** and **Decimal inches**. Areas can be in **Square feet**, **Square inches** or **Acres**, volumes in **Cubic feet**, **Cubic inches** or **Cubic yards**, slopes as **Rise/12"** or as a ratio. With fractions, **Rounding** is to the nearest inch or to the nearest 1/2" to 1/256"; a value that rounds up to 12" is carried into the feet: at 1/16", `11 63/64"` reads `1' - 0"`. **Suppress 0 feet** writes `4"` for `0' - 4"`, **Remove spaces** writes `1'-2"` for `1' - 2"`, and **Unit symbol** chooses between `'` and `ft`, or `ft²` and `SF`. **Other values** shows the format on a few more values.

A schedule field's **Field Format...** and a dimension type's **Units Format** have the same fields, besides **Use project settings**: a takeoff can give volumes in cubic yards in a project in cubic feet, and a dimension type for details can round to 1/32" where the plans round to 1/16".

The model is kept in metres: changing the units changes how values are shown and typed, never the geometry. Switching a project from metric to imperial and back leaves every element where it was.

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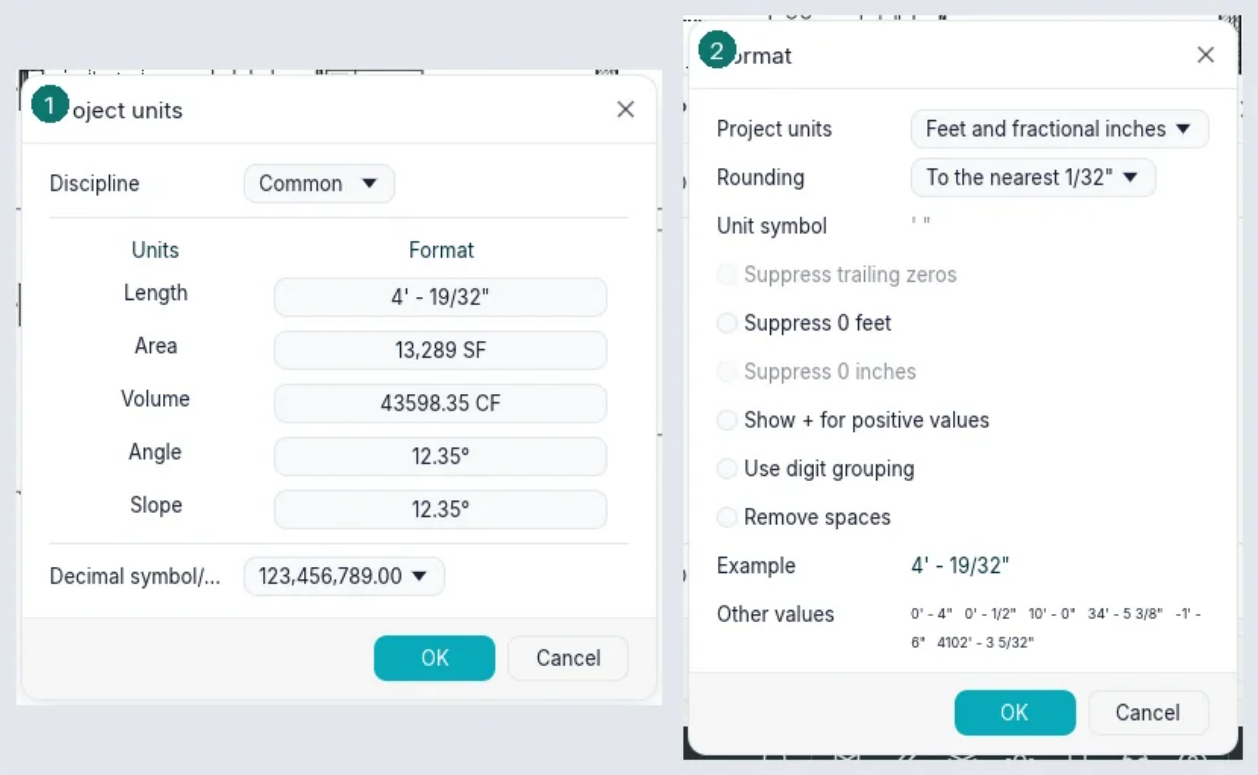

Project units in feet and inches

In a project in feet and inches the view scales are the architectural ones, from `12" = 1'-0"` to `3/64" = 1'-0"`, and the engineering ones, from `1" = 10'-0"` to `1" = 100'-0"`: in the view bar, in the Properties and on sheets. A scale that is neither shows as a ratio.

Scale	Ratio	Typical use
`12" = 1'-0"`	1:1	full-size details
`6" = 1'-0"`	1:2	complex joints and nodes
`3" = 1'-0"`	1:4	masonry and joinery details
`1 1/2" = 1'-0"`	1:8	component and window details
`1" = 1'-0"`	1:12	standard construction details
`3/4" = 1'-0"`	1:16	structural nodes
`1/2" = 1'-0"`	1:24	wall sections and partial elevations
`3/8" = 1'-0"`	1:32	kitchen and bathroom plans, enlarged details
`1/4" = 1'-0"`	1:48	the architectural standard for plans and sections
`1/8" = 1'-0"`	1:96	the engineering and MEP standard, general plans
`3/32" = 1'-0"`	1:128	large commercial plans
`1/16" = 1'-0"`	1:192	key plans and general layouts
`3/64" = 1'-0"`	1:256	very large site plans
`1" = 10'-0"`	1:120	civil engineering and site plans
`1" = 20'-0"`	1:240	subdivision plans
`1" = 30'-0"`	1:360	civil works and utilities
`1" = 40'-0"`	1:480	topography and road surveys
`1" = 50'-0"`	1:600	urban planning and context plans
`1" = 60'-0"`	1:720	territorial surveys
`1" = 100'-0"`	1:1200	cadastral maps

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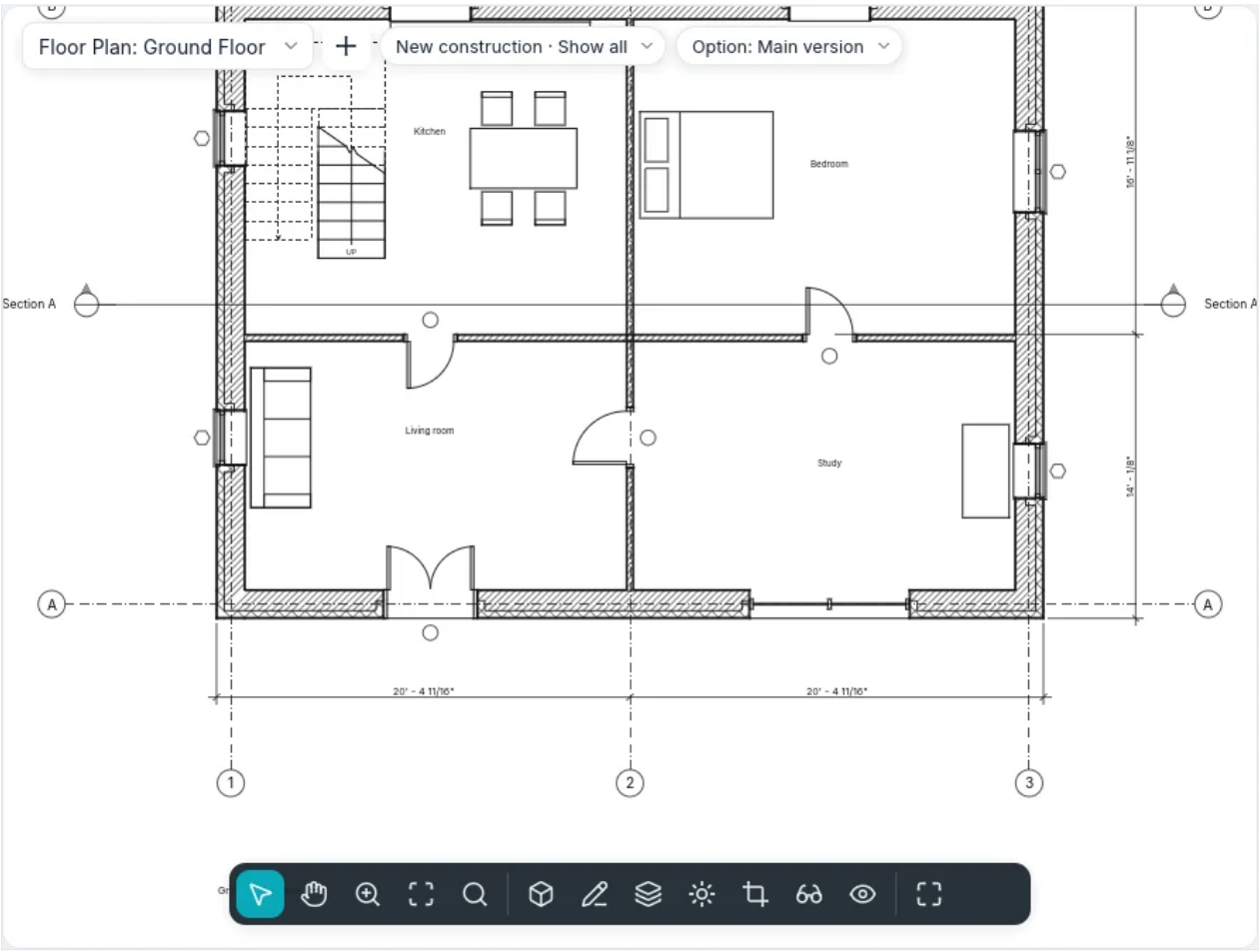
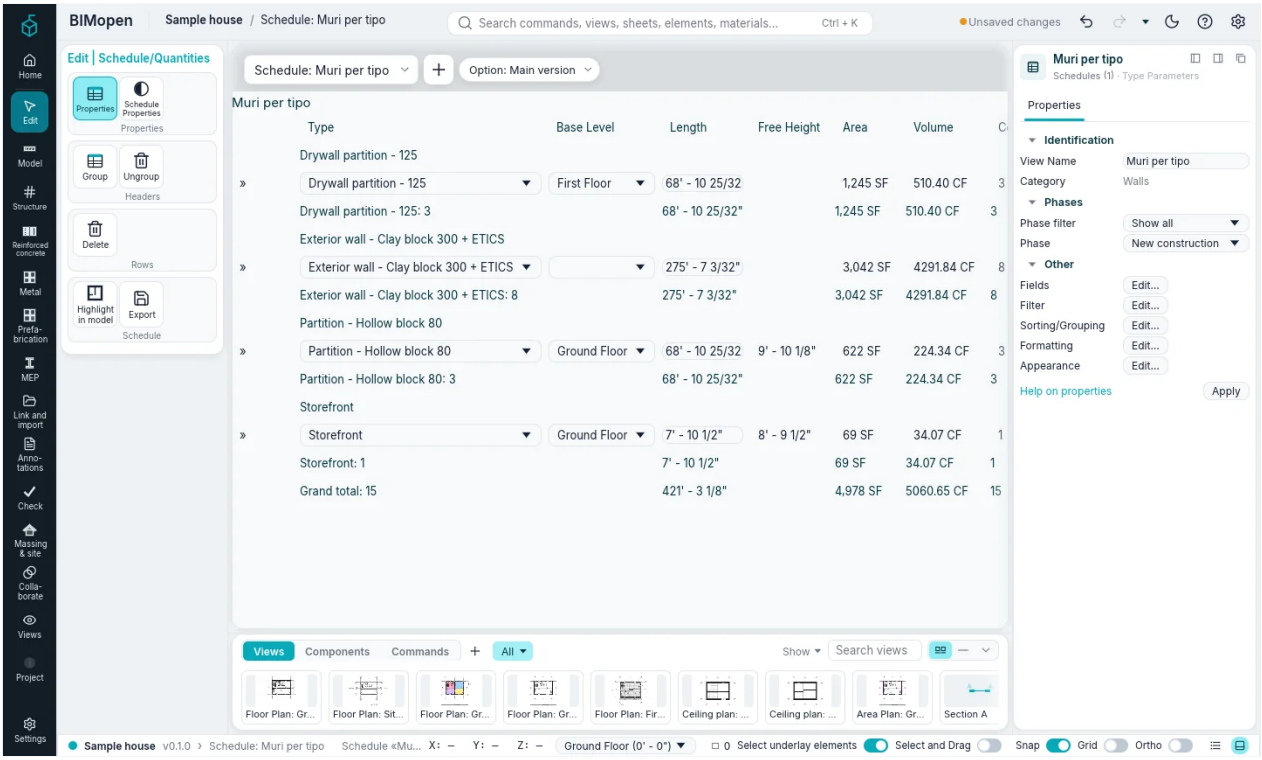


A plan at 1/4" = 1'-0", dimensioned to 1/16"



A schedule in feet and inches

The **BIMopen imperial template** has the same set of types as the metric one, in US trade sizes: walls of 2x4 and 2x6 wood studs, of 8" blocks and brick, of metal studs; doors from 30" to 36" x 80"; windows with a 3'-0" sill; stairs with risers of at most 7 3/4" and treads of at least 10"; railings 36" and 42" high. Lengths and dimensions are rounded to 1/32", the length dimension

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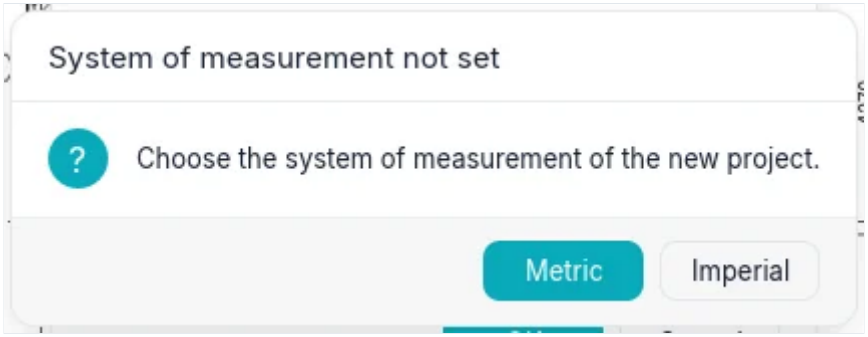
snap increments are 4', 1', 6" and 1", the work plane grid is 4', plans are at 1/8" = 1'-0" and text is 3/32" and 1/8" high. These are starting points: each one can be changed in the project. Changing the units of a project does not rename its types.

Tip: to keep drawings readable on site, give the dimension types a coarser rounding: in a dimension type's **Type Parameters**, **Units Format** opens **Format**; untick **Use project settings** and round to 1/16" or 1/8".



A new project from no template asks for the system of measurement

Typing lengths

A length typed in the Properties, on a temporary dimension, in the options strip, in a dialog or at the command line is read this way:

- with a unit, in that unit, in any project: `300mm`, `1.2 m`, `1 m 20 cm`, `6"`, `5.5'`, `5.5 ft`;
- feet and inches: `5'6"`, `5' 6 1/2"`, `5'-6 1/2"`; after the feet the last `"` can be left out (`5'6`), and inches from 12 up are carried into the feet (`5' 14"` is `6'-2"`);
- a number without a unit is in the project's unit of length; in a project in feet, two numbers are feet and inches: `8 6` is `8'-6"`, `8 6 1/2` is `8'-6 1/2"`;
- fractions are of an inch; a fraction of a foot is written as a decimal (`5.5'`);
- after `=`, a formula: `=10' + 6"`, `=12'/5`, `=3000mm/2`; its numbers without a unit are in the project's unit.

The decimal symbol is the project's. While a value is being edited it is shown at full precision (to 1/256" in feet and inches), so confirming it unchanged changes nothing. At the command line a number without a unit is in metres in a metric project and in feet in a project in feet and inches; a length can always carry its unit: `wall 0,0 20',0`.

Sun Settings

Settings ➤ Additional settings ➤ Sun Settings opens the sun settings of the active view, the same as the sun button of the view control bar (see Views and graphics).

Options

File ➤ Options has four tabs; the choices are kept between sessions.

Tab	What it sets
General	The save reminder interval
User Interface	The theme (**Light** or **Dark**), the language, the interface style and size; **Enable Recent Files page at startup**; in the **Classic** style, the ribbon tab after a selection ends or a tool closes (**Stay on the Modify tab** or **Return to the previous tab**) and **Display the contextual tab on selection**; **Tooltip Assistance** (**None**, **Minimal**, **Normal**, **Extra High**)
Graphics	**Hardware acceleration**, **Graphics card** and the **In use** line, **3D view ▸ Edges**, **Use anti-aliasing for 3D views**, **Invert background color**, **Selection color**, **Highlight color**, **Temporary Dimension Text Appearance** (size and background)
File Locations	**Default template file path** (the template of **New**) and **Default path for user files** (where the file dialogs open)

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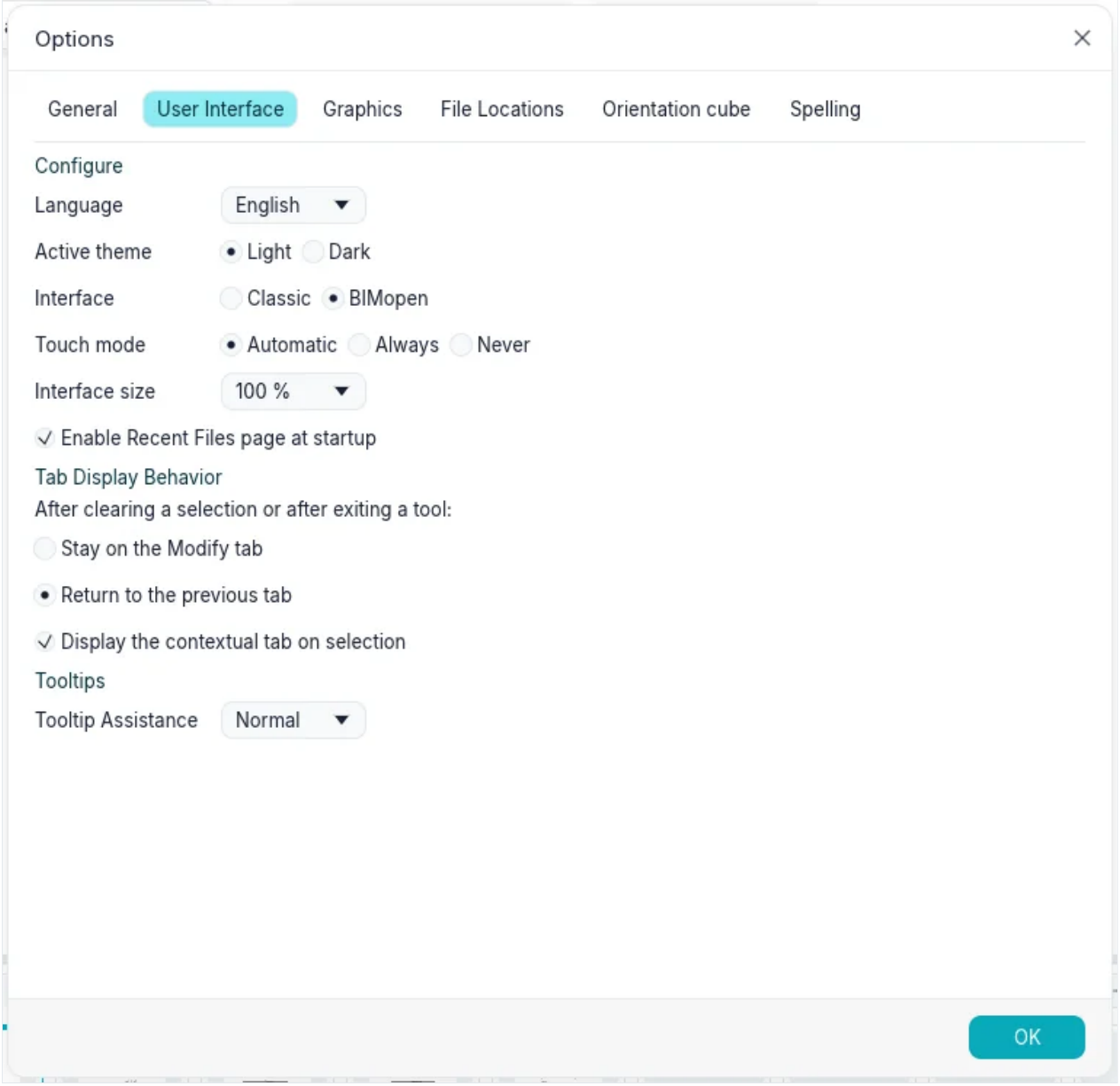

Options, User Interface tab

Graphics: graphics card and edges

- **Hardware acceleration**: **Automatic** uses the graphics card where there is one and, if it is missing or does not start, draws on the processor; **Always on** never falls back to the processor (without a graphics card the 3D views say so instead of drawing: useful to find a faulty driver); **Off (processor only)** draws the 3D views on the processor, for faulty drivers (black screen, flickering), more slowly on large models.
- **Graphics card**: **Automatic** (the dedicated one, if there is one), the dedicated or the integrated card, by name; on laptops the integrated one uses less power. The card is chosen when BIMopen starts: after a change the tab says to restart BIMopen. A card that cannot draw in the window is greyed, with the reason; if the chosen card is missing or cannot draw, BIMopen uses the automatic one, says so and goes back to **Automatic**. In a browser and on Android the browser or the system chooses it.
- **In use** shows the card and the graphics API really in use (for example Vulkan, Direct3D 12, Metal, WebGPU, WebGL2), or the processor and why.
- **3D view ▸ Edges**: **Automatic** edges follow the view instantly, also right after a change; **Exact** edges are the same as in print, slower on large models. Printing and export always use the exact ones, as do the **Wireframe** style, **Sketchy Lines** and views without acceleration or on WebGL2. Each 3D view also has its own **Edges** property: **As in Options**, **Automatic** or **Exact**.

At the command line: `graphicsettings {"acceleration":"automatica|sempre|disattivata","adapter":"automatica|dedicata|integrata","edges":"automatici|esatti","antialias":true}` (without arguments it reports the settings in use); a 3D view's edges with `graphicoptions {"edges3d":"opzioni|automatici|esatti"}`.

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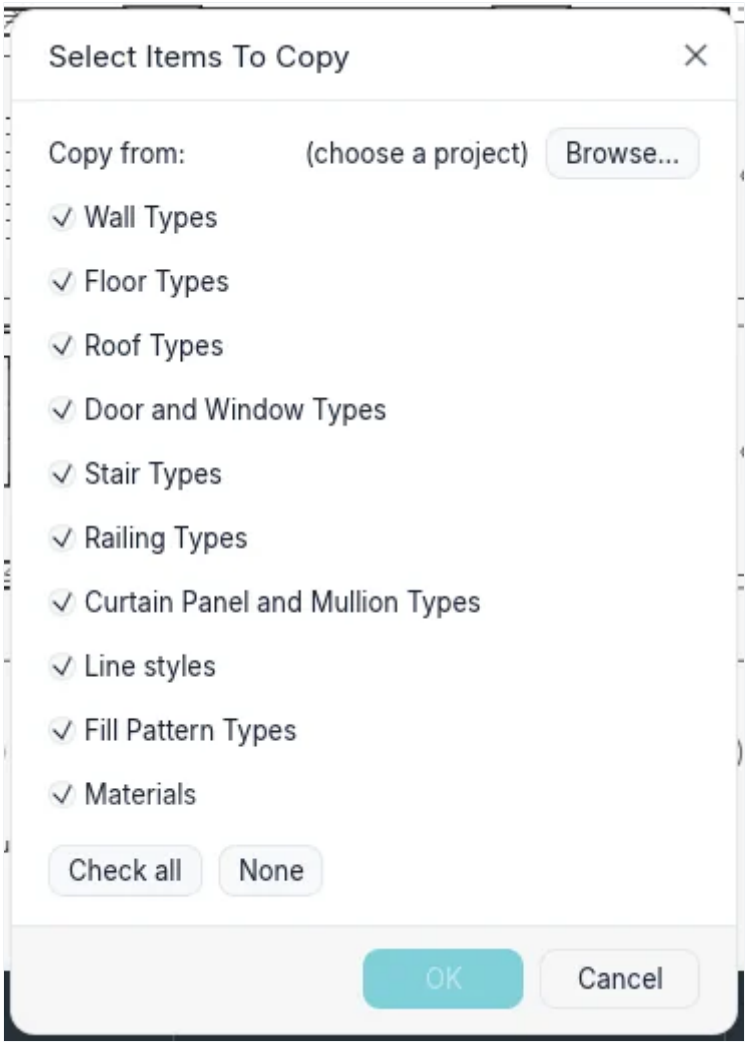
Copying settings from a project

Settings ➤ Settings ➤ Copy Settings from Project copies settings from another project into this one. In **Select Items To Copy**, choose the project with **Browse…** (**Copy from:**), then tick what to copy: the types of walls, floors, roofs, doors and windows, stairs, railings, curtain panels and mullions, line styles, filled region types, materials, fill and line patterns, line weights, styles by category, view templates, filters, viewport types, title blocks, the project units, the print settings and the project information (**Check all**, **None**).



Copying settings from another project

What is copied brings along what it uses: a wall type its materials, a material its patterns, a view template its filters. When names are already in the project, **Duplicate Types** asks whether to **Overwrite** them or to transfer the **New Only**. The whole transfer is one step of undo.

Exchanging files

Exporting to IFC

File ➤ Export ➤ IFC… writes the model in IFC4: walls with their axis and body, openings, doors, windows, floors, roofs, stairs, railings, curtain walls, furniture and rooms as spaces, on their storeys.

Export ➤ Options ➤ IFC Options… opens **IFC Export Classes**: the IFC class of each category; **Not Exported** leaves a category out, and **Default** goes back to the standard classes. The classes are kept with the project.

Opening IFC files

File ➤ Open… ➤ IFC… makes a new project from an IFC2x3 or IFC4 file, in metres or millimetres: storeys become levels, walls are rebuilt from their axis (or from the long side of their body), slabs become floors, doors and windows go in the walls their openings cut, and spaces become rooms. The messages count what was not read.

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Open ➤ IFC Options… opens **Import IFC Options**: what each IFC class becomes (walls, floors, doors, windows, rooms or **Not Imported**), and **Default**. The choices are kept between sessions.

Objects with a shape that are not walls, floors, doors, windows or spaces are kept as imported objects, to convert: see Importing and converting.

Exporting to USD

File ➤ Export ➤ USD… saves the active 3D view as a USD scene, an open format read by rendering, animation and augmented reality programs. When the active view is not 3D, **{3D}** is exported. Elements and categories hidden in the view are left out, and with the section box on, so are the elements wholly outside the box. The scene is arranged by level, category and element, in metres with the Z axis up.

- **.usdz** is a single file holding the scene and the materials' textures; **.usda** is the scene as text, without textures.
- **Materials**: colour, roughness, metallic, transparency and the appearance's images, as in the **Realistic** view.
- **BIM properties**: category, type, level, mark, IFC identifier and parameters of every element.
- **Identical shapes as instances**: identical doors, windows and components are written once.
- **The view's camera** and **The view's sun and sky**.
- **Naming**: **Manual** uses the name typed in **Name/prefix** (or the project's), **Automatic - long** adds the view's type and name to the prefix, **Automatic - short** uses only the view's type and name.

The choices stay for the next time.

Project bundles

File ➤ Export ➤ Share ➤ Project bundle… gathers the project and the files it uses into a single **.zip** file: linked models (with the files they use), the source files of imported objects, type catalogues. The dialog lists every file with its state: **not found** when it is missing beside the project, **to add** on the web, where the browser cannot see the folder. **Add files…** puts other documents in the bundle, such as a report or a specification. The bundle takes the project as it is now, unsaved changes included. Components and material images are already inside the project.

Inside the .zip all files sit at the same level and the project refers to them by name, so the bundle works wherever it is unpacked. The **Bundle report.txt** says what was included and what was missing. **Purge unused types**, **Remove worksets** and **Remove sheets** change only the copy in the bundle. With **Save settings for next use** the choices stay for the next time.

File ➤ Open… ➤ Project bundle…, or **Project…** choosing the .zip, unpacks the bundle into a folder beside it with the same name and opens the project. On the web the project opens from memory.

Other exports

Printing to PDF and exporting images are described in Sheets, schedules and printing; schedules export as delimited text from the same chapter. `bimopen-cli convert` exports from the command line (see Command line and MCP).

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Importing and converting

A model made with other software arrives as an IFC file. BIMopen reads its walls, floors, doors, windows and spaces as its own elements. Everything else with a shape (roofs, beams, columns, stairs, railings, curtain walls, furniture, fittings) is kept as an **imported object**: its geometry as it was read, as a generic model. **Convert** then makes each imported object a BIMopen element of any category, when and as far as you need it.

Opening an IFC file

1. Open the **File** panel (BIMopen's mark at the top of the rail) and choose **Open… ➤ IFC…**.
2. Pick an IFC2x3 or IFC4 file.

BIMopen makes a new project: storeys become levels, walls are rebuilt from their axis, slabs become floors, doors and windows go in the walls their openings cut, spaces become rooms. Every other object with a shape becomes an imported object. Each one remembers:

- the file it came from and its format (`casa.ifc · IFC4`);
- its IFC class (`IfcRoof`, `IfcFurniture`…) and its identifier;
- the object it is a part of: the flights of a stair, the panels of a curtain wall;
- its property sets and quantities, shown in **Properties** under **IFC Properties**.

The messages count what was read, what was approximated and what was left out.

Import IFC Options

Coordinate ▸ IFC Options… opens **Import IFC Options**: what each IFC class becomes when a file is opened.

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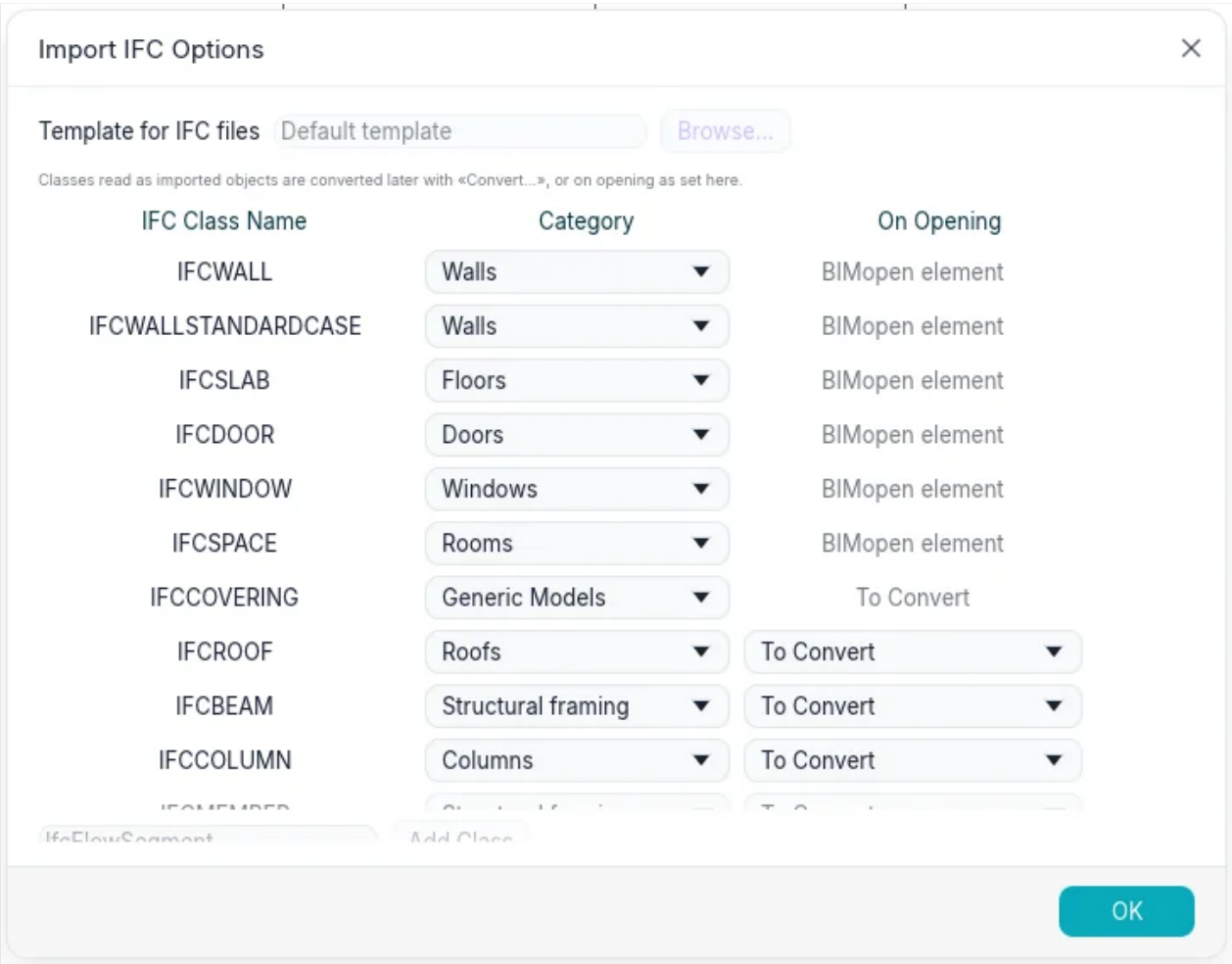

Each IFC class, its category and what happens on opening

- Walls, slabs, doors, windows and spaces are read as **BIMopen elements**; any of them can be read as an imported object instead (**Generic Models**) or left out (**Not Imported**).
- The other classes are read as imported objects. **Category** is the category each class is proposed as when it is converted, any category BIMopen has.
- **On Opening** says what happens to them as the file opens: **To Convert** leaves them for later; **Keep the geometry**, **Rebuild as a BIMopen element** or **Create a component** converts them at once (see below).
- **Add Class** adds a class the list does not have.
- **Load...** reads a class mapping file, **Save As...** writes one: a text file, one class a line. **Default** puts the standard mapping back.

The options are kept between sessions.

Importing an IFC file into the project

Open ➤ IFC... makes a new project. **Link and import ▸ Import IFC** brings the file **into the open project** instead, which already has its levels and types: converting is faster, because **Convert...** proposes the project's types first.

1. **Link and import ▸ Import IFC** (or **File ▸ Open ▸ Import IFC...**, or Ctrl+K, «import ifc»), then pick the file.
2. In the **Import IFC** dialog:
3. **Placement**: **Auto - Origin to Origin** (the usual case), **Center to Center**, or **Manual**: after **Import** you click the point in a plan where the file's origin goes.
4. **Levels**: each storey of the file goes to a project level, by name and then by elevation (±100 mm), or to a new level; the choice can be changed row by row.
5. **Classes to import**: the file's classes with their objects and the category proposed; untick those not needed. Spaces become rooms only if you choose so.
6. **Project types**: **Propose the project's types first** (on) and **Convert at once when the match is high**.

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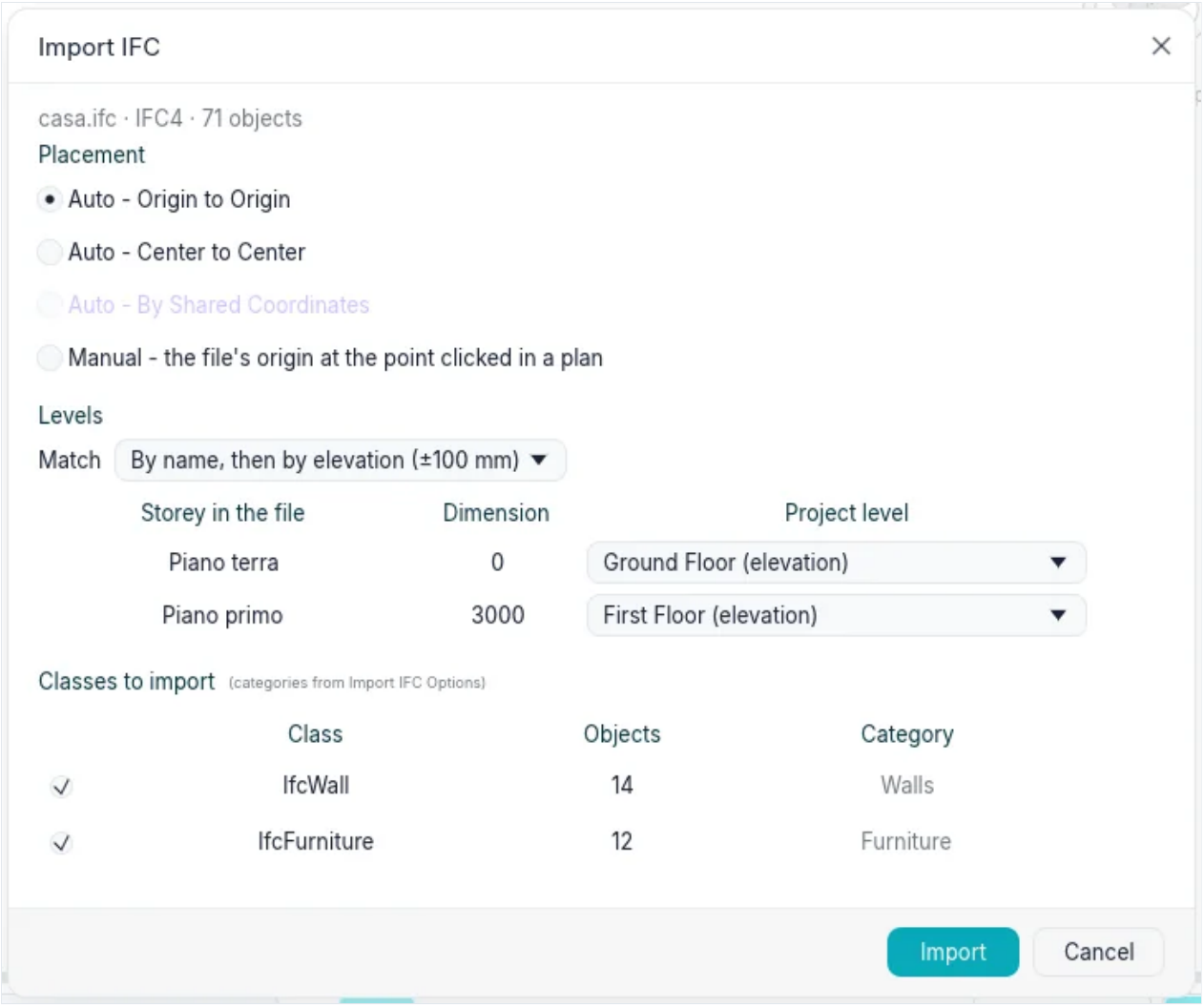
7. **Already imported from this file**: on a second import of the same file, **Update** compares the identifiers (GUIDs): it adds the new ones, updates the changed ones, and reports those no longer there without removing them; or **Import as a copy**.
8. **Import**. It is all one Undo step; **Conversion Status** turns on in the view.



Import IFC: the file's storeys go to the project's levels

Walls, floors, doors and windows come in as imported objects too, to be converted into the project's types. In **Convert…**, **Type** lists the project's types **ranked by match**, the first marked ★: the category, the thickness or sizes (the same within 5 mm) and the name (the same, or words in common). Each row says how well it matches (**high** from 85, **medium** from 60, **low**) and why; the first is already chosen if the match is at least medium, otherwise **New type from object**. Converting a whole class, one row for each type of the file says which project type it becomes.

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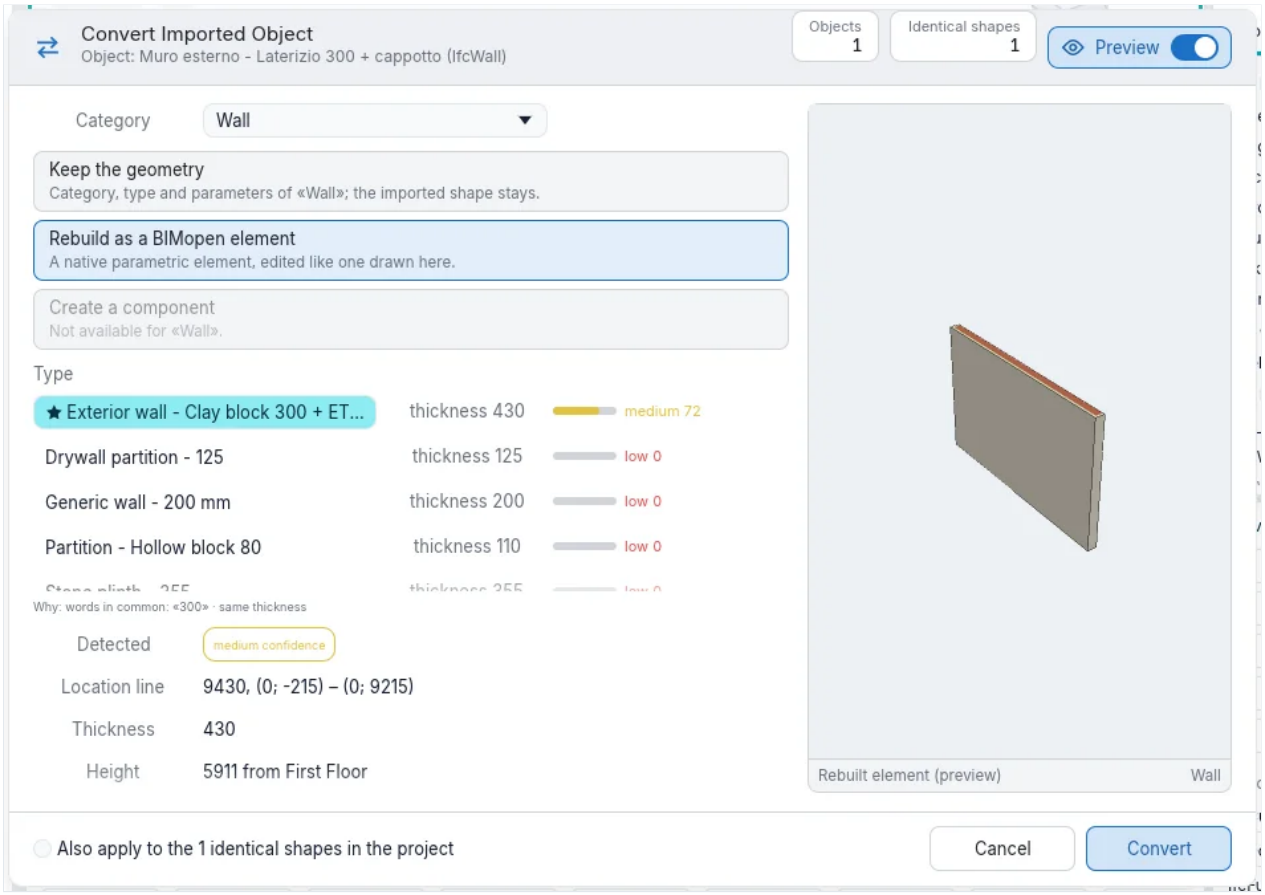

The project's types, the closest first

Conversion Status

While the project holds imported objects still to convert, the view bar shows **Conversion Status**, with their number. Its tooltip says how many there are and in how many IFC classes.

Conversion Status in the view bar

Click it: the view shows each object in the colour of where it stands, inside a teal frame.

Colour	Status	Meaning
Orange	**To Convert**	imported, still a generic model
Yellow	**Category Assigned**	the imported geometry kept, with a category
Green	**Converted**	a BIMopen element or a component instance
Red	**Not Recognised**	a rebuild did not recognise the shape: geometry only
Grey	**Born in BIMopen**	not imported

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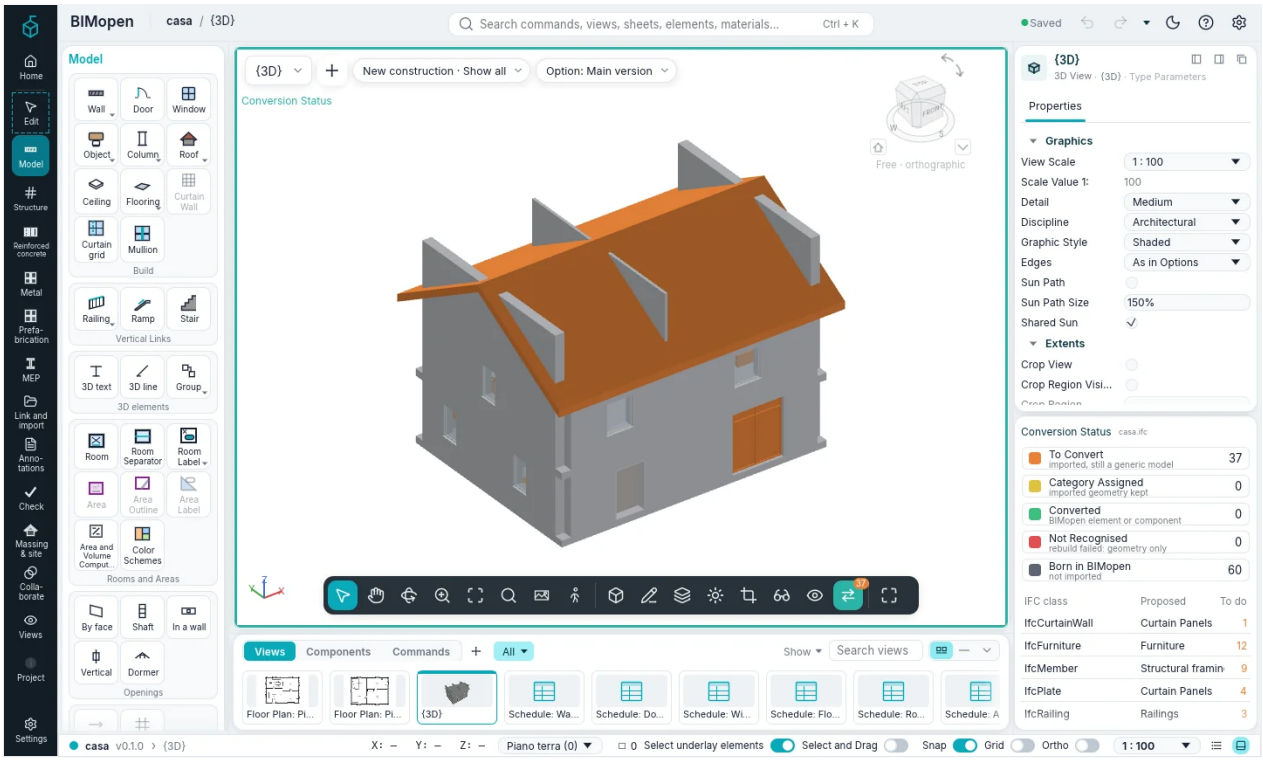
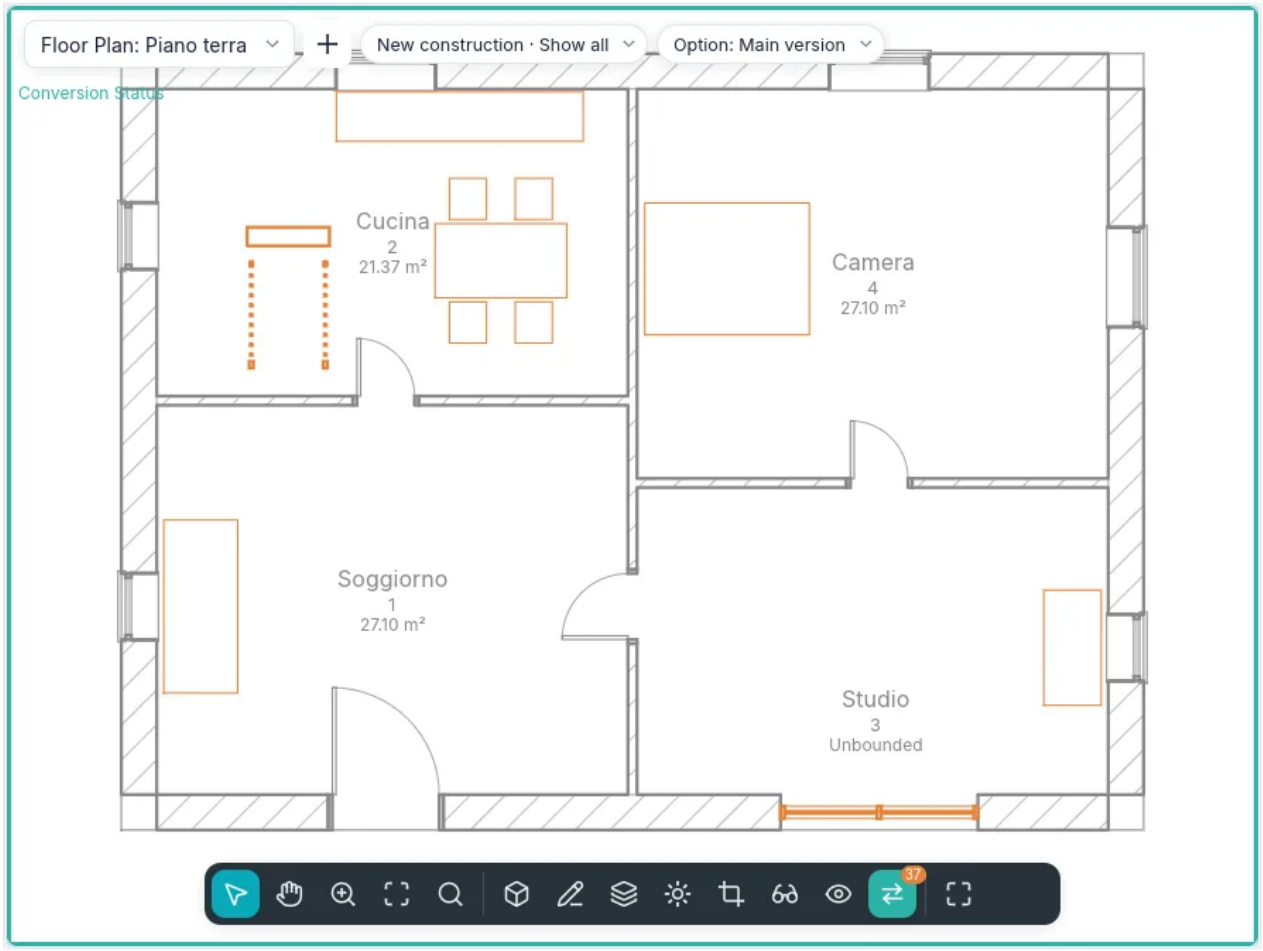


Conversion Status in the 3D view, with its panel

It works in plans, sections, elevations and 3D views, and only changes the window: the model and the other views stay as they are. While it is on, only imported objects can be selected.



The same in a plan

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The panel **Conversion Status**, under **Properties**, has:

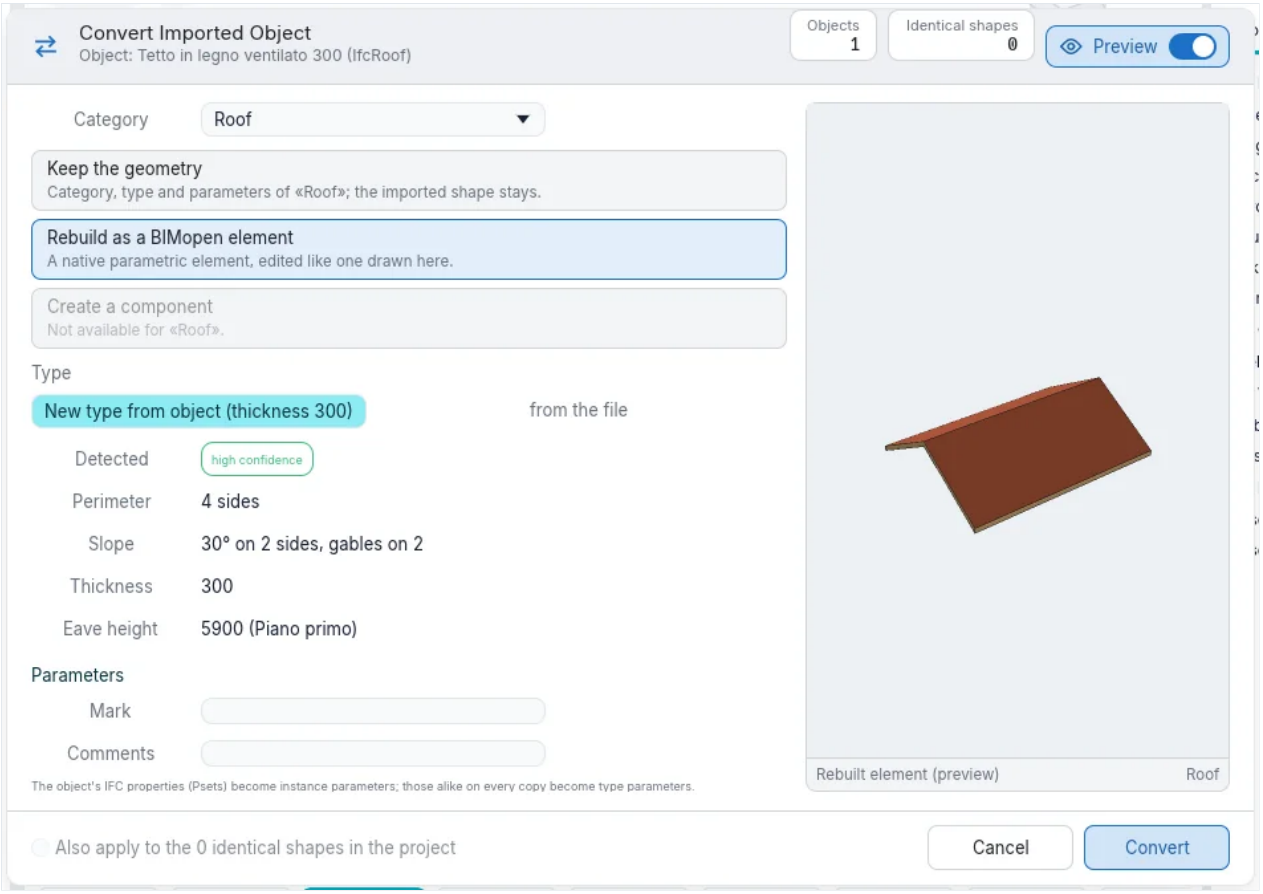
- the legend with the count of each status: a click on a status hides it in the view, or shows it again;
- the IFC classes with objects still to convert, the category each is proposed as and how many are left. A click on a class selects its objects to convert and opens **Convert...**.

When nothing is left to convert, the panel says so. Turn **Conversion Status** off and its button leaves the view bar.

In the Classic style the panel is a tab beside **Properties**; in touch mode it is a sheet, opened from the view's settings.

Converting an imported object

Select one or more imported objects: the contextual card **Edit | Imported Objects** opens. **Convert...** opens **Convert Imported Object**.



The roof, rebuilt as a BIMopen roof

1. **Category**: what the object becomes. The class's proposal comes first.
2. One of the three ways (below). A way that does not apply to the category is greyed.
3. **Type**: one of the category's types (the project's first, ranked by match), or a new type made from the object.
4. **Detected**: what BIMopen reads in the shape and how sure it is.
5. **Parameters**: **Mark** and **Comments**, and the object's IFC properties.
6. **Convert**.

The preview shows what the object will be. The **Preview** switch in the header hides it; the dialog keeps its size.

The three ways

Way	Categories	What happens
Keep the geometry	all	The object takes the category, a type and parameters. It counts in schedules, filters, visibility and the IFC export as an element of that category; its shape stays the imported one. It can be moved, rotated and mirrored.
	walls, floors, roofs, doors, windows	BIMopen reads the shape and makes the native element, edited like one drawn: a wall's axis, thickness and height; a floor's outline, thickness and top; a roof's outline,

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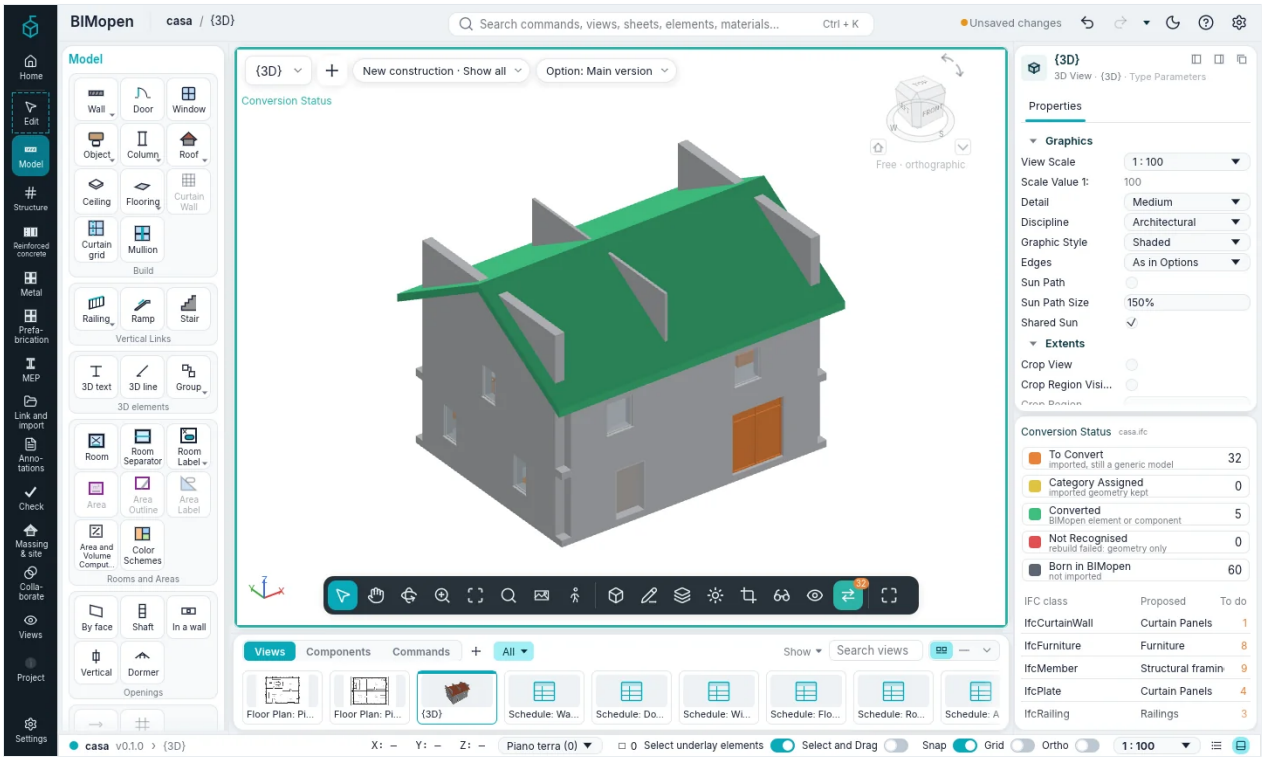
Way	Categories	What happens
Rebuild as a BIMopen element		slopes, thickness and eave height; a door or window in the wall it stands in, with its width, height and sill.
Create a component	furniture, plumbing fixtures, generic models, doors, windows, columns, beams	The geometry becomes a component type with **Width**, **Depth** and **Height** from its overall size; changing them stretches the shape. The object becomes its instance.

A shape that cannot be rebuilt as the chosen category turns red (**Not Recognised**) and stays as it was: convert it again keeping its geometry. Rebuilding columns, beams, stairs and railings comes in a later phase.

When it rebuilds, BIMopen carries the file's materials into the native element: the layers of a wall, floor or roof with their thickness, the frame and glass of a window. A material the project already has with the same name is used as it is; a new one takes the file's colour and the hatch patterns of a similar material. A wall that turns a corner becomes one wall per stretch; a roof with planes at different heights, one roof per plane. The covering, ridges and gutters on a roof go with the new roof, and the walls the roof cuts are attached to it. **Restore Original** puts everything back as it was.

Category in **Properties** does the same as **Keep the geometry**: choosing a category there converts the selected object at once.



Converted objects turn green

Components from imported objects

Create Component on the contextual card opens the dialog with **Create a component** chosen.

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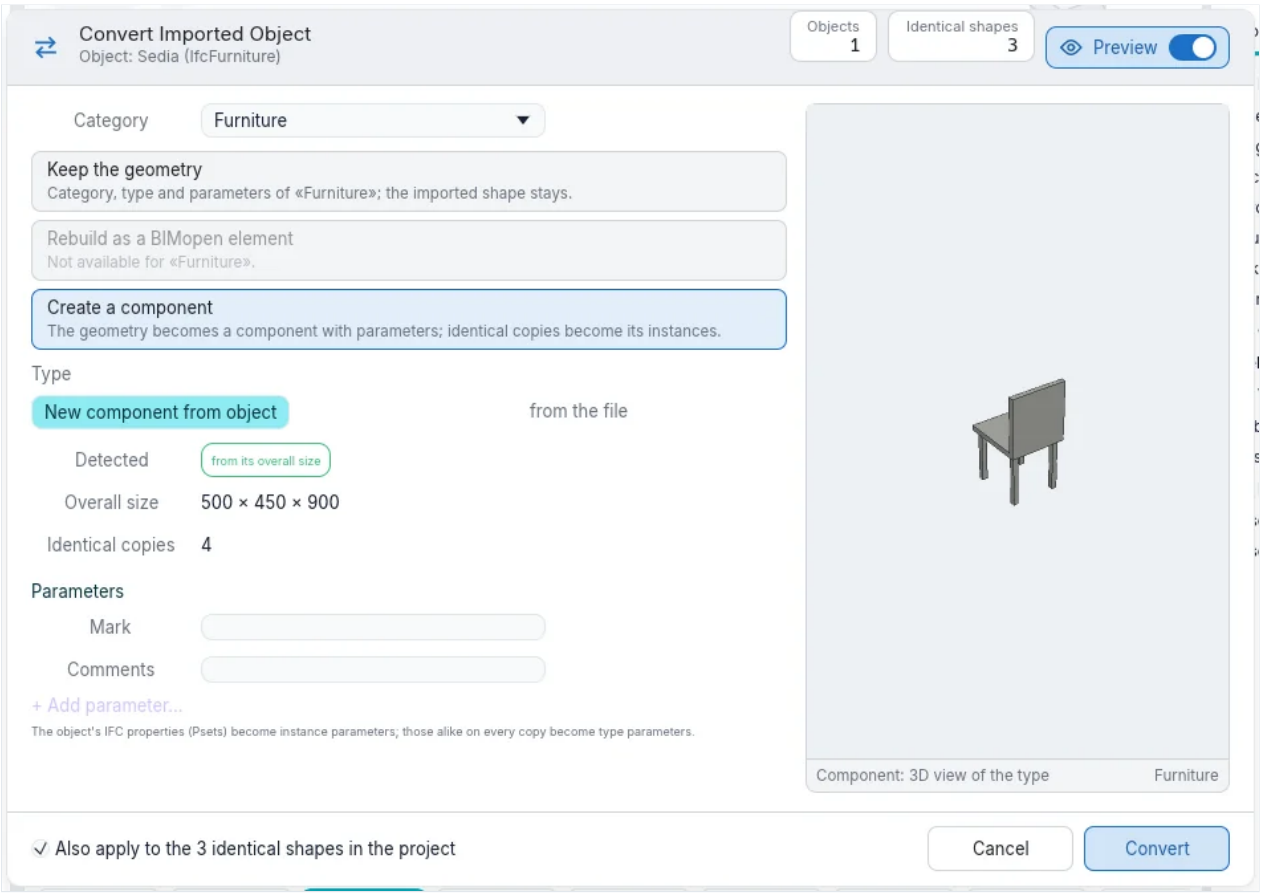

A chair becomes a component; its three copies become instances

The IFC properties that every copy has alike become type parameters; those that differ stay with each instance. The component appears in the Project Index under **Imported Objects**, and its **Type Parameters** change every instance at once.

Identical shapes

Copies of one shape, moved and rotated in plan, are recognised as identical. **Select Identical Shapes** selects them; in the dialog, **Also apply to the N identical shapes in the project** converts them together (on by default when making a component). As instances of one component they take less memory and draw faster on large models.

Selecting imported objects

- **Select Class** selects the imported objects of the same IFC class.
- A class in the panel **Conversion Status**, or **Convert All (N)...** on a class in the Project Index, selects its objects to convert and opens the dialog.
- An object that is a part of another one (a flight of a stair) goes with it: converting the whole takes its parts.

Imported objects in the Project Index

Imported Objects in the Project Index lists each file read, with its format and how many objects are left to convert; under it the IFC classes, under each class its objects, with the colour of their status and, once converted, the category they became. Objects made of parts list their parts under them.

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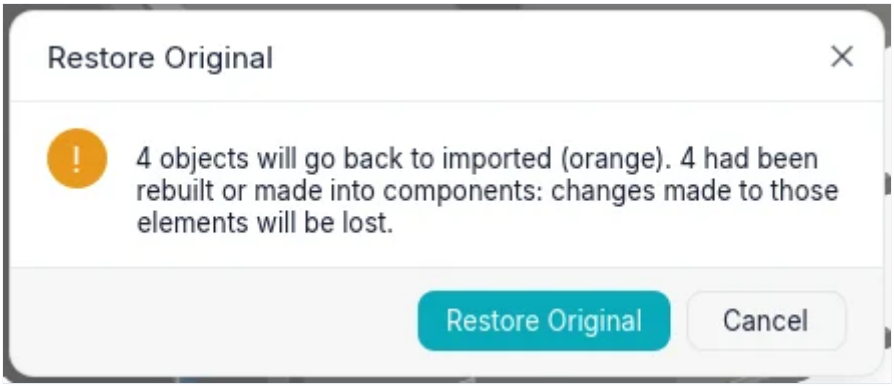
Imported Objects in the Project Index

- **Only to convert** hides what is already converted.
- A click on an object selects it; a double click shows it in a view.
- The right button on a class: **Convert All (N)...**, **Select All to Convert (N)**, **Restore Originals (N)**.
- The right button on a file: the same, and **Remove Source References...**.
- The right button on an object: **Convert...**, **Select Identical Shapes**, **Show in View**, **Conceal in view**, **Restore Original**.

Restoring the originals

What a conversion replaces is kept in the project, in the `.bpr` file: the original object, with the parts it took along. **Restore Original** (on the contextual card, in the Project Index or in the right button's menu) puts it back as it was imported, orange again.



Restoring asks first

It asks first, and says what will be lost: changes made to rebuilt elements or components, and doors and windows hosted in converted walls. A component made by the conversion goes with its last instance. Undo takes back a conversion or a restore like any other change.

The originals' geometry is stored once for every identical mesh, so keeping them costs little. When the conversions are final, **Remove Source References...** on the file in the Project Index deletes the originals: after it, **Restore Original** is no longer possible.

Coming next

- **Link IFC**: the file as a read-only reference, reloaded when it changes; placement by shared coordinates.
- **Update from File**: reading the source file again and updating its objects.
- Tools by face, to place walls, floors and roofs on free and curved forms.

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- Rebuilding columns, beams, stairs and railings.
- Converting objects from OBJ and glTF files with the same dialog.

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DWG and DXF

BIMopen writes and reads AutoCAD drawings, DWG and DXF, of every release from 2000 to 2018. A view or a sheet goes out as a drawing with a layer per category; a drawing comes in as a **CAD file** that stays one element of the project, imported or linked.

Exporting a view or a sheet

File ▸ Export ▸ DWG / DXF... opens **Export CAD Formats**:

Option	What it does
Name	The file's name, the view's or sheet's by default
Current view or sheet / **Selected views/sheets**	The open view or sheet, or those ticked with **Select...** (a file each, named after them)
Format	DWG or DXF
Version	AutoCAD 2000, 2004, 2007, 2010, 2013 or 2018
One DWG unit is	millimetre (the default), centimetre, metre, inch or foot
Layers and properties	**Category on layer, overrides by entity** (the default: a colour other than the layer's stays on the object), **All on layer, no overrides**, or **All on layer, new layers for overrides** (a layer `NAME-RRGGBB` for each other colour)
Colours	**RGB as in the view**, or **Index colour (255)**: the nearest colour of the 255-colour table
Coordinates	**Project Internal**, or **Shared**: plans and views with their origin at the survey point and their axes on true north (only when the project has shared coordinates; sheets keep paper coordinates)
Naming	**Automatic: short** (the view's name) or **Automatic: long** (the project's name before the view's)

What the view draws goes out: cut walls and the other elements as closed polylines with their solid fill, lines, arcs, circles and texts (dimensions and tags as texts and lines). Each element goes on the layer of its category, after the AIA names (A-WALL, A-DOOR, A-GLAZ, A-FLOR, S-COLS, S-GRID, A-ANNO-DIMS, A-ANNO-TEXT...), fills on the same name with `-PATT`; the colours are the ones of the view. Plans, sections, elevations, drafting views and sheets can be exported. On the web the file is downloaded.

The choices stay for the next export of the same session.

At the command line: `export pianta.dwg {"version":"2013","unit":"mm","layers":"bylayer","colors":"aci","coordinates":"shared"}`.

Importing or linking a drawing

Insert ▸ Import CAD and **Insert ▸ Link CAD** (in the BIMopen style: **Import CAD...** and **Link CAD...** in the File panel and among the commands) ask for a DWG or DXF, then show its options:

Option	What it does
Current view only	The drawing belongs to the open view only (a plan, a section, a 2D view)
Colours	**Invert** (light lines drawn dark: for drawings made on a black background), **Preserve**, **Black and white**
Layers	**All**, **Visible** (the layers off in the file are left out), or **Specify...**: the layers to import are ticked
Import units	**Auto-detect** (the unit the file declares, else millimetres), mm, cm, dm, m, inches, feet, or **Custom**: how many metres one unit of the file is
Positioning	**Auto – Center to center** (the drawing's centre on the centre of the level's walls), **Auto – Origin to origin**, or **Auto – By shared coordinates** (the drawing's origin on the survey point, its axes on true north)
Place at	The level whose plans show the drawing
Correct lines that are slightly off axis	On (the default): lines within 0.1° of the axes become horizontal or vertical
Orient to View	On (the default): the drawing follows the view's axes; off, its north is the project's true north

What is read: lines, polylines with their arcs, arcs, circles, ellipses, splines, texts, solid hatches and blocks (expanded) of model space; heights are ignored, the drawing is 2D.

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The CAD file in the project

A CAD file is one element, named after its file: it is selected, moved, turned, copied and deleted as the others, and its lines are snap points. It is drawn in the plans of its level, or only in its view. Its **Properties** show its level, its colours and, linked, its path, and its **layers**, each with a tick that shows or hides it.

The contextual panel **Import Instance**:

- **Explode** (imports only): its lines, texts and solid hatches become detail lines, text notes and filled regions of the open view;
- **Delete Layers**: the layers ticked leave the project with their lines;
- **Reload** and **Manage Links** (links only).

Properties also change the drawing's **Scale Factor** and **Rotation**. **Explode** takes drawings of up to 10,000 objects.

Manage Links

Insert ▸ Manage Links lists the linked CAD files with their state (Loaded, Not loaded, Not found), their saved path and their objects. **Reload** reads the file again from its path (the hidden layers and the position stay), **Reload From...** takes another file, **Unload** keeps the link without drawing it, **Import** turns it into an import and **Remove** deletes it. On the web there is no path: **Reload** asks for the file again.

A linked drawing is kept in the project too: the project opens without the file.

The list also shows each file's size.

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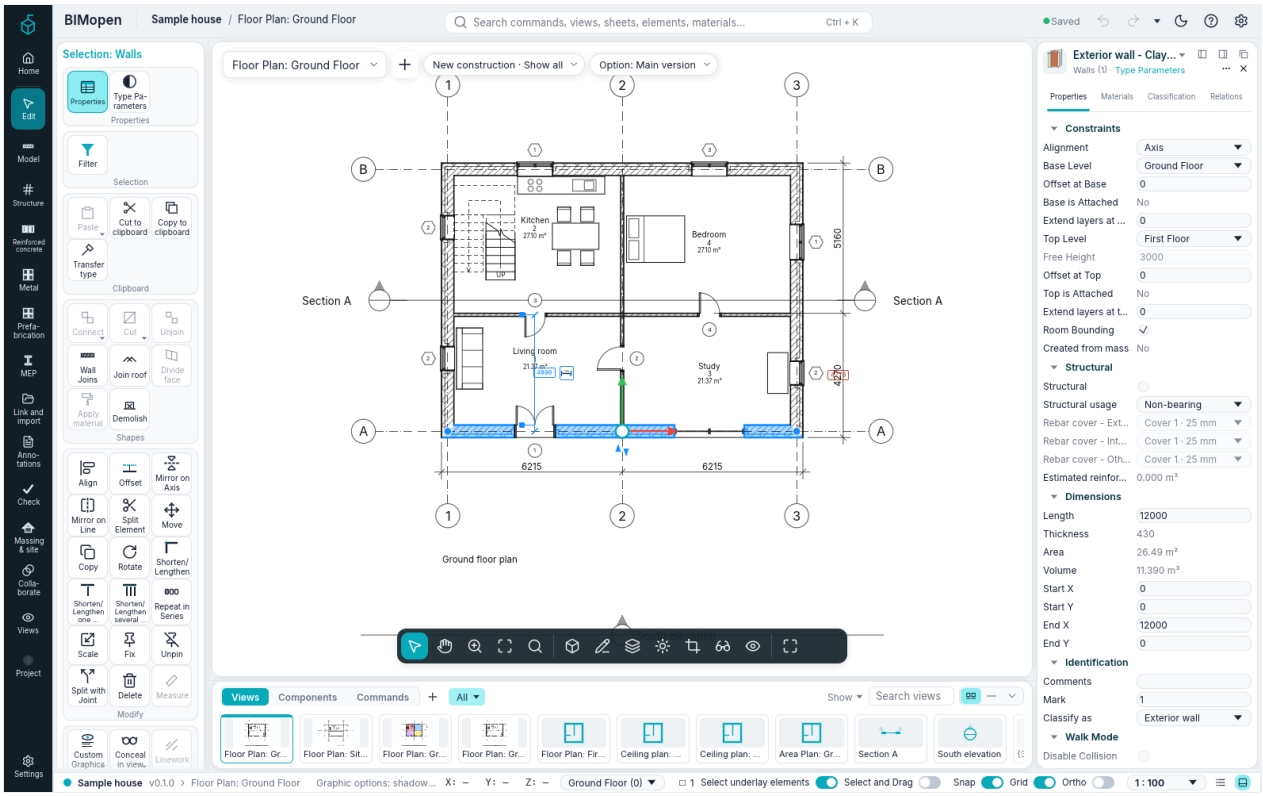
Selecting and modifying

Before most changes you select what to change. This chapter describes how elements are highlighted and selected, how a selection is refined, how selected elements are dragged, and the tools of the **Modify** group. Everything here works the same in the desktop app and in the browser.

Highlighting and selecting



A wall selected in the plan: its contextual card on the left, its properties in the element panel on the right

Move the cursor over an element: its lines are drawn wider, the status bar at the bottom of the window names it (category and type, as **Walls : Exterior wall...**), and after a moment the same words appear under the cursor. Click to select the highlighted element. A selected element is drawn in the selection colour in every view where it shows, and the count at the right end of the status bar says how many elements are selected.

A right click on an element selects it and opens the context menu. `Esc`, or a click on an empty spot, clears the selection.

To...	Do this
add an element to the selection	`Ctrl` + click
take an element out of the selection	`Shift` + click
select what is wholly inside a window	drag from left to right
select what a window touches	drag from right to left (the window is dashed)
add, or take away, what a window catches	drag with `Ctrl`, or with `Shift`
select again what was selected before	`Ctrl+Left`, or right click ▸ **Select Previous**

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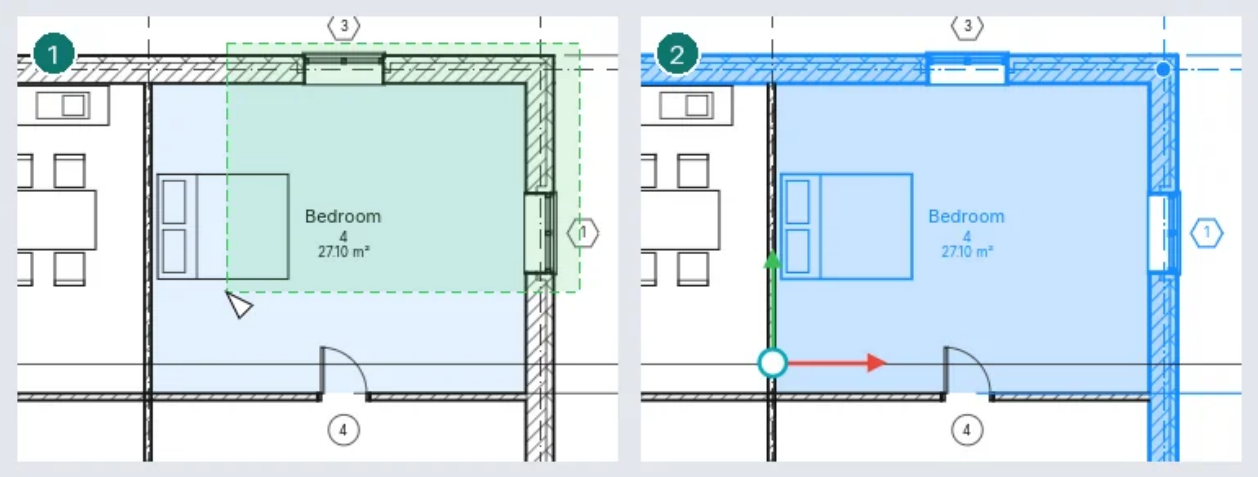


A window from right to left selects what it touches

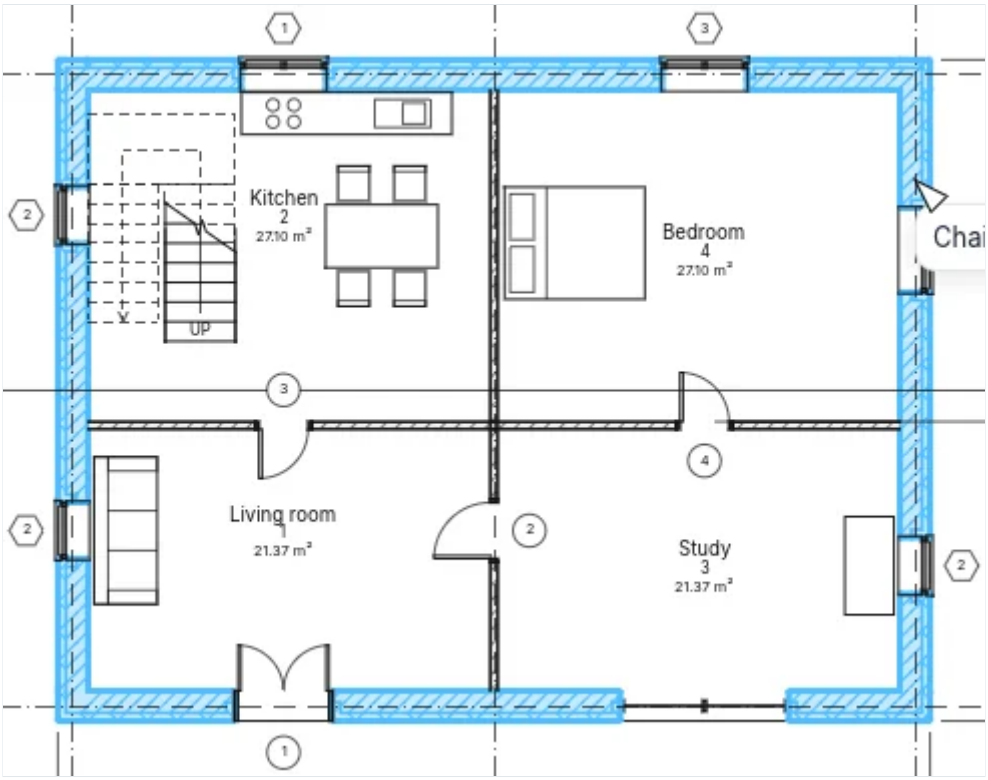
Windows work in plans, sections, sheets and 3D views. In a 3D view the left button selects; the right button orbits and the middle button pans (see The interface).

Elements close together: TAB

When elements lie close together or one over the other, the one highlighted may not be the one you want. Keep the cursor still and press `Tab`: the next element under the cursor is highlighted and the status bar names it. `Shift+Tab` goes back. Click to select the highlighted element. In a 3D view `Tab` goes from the nearest element to the ones behind it; on a sheet, through views placed one over the other.

Chains of walls and lines

Walls and lines joined end to end form a chain. With the cursor on one of them, `Tab` highlights the whole chain (the status bar says **Chain of n joined elements**); a click selects it. A chain can mix walls with room separators and area outlines; detail lines chain with the detail lines of their view. Where three or more meet, the chain goes on along the one that makes the smallest angle with the last; a wall that stops against another one between its ends takes the chain on along that wall.



TAB: the whole chain of walls

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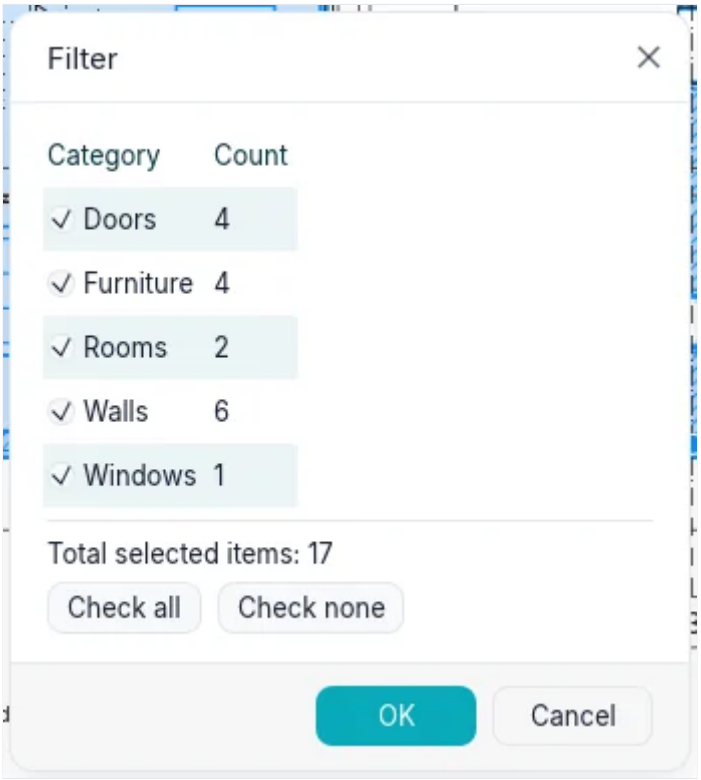
Part of a chain. Select the first element of the part, move the cursor over the last one and press `Tab`: the elements between them are highlighted. `Tab` again highlights the whole chain, a third time the last element alone. Click to select what is highlighted. On a closed chain the part goes round by the side where the cursor is: near the end of the last element that one way round reaches first.

Elements joined at a point. Select an element and keep the cursor on it, near an end where others are joined. `Tab` highlights it with the nearest one joined there; `Tab` again, every element joined there. Click to select them.

A selected chain. Where two selected walls or lines meet, a round control appears. Drag it to move the joint: the selected ends there move together and stay joined; elements not selected are left where they are. Drag the chain with `Shift` to move it whole, horizontally or vertically; drag it with `Ctrl` to place a copy.

Filtering a selection

Click the count in the status bar, or **Filter** on the contextual card, to open **Filter**. It lists the categories in the selection, how many elements each has, and the total. Untick a category to leave its elements out: the selection and the count in the status bar follow at once. **Check all** and **Check none** tick or untick every category. **OK** keeps the selection, **Cancel** brings back the one you started with.



Filter

Select All Instances

Right-click an element ▸ **Select All Instances**, then **Visible in View** (the elements of the same type shown in the active view) or **In Entire Project**. Typing `SA` and `Enter` does the same as **Visible in View**. The element must have a type: rooms, for example, have none.

Dragging elements

Drag a selected element to move it; the whole selection moves together, and one drag is undone by one `Ctrl+Z`. A temporary dimension shows the move, and the status bar its length (**Move: 1500**). With **Select and Drag** turned on in the status bar, an element can be dragged without selecting it first; with it off, a drag that starts on an element not selected draws a window.

- A wall joined at its ends to walls that are not selected moves square to itself only, and the joined walls stretch to stay joined. Walls that stop against it follow too. Hold `Shift` to move it freely; the joined walls still follow.
- Any other element moves freely; hold `Shift` to keep the move horizontal or vertical.

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- Doors and windows move with their wall and keep their place along it. Dragged on their own, they slide along the wall; dropped on another wall, they move into it.
- Hold `Ctrl` when you start dragging to place a copy instead.

Elements are dragged in plans, sections and elevations (along the view), on sheets (views), and in 3D views, where they move on the horizontal plane where you grabbed them. The arrow keys move the selection a small step, rounded to a tidy value at the zoom of the view; with `Shift`, ten steps.

The Modify tools

The **Modify** group of the contextual card has three rows: **Align**, **Offset**, **Mirror on Axis**, **Mirror on Line**, **Split Element**, **Split with Joint**; **Move**, **Copy**, **Rotate**, **Shorten/Lengthen**, **Shorten/Lengthen one element**, **Shorten/Lengthen several elements**; **Repeat in Series**, **Scale**, **Fix**, **Unpin**, **Delete**. The status bar says what each tool expects next, and the options strip shows its options. `Esc` cancels the step in progress; pressed again, it ends the tool.

Move, **Copy**, **Rotate**, the two **Mirror** tools, **Repeat in Series** and **Scale** work on the selection. Started with nothing selected, they let you select first (clicks, `Ctrl`, windows, `Tab`), then go on when you press `Enter` or **End** on the options strip.

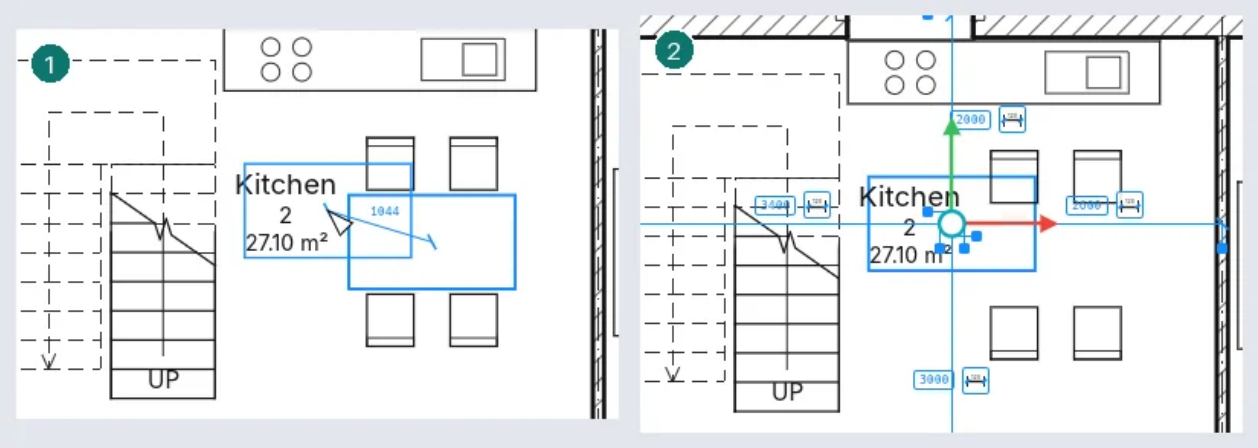
Move, **Copy** and **Rotate** also work in 3D views: their points are taken on the plane of the level clicked.

Before each click the tool shows what the click would do: the selection drawn where it would go (moved, copied, turned, mirrored about the axis under the cursor, offset to the cursor's side, aligned to the reference), following the cursor, with the distance or the angle beside it.

Where a tool asks for a second point, you can type a distance (or an angle) instead and press `Enter`: it shows next to the cursor while you type, and is measured towards the cursor.

Move

Click a base point, then the destination. The tool ends and the elements stay selected. Options:



Move: the preview, then the result

- **Constrain**: the move is along or square to the selected wall (horizontal or vertical for other elements);
- **Disjoin**: walls joined to the moved ones do not follow, and a door or window can be moved onto another wall;
- **Copy**: copies are placed instead; with **Multiple**, one at each click until `Esc`.

Copy

As **Move**, but a copy is placed. With **Multiple**, a copy at each click until `Esc`; **Constrain** as for Move.

Rotate

The centre of rotation shows at the middle of the selection; drag it to put it elsewhere. Click the first ray, then the second: the selection turns by the angle between them, shown by a temporary angular dimension. Instead of the second ray, type the angle and press `Enter`, or type it in **Angle** on the options strip and press `Enter`. **Copy** turns a copy; without **Disjoin**, walls joined to the turned ones follow.

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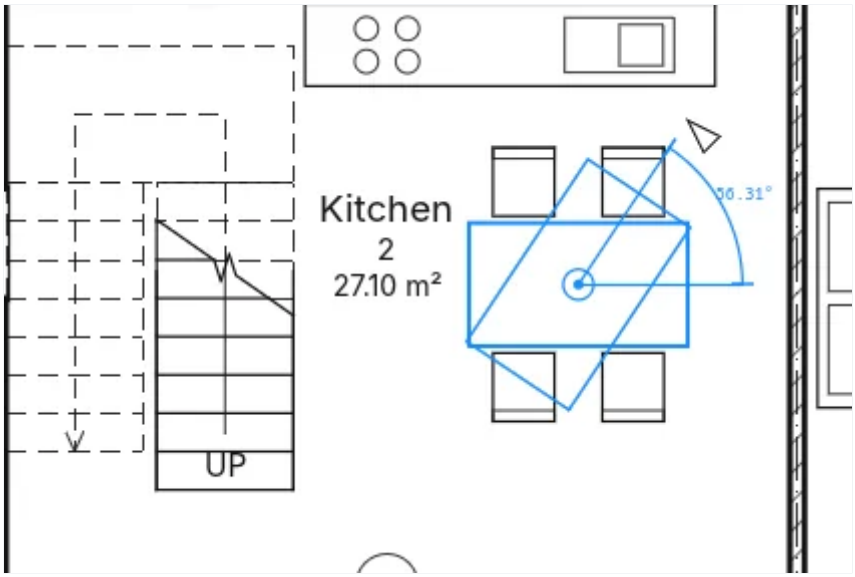

Rotate: the angle before the second click

Mirror

Mirror on Axis: click a wall, a grid or a line as the axis. **Mirror on Line**: click two points. **Copy** is ticked: a mirrored copy is placed; untick it to mirror the elements themselves.

Repeat in Series

Linear: click two points. **Number** is how many elements in all. With **Move To: 2°** the two points give the spacing; with **Last**, the last element goes to the second point and the others are spread evenly. **Constrain** keeps the direction along or square to the selection.

Radial: drag the centre where the copies turn about, click the first ray, then the second (or type the angle and press `Enter`, also in **Angle**). With **2°** the angle is between neighbours, with **Last** it is the whole span.

Offset

Moves a wall or a line square to itself, or places a copy (**Copy** is ticked).

- **Numerical**: set **Offset**, then click the element on the side it should go to. Press `Tab` first to offset the whole chain (of the same category) as one: its corners are kept.
- **Graphical**: click the element (`Tab` for the chain), then two points: the distance between them, square to the element, is the offset. You can type it instead. `Ctrl` copies.

Align

Click the reference, then the element to align: the face of the element nearest to your click goes onto the face of the reference nearest to its click. **Prefer** chooses **Wall faces** or **Wall centerlines**. After aligning, a padlock shows on the aligned face: click it to keep the two aligned, so that moving the reference moves the other one too. Without **Multiple Alignment** the next click picks a new reference; with it (or holding `Ctrl`) you can align several elements to the same one. `Esc` once starts a new alignment, twice ends the tool.

A wall aligned and locked to a grid follows it when you move the grid, in 3D too, and the walls joined to its ends stretch or shorten and stay joined. With **Moves with nearby elements**, in a component's properties, the component moves with the nearest wall, at the same distance.

Shorten/Lengthen

Walls and lines are trimmed or extended; click each one on the part to keep.

- **Shorten/Lengthen**: two elements, which then meet at a corner;
- **Shorten/Lengthen one element**: the boundary, then the element;
- **Shorten/Lengthen several elements**: the boundary, then each element in turn.

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Split Element and Split with Joint

Split Element: click a wall or a line where it should become two. With **Delete Inner Segment**, click two points: the part between them is removed, with the doors and windows in it. **Split with Joint**: set **Joint Gap**, then click: the two parts are that far apart.

Scale

Walls and lines (and the position of dimensions) can be scaled; with anything else selected, **Scale** is not available. **Graphical**: click the origin, the end of a first vector, the end of a second one: the elements are scaled by the ratio of the two lengths (you can type the second one). **Numerical**: set **Scale Factor**, then click the origin. Doors and windows keep their distance from the middle of their wall.

Fix and Unpin

Fix keeps the selected elements where they are: they cannot be dragged, moved by a tool or deleted, and a message says so. A pinned element shows a pushpin when selected; click the pushpin, or use **Unpin**, to free it.

Delete, undo and redo

Delete (or the `Del` key) removes the selection and what it hosts; pinned elements stay. `Ctrl+Z` undoes the last change and `Ctrl+Y` redoes it: each tool, drag or nudge is one step.

Copy, cut and paste

Copy to clipboard (`Ctrl+C`) puts the selection on the clipboard; **Cut to clipboard** (`Ctrl+X`) puts it there and takes it out of the project. **Paste** (`Ctrl+V`) brings it back. The three buttons are in the **Modify** group of the contextual card, next to the other ways of **Paste**; in the Classic style, in the **Clipboard** panel. The keys act while no tool is active and no field is being edited.

In a plan, **Paste** shows the lines of what was copied at the cursor: they follow it, and a click drops them there. The elements pasted stay selected. Pressed outside a plan, `Ctrl+V` pastes in the same place.

The other ways of **Paste** (in the Classic style, in its drop-down) keep the elements where they were and need no point:

Way	Where the elements go
Paste on the chosen levels	The model elements, in the same place, on each level ticked
Paste in the chosen views	The elements of a view (lines, texts, dimensions...), in each view ticked, of the same kind as the one they come from
Paste in the active view	In the same place, in the open view and at its level
Paste in the same place	Where they were, also in another project
Paste on the clicked level	In a section or an elevation: the level nearest the click is highlighted with its name; the click pastes on it

The two ways with "chosen" open a window with a **Search** field over the list of levels or views; tick the ones you want and click **OK**. A way that cannot paste what the clipboard holds in the open view is greyed, and the status bar says why.

- Each paste is one step: one `Ctrl+Z` takes it all away.
- The clipboard stays when another project is opened, so elements can be pasted from one project to another. Their types come along; where that project already has a type with the same name, that one is used. Levels are found by name, then by elevation.
- Pasted on another level, an element whose top is tied to a level is tied to the level as many levels higher; where there is no such level, it gets a **Free Height** of the same value.
- The elements of a view are pasted only into views of the same kind; the model elements are left out of **Paste in the chosen views**, the elements of a view out of **Paste on the chosen levels**. The message says how many were left out, and why.
- A door or window pasted alone goes into the wall it lands on; where there is none, it is left out and the message says so.
- A copy of a group is copied with its members (see Groups).
- **Paste in the active view** is greyed in the view the elements were copied from: there it would not move them. Open another view, or use **Paste in the same place**.
- Copied alone, a copy of a model group follows the cursor by its origin.

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Edit pasted

After every paste, **Edit pasted** opens: a bar over the drawing with the number of elements pasted, while the rest of the project is in halftone and cannot be selected. The pasted elements can still be moved and changed with the usual tools.

Button	What it does
Finish selected	The selected pasted elements stay where they are and leave the edit
Select all	Selects all the pasted elements
Finish	The pasted elements stay: the paste and the changes made to them are a single `Ctrl+Z` step
Cancel	The pasted elements go, as before the paste

When a paste left elements out (a door that finds no wall, for example), the message says so and points to the bar.

Transfer type

Transfer type (`MA`), in the same group, gives the type of one element to others. Click the element whose type to take: the brush next to the cursor takes colour, and the status bar names the type. Then click each element that should take it; only elements of the same category change. To give it to many at once, drag a window around them and click **Finish selection** in the options bar.

Between two walls, besides the type, the base offset, the layer extensions and, when both have one, the unconnected height go across; when they stand on the same base level, also the top constraint and its offset. The other instance parameters stay as they are.

A click on nothing, or `Esc`, empties the brush to take another type; `Esc` again ends the tool.

Groups

A group keeps elements together as one, and places copies of them elsewhere: every copy of a group is the same, and a change made to one reaches the others.

Group (`GP`) makes a new group of the selection and its first copy. Doors and windows join with their walls. What the group is depends on what is selected:

- model elements only: a model group;
- elements of a view only (lines, texts, dimensions...): a 2D group, which belongs to that view;
- both: a model group, with an attached 2D group (**Attached 2D group**) that goes with it.

An element is in one group at most: groups inside groups are not made. Moved, rotated, mirrored or deleted, a copy of a group takes its members with it; copied, it makes a new copy of the same group.

Place group places a new copy of a group at the point clicked, on the active level. Placed on another level, the copy moves its levels up by as many levels: an element whose top is beyond the last level gets a **Free Height** of the same value. A 2D group is placed in a view.

Edit group opens the group editor on the selected copy: **Add** and **Remove** add elements to the group and take members out of it (a wall leaves with its doors and windows; a member removed becomes an ordinary element), **Finish** ends the edit and every other copy of the group is made again from this one, **Cancel** leaves the group as it was.

Ungroup turns the members of the selected copy back into ordinary elements; the group stays in the project, with its other copies.

Exclude from this copy takes the selected members out of their copy only: the other copies keep them. **Restore excluded** brings them back into the copy, taken from another copy.

Attached 2D groups shows in the open view, with the selected copy of a model group, an attached 2D group placed with it as it is with another copy (the dimensions of one copy, for example, then measure the walls of this one); it also hides it again.

In the **Project Index**, the **Groups** node lists the groups, under **Model** and **2D**. The menu of a group has **Place copy**, **Select all copies**, **Rename**, **Duplicate** and **Delete**; a group is deleted only when it has no copies left.

Duplicate gives the new group the next free number ("Group 1" becomes "Group 2"); `F2` renames the group you clicked in the index.

Group with nothing selected opens a window with the name and the kind of the group (**Model group** or **2D group**): the group opens empty in the editor, and what you draw or add there joins it. **Finish** with no members does not make the group.

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A double click on a copy, or on one of its members, opens the group editor on that copy, in a plan, a section, an elevation or a 3D view.

Move to project, in the **Group** panel or in the right-button menu on a member reached with `Tab` (next to **Exclude**), takes the member out of its copy and leaves it in the project as an ordinary element; a wall takes its doors and windows with it. The other copies keep it.

Disassociate from model group, on a copy of an attached 2D group with no dimensions or tags, makes it a 2D group of its own (of a new group, when other copies stay attached).

In the **Properties** of a copy of a model group, **Reference Level** and **Origin Level Offset** move the members: a new level takes them up by as many levels, a new offset raises or lowers them by the difference. Members that cannot follow (a hosted element, a wall attached to a floor) stay where they are, and a warning says so.

Typed shortcuts

Type the letters and press `Enter`: `MV` **Move**, `CO` **Copy**, `RO` **Rotate**, `MM` **Mirror on Axis**, `DM` **Mirror on Line**, `AR` **Repeat in Series**, `OF` **Offset**, `AL` **Align**, `TR` **Shorten/Lengthen**, `SL` **Split Element**, `SG` **Split with Joint**, `RE` **Scale**, `MA` **Transfer type**, `GP` **Group**. `PN` pins and `UP` unpins the selection, `SA` selects all its instances visible in the view. Keyboard shortcuts lists them all.

Not yet available

- `Tab` to reach a part of an element: a curtain panel or mullion, a part of a stacked wall;
- **Select All Instances** from a type in the Project Index;
- **Group and Associate** of **Repeat in Series**;
- splitting walls along a horizontal line in elevations and 3D views;
- groups inside groups, and **Load as group**.

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Walls, doors and windows

This chapter describes how walls are drawn, how they join and attach to roofs and floors, how their types and layers are set up, and how doors and windows are placed in them, flipped, moved and drawn at each level of detail.

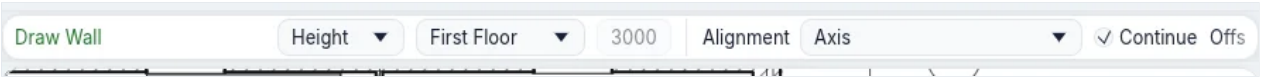
Drawing walls

Create ▸ Build ▸ Wall (or type `WA`) starts the wall tool, in a plan or in the 3D view. Pick the wall type in the type selector at the top of the element panel; **Type Parameters** changes the type, and with it every wall of that type. The contextual card **Draw Wall** shows the **Draw** group, and the options strip shows:

Option	What it does
Height or **Depth**	The wall rises from the level (**Height**) or goes down from it (**Depth**)
A level, or **Unconnected** with a value	The wall reaches that level, or is that high (or deep)
Level	In a 3D view only: the level the walls stand on; takes the level of the surface of the first click
Alignment	The plane of the wall that runs along the clicks: **Axis** (the default), the core centre line, or one of the finish and core faces, exterior or interior
Continue	On, each click ends a wall and starts the next one; off, each two clicks make a wall on its own
Offset	The alignment line runs this far from the clicks, towards the exterior
Radius	With **Line** and **Rectangle** every corner between two straight walls becomes a curved wall of that radius, also where a wall ends on the free end of another (where three walls meet the corner stays sharp; so it does with a radius too large, with a message); **Offset** then does not apply. With **Circle**, the arcs and the polygons: the radius of the next one, so a click places it
Join at the ends	Not available yet, shown greyed: the walls drawn always join; an end's join is prevented afterwards from its control
Automatic joining	Not available yet, shown greyed and ticked: the walls join those they meet

Structural wall, from the **Wall** drop-down or the **Structure** tab, is the same tool: every wall drawn is **Bearing** (with **Wall**, **Non-bearing**) and it starts with **Depth**; each of the two tools remembers its own choice of **Height** or **Depth**. **Structural usage** is changed afterwards, in Properties.

With a placement tool active (wall, door, window, object, column, pillar) **Properties** shows under the type selector the properties of the next element: the values typed there go to the elements the tool places, and the rows the options bar also holds stay in step with it.



The options strip of the wall tool

In the 3D view the wall being drawn follows the cursor as a solid, with its thickness and its height up to the top level (or the height typed), on the plane of the level, with its length and angle as a temporary dimension; **Shift** keeps it square and **Esc** takes it away. **Rectangle**, the polygons and **Pick Lines** work there too: the walls of the whole shape follow the cursor, and the line of the level under the cursor is picked as in plan.

While drawing:

- the wall's exterior is on the left of the drawing direction: drawn from left to right, the exterior face is at the top. The preview shows the exterior face thicker;
- the **space bar** puts the exterior of the next wall on the other side;
- after the first point, type a length and press Enter: the value shows in a box at the cursor, and the wall ends that far, in the direction of the cursor (grids, room separators and area outlines, sections and elevations take a typed length the same way);
- **Esc** ends the chain; **Esc** again, or **Modify**, ends the tool.

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Drawing a chain of walls

The **Draw** group picks the shape:

Shape	How
Line	Click the start and the end of each wall
Rectangle	Click two opposite corners: four joined walls, with the exterior outside
Inscribed Polygon	Set **Sides** in the options strip, click the centre and a corner: the corners are on the circle
Circumscribed Polygon	The same, but the sides touch the circle
Circle	Click the centre and a point of the circle (or type the radius): two half circles of walls, with the exterior outside
Start-End-Radius Arc	Click the start, the end, then a point of the arc (or type the radius)
Center-ends arc	Click the centre, the start (the radius), then the end: at most half a turn
Tangent End Arc	Click the end of a wall: the arc starts there, tangent to it, and ends at the next click
Fillet Arc	Click two straight walls, then where the arc passes (or set **Radius**): the walls are trimmed and joined by a curved wall
Ellipse, **Partial Ellipse**	Elliptical walls: see below
Pick Lines	Click a line of the plan (a line or an edge of an element): a wall along it. Press **TAB** over a line to take the whole chain of lines joined to it, then click

Curved walls

A curved wall is one wall along an arc: it is selected, scheduled and tagged as one, and its **Length** is measured along the arc. In plan, section and 3D it is drawn as an arc, made of short straight pieces no farther than 2 mm from it. Its properties add **Radius** (changing it keeps the ends where they are), **Center X** and **Center Y** (moving the centre keeps the radius), and **Center Mark Visible**, the «+» shown in plan at the centre; a curved wall has it on when it is drawn.

- A door or a window clicked on a curved wall stands on a straight piece as wide as it, with a margin each side, so its frame stays flat; moved along the wall, it keeps its place on the arc.
- Curved walls join each other and straight walls at their ends, as straight walls do.
- A selected curved wall has a third blue control at the middle of its arc: drag it and the arc passes through the pointer, its ends where they are (`wallarc` at the command line).
- **Edit Profile** is greyed on a curved wall: its contour has no plane to be drawn in.
- In IFC a curved wall is one `IfcWall` made of its pieces.

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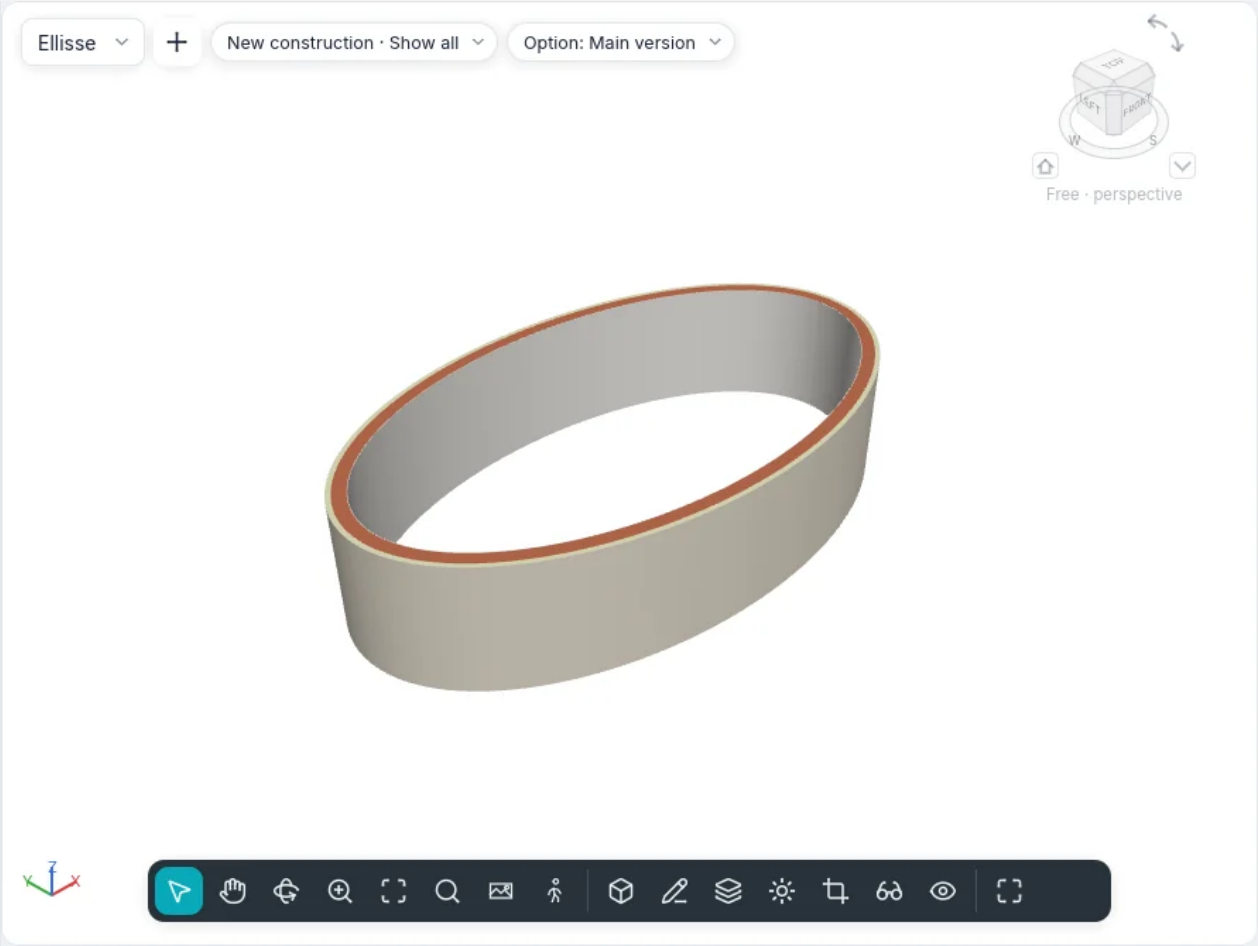
Elliptical walls

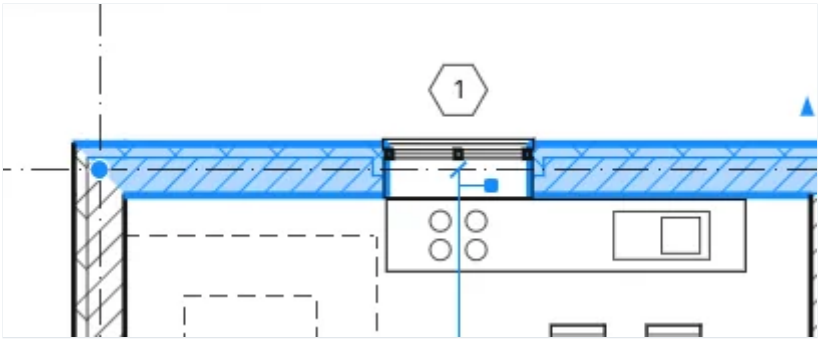
Ellipse (centre, major radius, minor radius) draws a ring of two half-ellipse walls with the exterior outside; **Partial Ellipse** (the two ends of the axis, then the shape) draws one half-ellipse wall on the side of the cursor, Space flips its exterior. Radii can be typed; two equal radii are refused (use **Circle**). An elliptical wall stands on its centre line: **Alignment** is locked, and **Offset**, **Radius** and **Continue** are hidden. A selected elliptical wall has the controls of an ellipse (quadrants, ends with Tab, temporary dimensions of its radii, **Editing Retains Proportions**); a ring's two halves change together. Its properties add **Major Axis Radius**, **Minor Axis Radius** and **Center X/Y**. Doors, windows, openings, joins, rooms, sections, 3D, schedules and IFC work as on curved walls. **Offset**, **Fillet Arc** and **Split Element** are refused; a stacked type is accepted only with every sub-wall centred.



An elliptical wall: one wall with all its layers

The alignment and flipping

Select a wall: blue dots mark the ends of its alignment line, and two arrows stand on its exterior side. Clicking the arrows turns the wall over about its alignment line: the first and the last layers swap, and the doors and windows keep their orientation. Changing **Alignment** of a placed wall does not move it; the next flip turns it about the new plane. Changing the type of a wall, or its layers, keeps its alignment line where it was.



A selected wall: ends and flip arrows

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Attaching walls to roofs and floors

A wall can follow a roof or a floor instead of a fixed height.

1. Select the walls.
2. **Selection: Walls ▸ Modify Wall ▸ Link Top/Base**.
3. In the options strip, **Attach Wall: Top** or **Base**.
4. Click the roof or floor, in a plan, a section or the 3D view.

With **Top**, the walls rise or drop to the underside of the roof (following its slopes: the gable walls go up to the ridge) or of the floor above. With **Base**, they stand on the top face of the floor below. From then on they follow it: a steeper roof, a roof raised by its offset, a moved floor carry the walls with them.

To detach them: select the walls, **Selection: Walls ▸ Modify Wall ▸ Unlink Top/Base**, then click the element to detach them from, or press **Detach All** in the options strip. The walls keep the height they have at that moment, and can be attached again at any time. When a roof or floor with walls attached to it is deleted, the walls keep their height too.

The properties **Top is Attached** and **Base is Attached** show the state (read-only); while the top is attached, **Free Height** is read-only.

Editing a wall's profile

A wall need not be a rectangle: its outline can be sketched in an elevation or a section parallel to it, with holes, as for a gable or a stepped wall.

1. Select the wall and click **Mode ▸ Edit Profile** in its contextual card. From a plan or a view that is not parallel to the wall, **Go to View** lists the sections and elevations parallel to it: choose one and click **Open View**.
2. Sketch mode opens with the wall's outline in magenta. Change it with **Line** and **Rectangle** (see sketch mode): pick a line and press `Del`, draw new ones, or a **Circle**. Every closed loop inside the outline is a hole through all the layers; a loop inside a hole is solid again.
3. Click **Finish Edit Mode** (or **Cancel Edit Mode** to leave the wall as it was).

The wall takes that outline in 3D, in sections and in plan, where it is cut only where the outline reaches the cut plane; its doors and windows still cut it, also over a hole, which stays a hole. A top attached to a roof or a floor still clips it. Plan, sections, elevations and the 3D view all show one solid, cut from the same outline and holes. The layers that wrap line the holes as they line doors and windows (see Wall types and layers). A curtain wall keeps the panels and mullions inside its new outline. **Mode ▸ Reset Boundary** makes it a rectangle again; `Ctrl+Z` undoes either.

The boundary can also be edited in the 3D view: the sketch lies on the wall's plane and the view stays 3D. Only from a plan does **Go to View** appear, and nothing changes until a view is opened with **Open View**. While sketching, the wall's original outline is drawn as dashed reference lines; the sketch snaps to their ends and onto them.

Wall joins

Walls whose ends meet join by themselves, and so do a wall ending on the side of another and two walls crossing each other. At **Medium** and **Fine** detail each layer is resolved against the layers of the other wall:

- at a corner (an L), each layer meets the same layer of the other wall: the finishes wrap round the corner and the cores meet;
- at a T, the core of the wall that ends runs through the finishes of the other wall up to its core; its own finishes stop at the face, which is open there;
- where two walls cross (an X), a core runs through the finishes of the other wall;
- pieces of the same material merge with no line between them; a thin line separates two different materials; the free outline is drawn thick.

At **Coarse** detail joined walls are one fill, with no line between them. Sections and the 3D view use the same pieces, so nothing overlaps.

The Wall Joins tool

Shapes ▸ Wall Joins in the contextual card, then click the grey square that appears on a corner. The options strip offers:

- **Configuration: Butt** (one wall runs through, the other stops against it), **Miter** (both end on the bisector) or **Square off** (the wall that stops ends square); **Previous** and **Next** change which wall runs through;
- **Display: Clean Join**, **Don't Clean Join** (the walls butt, with a thick line between them) or **Use View Setting**.

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Each change can be undone. **Esc** ends the tool.

The plan's property **Wall Join Display** (in **Graphics**) is the view setting: **Clean all wall joins**, or **Clean same wall type** to keep a thick line where walls of two types meet, unless their corner is set to **Clean Join**.

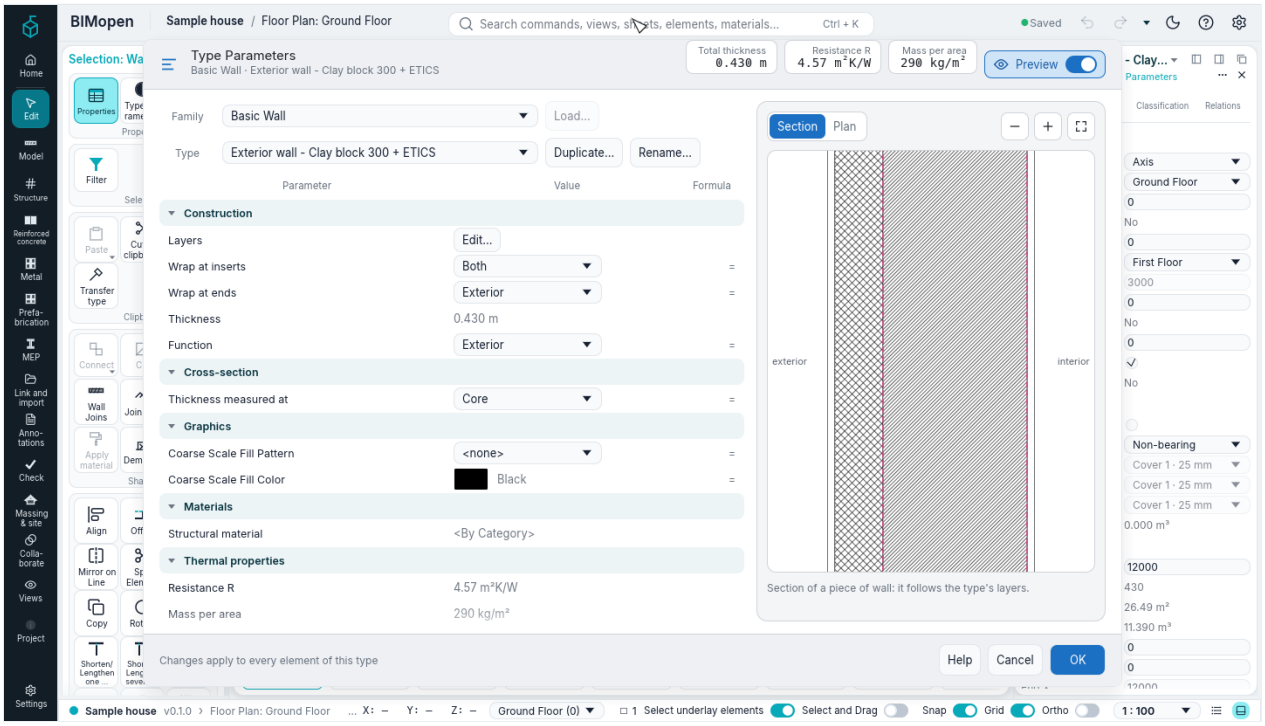
Disallowing a join

To keep an end of a wall from joining (an expansion gap, say), select the wall, right-click the blue dot of that end and choose **Disallow Join**: both walls then end square, apart. A small mark beside the end shows it; click it, or right-click the dot and choose **Allow Join**, to join it again.

Wall types and layers

Type Parameters (the tile of the tool panel while a wall is selected, the button under the type selector of **Properties**, or a type's menu in the Project Browser) shows the parameters of a wall type; a change applies to every wall of that type.

Type parameters



The Type Parameters of a wall type, with its section

At the top are the family and the type and, for a type with layers, the three figures of **Type layers**: **Total thickness**, **Resistance R** and **Mass per area**. Below, **Family** with **Load...** (a family from the library; greyed for walls and the other system families, which are duplicated rather than loaded) and **Type** with the other types of the family, **Duplicate...** (a copy, which becomes the type of the selected walls; its name does not follow its layers) and **Rename...**. The parameters follow in a table **Parameter · Value · Formula**, in groups that open and close with a click on their title:

- **Construction**: **Layers** with **Edit...**, which opens **Type layers**; **Wrap at inserts** (**No**, **Exterior**, **Interior**, **Both**); **Wrap at ends** (**None**, **Exterior**, **Interior**); **Thickness** (the sum of the layers in metres, read-only); **Function** (**Exterior**, **Interior**, **Foundation**, **Retaining**, **Cavity**, **Core-shaft**).
- **Cross-section**: **Thickness measured at** (**Core**, **Exterior face**, **Interior face**).
- **Graphics**: **Coarse Scale Fill Pattern** and **Coarse Scale Fill Color** (a swatch that opens the colour picker), how cut walls of the type are filled at **Coarse** detail.
- **Materials**: **Structural material**, read-only: the material of the layer marked structural in **Type layers**, «» when none.
- **Thermal properties**: **Resistance R** and **Mass per area**, read-only and computed from the layers (they show «—» while a material lacks its thermal values); **Solar absorptance** (0 to 1, 0.7 at first) and **Roughness** (from **1 · Very smooth** to **5 · Very rough**).
- **Identification**: **Keynote** (typed, or **...** opens the project's **Keynotes**: codes and descriptions to add, change or delete; a row picked and **OK** give its code to the type), **Assembly Code**, **Assembly Description** (read-only), **Model**, **Manufacturer**, **Type Comments**, **URL**, **Description**, **Type Mark** (what a wall tag shows), **Fire resistance** and **Cost**.

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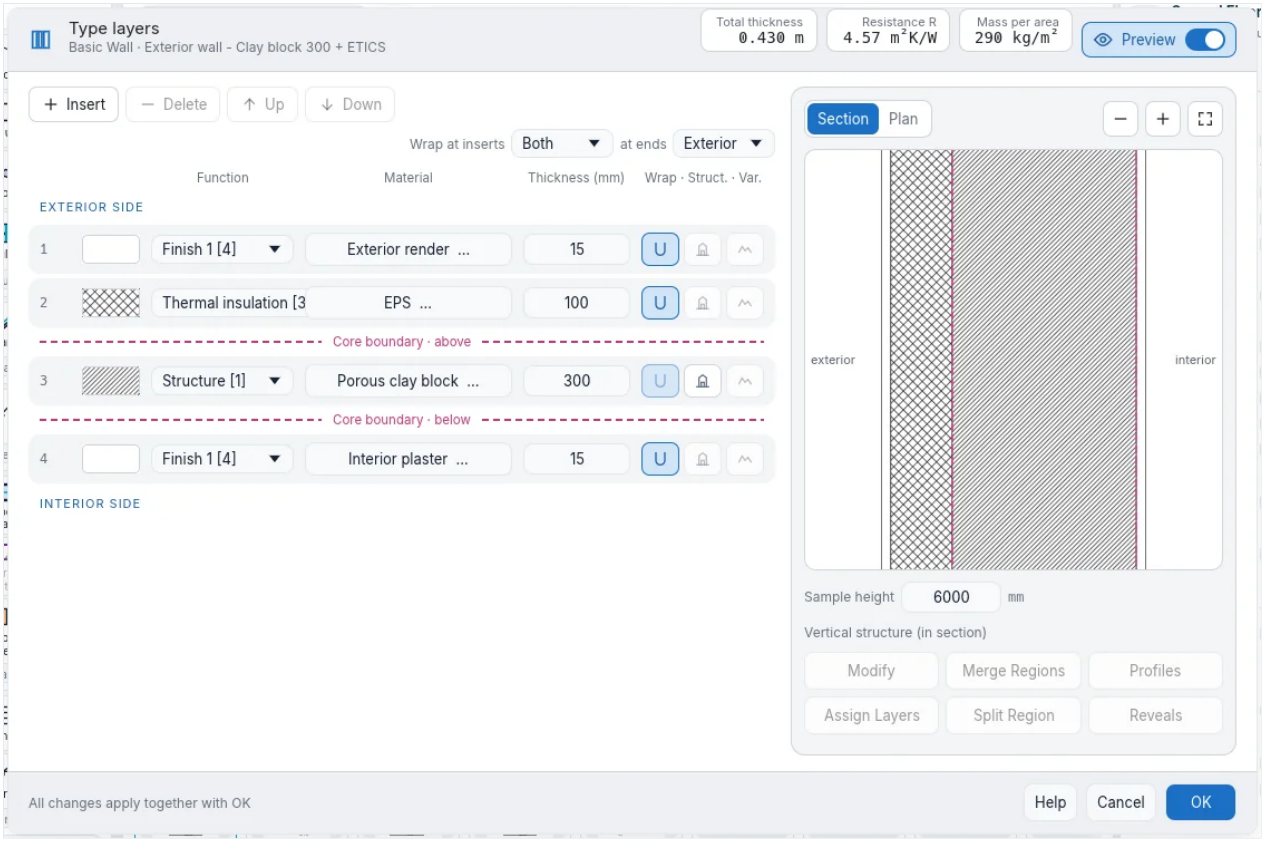
The **Formula** column marks with «=» the values that can be changed. Every change applies at once and Ctrl+Z undoes it, one step each; **Cancel** undoes all the changes made since the dialog opened, **OK** keeps them, **Help** opens this page.

On the right the preview draws the type as **Type layers** does: **Section** (the wall cut through, its layers side by side) or **Plan** (the end of the wall), with the buttons to zoom out, zoom in and fit; floors and roofs show their layers stacked; doors, windows, components and curtain walls a picture of one of them. The **Preview** switch in the header, beside the figures (an eye and an on/off switch), hides it and the table widens into the freed space, while the dialog keeps its size and place; switched on again it shows the preview, and the choice is kept. Floors, roofs and the other families use the same dialog, with their own parameters.

Type layers

Layers ▸ Edit... opens **Type layers**, the same dialog for walls, floors, roofs and ceilings. At the top are the family and the type, and three figures updated at every change: **Total thickness** (in metres), **Resistance R** (m^2K/W: the sum of thickness / thermal conductivity of the layers, plus 0.17 for the two surfaces) and **Mass per area** (kg/m^2: the sum of thickness × density). R and mass show when every material has its **Thermal conductivity** and **Density** (the template's materials have them, others are completed in **Materials**); otherwise they show «—».



Type layers

On the left is the list of the layers, from the **EXTERIOR SIDE** at the top to the **INTERIOR SIDE** at the bottom. Each layer is a card with its number, a sample of its material (colour and cut pattern), its **Function** (**Structure**, **Substrate**, **Thermal insulation**, **Air space**, **Finish 1**, **Finish 2**, **Membrane**), its **Material** (a click opens **Materials** to choose it), its **Thickness** in millimetres and three toggles:

- **Wraps**: the layer may wrap at openings and ends (never a layer of a wall's core);
- **Structural material**: the layer's material is the type's structural material (only one, in the core);
- **Variable**: the layer's thickness may vary (only one, in floors and roofs).

Two dashed lines, **Core boundary · above** and **Core boundary · below**, delimit the core. A click on a card selects it; **Insert** adds a layer under the selected one; **Delete**, **Up** and **Down** act on the selected layer and are greyed without a selection. In a wall a core boundary is also selected with a click and moved with **Up** and **Down**. A membrane layer has no thickness, and every other layer is at least 4 mm thick: other values are refused, the dialog stays open and the status line at the bottom says why.

For walls, on the row of the buttons:

- **Wrap at inserts** (**No**, **Exterior**, **Interior**, **Both**): the wrapping layers outside the core on that side turn into the jambs of doors and windows; the layers behind them are pulled back;
- **at ends** (**None**, **Exterior**, **Interior**): the same at the free ends of the walls.

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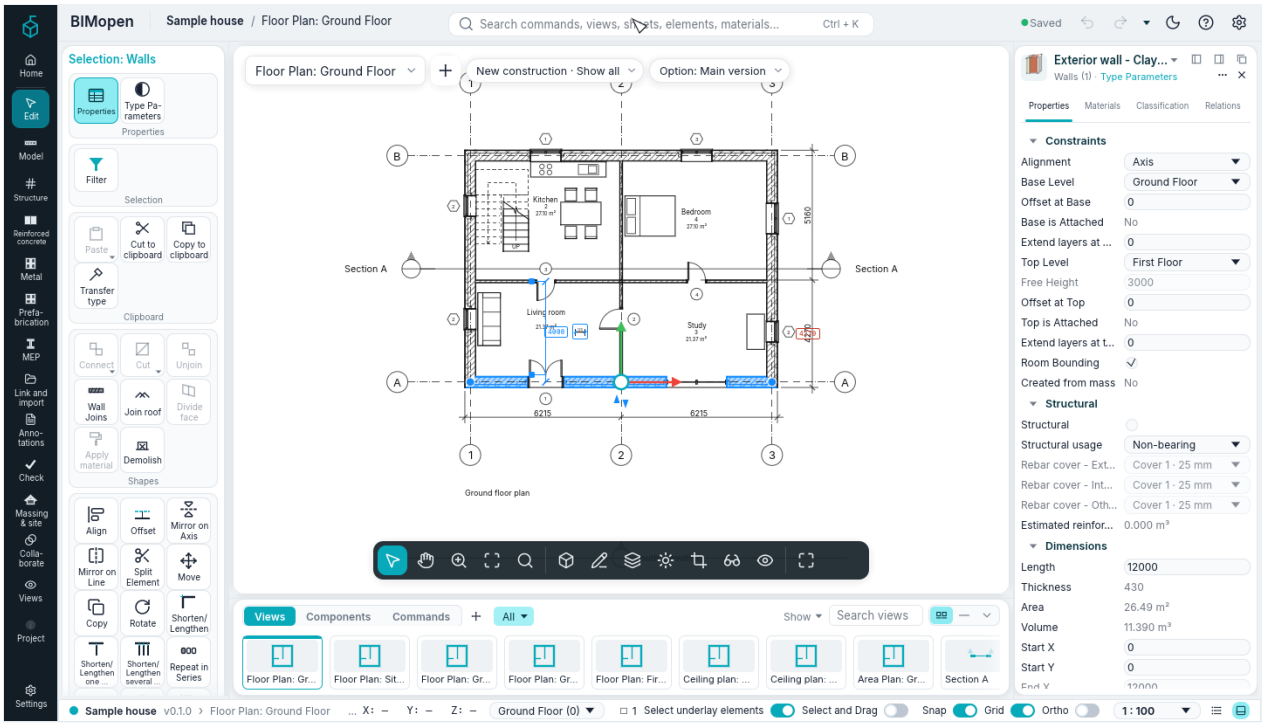
On the right the preview follows every change. **Section** shows the wall cut through, its layers side by side and as tall as **Sample height** (6000 mm at first), or the floor, roof or ceiling with its layers stacked; **Plan** (walls only) shows the end of the wall with the layers that wrap round it. The selected layer is outlined and the core lies between two dashed lines. The buttons above the drawing zoom it out, zoom it in and fit it; the wheel, a pinch and a drag zoom and move it too, and a click on a layer selects it in the list. Below, for walls, **Vertical structure (in section)** gathers **Modify**, **Merge Regions**, **Profiles**, **Assign Layers**, **Split Region** and **Reveals**, not available yet (greyed).

The preview draws each layer exactly as plans and sections draw it at **Medium** detail: the material's cut pattern in its colour, on the view's background in **Lines** or over the material's shading colour in the shaded styles, following the active view's graphic style and the theme. A material changed in **Materials** changes the preview, the plans, the sections and the 3D view alike; the overrides of a view (**Visibility/Graphics**, filters, an element's overrides) change that view only, never the preview.

The **Preview** switch in the header, beside the figures, hides the preview and the list takes the whole width, while the dialog keeps its size and place; switched on again it shows the preview, and the choice is kept. At the bottom, **OK** applies all the changes in one step, which Ctrl+Z undoes; **Cancel** closes without changing anything; **Help** opens this page. In a narrow window the preview stands above the list.

Wrapping shows in every view: in plan, in the 3D view, in sections and in elevations the ticked layers line the jambs, the head and the sill of doors and windows, the sides of the holes of an edited boundary (see Editing a wall's profile) and the free ends. A door or window type overrides the wall's wrapping with its **Wall Closure**.

Wall properties



A wall selected, with its Properties

With a wall selected, **Properties** shows at the top the type selector: the picture of the type, its family and its name. A click opens the list of the types grouped by family (**Basic Wall**, **Curtain Wall**), each with a small picture — a wall's layers side by side in their materials' colours, a curtain wall's grid — and the current type highlighted; typing in **Search a type...** keeps the types whose name contains the text, the arrows and Enter pick one. Under it **Walls (1)** says what is shown and **Type Parameters** opens the dialog described above.

The parameters come in groups that open and close with a click on their title:

- **Constraints**: **Alignment** (**Axis**, **Core Centerline**, **Finish Face: Exterior**, **Finish Face: Interior**, **Core Face: Exterior**, **Core Face: Interior**), **Base Level**, **Offset at Base**, **Base is Attached** (read-only), **Extend layers at base**, **Top Level** (a level, or **Unconnected**), **Free Height** (typed only with **Unconnected**, greyed otherwise; at the command line it is refused, saying why, while the wall has a top level), **Offset at Top** (only with a level, greyed otherwise), **Top is Attached** (read-only), **Extend layers at top**, **Room Bounding** (untick it and the rooms around the wall ignore it) and **Created from mass** (read-only).
- **Structural**: **Role** (**Non-bearing**, **Bearing**, **Shear**, **Combined**). A wall other than **Non-bearing** is structural, and its **Rebar cover - Other faces**, **Rebar cover - Interior face** and **Rebar cover - Exterior face** can be chosen (25, 35 or 40 mm); they are greyed otherwise. **Estimated reinforcement volume** is read-only.

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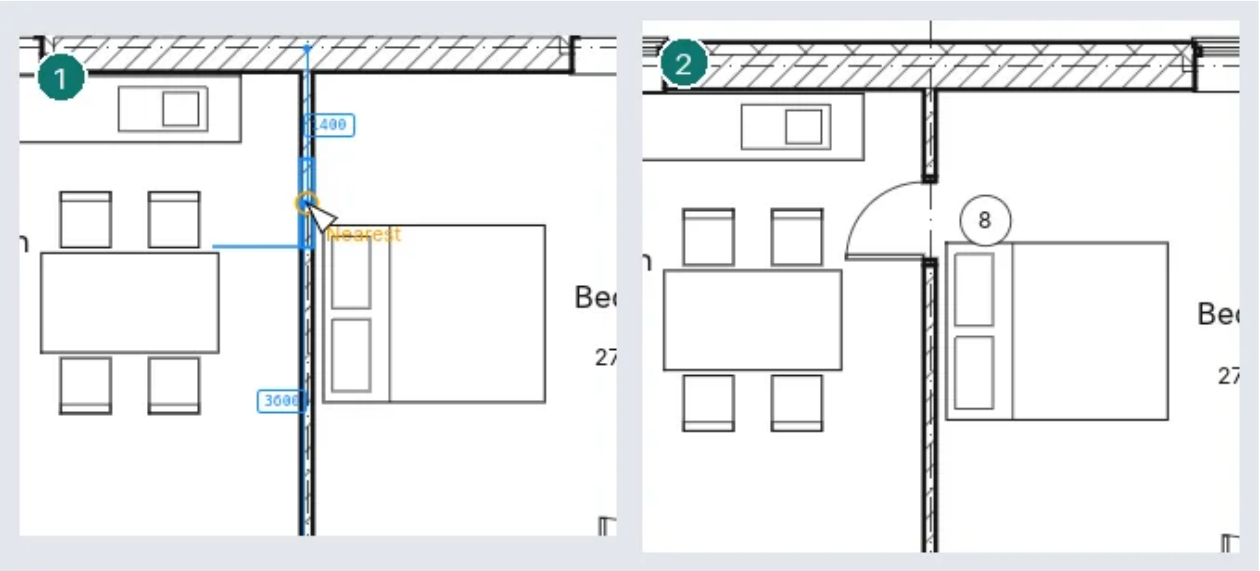
- **Dimensions**: **Length**, **Thickness**, **Area**, **Volume** and the ends of the wall's line.
- **Identification**: **Comments**, **Mark** and **Classify as** (**Exterior wall**, **Interior wall**, **Partition**, **Foundation wall**; until one is chosen, after the type's function).
- **Phasing**: **Phase created** (**Existing**, **New construction**) and **Phase demolished** (**None**, **Existing**, **New construction**). They are kept with the wall; views do not draw phases yet.

Extend layers at base and **at top** carry the layers outside the core below the base and above the top by that distance (a negative one shortens them), in the 3D view, in sections and in elevations; the core stays between the base and the top. At the top they extend where the top is flat, not where the wall is attached to a roof or a sloped plane.

With several walls selected the filter says **Walls (n)**, the type selector shows their type (or «‹varies›») and each row shows the value the walls share, or «‹varies›» where they differ; a change goes to all of them in one step. **Help on properties** at the bottom opens this page; **Apply** confirms the value being typed. Every change is one step that Ctrl+Z undoes.

Placing doors and windows

Create ▸ Build ▸ Door (`DR`) or **Window** (`WN`), in a plan, a section, an elevation or the 3D view. Pick the type, then move over a wall: the opening follows the cursor along it, and temporary dimensions measure from its centre line to the centre lines of the nearest walls across the host on each side. A door swings to the side of the wall the cursor is on; the **space bar** puts its hinge on the other jamb. Click to place it: the wall is cut in every view. A window takes its type's **Default Sill Height**.



A door: along the wall, then placed

Doors and windows get a **Mark** that counts up by category (1, 2, 3…), from the highest.

Tags

Tag on Placement, on the options strip of the door and window tools, tags each opening as it is placed. The options strip then offers **Horizontal** or **Vertical**, and **Leader** with its length. A door's tag shows its **Mark**; a window's tag shows its type's **Type Mark**. **Document ▸ Tag All** tags the openings not tagged yet.

Flipping a door or window

Select a door or window in a plan: arrows across the wall, on the side it opens to, turn its swing over; arrows beside it, along the wall, put its hinge on the other jamb. The context menu does the same: **Flip Hand** (the hinge) and **Flip Facing** (the swing).

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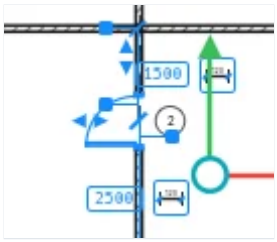


A selected door and its flip arrows

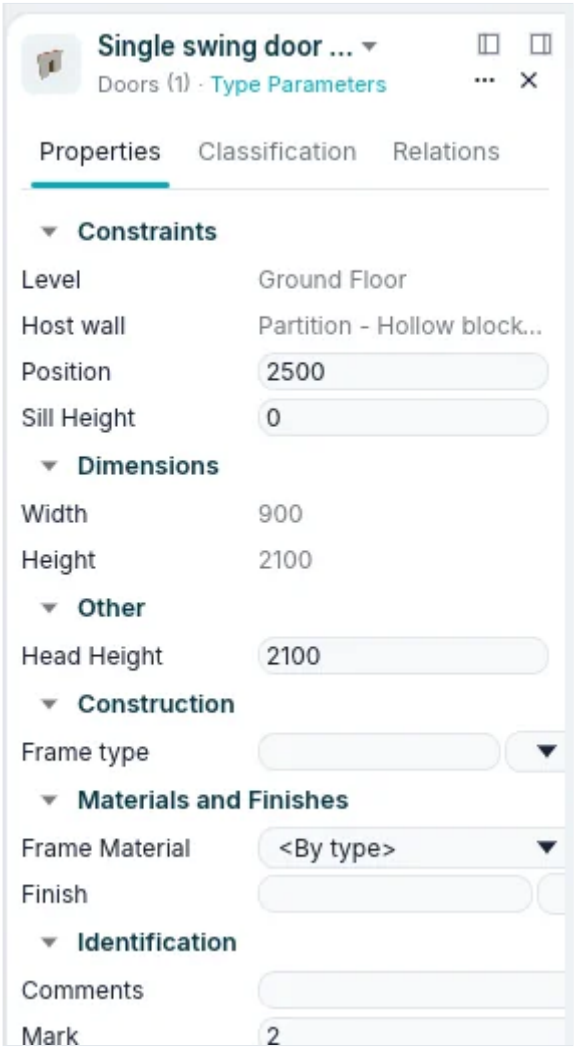
Moving a door or window to another wall

Select it, **Selection: Doors ▸ Host ▸ Pick New Host** (or **Selection: Windows**), then click another wall: the opening moves there, where you clicked, with its type and sill. It stays selected.

Dragging a door or window keeps it in a wall: the pointer is projected onto the wall's alignment line, so the opening slides along the wall, between its ends, with the temporary dimensions to the nearest walls, in a plan, a section, an elevation or the 3D view; there a click on a window's glass picks the window, not what is seen through it. Dropped over another wall, it moves into that wall; released over empty space, it stays at the last place on its own. The arrow keys move it along its wall the same way.

Door and window properties

Instance: **Level**, **Sill Height**, **Width** and **Height** (of the type), **Head Height**, **Comments**, **Mark**. Changing **Sill Height** or **Head Height** moves the opening up or down; its size stays.



The properties of a door

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Type: **Wall Closure** (**By host**, or a wrapping of its own), **Function** (**Interior** or **Exterior**), **Door Material** (**Glass Material** for windows) and **Frame Material**, **Trim Width**, **Thickness** (of a door's leaf, or the depth of a window's frame), **Default Sill Height** and **Window Inset** (windows: how far the frame sits from the exterior face), **Trim Projection Ext** and **Trim Projection Int** (doors), **Leaves** and **Type Mark**. A door cannot become a window by changing its type.

How walls, doors and windows are drawn

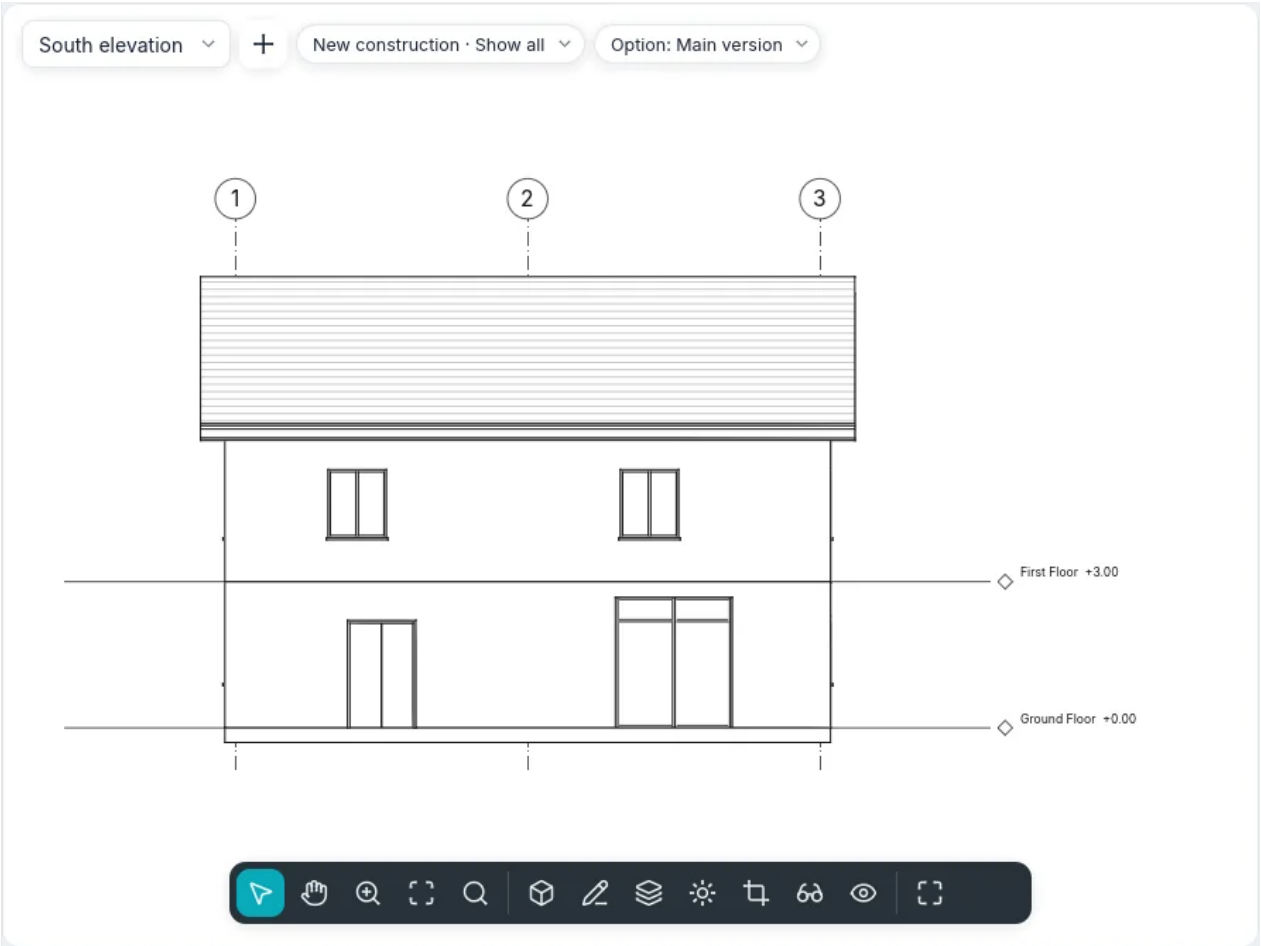
The **Detail** of the plan decides:

	Coarse	Medium	Fine
Walls	One fill (or the type's coarse pattern), joined walls merged	Each layer with its material's cut pattern	As **Medium**
Windows	The frame as one outline, the glass along it, the wall below seen as lines	The frame cut at each jamb and at the mullion of a two-leaf window, at its inset; one line of glass; the exterior sill and the interior sill board	As **Medium**, with double glass
Doors	The leaf as a line and its swing	The frame at each jamb, the leaf with its thickness, its swing	As **Medium**, with the trims out of the faces

In every case the layers end at the jambs of the opening, closed by the cut line, or wrap into them as the types ask. In sections and in 3D, doors and windows show their frame, leaves, glass and sills, in the type's materials.

Curtain walls

A curtain wall is a wall of a curtain wall type, drawn with the **Wall** tool: pick **Curtain Wall** (one glass panel the length of the wall), **Exterior Glazing** (grid lines by the type) or **Storefront** (grid lines and mullions) in the type selector, then draw as for any wall, in plan or in 3D, with **Continue** and the **Draw** shapes. A basic wall changes into a curtain wall by picking a curtain wall type in the type selector; the doors and windows it hosts are deleted, because a curtain wall does not host them, and a message says so (undo brings them back).



A storefront in the south elevation

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In plan each glass panel is drawn with its thickness and the mullions are cut. A type with **Automatically Embed** ticked (the **Storefront**) drawn along a wall cuts its own hole in it, in plan, sections, 3D and IFC. The wall stays one wall: beside and above the hole it goes on with its layers, its net area leaves the hole out, and moving or resizing the curtain wall takes the hole with it.

- The type's **Vertical Layout** and horizontal layout (**Fixed distance**, **Fixed number**, **Maximum spacing**) with their spacing, and the instance's **Number of Vertical Grids** and **Vertical Justification**, place the grid lines by themselves.
- **Create ▸ Build ▸ Curtain Grid**: click near the top or bottom edge for a vertical line, near a side for a horizontal one, in an elevation, a section, a plan or the 3D view (on the face under the cursor); it snaps to the middle and the thirds of the panel.
- **Create ▸ Build ▸ Mullion**: **Grid Line**, **Grid Line Segment** or **All Grid Lines**, with the mullion type picked in the type selector; click the grid line in an elevation, a section or the 3D view.
- Select the curtain wall, then **Panel** in its contextual card: pick a panel type (a **Curtain Wall Door**, a glazed or solid panel, or none) and click the panel to change, in an elevation, a section or the 3D view.

Panels and mullions have their own schedules, and the IFC export writes the curtain wall with its plates and members.

- **Curtain Grid ▸ Placement**: **All Segments** (the whole line), **One Segment** (only across the panel clicked) or **All Except Picked** (click to place the line, click the segments to leave out, then click away from the line).
- With a curtain wall selected, **Add/Remove Segments**: click a segment of a grid line to take it off (the two panels on either side become one, and its mullion goes) or to put it back. Panels stay rectangular: a segment that would make an L-shaped panel is refused.
- Hover over a curtain wall and press TAB: the panel under the cursor is outlined; click to select it alone. The element panel shows its **Width** and **Height**, and the type selector changes its type.

Grid lines and mullions

TAB over a curtain wall goes from the wall to the mullion under the cursor, then to its grid line, then to the panel; the status bar says what is highlighted and a click selects that alone (CTRL+click adds more of the same curtain wall). It works in plan (vertical lines and mullions), in sections, elevations and the 3D view.

- **A selected grid line**: the lines of the type's layout show a pushpin and do not move; a click on the pushpin (or **Tie to Type ▸ Independent** in Properties) makes the line independent: it stays where it is with its mullions when the wall changes size or the type changes its layout. An independent line, or one added by hand, drags between the lines beside it (the panels follow), moves when one of its temporary dimensions is typed, and is deleted with **DELETE**. Its pushpin ties it to the type again.
- **A selected mullion**: one per segment of a grid line. Properties shows **Comments**, **Length** (the segment, before the cut at the joints) and **Mark** (a mark another mullion uses is kept, with a warning). The type's mullions have a pushpin: unpinned, they take another type from the type selector or are deleted with **DELETE**.
- **Make Continuous** and **Break at Join** in the contextual tab: the selected mullion runs through the joints at its ends (those crossing it stop), or stops there. The cross control on each joint swaps that joint only.
- The curtain wall's context menu selects **the curtain wall's panels** or its **mullions** (on the vertical or horizontal grid, inner, border, all); on a panel those along its grid, on a mullion those on its grid line or across it. Properties changes their type together.
- Placing a grid line, besides the panel's middle and thirds it snaps to the levels, grids and reference planes shown and, on the edge two curtain walls share at a corner, to the lines of the other one. With **Mullion**, SHIFT held places the mullion on the segment clicked only, CTRL on every segment that has none.

Mullion and panel types

Besides rectangular mullions (**Width on side 1** and **2**, **Thickness**, **Angle** between −90° and 90°, **Position**), the template has a **Circular Mullion** (**Radius**) and four corner mullions for the edge where two curtain walls meet: **L** (two legs along the curtain walls, **Leg 1** and **2**), **V** (two legs square to the curtain walls), **trapezoid** (**Center Width**, **Depth**) and **square** (**Depth 1** and **2**). A corner mullion is drawn once at the corner and the panels of both curtain walls stop at its sides; at a free end it is straight. The **Offset** of a mullion or panel type moves it across the curtain wall, towards the outside.

The curtain wall type and the mullion type have all the identity data (keynote, manufacturer, type mark, cost...). In the curtain wall's Properties, **Vertical Offset** and **Horizontal Offset** start the type's lines that far from their justification point, and **Structural Usage** says how it is used.

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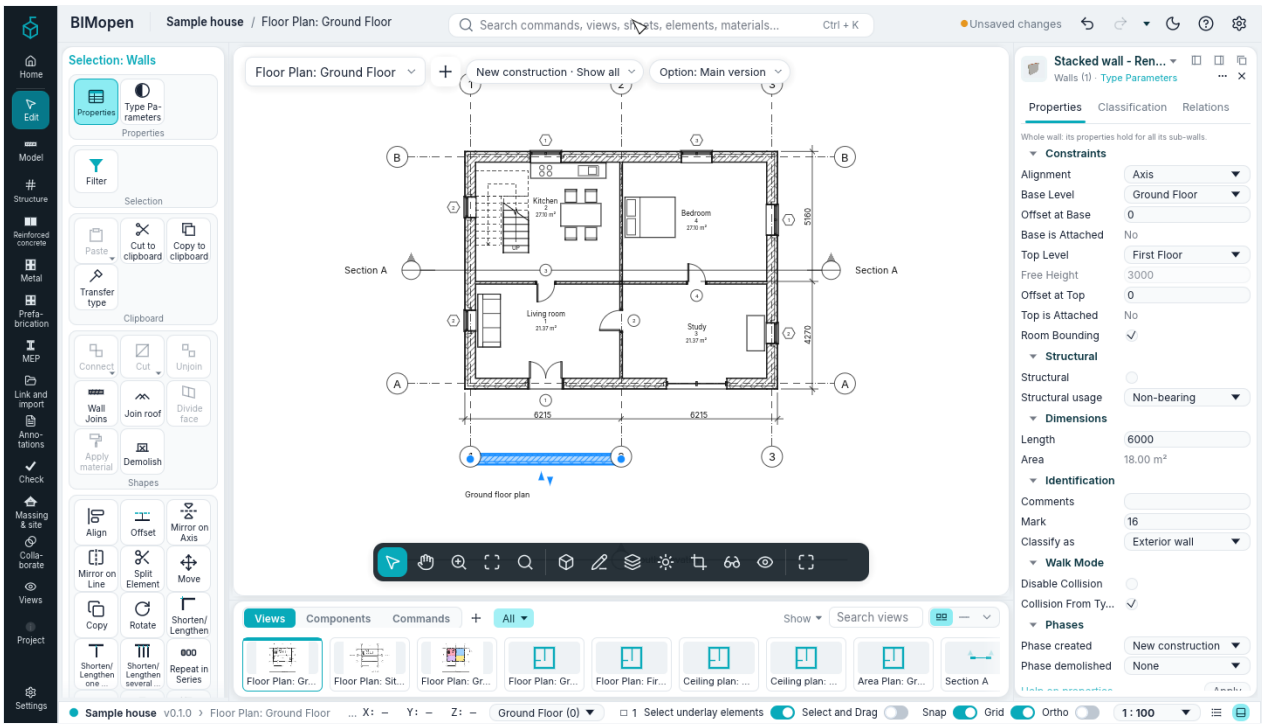
Stacked walls



A stacked wall selected, with its Properties

A **stacked wall** is made of sub-walls one above the other: a rendered wall over a stone plinth, for instance. Each sub-wall is a basic wall type, with its own layers and materials. The template has one, **Stacked wall - Render over stone plinth** (the exterior wall over a 0.9 m stone plinth): pick it in the type selector of **Wall** and draw it as any wall.

The stacked wall holds the line, the levels and the height; its sub-walls follow them. In plans, sections, elevations and the 3D view each sub-wall is drawn with its own type's layers, in the colours of their materials. The sub-walls join the walls they meet, each at its own height: at a corner of two stacked walls plinth meets plinth and the wall above meets the wall above, a basic wall meets each sub-wall in turn, and two stacks of different heights or thicknesses meet over each height they share; every layer closes round the corner, in plan and in 3D. The **Wall Schedule** lists the sub-walls, one row each. A door or window placed on a stacked wall cuts every sub-wall it passes through, its sill measured from the stacked wall's base.

A click selects the whole wall; **Tab** goes on to each sub-wall, from the bottom up (the status bar says «Tab · next element»). The Properties of the whole wall hold for all its sub-walls; its **Alignment** is the plane it is drawn on, as for any wall: the faces of the band its sub-walls fill, or the core of the variable sub-wall. Giving a basic wall a stacked type, or a stacked wall a basic one, keeps it on its alignment. Of a sub-wall only **Room Bounding** and **Role** change: the rest comes from the stacked wall and its type, and its type is changed in the stacked type. **Break up into single walls**, in the context menu and with the wall selected, makes each sub-wall an independent basic wall where it stands; they do not go back together (Ctrl+Z undoes it). The boundary of a stacked wall or of a sub-wall cannot be edited yet.

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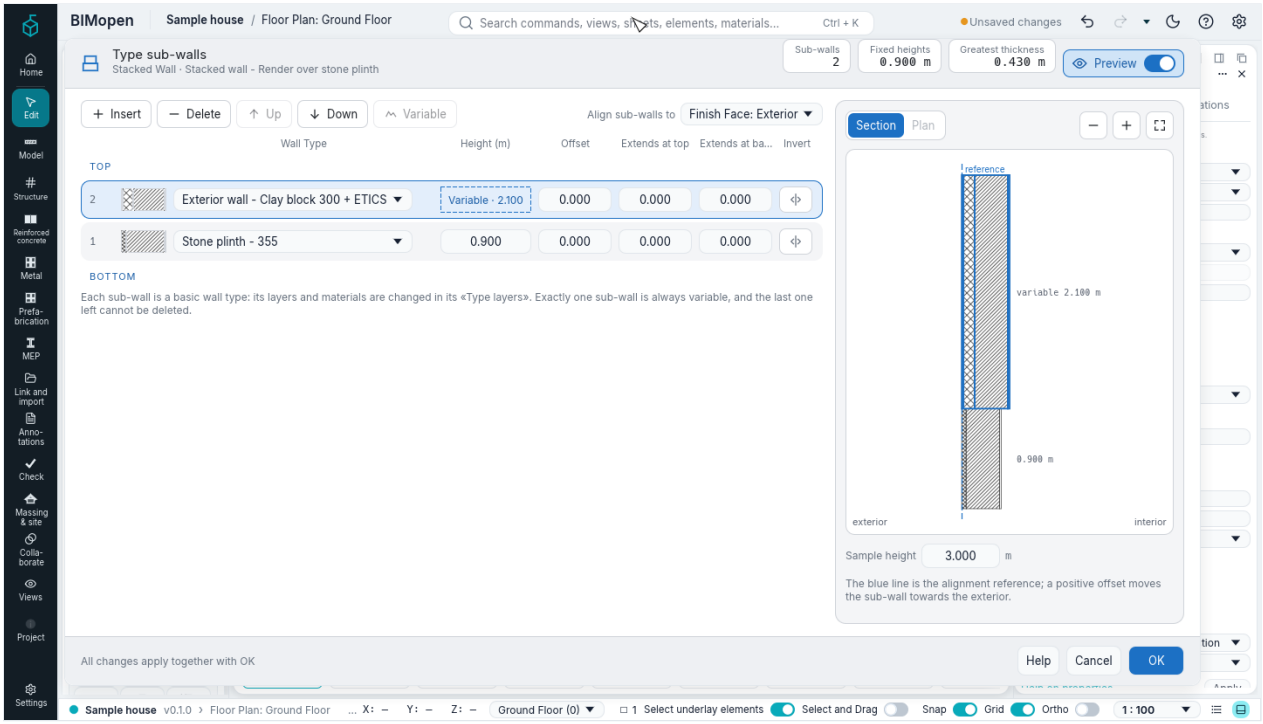
The sub-walls of the type



The Type sub-walls of the sample stacked type

Type Parameters of a stacked wall type shows **Sub-walls** with **Edit...**, which opens **Type sub-walls**. At the top are the type and three figures: how many sub-walls, the sum of the fixed heights and the greatest thickness. The table lists the sub-walls from **TOP** to **BOTTOM**, numbered from the bottom: **Wall Type** (a basic wall type), **Height** in metres, **Offset** from the reference line (positive towards the exterior), **Extends at top** and **Extends at base** (how far the layers outside the core go beyond the sub-wall; negative, they stop short) and **Flip**. Exactly one sub-wall is **Variable**: its height is what remains of the wall's height.

Insert adds a sub-wall below the selected one, **Delete** removes it (never the last one; deleting the variable one makes the one above variable), **Up** and **Down** move it, **Variable** makes it the variable one. A positive **Extends at base** hooks the top of the sub-wall below («Hooked»): its layers stop where those of the one above begin. **Align sub-walls to** chooses the plane of each sub-wall that lies on the reference line, among those of **Alignment** (**Axis**, **Finish Face: Exterior**...).

On the right is the section of the type, as tall as **Sample height**, the reference line in blue, the exterior on the left; a click on a sub-wall selects its row. The slim **Preview** strip hides and shows it, and the dialog keeps its size and place. **OK** applies the changes in one step (Ctrl+Z undoes it); **Cancel** leaves the type as it was.

Levels

Create ▸ References ▸ Level draws a level in a section or an elevation with two clicks, at the height of the first one; while drawing, a temporary dimension gives the distance to the nearest level. The options strip shows:

- **Make Plan View**: on, the new level gets its floor plan; off, it is a reference level with no plan, which walls and other elements can still take as their base or top;
- **Plan View Types...**: the kinds of plan the level gets (**Floor Plan**; ceiling plans are not available yet);
- **Offset**: how far above the clicks the level goes.

Work planes

Every view draws and places on a work plane. A plan's work plane is its level, a 3D view's the lowest level; a section or an elevation has none until one is set. The plane is kept with the view.

Setting the work plane

Create ▸ Work Plane ▸ Define opens the **Work Plane** dialog. **Current Work Plane** shows the plane the view uses now. Under **Specify a new Work Plane**:

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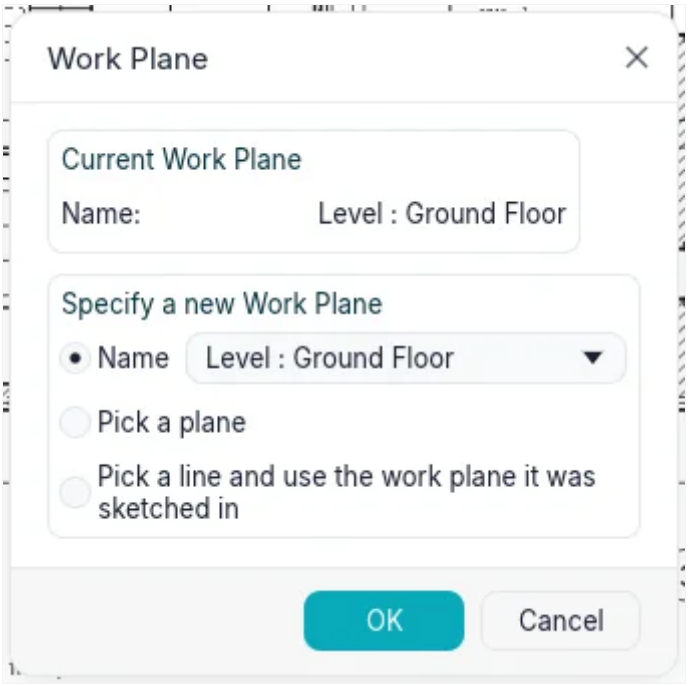

Work Plane

- **Name**: pick a level, a grid or a named reference plane in the list (picking one selects **Name**), then **OK**;
- **Pick a plane**: after **OK**, click a wall (its face on the side of the cursor in a plan, the face turned to you in a section or a 3D view), a grid, a reference plane or, in a section, a level;
- **Pick a line and use the work plane it was sketched in**: after **OK**, click a wall or a room separator: its level becomes the work plane.

A plane that does not lie in the view, such as a grid picked in a plan or a level in a section, is not set there: **Go to View** lists the views it lies in (the sections and elevations facing a vertical plane, the plans of a horizontal one) and the 3D views, and **Open View** opens the chosen one with that work plane. **Undo** takes a new work plane back.

Showing the work plane

Create ▸ Work Plane ▸ Display draws the work plane's grid in the view, over the model, when the plane lies in the view (in a 3D view always); **Show** again hides it. In a section with no work plane it opens the **Work Plane** dialog first. Click the border of the grid to select it: the options strip shows **Spacing**, the distance between its lines (1 m to start with). While the grid is shown, the points of walls and other lines drawn in the plan snap to the crossings of its lines.

In a 3D view, a click where there is no element under the cursor lands on the view's work plane when one has been set for it: a component placed east of the house with the work plane set to the first floor stands on the first floor.

Reference planes

Create ▸ Work Plane ▸ Reference Plane (or type `RP`) draws a reference plane with two clicks, like a grid. Drawn in a plan it is the vertical plane through the line, shown as a green dashed line in every plan and in the sections and elevations that cross it. Drawn in a section or an elevation it is the plane through the line across the view's depth (a horizontal line makes a horizontal plane), shown in the views facing the same way. Select one to give it a **Name** in the element panel: a named reference plane is listed in the **Work Plane** dialog and its name is written at the start of its line. Reference planes are snapped to, moved, copied, rotated, mirrored and deleted like grids.

3D lines and 3D text

Two elements of the model stand on a work plane: the **3D line**, a line that exists in space and is seen in every view that reaches it, and the **3D text**, letters cut out of the plane and extruded, for signs and lettering on walls. Both are in **Model ▸ 3D elements**.

3D lines

Model ▸ 3D elements ▸ 3D line (or type `LM`) draws on the view's work plane with the same **Draw** panel as detail lines (lines, rectangles, polygons, circles, arcs, fillets, splines, ellipses, **Pick Lines**) and the **Line style** list. The options strip has **Placement**

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Plane (the levels, grids and named reference planes, or **Select…** for the **Work Plane** dialog), **Continue**, **Offset**, **Radius** and **Move With Nearby Elements**; lengths and radii are typed while drawing, as for detail lines. A plane that does not lie in the view asks **Go to View**; an elevation or a section with no work plane opens the **Work Plane** dialog first. In a 3D view the line is drawn where the cursor meets the work plane.

A 3D line is seen in every plan whose view range reaches it (from the view depth up to the cut plane; in a ceiling plan from the cut plane up), in the sections and elevations whose depth holds it, in 3D views and on sheets, always in its line style. Its properties: **Work Plane** (read-only; once moved off it), **Move With Nearby Elements** (a line parallel to a wall, or concentric with a curved wall, follows it when the wall moves), **Line style**, **Center Mark Visible** for arcs and circles, **Detail Line** (read-only, unticked), **Length** and the phases. It is moved, copied, rotated, mirrored, arrayed, trimmed, split, offset and aligned like a detail line; in a section or an elevation **Move** and **Copy** also go up and down and **Rotate** turns it about the axis across the view.

With lines selected, **Convert Lines** turns 3D lines into detail lines of the active view, and detail lines into 3D lines on its work plane; a selection of both kinds asks which to convert. **Edit Work Plane** moves the selected lines (or texts) onto another plane, straight across onto a parallel one. In DXF and DWG a 3D line is on the model layer of its style (`A-MODL-…`); in IFC it is an annotation with its curve.

3D text

Set the work plane (a wall's face, for lettering on it), then **Model ▸ 3D elements ▸ 3D text**: write the text in **Edit Text** (several lines) and click **OK**. The outline of the letters follows the cursor on the plane; a click places them, extruded by their **Depth** out of the wall's face (up from a level). **Edit Text** (or a double click, or **Text** in the properties) opens the dialog again; dragging moves the text on its plane, **Edit Work Plane** and **Pick New Host** take it to another plane or wall face.

The instance has **Work Plane**, **Text**, **Horizontal Align** (left, centre, right), **Material** (the colour comes from it), **Depth**, **Comments**, **Mark** (one used by another text is allowed, with a warning), **Subcategory** and the phases; the type has the **Text Font** (the fonts BIMopen ships), **Text Size**, **Bold**, **Italic** and the identity data. The first 3D text makes the type «3D text 300 mm». A plan whose cut plane crosses the letters shows them cut; below it, in projection. Realistic views, PDF and DXF draw the text with its material; IFC exports it as a building element proxy with its solid.

At the command line

The same operations, typed:

```
attach                      attach the selected walls to the roof over them
attach 412                  attach their top to element 412 (a roof or a floor)
attach base 398             attach their base to the floor 398
detach all                  detach them from everything
walljoin 0,0 {"kind":"Giunto d'angolo","cleanup":"Non calcolare unioni"}
walljoin 57 1 disallow      wall 57 does not join at its end
rehost 120 12,4             move door or window 120 into the wall at 12,4
door {"point":[3,0],"tag":{"vertical":true,"leader":true,"length":1}}
refplane 3,-2 3,11 Entrance axis   a reference plane in the open plan, named
workplane Entrance axis     the open view's work plane, by name
showworkplane on 0.5        show the work plane's grid, 0.5 m apart
modelline 20,0 24,0         a 3D line on the view's work plane
modeltext {"text":"CASA 12","point":[1,1],"plane":{"wall":57},"coords":"plane","depth":0.05}
convertlines                3D lines of the selection into detail lines, or the reverse
```

Not available yet

Walls along ellipses, **Pick Faces**, structural walls, stacked walls, wall sweeps and reveals, vertically compound layers (regions), **Join Geometry**, phases, and skylights in roofs.

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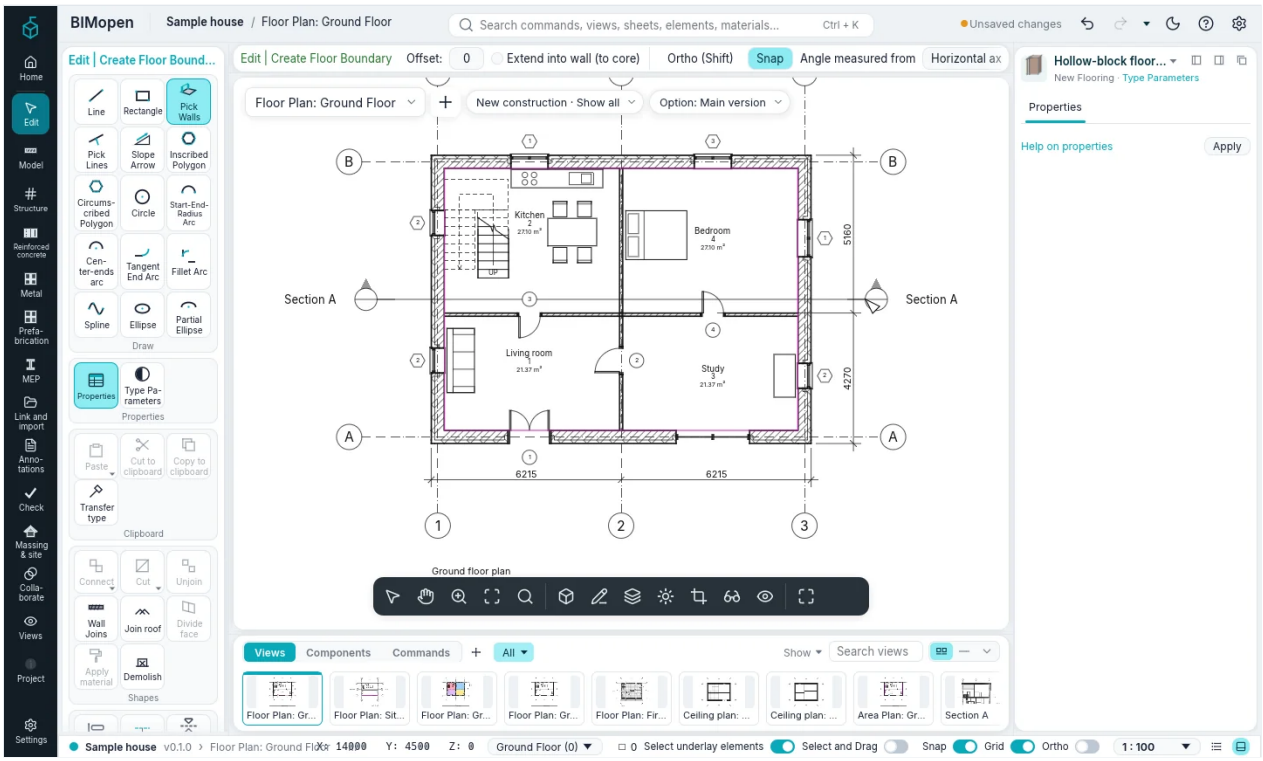
Floors, roofs, stairs and components

Floors, roofs, stairs and railings are drawn in **sketch mode**: their lines are sketched first, and the element is made only when the sketch is finished. Components (furniture, casework and plumbing fixtures) are placed with a click. Curtain walls are described in Walls, doors and windows.

Sketch mode

Create ▸ Build ▸ Flooring, **Roof**, **Stairs** and **Railing** open sketch mode. The contextual card is named after what is sketched (**Edit | Create Floor Boundary**, **Create Roof Footprint**, **Create Stairs Sketch**, **Create Railing Path**) and has:

- the **Mode** group: **Finish Edit Mode** makes the element; **Cancel Edit Mode** drops the sketch and leaves the model as it was;
- the **Draw** group, with the drawing tools of the element;
- for stairs and railings, the **Tools** group.



Sketch mode: the contextual card and the magenta lines

Nothing is made while sketching: the lines are magenta, and the model changes only with **Finish Edit Mode**. The whole element is then one step of the undo history: one `Ctrl+Z` takes it away.

Drawing lines

Tool	How it draws
Line	Click point after point: each click ends a line and starts the next. The length and angle of the line under the cursor are shown beside it; `Shift` keeps it horizontal or vertical. With **Continue** off on the options strip, each line is drawn on its own. Clicking the start of the chain again closes the loop Type a length and press `Enter`: the line ends that far along the cursor's direction (a stair's run too).
Rectangle	Two opposite corners: four lines
Pick Walls	Click a wall: its line is added to the sketch (see below). Press `Tab` over a wall, then click, to pick the whole chain of walls joined end to end
Pick Lines	Move over a wall, a grid, a line or the edge of a floor or roof: the line under the cursor is highlighted and named (a wall's faces, each layer boundary, its two **Core Boundary** lines and its axis); `Tab` goes to the next one there. Click to add a sketch line on it (set off by **Offset** towards the cursor)
Slope Arrow	Click the tail, then the head of the arrow (floors and roofs)

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Offset on the options strip moves the drawn lines (and the lines of picked walls) by that distance.

The arcs of the **Draw** group (and a circle, as two halves) are one sketch line each: picked, measured and dragged as arcs, kept by the floor, roof, stair, wall boundary or filled region, and given back as arcs by **Edit Boundary** or **Edit Sketch**. In every view a sketch point snaps to the ends and middles of the sketch's lines and arcs, to arcs' centres, where lines cross and onto a line or an arc; from the point being drawn from, to the foot of a perpendicular and to the point where a line touches an arc. The marker and its name show where the point snaps; `S C` and `S P` ask for the centre or the perpendicular for the next point.

Floors and roofs need closed loops. A loop drawn inside the outer one is an **opening**: the floor or the roof has a hole there. When **Finish Edit Mode** finds lines that do not close or that cross, a message says so and the sketch stays open to be fixed (**Continue**).

Picking and changing sketch lines

`Esc` ends the chain being drawn; a second `Esc` puts the drawing tool down (so does **Modify**). While the sketch is open its lines are elements of the view, and **Modify** works on them as on other lines; nothing else can be picked until the sketch is finished:

- the line under the cursor is highlighted; a click selects it, `Ctrl` adds lines, `Shift` takes them away, a window selects several, `Tab` the chain of joined lines;
- the end of a line selected on its own drags alone: the line next to it stays where it is and the join is broken; with a chain selected (`Tab`), the ends where its lines meet drag together; a dragged line takes the lines joined to it along;
- **Move**, **Copy**, **Rotate**, **Mirror on Line**, **Repeat in Series**, **Align**, **Offset**, **Shorten/Lengthen** and **Split Element** work on them, and the sketch stays open;
- one selected line shows temporary dimensions to the nearest parallel line on each side (another sketch line, the wall's original outline, a wall, a grid, a level): click the value and type a distance to move the line; the symbol beside it makes a permanent dimension, kept while the sketch is open;
- its properties are in the element panel: **Defines Slope** and **Slope**, for a roof line picked from a wall its **Overhang**, and its **Length**.

Align takes as its reference the line under the cursor, and `Tab` goes to the next one there, so a sketch line can be aligned to a wall's layer or core boundary. While drawing or dragging an end, the points snap to the corners of the walls' faces, layers and core boundaries, then onto those lines. `Del` deletes the selected lines. `Ctrl+Z` while sketching takes back the last change to the sketch, not to the model. All this works in plans, in sections and elevations (a wall's boundary) and, for selecting, dragging ends and deleting, in the 3D view.

Editing the sketch of an element

Select a floor, a roof, a stair or a railing: the **Mode** group of its contextual card has **Edit Boundary** (floors), **Edit Footprint** (roofs), **Edit Sketch** (stairs) or **Edit Path** (railings). Sketch mode opens with the element's lines, openings, slopes and arrow; **Finish Edit Mode** changes the same element, **Cancel Edit Mode** leaves it unchanged.

The sketch opens with **Modify** active: its lines can be selected and changed at once.

In a plan the sketch is drawn on the level; in the 3D view it lies on the element's own plane (the floor's top, the roof's base) and the 3D view stays open.

A new floor can be sketched in the 3D view too: **Flooring** started there puts the sketch on the plane of the level, the line being drawn follows the cursor with its length, and **Finish Edit Mode** makes the floor without leaving the 3D view. In the 3D view the options strip's **Level** chooses the work plane: the level whose plane the sketch lies on and the floor goes on. **Pick Walls** takes the wall whose face is under the cursor. Roofs by footprint, stairs and railings are sketched in a plan.

Floors

1. Click **Create ▸ Build ▸ Flooring**. Sketch mode starts with **Pick Walls**.
2. Click the walls around the floor, or press `Tab` over one of them and click to pick them all. Each line goes on the **interior face** of its wall; the lines of walls that meet are trimmed or extended to meet at the corner.
3. Or draw the boundary with **Line** or **Rectangle**; draw a loop inside it for an opening.
4. Click **Finish Edit Mode**.

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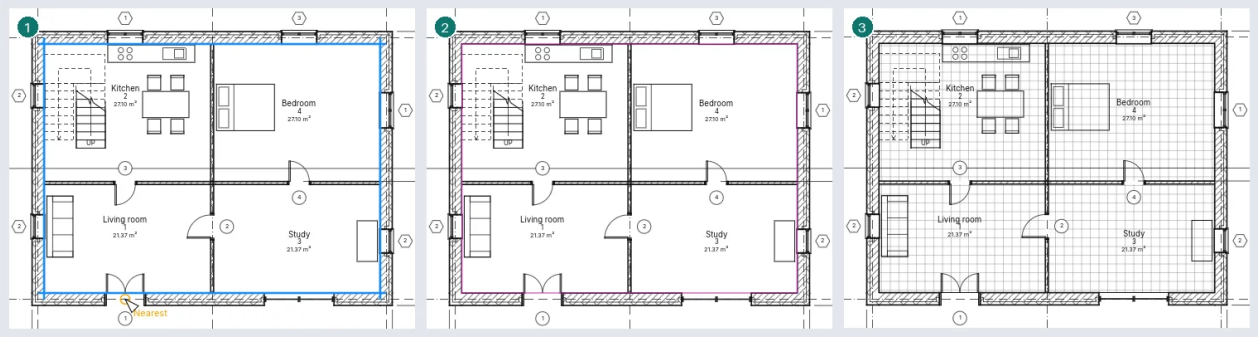

A floor: TAB on the walls, click, Finish Edit Mode

On the options strip, **Extend into wall (to core)** puts picked lines into the wall up to the face of its core, and **Offset** moves them (a negative offset moves them away from the walls). The type of the new floor is the one in the type selector at the top of **Properties**.

Sloped floors

A floor slopes in one direction, set in one of two ways while sketching:

- a **Slope Arrow**: its properties are **Specify** (**Height at Tail**, from the heights of its two ends, or **Slope**), **Height Offset at Tail**, **Height Offset at Head** and **Slope**. The heights are measured from the top of the floor;
- one boundary line with **Defines Slope** ticked and its **Slope** in degrees: the floor rises from that line, inwards.

The floor is sloped in 3D and in sections. Its **Slope Angle** in **Properties** changes the slope afterwards.

Floors over walls

When a floor crosses walls (an upper floor resting on the walls below, for instance), its slab is cut out of them: in 3D and in sections the wall stops under the floor and goes on above it, and walls go on through the floor's openings.

Floor properties

Property	What it is
Level, **Height Offset From Level**	The level and the height of the top face from it
Slope Angle	The slope of a sloped floor, in degrees
Thickness	From the type's layers (read-only)
Perimeter, **Area**, **Volume**	Without the openings (read-only)
Comments, **Mark**	Free text

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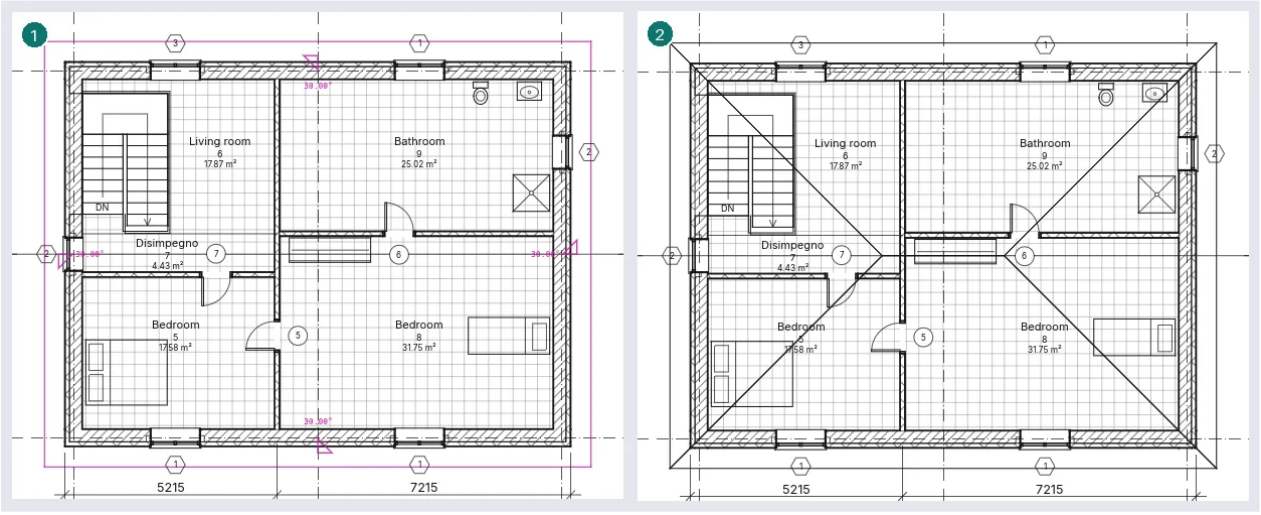
The properties of a floor

Type Parameters has the floor's **Structure** (its layers) and its default thickness.

Roofs

1. Click **Create ▸ Build ▸ Roof**. Sketch mode starts with **Pick Walls**; the options strip has **Defines Slope**, **Overhang** and **Extend into wall (to core)**.
2. Pick the walls (Tab for the whole chain). Each line goes on the **exterior face** of its wall, moved out by the overhang (from the face of the core with **Extend into wall (to core)**).
3. Or draw the footprint with **Line** or **Rectangle**; a loop inside is an opening.
4. Click **Finish Edit Mode**.



A roof: the footprint, then the roof

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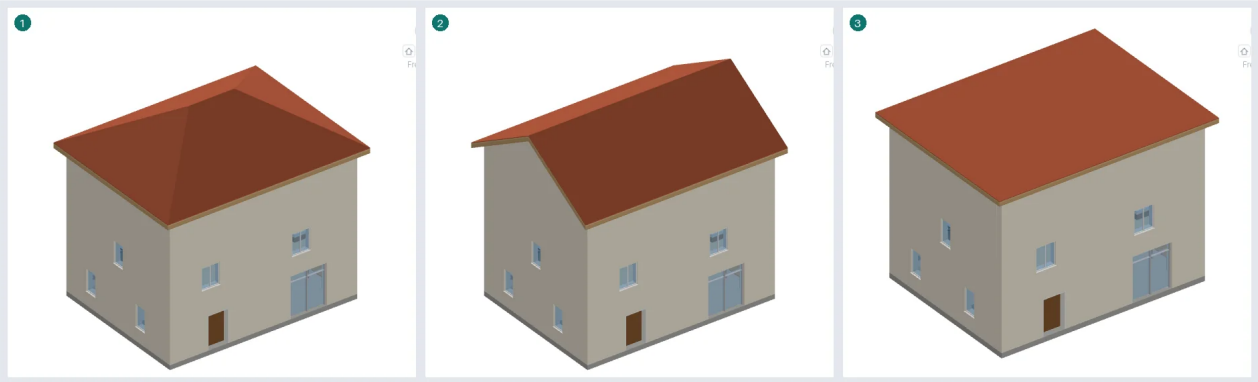
Every line with **Defines Slope** rises inwards at its **Slope** (30° by default); a small triangle with the slope is drawn beside it. The shape comes from those lines:

- all the lines define the slope: a **hip** roof;
- two opposite lines do not: a **gable** roof, the walls under the gables rise to the ridge when attached;
- no line defines the slope: a **flat** roof, or a roof sloping along a **Slope Arrow** (then no line may define a slope).



Hip, gable and flat

To change one line, press Esc twice, click the line and change **Defines Roof Slope**, **Slope** or (lines picked from walls) **Overhang** in **Properties**.

Select walls, click **Modify Wall ▸ Link Top/Base**, then the roof: they rise to its underside, gables included (Walls, doors and windows tells more).

Roof properties

Property	What it is
Base Level, **Base Offset From Level**	Where the eaves are: by default on top of the walls of the level
Slope	The slope of every line that defines one (or of the arrow)
Thickness, **Maximum Ridge Height**, **Area**, **Volume**	Read-only
Comments, **Mark**	Free text

In plan a roof shows its outline, ridges and hips; in 3D its slopes, overhangs and layers.

Extrusion roofs

An extrusion roof is an open profile drawn on a vertical work plane and pushed along the plane's normal: a canopy over a door, a barrel roof, a dormer roof drawn on the front wall.

1. Open an elevation, a section or a 3D view (started in a plan, **Go to View** asks for one).
2. Click **Create ▸ Build ▸ Roof ▾ ▸ Extruded roof**. The **Work Plane** dialog opens: pick a named plane, or **Pick a plane** and click a wall's face, a grid or a reference plane. The plane must be vertical.
3. In **Roof Reference Level and Offset** choose the **Level** (the highest one by default) and the **Offset**: a dashed line marks that height.
4. Draw the profile as one **open** chain with **Line**, the arcs or **Pick Lines**. It is the roof's top: the layers of the roof type hang under it.
5. Click **Finish Edit Mode**. The roof runs from the plane (**Extrusion Start** 0) across the building, away from you (**Extrusion End**).

A closed profile, a branch or a profile that turns back over itself (a part over another, or a vertical line) is refused at **Finish Edit Mode** with the reason: an extruded roof must face up.

Selected, an extrusion roof has **Edit profile** (in an elevation or section facing its plane, or in 3D) and **Edit Work Plane**; in a plan and in 3D the arrows at its two ends drag **Extrusion Start** and **Extrusion End**. Walls attach to its underside as to any roof, and **Opening ▸ Vertical** cuts it. It follows its plane: when the wall it was drawn on moves, the roof moves with it.

Property	What it is
Work Plane	The plane of the profile (read-only; change it with **Edit Work Plane**)

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Property	What it is
Extrusion Start, **Extrusion End**	Where the extrusion begins and ends, along the plane's normal
Reference level, **Level Offset**	The height the profile is measured from: the whole roof moves with them
Rafter Cut, **Fascia Depth**	How the eaves at the two ends of the profile are cut
Thickness, **Area**, **Volume**	Read-only

Joining roofs

Modify ▸ Shapes ▸ Join roof (in the BIMopen style **Modify ▸ Join roof**) joins a roof to another roof's top face or to a wall: the dormer's roof meets the main roof.

1. Click **Join roof**. The roof edges highlight under the cursor, in a plan, a section, an elevation or a 3D view.
2. Click the edge to join: an edge that does not define the slope (a gable), or an end of an extrusion roof.
3. Click the roof or the wall to join it to. The edge moves out (or back) until the roof meets the other roof's top face, or the wall's face; what lies under that face is cut away.

A joined edge shows orange: clicking it again disjoins it. The join follows: when the main roof's slope or height changes the joined roof meets it again, and a join to a deleted roof goes.

Sloped glazing

A sloped glazing is a roof made of glass: a footprint or an extrusion roof whose type is **Sloped Glazing**. Every planar face of the roof carries curtain grids, mullions and panels.

1. Draw the roof as usual (**Roof** or **Extruded roof**).
2. Select it and pick **Sloped Glazing** in the type selector. The first time, BIMopen makes the type with the project's glazed panel and mullions.
3. Each face is divided by the type's two grids: **grid 1** runs up the slope, spaced along the eaves; **grid 2** runs along the eaves, spaced up the slope.

With the glazing selected, the curtain tools work on the face under the cursor, in a plan, a section, an elevation or a 3D view:

- **Curtain grid** places a line where you click: of grid 1 near the eaves or the ridge, of grid 2 near a side. Press TAB to switch to the other grid. **All Segments**, **One Segment** and **All Except Picked** work as on a curtain wall.
- **Mullion** puts mullions of the chosen type on a **Grid Line**, a **Grid Line Segment** or **All Grid Lines** of the face.
- **Panel** changes the panel you click to the type in the selector: glazed, solid, a door or empty.
- **Add/Remove Segments** takes off the segment you click (the two panels become one), or puts it back.

Property	What it is
Curtain Panel (type)	The panel of every cell
Join Condition (type)	Which mullions run through where the grids cross: grid 1, grid 2, or the borders with one of them
Grid 1/2 Pattern (type): **Layout**, **Spacing**, **Adjust for Mullion Size**	How each grid places its lines: none, a fixed distance, a fixed number, a maximum spacing
Grid 1/2 Mullions (type): **Interior Type**, **Border 1 Type**, **Border 2 Type**	The mullions on the lines and on the face's edges
Number, **Justification**, **Angle**, **Offset** of grid 1 and 2	This roof's own values: how many lines (a fixed number), where the remainder goes, how the lines turn on the face (−89° to 89°), how far they move
Panels, **Mullions**	Read-only counts

In a plan the glazing shows its panels and mullions seen from above; **Visibility/Graphics** hides curtain panels and mullions as for curtain walls. IFC writes a roof made of plates and members.

Soffits

A soffit closes the underside of the eaves, between the walls and the roof's edge.

1. In a plan or a 3D view click **Create ▸ Build ▸ Roof ▾ ▸ Soffit**. Sketch mode starts with **Pick Roof Edges**.
2. Click the roof's edges: their lines stay locked to the roof.
3. Click **Pick Walls** and the exterior faces of the walls under the roof (TAB picks the chain): these lines stay locked to the walls.

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4. Close the outline with **Line** and **Shorten/Lengthen**, then click **Finish Edit Mode**.

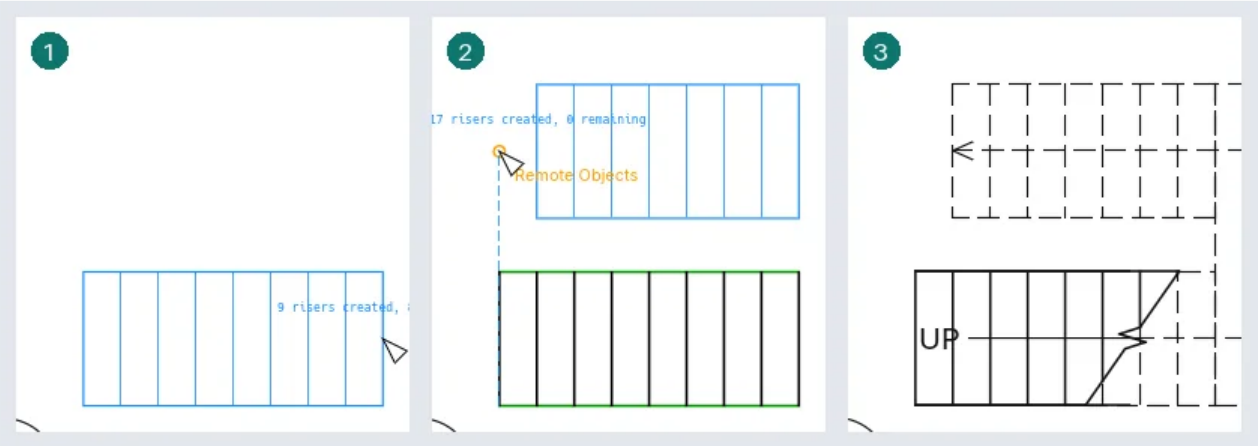
The soffit's top is under the roof's eaves; its layers (the type's **Structure**) hang below. When the roof or the walls move, the soffit follows them. A **Slope Arrow**, or a line that **Defines Slope**, slopes it. Selected, **Edit Boundary** opens its sketch again.

Property	What it is
Structure, **Thickness** (type)	The layers, top down, and their sum
Level, **Height Offset From Level**	Where its top is
Room Bounding	Whether rooms stop at it
Slope	Replaces the slope of its sloping line or arrow
Perimeter, **Area**, **Volume**	Read-only

Sections cut it with its layers, the ceiling plan of its level sees it from below; it has its own category in **Visibility/Graphics** and in schedules, and IFC writes it as a covering.

Stairs

1. Click **Create ▸ Vertical Links ▸ Stairs**. Sketch mode starts with **Run**.
2. Click where the first riser is, then where the run ends: the run snaps to whole treads, and the status bar counts the risers made and left (for example `9 risers created, 8 remaining`).
3. To turn, click again away from the end of the first run and draw the next one: a landing joins them. At any angle the landing is bounded by the two runs' risers, its outer corner where their outer sides meet; where the runs end out of line their inner sides also run on until they meet, a square corner rather than a slanting edge.
4. When every riser is placed, click **Finish Edit Mode**.



A stair by runs: the count of risers made and left

Dashed guides show where the next run can start: the last run's extension beyond its end and the lines of its first and last risers. A click near them snaps onto them. In a 3D view the stair is drawn the same way, on the plane of the active level.

A curved run (a spiral stair): with **Run** active, click **Center-ends arc**, then the centre, the start of the run line (its distance from the centre is the radius of the walking line) and the end, going round past half a turn if needed, but under one turn (further, the run stops short and the status bar warns that the runs would overlap). The risers radiate from the centre a tread apart on the run line; riser lines drawn radiating from a centre make a curved run too. A second curved run round the same centre and on the same run line (a click near them takes them) is joined to the first by a landing along the arc.

A stair can also be drawn by its lines. In its sketch, click **Draw ▸ Boundary** and draw the left and right edges (green), then **Riser** and draw the riser lines (black) between them, with **Line** or **Pick Lines**. Parallel, evenly spaced risers make a run, as wide as the risers; a landing joins two runs out to the boundary lines. Each riser is as high as the storey divided by the desired number of risers; with fewer riser lines the stair stops short of the top (**Properties** shows the actual number). **Edit Sketch** gives the same lines back. Each run drawn with **Run** is at once boundary and riser lines like these: they can be moved or deleted like the others. A landing follows its boundary lines: split its far edge with **Split Element** and drag a piece to notch it, or draw it wider than the runs.

Riser lines need not be straight or parallel: draw them with any drawing tool, an arc too, or slant one. Each tread is then the area between a riser line and the next one of its run, bounded by the boundary lines, and the landing the area between the last riser of one run and the first of the next; the stair is built that way in plan, section, elevation and 3D (a monolithic stair as one

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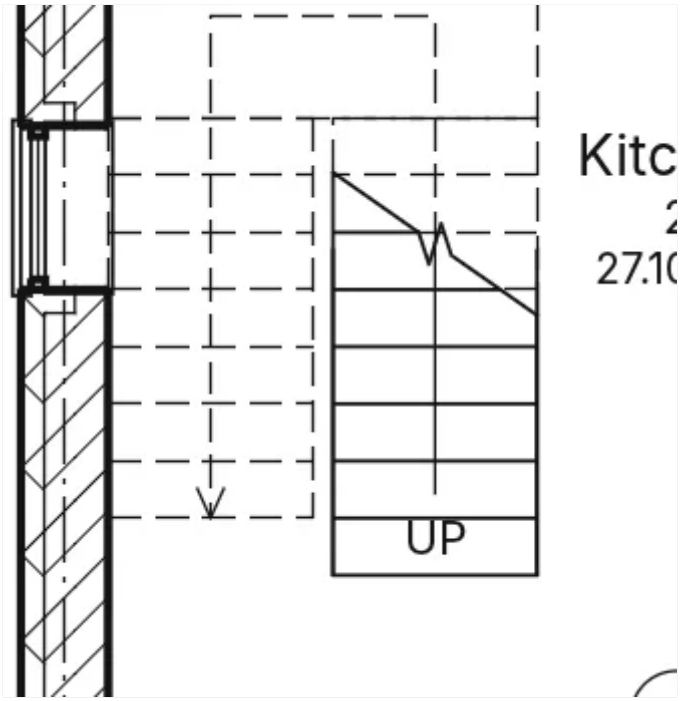
slab under its treads), and its railings follow the boundary lines over those treads. A riser line running past a boundary is cut there. A riser line that does not touch both boundaries, or more or fewer riser lines than the stair's height needs, gives a warning; the stair is made all the same. A stair drawn this way keeps its lines when **Width** or **Actual Tread Depth** changes: change them in **Edit Sketch**. The risers follow one another as the boundaries join them, whichever line was drawn first: the stair starts at the riser the first boundary line starts on, otherwise at the first riser drawn; a tread drawn deeper than the others (short of a landing) stays in its run. The type's **Nosing Length** and **Nosing Profile** run along each riser line, and along the edge of a landing over the riser that reaches it (in plan, a line in front of each riser); its **Middle Stringers** run under the treads from riser line to riser line, following them where the boundaries turn; a monolithic stair's **Underside of Stringer** is **Smooth** or **Stepped** here too.

The **Width** on the options strip is that of the new stair (by default the type's). **Tools ▸ Railing Type** chooses the railings of the new stair: a railing type, put on both sides, or **None**.

The number of risers comes from the height between **Base Level** and **Top Level** (with their offsets) and the type's **Maximum Riser Height**; changing a level or an offset in **Properties** changes the risers. **Edit Sketch** reopens its boundary and riser lines: change or delete them (click, `Del`) and draw runs again; the railings of the stair follow. **Multistory Top Level** repeats the stair, with its railings, storey by storey up to that level; a storey whose height is not the stair's gets no copy, and a warning names it. With the stair selected, the blue flip arrows beside its first riser make it go up from the other end; **Up Label**, **Up Arrow**, **Down Label** and **Down Arrow** in **Properties** show or hide the plan symbol's parts. The square grip under the **UP** text drags it elsewhere.

In the plan of its base level the stair shows its treads up to the cut line, the break line, the arrow and the **UP** text; in the plan of its top level it shows all of it, landings and railings included, with no break line, and the arrow and the **DN** text down to the first riser. On a level between two storeys of a multistory stair both parts show; in 3D its treads and risers, or a monolithic slab, its landings and railings.



A stair in plan

On the base level the part of the stair above the cut is drawn too, dashed. Each part has a subcategory of **Stairs** in **Styles by Category**: **Outlines**, **Riser Lines**, **Nosing Lines**, **Supports**, **Cut Marks** and **Up Arrows**, each also ; their weight, colour and pattern change there, and view by view in **Visibility and Graphics**.

Annotations ▸ Tag ▸ Tread Number: choose the **Reference** (**Left**, **Left Quarter**, **Center**, **Right Quarter**, **Right**), what to number (**Risers** or **Treads**) and the **Start Number** in the options bar, then click a stair in a plan. Its steps are numbered one after the other through all its runs, in that plan only. With the stair selected, its **Properties** show the numbering's settings.

How a stair is built

Type Parameters of the stair decide its construction:

- **Monolithic Stairs** ticked: each run is a concrete slab cut to the steps; its underside runs parallel to the slope, **Structural Depth** below the inner corners of the steps, and the landings are as thick: the sloping underside of a run meets the

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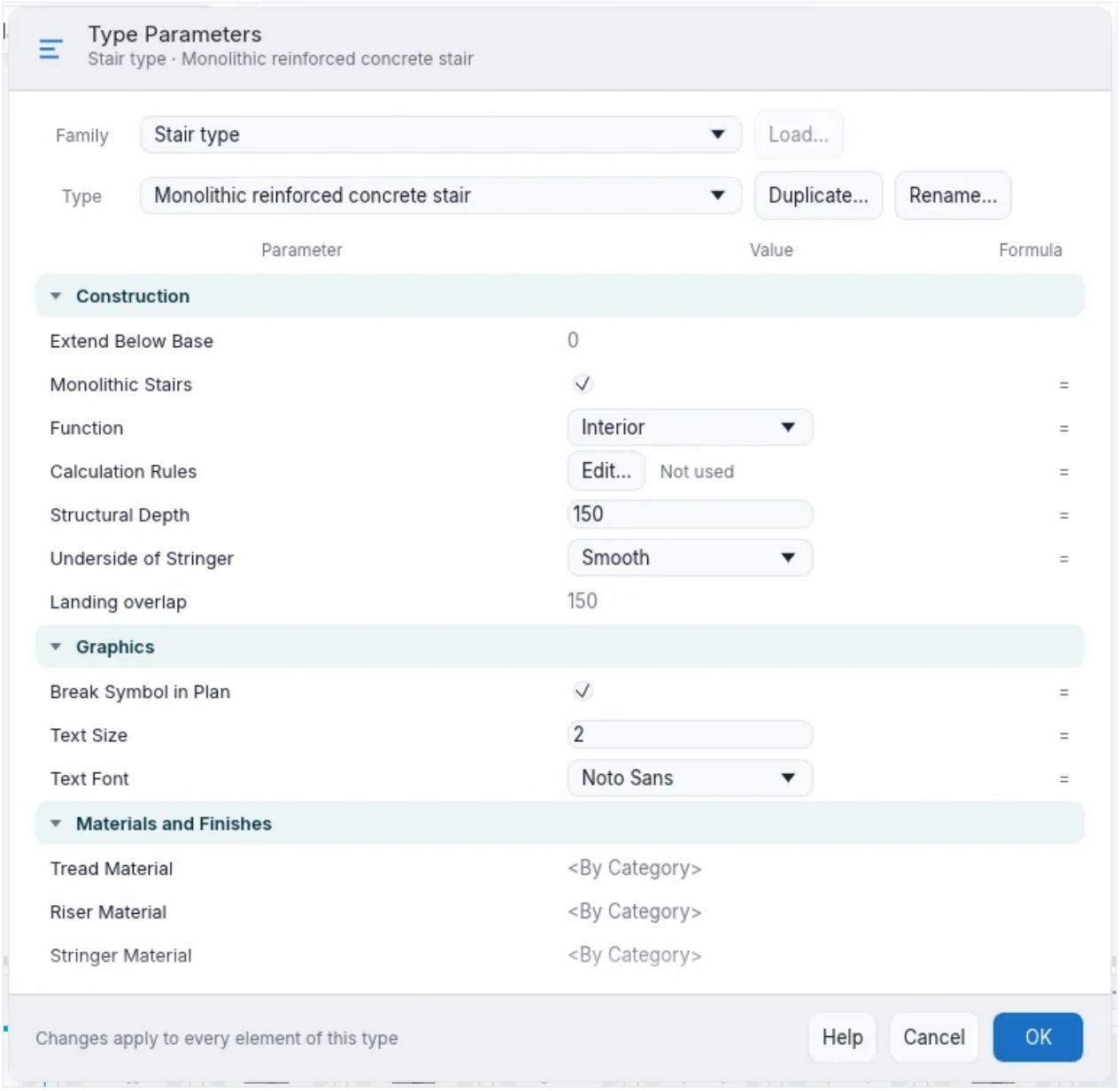
landing's underside along a line, without a step; each run ends with its last tread: the landing, flush with it, or the floor above makes the last riser, so steps and slab show as one solid with no line across the risers.

- otherwise treads (**Tread Thickness**) with risers (**Riser Type Straight**, **Riser Thickness**) or open (**None**), on stringers: **Right Stringer** and **Left Stringer Closed** (a board the treads are housed in), **Open** (a board cut to the steps, the treads on top) or **None**, and **Middle Stringers** under the treads, with their **Stringer Thickness** and **Stringer Height**.
- **Begin with Riser** and **End with Riser** cleared: the board of the stair's first or last riser is not built.
- **Riser Type Slanted** leans each riser board, its top under the front of the tread above; **Riser to Tread Connection** runs each tread under the riser behind it (**Extend Tread Under Riser**) or the riser down behind the tread (**Extend Riser Behind Tread**).
- **Nosing Length** sets how far a tread runs past its riser; **Extend Below Base** moves the foot of the stringers (negative: below the base level); **Tread Material**, **Riser Material**, **Stringer Material** and **Monolithic Material** colour the parts; **Break Symbol in Plan** and **Text Size** control the plan symbol.



The type parameters of a stair

The landing stops behind the last riser of the run that reaches it, following its back face; in a monolithic stair it starts at the riser's face. In a monolithic stair whose top floor has its edge on the last riser, the run's underside goes on under that floor until it meets the floor's underside, as under a landing.

A section across the stair shows it cut with its slab. **Volume** in **Properties** and in the schedule of stairs is the volume of its solids.

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The type's **Text Font**, **Nosing Profile** (**Default** or **Rounded**) and **Apply Nosing Profile** (front only, front and left, or front, left and right: past a side without a closed stringer), **Open Stringer Offset**, **Trim Stringers at Top** (**Match Level** stops the side stringers at the top level) and its identification (**Code**, **Model**, **Manufacturer**, **Type Comments**, **URL**, **Description**, **Assembly Code**) are in **Type Parameters**, as is a monolithic stair's **Underside of Stringer** (**Smooth**, a sloping slab, or **Stepped**: the slab follows the steps). Under **Calculation Rules**, **Use Stair Calculator** with a **Target**, **Maximum** and **Minimum Result** checks 2 risers + 1 tread: a stair outside the range is made, and a warning says so.

New in the realignment. In a stair's or a ramp's **Properties**, **Show Up arrow in all views** shows the whole arrow and **UP** in the top level's plan too. **Desired number of risers** can be typed: the risers become as high as the storey over that number (a number that would make them higher than the type's **Maximum riser height** is refused). When the risers drawn exceed the desired ones, or the **Actual tread depth** is under the type's minimum, the stair is made with a warning. With **Use stair calculator** on in the type, a new stair starts with the tread that gives the **Target result**. The **Railing type** dialog also offers **Default**, the Railing tool's type. In the sketch, each run's dashed run line has a grip at its end: dragged, it adds or removes risers. **Work plane ▸ Set** in a new stair's sketch puts it on another level. The **UP**/**DN** text is dragged by its grip also with the cursor over it, after Tab on the highlighted stair, and in the top level's plan. In the type: **Stringer carriage height** moves the side stringers under the treads, **Landing stringer height** is the height of the stringers that go on along the landings, **Trim stringers at top** also has **Match landing stringer**, and **Landing overlap** (stepped monolithic stairs) is how far the upright faces stand behind the risers; the type also has its **Cost**.

Stair calculator. In **Type Properties**, **Calculation Rules** is one row with its summary (for example «2 × A + 1 × P = 630 mm, 600–650») and the **Edit...** button, which opens **Stair calculator**: **Use Stair Calculator**, the riser's and the tread's multipliers (0.5 to 5, one decimal), **Target Result**, **Maximum Result** and **Minimum Result**. The selected stair's result updates at once, green within the range and red outside it; with a maximum under the minimum **OK** is off. **OK** gives the type the rules in one undo step; existing stairs keep their tread, new ones take it from the multipliers. In the BIMopen style the dialog has a preview.

Ramps

1. Click **Create ▸ Vertical Links ▸ Ramp**. Sketch mode starts with **Run**.
2. Click where a run starts, then where it ends: it slopes at most the type's **Ramp Max Slope (1/x)** and rises at most its **Maximum Incline Length** (while drawing, a longer run stops there); the length made and left show beside the cursor.
3. Draw the next run away from the end of the first: a landing joins them. **Boundary** and **Riser** draw a ramp by its lines, with a riser line at the start and the end of each run.
4. Click **Finish Edit Mode**.

The type's **Shape** is **Thick** (its **Thickness**) or **Solid** (down to the base), and **Ramp Material** colours it. Its railings, **Edit Sketch** and **Properties** work as for stairs.

With **Arc from Center and End** in the **Draw** group, click the centre, the start of the run line and then its end round the centre: the run curves round it, two lines radiating from the centre at its ends and two boundary arcs at its sides. The type's **Show Up Arrow in All Views** draws the whole up arrow and **UP** in the plan of the top level too; its **Text Font** and identification are as for stairs.

Railings

1. Click **Create ▸ Vertical Links ▸ Railing**.
2. Draw the path with **Line**: one chain of connected lines (a path in pieces, a closed path or one that goes back over itself is refused at **Finish Edit Mode**).
3. To put the railing on a stair, click **Tools ▸ Pick New Host** and then the stair: the path drawn on it climbs with the stair.
4. Click **Finish Edit Mode**.

Edit Path reopens a railing's path. **Create ▸ Vertical Links ▸ Railing ▸ Place on a host** puts a railing on each side of the stair or ramp you click: on the options bar **Position** chooses **On treads** (on the stair's edge) or **On stringer** (over the stringer, where the stair has one), **Offset from Path** moves it towards the inside of the stair and **Invert** switches between the two positions. The same two values are in the **Properties** of a stair's railing.

The railing's **Type Parameters** hold for every railing of that type: **Railing Height** (that of the top rail), **Baluster Spacing**, **Baluster Offset** (balusters and posts moved sideways) and the identity data; the preview shows the railing on a sample stair, seen from the side. **Rail Structure (Non-Continuous) ▸ Edit...** opens the list of the rails below the top one (**Name**, **Height** above the walking surface, **Offset** sideways, a square or round **Profile**) with **Insert**, **Delete**, **Up** and **Down**. **Baluster Placement ▸ Edit...** opens the **Main pattern**: the baluster or **None**, where it starts and ends (**Host**, **Top rail** or one of the rails) with their offsets, the distance between balusters, **Break pattern at** (at the end of each segment, at angles greater than a value, or never), **Justify**, **Excess length fill** and **Use baluster per tread** with how many on each tread; and the start, corner

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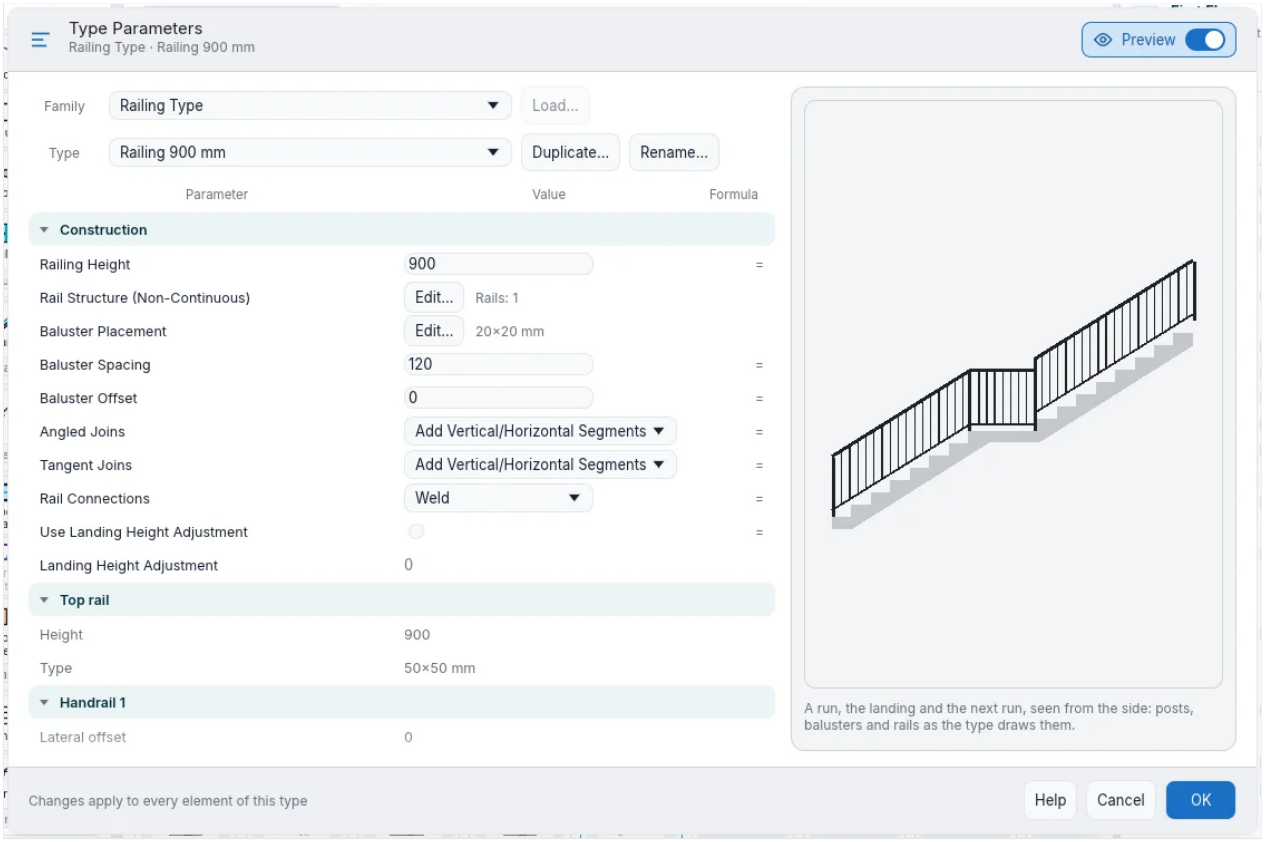
and end **Posts** (the post or **None**, its offsets, **Space** along the path, **Offset** sideways) with **Corner posts at**. The changes in these two windows apply with **OK** and are undone with a single `Ctrl+Z`. The **Top rail**, **Handrail 1** and **Handrail 2** rows are greyed: not available yet.



The type parameters of a railing, with the preview

The railings of a stair run on over its landings as one rail, at their height above the line of the nosings and above the landings. Where a run leaves a landing one riser higher, the type's **Tangent Joins** (the rail goes straight on) and **Angled Joins** (it turns) decide the join: **Extend Rails to Meet** (tangent joins only) carries the sloping rail on down to the landing, **Add Vertical/Horizontal Segments** rises upright at the first riser, **No Connector** leaves a gap.

The rails are mitred at every joint, where the path turns and where the slope changes: with **Rail Connections Weld** each rail is one piece, with **Trim** its pieces are cut on the joint's vertical plane. A post stands at the start and the end of each railing and at every change of direction in plan and wherever the slope changes, as where a run meets a landing. **Use Landing Height Adjustment** with **Landing Height Adjustment** raises (or lowers) the rail over the landings by that much.

A stair railing's path follows the stair's outline. Every post stands whole on the tread, landing or floor under it: where it would fall on a riser's line it moves by half its section, so no post sinks into the steps or hangs in front of the first one. Between two posts the balusters are evenly spaced, never wider apart than the spacing.

Place on a host works in plans and in the 3D view: the stair under the cursor shows where the two railings go. **On stringer** can be chosen only for a stair that has a stringer: on a monolithic stair it is greyed, on the options bar and in **Properties**. A new railing type has both **Angled Joins** and **Tangent Joins** set to **Add Vertical/Horizontal Segments**, **Weld**, no landing height adjustment, a bottom rail at 0.100 m (30×30), 20×20 balusters from the bottom rail to the top rail every 0.120 m, broken at the end of each segment and fitted to its length, and 50×50 posts standing on the host: where a run meets a landing the post stands on the landing at the riser's line, with the upright piece of the rail, and nothing runs over the treads. Where the rail rises upright at the start of a run, a low rail stays on the landing without entering the first tread.

New in the realignment. A railing's path is drawn with every tool of the **Draw** panel: an arc stays an arc in the railing, in 3D and in **Edit Path**, with posts only where it starts and ends. **Pick New Host** also takes a floor: the railing stands on its top. In **Edit Path**, with a line selected, the options bar offers **Height correction** (**By type** or **Custom** with its value) and **Slope** (**By host**, **Flat**, **Sloped**); **Tools ▸ Edit Joins** shows a box round the joint under the cursor, a click marks it with an X and **Join** in the options bar chooses how the rails meet there. In **Rail structure** each rail has its **Material**.

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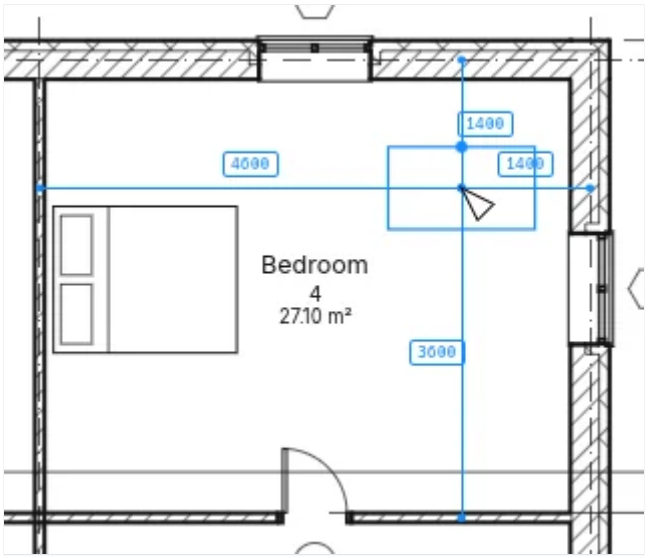
Components

1. Click **Create ▸ Build ▸ Component**. The type selector lists the components loaded in the project; a preview follows the cursor, in plan and in the 3D view.
2. Press Space to turn the preview by 90°.
3. Click to place the component on the level of the plan.



A component follows the cursor before the click

Mode ▸ Load component opens the library, by folder (**Furniture**, **Casework**, **Plumbing Fixtures**): choose a component to see its types with their sizes (width × depth × height) and click **Open** to add them to the project.

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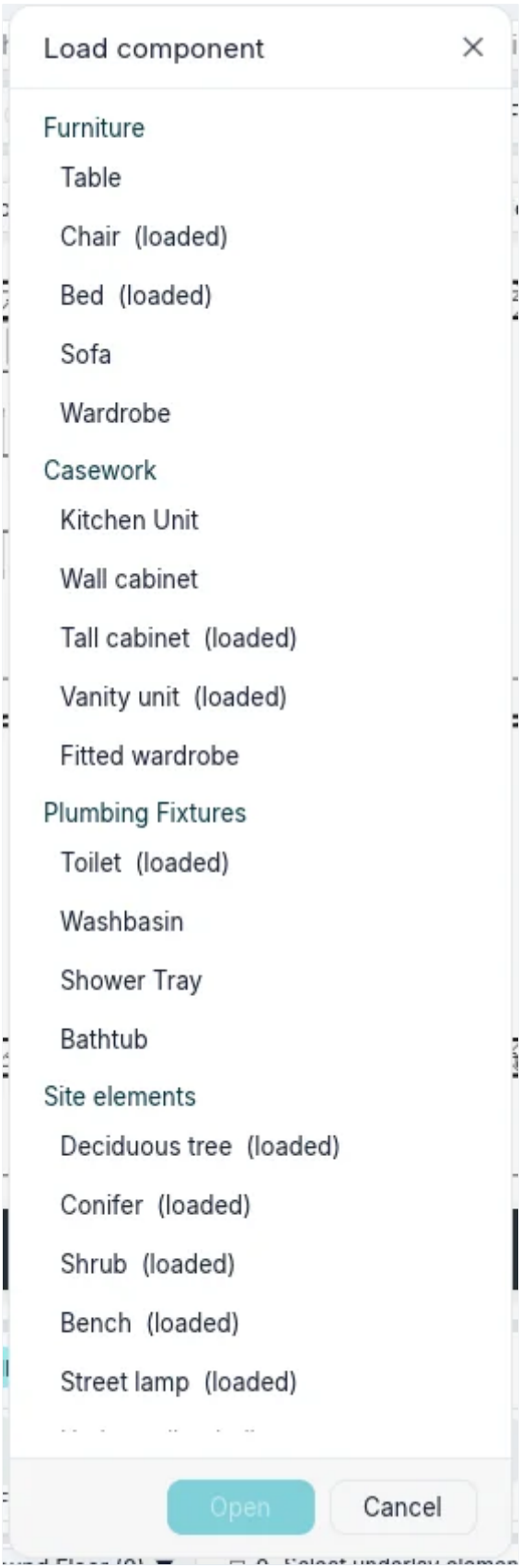

Load component

Select a component and click **Host ▸ Pick New Host**, then a floor: the component moves onto the floor and stays on its top (click an empty point to put it back on the level). Its properties are **Level**, **Host**, **Offset from Level**, **Rotation**, **Mark** and **Comments**; **Type Parameters** has its width, depth, height and material.

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Casework

Casework is a category of its own, next to **Furniture**: the units fitted in kitchens and bathrooms. Its library folder has the **Kitchen Unit** (3.00 and 2.40 m long), the **Wall Cabinet** (60, 90 and 120 cm wide, 35 cm deep, 70 cm high), the **Tall Cabinet 60** (60 × 60 cm, 2.10 m high), the **Vanity Unit 80** and the **Fitted Wardrobe**; the free-standing **Wardrobe** stays furniture. A wall cabinet is placed 1.45 m above its level: its type's **Default Offset** says how high new ones go, and each keeps its own **Offset from Level**.

Casework is drawn by the view range of the plan, as walls are:

- a unit the cut plane crosses (a tall cabinet, at the default 1.20 m) is **cut**: its outline with the cut line weight of the category, filled with its material's cut pattern;
- a unit under the cut plane (the kitchen unit) is seen from above, in projection;
- a unit over the cut plane and under the top of the view range (a wall cabinet) is seen from above too, **dashed**; over the top of the range it is not drawn.

Furniture and plumbing fixtures are never cut: they are drawn whole where the cut plane crosses them, and not drawn over it. So **Visibility/Graphics** and **Object Styles** give **Casework** its cut lines and cut patterns, and show «—» there for **Furniture** and **Plumbing Fixtures**. Casework has its own row in filters, schedules (**Schedule/Quantities ▸ Casework**), material takeoffs, project and shared parameters and legends. A project saved before casework had its own category opens with its kitchen units as casework, and the message of the opening says how many. **Export IFC** writes casework as furniture marked *Casework*; on import, furniture named as a cabinet, a counter, a vanity unit or a kitchen is proposed as casework.

Schedules and IFC

Floors, roofs, stairs, railings, furniture and casework have their schedules (**Views ▸ Create ▸ Schedules ▸ Schedule/Quantities**), one row per element. **Export ▸ Export IFC4…** writes floors as slabs (with their openings and slopes), roofs, stairs with their flights and landings, railings, furniture, casework and plumbing fixtures. A multistory stair is written once in each storey it is repeated in, with its railings.

Not yet in BIMopen

Slab edges, fascias and gutters come after this chapter's realignment. Ceilings have their own chapter, Ceilings and columns; shafts and openings cut in floors, roofs and walls are in Openings.

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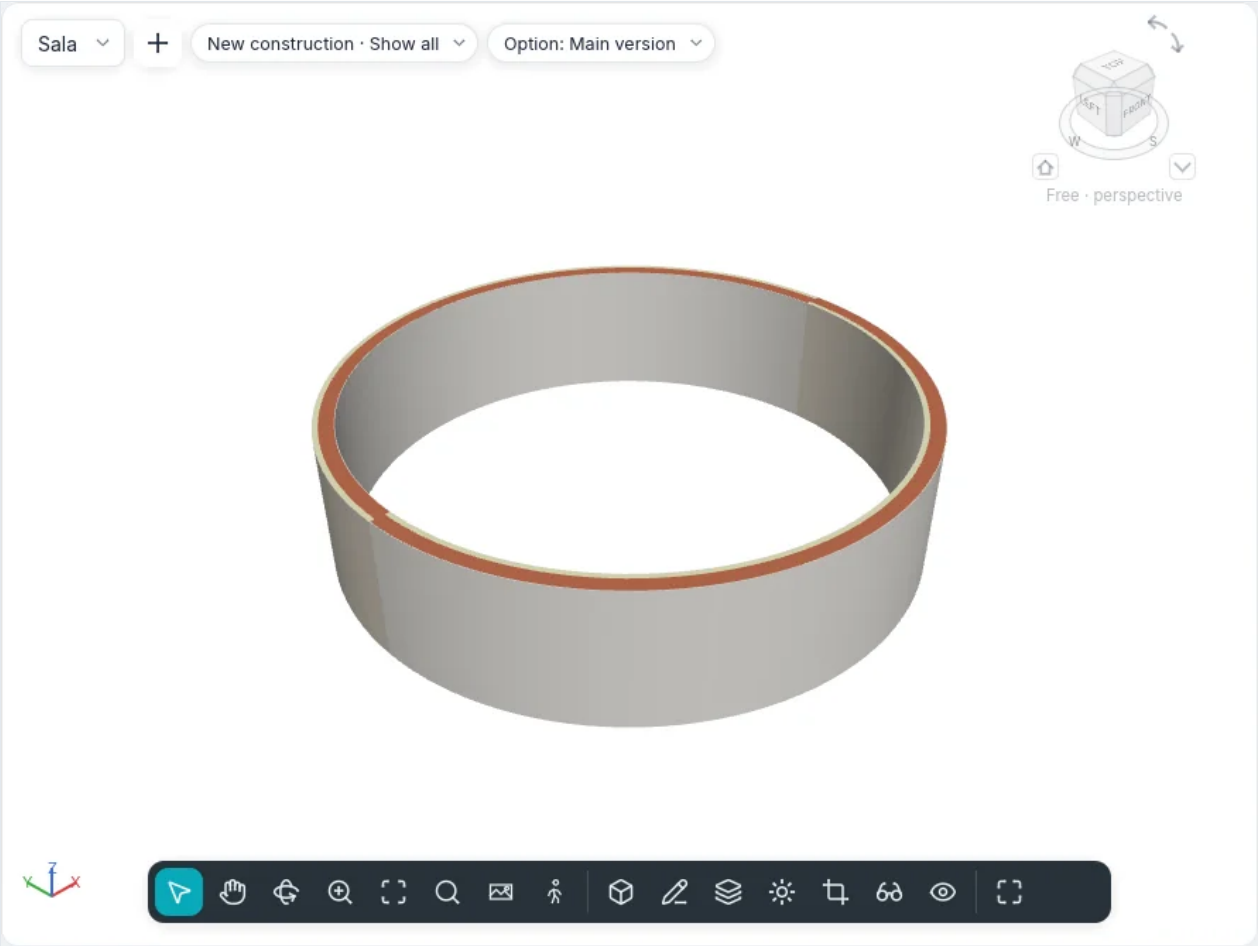
Ceilings and columns

Ceilings are made in a **ceiling plan**, the view that looks up from its cut plane. Columns and structural columns are placed with a click, in plans and in 3D views.



Ceilings

Model ▸ Build ▸ Ceiling works in a ceiling plan. Started in another view, **Go to View** offers the ceiling plans of the project: the tool never changes the view by itself.

The contextual card **Place Ceiling** has the **Ceiling** group:

Tool	How it works
Automatic Ceiling	The default. Move the cursor inside a room closed by walls: the region is lit; click to make the ceiling there. Its outline follows the faces of the walls; room separation lines do not bound a ceiling
Sketch Ceiling	Opens sketch mode (**Create Ceiling Boundary**), with the **Draw** group of floors: lines, rectangles, polygons, true arcs, **Pick Walls**, **Pick Lines** and **Slope Arrow**. A closed loop inside the outline is an opening. **Finish Edit Mode** makes the ceiling

The options strip sets the **Height Offset From Level** of the next ceiling (2.60 m in a new project) and whether it is **Room Bounding**.

A selected ceiling shows **Edit Boundary**, which opens its sketch again.

Types

The type selector lists two groups:

- **Basic Ceiling**: no thickness, only its material, whose pattern is drawn on it (**Generic**);
- **Compound Ceiling**: layers from the top down, edited in **Type layers** as those of walls and floors (**Plasterboard 12.5 on frame**, **Panels 600 × 600**). Its bottom layer is what is seen from below.

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The ceiling grid

The model pattern of the material seen from below is the ceiling grid: **Mineral panel 600x600** has a 600 × 600 grid. It is drawn in the ceiling plan and in 3D views, and a new ceiling's grid is centred in its outline, so that the border panels on opposite sides are the same. The command `ceilinggrid` moves the grid (`origin`, `by`), turns it (`angle`) or centres it again (`centre`).

Properties

Property	Meaning
Level	The level the ceiling belongs to
Height Offset From Level	The height of its underside above the level
Room Bounding	Whether rooms below end at the ceiling
Slope	The slope, for a ceiling sloped by a line or an arrow
Thickness, **Perimeter**, **Area**, **Volume**	Read-only

Ceiling plans

A ceiling plan shows the level from below, the same way round as its floor plan so that the two lie one over the other: the walls, columns and curtain walls its cut plane meets (2.30 m, over doors and windows, so a wall is whole above them), the ceilings with their grids, the grids and the view's own annotations.

- **Views ▸ Create ▸ Plan ▾ ▸ Ceiling plan** opens **New Plan** with **Type** set to ceiling plans: pick the levels; with **Do not duplicate existing views** a level that has one is not offered.
- A new level gets its ceiling plan when **Ceiling Plan** is ticked in **Plan View Types**, on the options strip of **Level**.
- In the **Project Index** the ceiling plans have their own branch, **Ceiling plans**. A ceiling plan has its level's name; its window title, and the command line, call it **Ceiling plan: Ground floor**.

In floor plans ceilings are not drawn: they are above the cut plane. In sections they are cut with their layers; in 3D views they are solids, one per layer. Walls can be attached to the underside of a ceiling with **Link Top/Base**. Ceilings are listed in schedules and material takeoffs, and exported to IFC as `IfcCovering` of type `CEILING`.

Columns and structural columns

Model ▸ Build ▸ Column ▾ offers **Architectural column** and **Structural column**; **Structure ▸ Structural elements ▸ Structural column** starts the second directly. Both work in plans and in 3D views (on the level under the cursor).

- **Click** to place it. Its temporary dimensions measure to the nearest references; type a value to place it at that distance.
- The **space bar** aligns it to the grids that cross under the cursor, or turns it by 90°.
- The options strip: **Rotate after inserting** (it is then selected and turned with **Rotate**), **Height** or **Depth**, the level it reaches or **Unconnected** with a height, and **Room Bounding**.

A structural column also goes on several places at once, from the **Multiple** group:

- **At Grids**: click the grids (again to leave one out); a structural column waits at each of their intersections; **End** or `Enter` places them;
- **At Columns**: click the architectural columns; one goes at the centre of each.

A structural column placed on two grids takes its **Column Location Mark** from them (**A-2**) and, with **Moves With Grids**, follows them when they move.

How they are drawn

- In plans a column cut by the cut plane is filled as a cut element: with its material's cut pattern, or with its type's low-detail pattern. A structural column keeps its whole outline; an architectural column inside a wall merges with it, and its outline shows only where it stands out of the wall.
- Below the cut plane a column is drawn in projection.
- In 3D views a round column is smooth, with no lines along its side.
- A **Room Bounding** column makes an island in the room around it, or a notch where it stands against a wall.

Link Top/Base and **Unlink Top/Base**, with columns selected, attach their top to a floor, roof, ceiling or reference plane above, or their base to a floor below, as for walls.

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With columns or pillars selected the options bar shows **Attach column:** with **Top** or **Base**, and three options: **Attachment style** (**Cut column**: the column stops at the element's face and follows its slope; **Cut target**: the column stays flat and a roof or floor it passes right through loses its section; **Do not cut**: they overlap), **Attachment justification** (**Minimum intersection**, **Intersect column midline**, **Maximum intersection**: where the section meets a sloped element) and **Offset from attachment** (up when positive). In Properties, under **Constraints**, **Top justification**, **Top attachment offset**, **Base justification** and **Base attachment offset** can be changed while that end is attached. **Unlink Top/Base** and a click on the element detach from that one only; **Detach All** from all.

An **architectural column** placed in a wall or against its face joins the wall: the wall's core material fills it, the layers outside the core wrap round the sides that stand out, no line runs between them, and at low detail it takes the wall's coarse pattern; at the corner of two walls it joins the one it overlaps most. A **structural column** always keeps its own material and cuts the wall it stands in: the wall loses its section in plan, section and 3D (over the pillar's height only), and the wall's area and volume, schedules included, are less by it. Moving the wall or the column works the join out again.

Types

Columns have our own parametric sections, rectangular (**Width** and **Depth**; a structural column's **b** and **h**) or circular (**Diameter**), with a material: **Rectangular column 400 × 400**, **Round column Ø 300**, **Reinforced concrete column 300 × 300**, **300 × 500** and **Round reinforced concrete column Ø 400**. Columns and structural columns are exported to IFC as `IfcColumn`.

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Openings

The **Openings** group of the **Model** and **Structure** tabs cuts holes that nothing fills: in floors, roofs and ceilings (**By face**, **Vertical**, **Dormer**), through several levels at once (**Shaft**) and in walls (**In a wall**). The five tools have the same names in both interface styles and in touch mode, and the command line runs them too.

An opening is an element of its own: it is selected, edited, moved and deleted like any other, and it goes where its host goes. A loop drawn inside a floor's, roof's or ceiling's own sketch is still an opening of that element's boundary, as before.

Openings in floors, roofs and ceilings

Tool	How it cuts
By face	Square to the face you pick: on a sloped roof the sides of the hole lean with the slope, through every layer
Vertical	Plumb, through the whole element, whatever its slope
Dormer	Plumb through the main roof, inside the outline picked from the dormer's walls and roof

1. Start the tool in a plan (a ceiling plan for ceilings) or in a 3D view. Started in a section or an elevation, **Go to View** offers the plans.
2. Move over the element: it is outlined in orange. With **By face**, press TAB to pick its underside instead of its top face (in 3D, the face you point at from below is the underside). **Dormer** offers only roofs.
3. Click it. Sketch mode opens, **Create opening boundary**, with the **Draw** group: **Line**, **Rectangle**, **Pick Lines** and the other shapes. **Dormer** offers only **Pick Roof/Wall Edges**: only what can bound a dormer lights up (a wall's side face, the dormer's roof, a floor, an edge of the main roof). Pick the inner faces of the dormer's walls and its roof: the lines need not meet, the opening they bound shows dashed, cut plumb along the walls and back to where the main roof rises above the dormer roof's underside. It follows them when the walls move or the roofs change.
4. Draw closed loops, then **Finish Edit Mode**. Open or crossing lines are refused with a message, and the sketch stays open.

A loop that crosses the element's edge notches it; a loop wholly outside cuts nothing and is left out, with a message.

A selected floor, roof or ceiling also has **Opening ▸ Vertical** in its contextual tab: the sketch opens on it at once, without picking it again.

Shafts

Shaft cuts every floor, roof and ceiling between two heights.

1. Start **Shaft** in a plan (or in 3D, on the active level). Sketch mode opens, **Create shaft opening sketch**.
2. With **Boundary line** chosen, draw the shaft's outline as closed loops (**Rectangle**, **Line**, **Pick Walls**...).
3. With **Symbolic line** chosen, draw the lines shown with the shaft in plans, for example its two diagonals.
4. Click **Finish Edit Mode**.

The shaft starts at the level of the plan it was drawn in and reaches the next level up (with no level above it, it is 3 m tall). It cuts the slabs only within its height: a slab it stops in keeps the part above or below it. Walls, stairs, columns and components are not cut.

Every plan whose cut plane is inside the shaft's height shows its outline and its symbolic lines; the plan of a floor it cuts shows the hole. It is one element: moved in one plan, it moves on every level.

Selected in a section, the shaft shows as a see-through box: drag the arrows at its top and its bottom to change its height; near a level they stop on it.

Property	
Base Level, **Offset at Base**	Where the shaft starts
Top Level	A level, or **Unconnected**
Offset at Top	Only with a top level
Free Height	Only when unconnected

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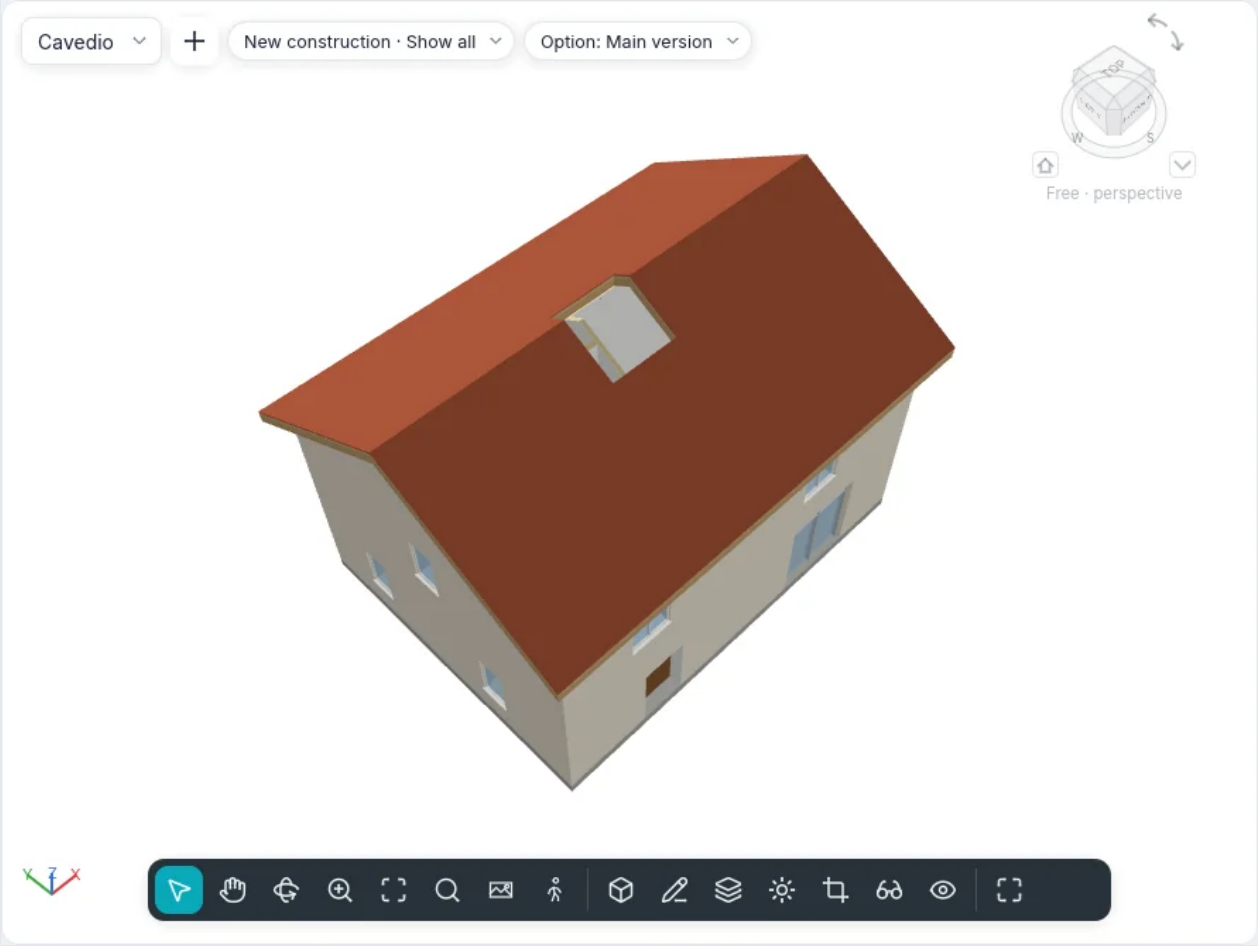

Openings in walls

In a wall cuts a rectangle through every layer of a wall, a curved wall included.

1. Open an elevation or a section that sees the wall, or a 3D view. From a plan, **Go to View** offers the elevations and sections.
2. Click the wall: it is outlined in orange. Curtain walls are not offered.
3. Click one corner, then the opposite one. The rectangle follows the cursor with its size and its distances from the start and the base of the wall; the opening is made at the second click and selected.

On a curved wall the hole follows the arc. In plan the wall is cut where the cut plane passes through the opening, with no layer returns.

Selected in an elevation or a section, the opening has four handles on its edges: drag them to resize it; drag inside it to move it along its wall (it never leaves it). Moving, copying or deleting the wall takes its openings with it. For a round or polygonal hole use **Edit Boundary** of the wall instead.

Its properties are those of a shaft, plus **Width** and **Height**.

Properties, schedules and files

Every opening has **Comments** and **Mark**, and shows **Phase created** and **Phase demolished** of its host, read-only. **By face**, **Vertical** and **Dormer** also show their **Host** and how they **Cut**.

Area and **Volume** of floors, roofs, ceilings and walls leave the openings out, in Properties, schedules and the material takeoff. IFC export writes each cut as an `IfcOpeningElement` voiding its host (a shaft gives one to each element it cuts); IFC import turns the openings no door or window fills back into openings of their floors and walls.

Visibility and Graphics lists **Shaft openings**, with **Symbolic lines**, and **Openings** among the model categories.

At the command line: `shaft`, `wallopening`, `faceopening`, `vertopening` and `dormeropening` (see Command line and MCP).

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Not yet in BIMopen

Openings in beams, braces and structural columns, a strip foundation interrupted under a wall opening, shafts through toposolids and openings in roofs by extrusion come with those elements.

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Phases and demolition

A renovation is drawn in **phases**: the existing state, the new construction and, when needed, phases in between. Every model element is created in a phase and can be demolished in a later one; every view shows the model as it is in its phase, with a **phase filter** that decides how new, existing, demolished and temporary elements look. One model then gives the plan of the existing state, the demolition plan, the design plan and the comparison drawing in yellows and reds.

Phase statuses

The view's phase and an element's two phases give the element its **status** in that view:

Status	The element...
New	is created in the view's phase
Existing	is created in an earlier phase and not demolished up to the view's phase
Demolished	is created in an earlier phase and demolished in the view's phase
Temporary	is created and demolished in the view's phase

An element created after the view's phase, or demolished before it, is not there in that view.

Project phases

Project ▸ Phases ▸ Phases (BIMopen style: **Project ▸ Phases...**) opens the **Phases** dialog, with three tabs. A new project has **Existing** and **New construction**.

In the **Project phases** tab each phase is a numbered row, from the past (top) to the future (bottom), with **Name** and **Description**. A click on the number selects the row.

- **Insert before** and **Insert after** add a phase next to the selected one, named in sequence (**Phase 1**, **Phase 2**...). Phases cannot be reordered: insert them in the right place.
- **Combine with ▸ Previous** or **Next** removes the selected phase: the elements, views, schedules and rooms that used it move to the phase it is combined with.

Names cannot be empty or repeated. **OK** applies the three tabs together, as one step that **Undo** (Ctrl+Z) takes back; the dialog's **Cancel** changes nothing. The **Preview** switch in the header shows the active view with the changes not yet confirmed, with the filter chosen under the preview.

Phase filters

In the **Phase filters** tab each filter says, for each status (**New**, **Existing**, **Demolished**, **Temporary**), whether its elements are drawn **By category** (as the view's styles and overrides say), **Overridden** (with the status's graphic overrides) or **Do not display** (not drawn, not selectable).

Default filter	New	Existing	Demolished	Temporary
Show all	By category	Overridden	Overridden	Overridden
Show demolished + new	By category	Do not display	Overridden	Overridden
Show new	By category	Do not display	Do not display	Do not display
Show previous + demolished	Do not display	By category	Overridden	Do not display
Show previous + new	By category	By category	Do not display	Do not display
Show previous phase	Do not display	Overridden	Overridden	Do not display
Yellows and reds	Overridden	Overridden	Overridden	Do not display

New adds a filter (everything by category), **Delete** removes it: when a view uses it the dialog asks first, and those views go to **None**. In the first phase **Show previous phase** shows nothing. To see everything without overrides, choose **None** as the view's filter.

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Graphic overrides

The **Graphic overrides** tab has a row per status: **Projection lines** and **Cut lines** (weight, colour, line pattern), **Surface Patterns** and **Cut patterns** (visible, pattern, colour; **Clear overrides** returns to the category's), **Halftone** (lines, patterns and fills blended towards the background, not the material colours of shaded views) and **Material** (its shaded colour and appearance in shaded and realistic views).

They apply where a filter says **Overridden**. The view's filters and the graphics by element win over them. The defaults follow the Italian practice: existing grey, demolished yellow and dashed, new red, temporary dotted in halftone.

The phase of views

Plans, ceiling plans, sections, elevations, 3D views, area plans and schedules have **Phase filter** and **Phase** in the **Phases** group of their properties. A new view takes the last phase and **Show all**; **Duplicate view** keeps phase and filter, then each can change: the same floor at two moments. A **view template** can set the filter (**Phase filter** among the included properties), never the phase.

In the BIMopen style the **Phase · Filter** chip at the top of the view opens a menu that changes either.

The phases of elements

Every model element (walls, doors, windows, components, floors, roofs, ceilings, columns, stairs, railings, ramps, curtain walls) has in the **Phases** group of its properties:

- **Phase created**: that of the view it was created (or pasted) in;
- **Phase demolished**: **None**, or the phase it is demolished in; only the phase created and the later ones are offered, and **None** undoes the demolition.

Annotations, dimensions, texts and detail items have no phases: they belong to their view. The sub-walls of a stacked wall and the pieces of a curved wall follow the whole wall. Earlier projects, with the two fixed phases of walls, open with the phases of the same names.

Demolish

Modify ▸ Shapes ▸ Demolish (BIMopen style: **Modify ▸ Demolish**) works in plans, sections, elevations and 3D views. A hammer appears beside the cursor and what can be demolished in the view's phase is highlighted. A click demolishes the element in that phase, in every view of the same phase:

- a demolished wall takes its doors and windows with it;
- an element created in the same phase becomes temporary;
- when the view's filter does not show demolitions, the element disappears at once.

With elements already selected, **Demolish** demolishes them all at once, then waits for more clicks. Esc or **Modify** close the tool; each click is one undo step.

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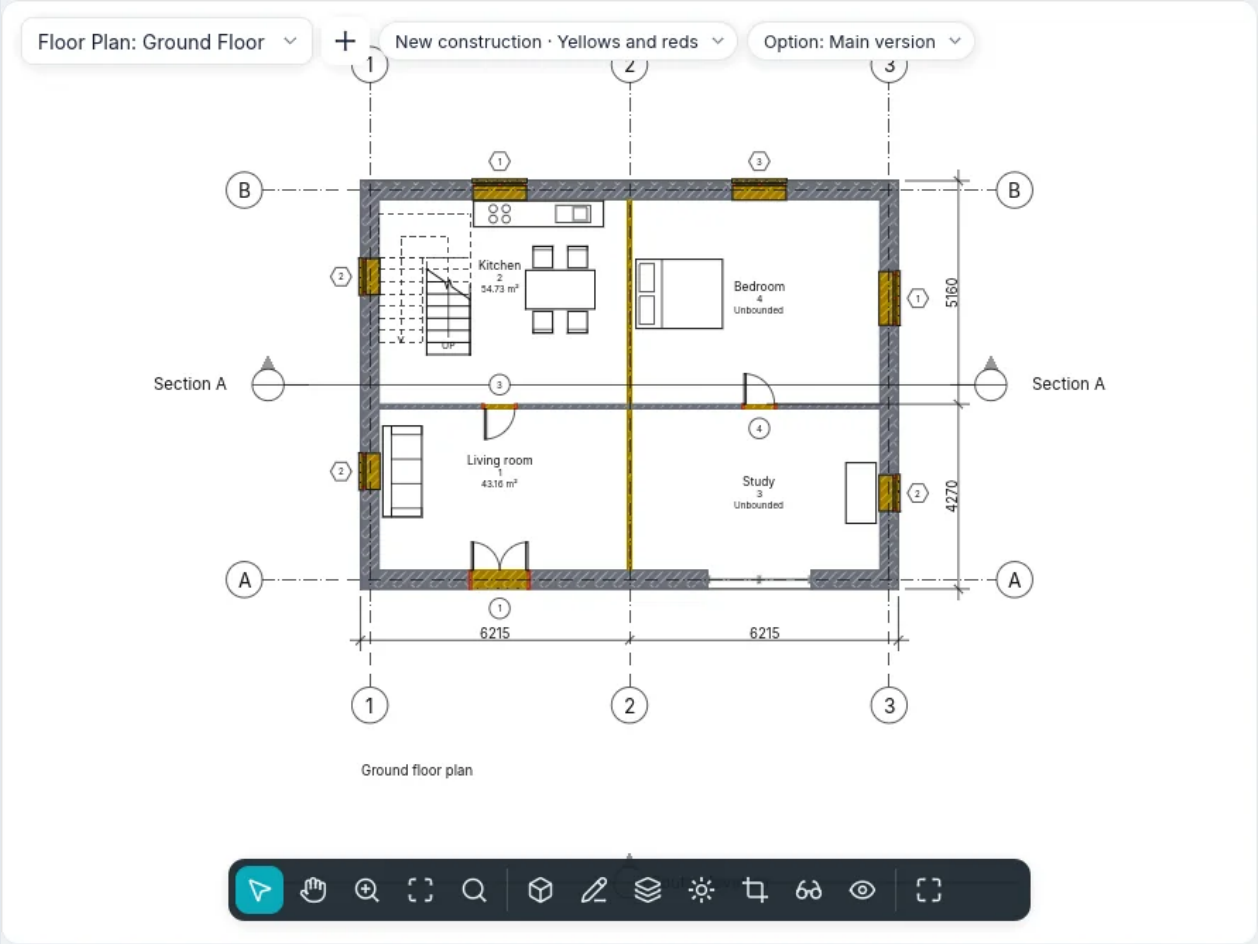

Infills

Where a door or window is demolished and its wall stays, the hole is filled from the demolition's phase on, with the wall's layers: it shows in plan, section, elevation and 3D. Where an opening is made in a later phase than its wall, the wall is whole in the earlier phases.

An infill is selected with a click, for instance in a section with **Show previous + new**: its **Type** can be another basic wall type, to draw a different infilling; the infill of the phases before the opening always has the wall's structure. Infills cannot be moved, rotated, mirrored, copied or pasted: they follow their openings.

Rooms and schedules by phase

A **room** belongs to the phase of the view it is placed in (its **Phase** is read only): a plan shows, tags and selects only the rooms of its phase, and the same number can exist in different phases. A room is bounded by what exists at the end of its phase: demolishing a partition changes the rooms of that phase and the later ones, not those of the existing state. **Copy to clipboard** a room from a view and **Paste in the active view** in a view of another phase: the room goes to that phase with the same properties.

A **schedule** has **Phase** and **Phase filter** as views do (in **New schedule** the last phase is proposed): its rows are the elements as they are in that phase, those the filter shows. **Show previous + demolished** lists the doors to remove, **Show new** those to buy. The fields **Phase created** and **Phase demolished** can be shown, sorted, filtered and edited in the cell.

View filters have rules on **Phase created** and **Phase demolished**: equal or not, and **less than** / **greater than** for earlier or later in the phases' order.

Phase legends

A **legend component** has the **Phase status** property: with **Existing**, **Demolished**, **New** or **Temporary** the sample is drawn with that status's overrides, beside its text.

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Exchange

- **IFC**: with a phased project, **Export ▸ IFC** asks the **Phase to export** (the last one is proposed). The elements that exist in that phase are written, each with its status (**Status**: NEW, EXISTING, DEMOLISH, TEMPORARY) in its class's common property set and its two phases by name. When an IFC file is opened, objects with this status take the matching phases of the project.
- **DWG and DXF**: in **Export CAD formats** the option **Phase status in the layer name** adds the status's suffix to the category's layer (-EXST, -DEMO, -NEWW, -TEMP): a demolished wall goes on **A-WALL-DEMO**.

Phase mapping of linked models will come with linked models.

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Structure and foundations

The **Structure** tab (in the BIMopen style: **Create ▸ Structure**) gathers the elements that carry the building: beams, braces, beam layouts, trusses, structural walls and floors, structural columns, pad footings, strip footings and raft foundations. Beams, braces and the members of trusses are one category, **Structural framing**; pad footings, strip footings and raft foundations are **Structural foundations**. The same tools work in plans, sections, elevations and 3D views, where the view allows it.

Structural framing types

Beam and brace types are BIMopen parametric sections, like those of structural columns. **Section shape** picks the shape: **Rectangular** (concrete or timber), **Round**, **I-shape** (also with unequal and sloped flanges), **Rectangular hollow section**, **Circular hollow section**, **Angular**, **C-channel** (also with a lip) and **T-shape**. **Type Parameters** shows only the dimensions of the chosen shape and computes, read-only, the **Structural analysis** properties: area, paintable perimeter, nominal weight (from the material's density), moments of inertia, elastic and plastic moduli, torsional and warping constants, shear areas and the angle of the principal axes. **Section name key** keeps the name of the commercial profile.

A new project has reinforced concrete beams, IPE, HEA, HEB and UPN profiles, angles, T-sections, hollow sections and glued laminated timber beams.

Beam

Structure ▸ Beam opens the **Place beam** card. Two clicks make a beam: the ends snap to the centre of structural columns, to the axis of bearing walls, to the ends and the line of other beams and to grid intersections, and the status bar says to what. The options strip has:

Option	What it does
Placement plane	The level of the view, on which the beam is laid
Structural usage	derives it from the supports: between columns or walls it is a **Girder**, between girders a **Joist**, between joists a **Purlin**; or pick one
3D Snapping	The ends snap to structural elements off the work plane: a sloped beam
Chain	The end of one beam is the start of the next; `Esc` ends the chain
Tag on Placement	Every beam is placed with its tag

Multiple ▸ On Grids: click the grids and **End** (or `Enter`); on each grid a beam is made for every span between the supports that cross it. A wall standing on the grid gets no beams; beams already there are not doubled.

In a section or an elevation the beam is drawn in the plane of the view, with its ends on the structural elements clicked; in 3D on the level under the cursor, or with **3D Snapping** on the element.

Beam properties

Group	Properties
Constraints	**Reference level**, **Work Plane**, **Start level offset** and **End level offset** (a sloped beam), **Orientation** (Normal, Horizontal), **Cross-section rotation**
Geometric position	**Start extension**/**End extension** (free ends), **Start join cutback**/**End join cutback** (joined ends, steel and timber), **yz Justification** (Smooth or Independent), **y Justification**/**z Justification** with their **Offset value**, also for start and end
Materials and Finishes	**Structural material**
Structural	**Structural usage**, **Stick symbol location**, **Start attachment type**/**End attachment type**, **Start attachment distance**/**End attachment distance** and reference column (above or below) for each end, **Start connection**/**End connection** (Moment frame, Cantilever moment), **Camber size**, **Number of verticals**, **Beam seated at start**/**end** on a structural wall, **Maximum weight**, **Exact weight**, **Painted area**
Dimensions	**System length**, **Cut length**, **Volume**, **Elevation at top** and **Elevation at bottom** («Varies» on a sloped beam)
Structural analysis	**Start release**/**End release** (Fixed, Pinned, Bending moment, User defined with the six constraints), **Analyze As**

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A selected beam has **Reverse Ends** and, for each end, **Disallow Join**.

How it is drawn

- **Coarse**: a line, with the line style of its usage (Styles by Category ▸ Structural framing ▸ Girder, Joist, Purlin...), held back from the columns and beams it joins by the **Cut symbol distance**. In sections and elevations beams are lines too, with the moment symbols.
- **Medium** and **Fine**: the outline of the section. In a plan whose **Discipline** is «Structural» the beams covered by a floor are dashed; in an «Architectural» plan they are not shown.
- In sections the beam is cut with its material's pattern.

Joins

A beam stops at the face of the column and of the bearing wall it rests on (never at a non-bearing wall); between beams the join is a **tee** (the beam that runs through stays whole) or a **corner** between two beams of the same type in the same plane. Concrete becomes a single mass: the column wins over the beam, floors and walls keep their shape, and a beam under a concrete floor stops at its underside.

Structure ▸ Framing ▸ Beam/Column Joins shows an arrow at each joined end: towards the join, that beam is cut; a click reverses it. The options strip filters by material (Steel, Wood, Precast concrete, Other).

Brace

Structure ▸ Brace:

- in a **Framing Elevation** (or in a section), two clicks on structural elements, for example the base of a column and a beam;
- in plan, with **Start level**, **Offset**, **End level**, **Offset** on the options strip;
- in 3D, on the points of the elements clicked.

An end laid on a beam stays attached at a **Ratio** (from 0 to 1) or a **Distance** along it, from its start or its end: if the beam moves or gets longer, the brace follows it. **Structural usage**: Vertical bracing, Kicker bracing, Web, Other.

Plan symbols and Structural Settings

Structure ▸ Framing ▸ Structural Settings sets, on paper:

- the **Cut symbol distance** of braces, beams and trusses, and columns;
- the **Brace symbols**: **Parallel line** (with its offset) or **Angled line**, **Show brace above** and **Show brace below** (also dashed), **Kicker brace symbol**;
- the **Connection symbols** of beams and braces.

The symbols show at the Coarse detail level.

Beam layout

Structure ▸ Beam layout fills a bay with parallel beams:

- **Automatic beam layout**: one click in a bay closed by beams or bearing walls;
- **Sketch beam layout**: the boundary in sketch mode (**Create beam layout boundary**) with **Pick Supports** (the boundary stays tied to the beams and walls clicked, and the first gives the direction), **Line**, **Rectangle**, **Pick Lines** and **Beam direction**. An inner loop is an opening.

Layout rule: **Fixed distance**, **Fixed number**, **Maximum spacing**, **Clear spacing**; with the first and the last, **Justification** (Start, End, Center, Direction line). **Beam type**, **Elevation** and **Tag on Placement** (one tag for the layout or one per beam) complete the options.

The beams follow the supports when they move. A selected beam layout has **Reset System** (beams that were changed go back to the layout) and **Remove Beam Layout** (the layout goes, the beams stay).

The layout rules space the beams as **Array** spaces its copies: **Fixed number** like **Last**, **Fixed distance** like **2°** from the **Justification**; with **Clear spacing** the gap is between the faces, and a beam of a bigger type pushes its neighbours away. **3D** in the options bar makes the system non-planar: the beams' ends rise to the slope-defining supports (beams always, walls with **Defines Slope** or **Defined slope walls**). **Edit Boundary** opens the sketch again; **Beam direction** is drawn or picked with **Pick Lines**. A beam deleted by hand stays deleted until **Reset System**. Outside a closed bay, **Automatic beam layout** opens the boundary sketch.

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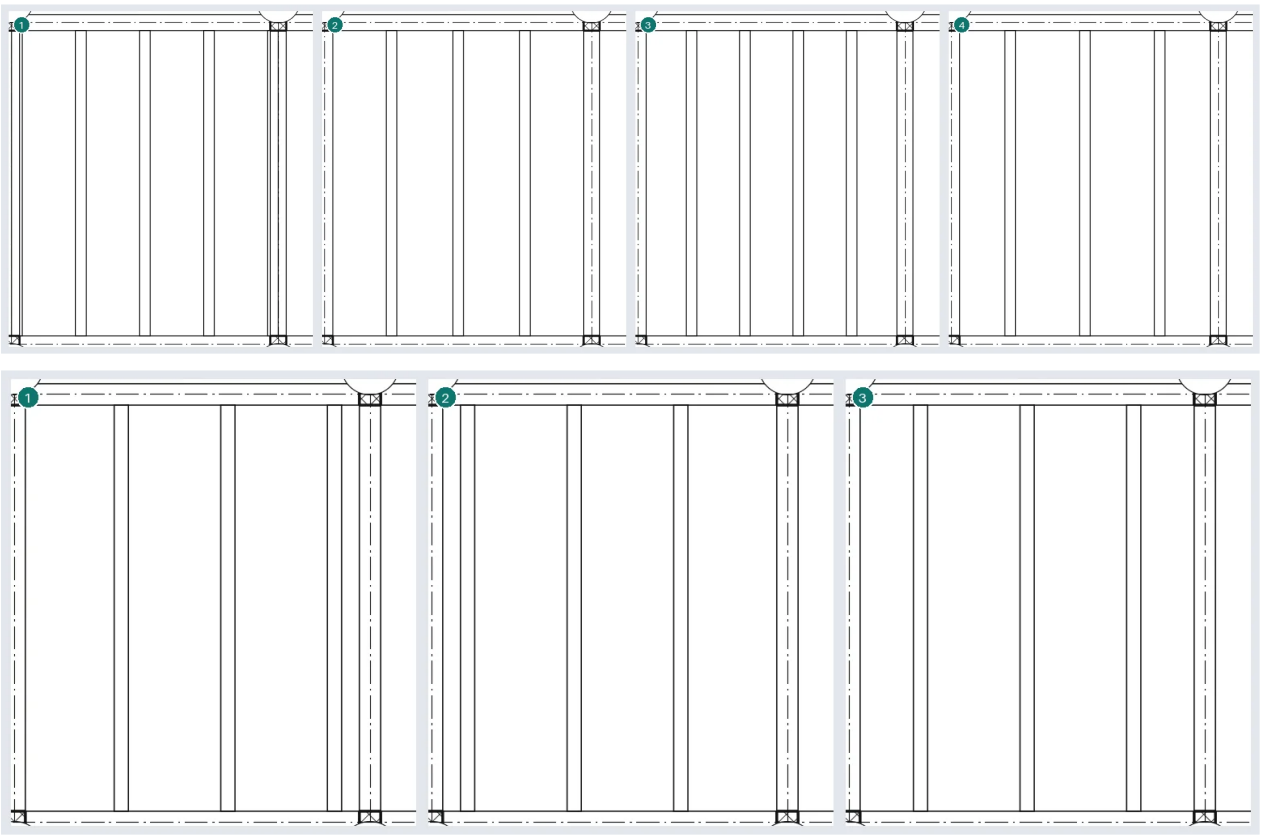

A beam system copied, mirrored or pasted associates itself with the beams and bearing walls its edges fall on in the new place; when an edge finds no support the command warns, and **Edit Boundary ▸ Pick Supports** associates it again. Copy it only between bays with the same shape and the same supports.

Curved beams and curved bearing walls also close a bay for an automatic beam system, but the beams' direction always comes from a straight support. A beam system sketched over more than one bay is still made, with a warning: the beams and walls across it do not divide it, so one beam system per bay is better.

Truss

Structure ▸ Truss: two clicks in plan lay the type's pattern (Pratt, Howe, Warren, Warren with verticals, Fink) over the span. The chords are beams, the webs are braces, each of the type of its group (**Top chords**, **Vertical webs**, **Diagonal webs**, **Bottom chords**). The properties include **Truss height**, **Bearing chord**, **Maximum panel width**, **Number of panels** and the rotation.

A selected truss has **Attach Top/Bottom** (the chord follows the face of a roof or a structural floor; with `Shift` the bottom chord), **Detach Top/Bottom**, **Reset Truss** and **Remove truss** (the members stay as beams and braces).

Edit Profile opens a section or elevation parallel to the truss (when the open view is not, **Go to View** asks which) with the chords' lines in sketch mode: the lines above the middle are the top chord, the others the bottom chord, and the webs follow them. **Reset Profile** brings the chords back to the layout's line. In a 3D view the lines are drawn straight on the truss's plane, without changing view.

With **Pick Lines** the truss is placed along a beam, a wall or a grid. A click on a member selects the whole truss; `Tab` goes on to the single member. An attached chord follows the host's face along the whole span; with the bearing chord attached, the other keeps the **Truss height** from it, and the drawn profile stays and comes back with **Detach Top/Bottom**. **Bearing vertical justification**, **Non-bearing offset**, **Web orientation** (**Orthogonal**: the verticals square to the bearing chord) and **Rotation angle** (the truss's plane turns about the location line) change the members. A member deleted by hand stays deleted until **Reset Truss**; the members are cut only in **Structural Settings**. New layouts drawn in their own component will come with the component editor.

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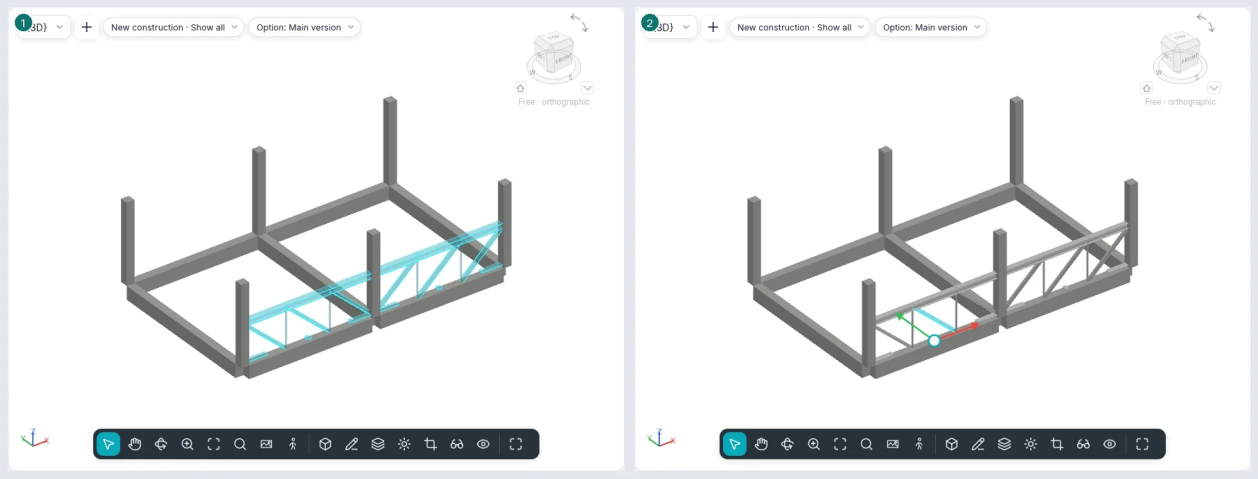

With **Webs have cut symbols** in the truss type, at coarse detail the webs stop short of the chords by the braces' **Cut symbol distance** from **Structural Settings**.

Structural wall and structural floor

Structural wall is the Wall tool with the same types and the same options strip: the wall is made **Bearing**. Every wall has **Structural** and **Structural usage** (**Non-bearing**, **Bearing**, **Shear wall**, **Structural combined**).

Structural floor is the floor with **Structural** ticked. In the sketch, **Span direction** draws the direction of its span; at the end the **Span direction symbol** is placed at the centre of the floor. **Annotations ▸ Span direction** places it again (with **Automatic placement** at the centre); selected, it can be turned, and **Align perpendicular** makes it perpendicular to an edge, a beam or a grid.

Shape editing

A selected floor (structural, a slab foundation or any floor) has the **Shape Editing** panel:

- **Modify Sub Elements**: click a corner or a point, then type its **Elevation** in the options bar; `Del` removes the point.
- **Add Point**: a point inside the floor at the elevation given; with **Relative** it is added to the surface's height there.
- **Add Split Line**: two clicks; the surface creases along the line.
- **Pick Supports**: click a beam under the floor; a split line follows the beam, at its top.
- **Reset Shape**: removes the points and lines; the floor is flat again (or keeps its slope).

The top becomes the triangulated surface through these points; the whole slab follows it and keeps its thickness. **Elevation at Top** and **Elevation at Bottom** then show **Varies**.

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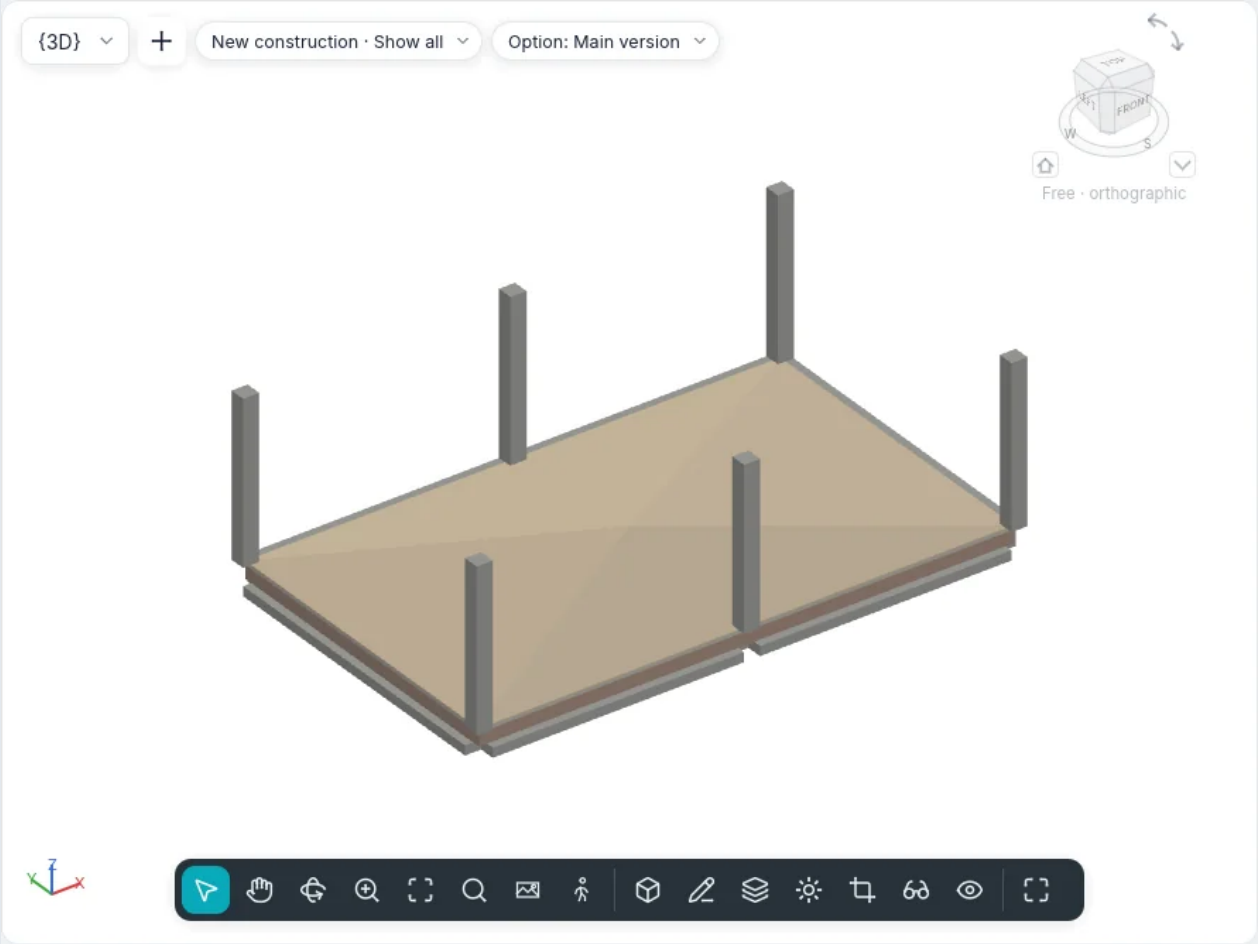
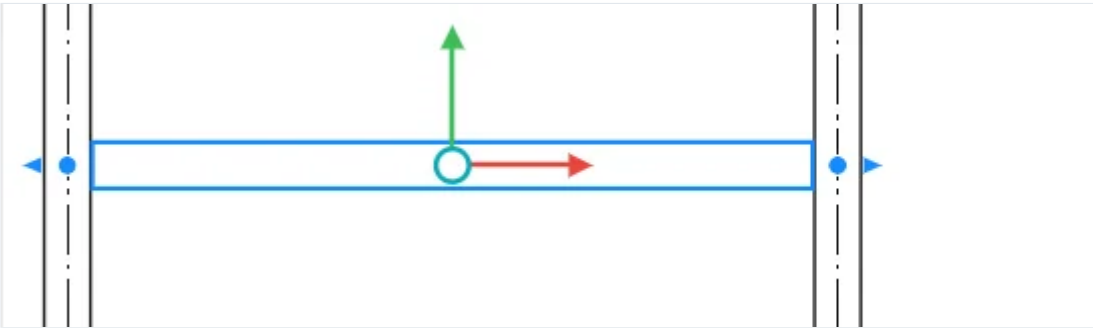


Foundations

Tool	How it works
Pad footing	One click in plan or in 3D (space bar: turn), or **At Grids** or **At Columns** and **End**. The top face is at the level plus **Height Offset From Level**; the column above rests on the footing. **Moves with grids** ties it to the grids
Strip footing	One click on a structural wall: the footing runs under its base and follows it. **Bearing**: **Width** and **Eccentricity**; **Retaining**: **Outer side width** and **Inner side width**. It breaks under doors and windows that touch the base, unless **Do not break at inserts**; at free ends it projects by the **Default end extension length**
Raft foundation	The boundary in sketch mode (**Create raft boundary**), like a floor: it lies under its level, down by the thickness of its layers. **Structural usage**: Foundation or Foundation slab

With a wall foundation selected, the filled circles at its ends lengthen or shorten it along the wall, snapping to what is near; the circles at the ends of a beam or a brace do the same. On steel, precast concrete and timber beams, the arrows beyond the ends change **Start/End extension** at a free end and **Start/End join cutback** at a joined one. Concrete pads and wall foundations that touch are one mass: the pad keeps its shape and cuts the wall foundation (of two pads, the earlier cuts the other), and no line is drawn between them in plan.



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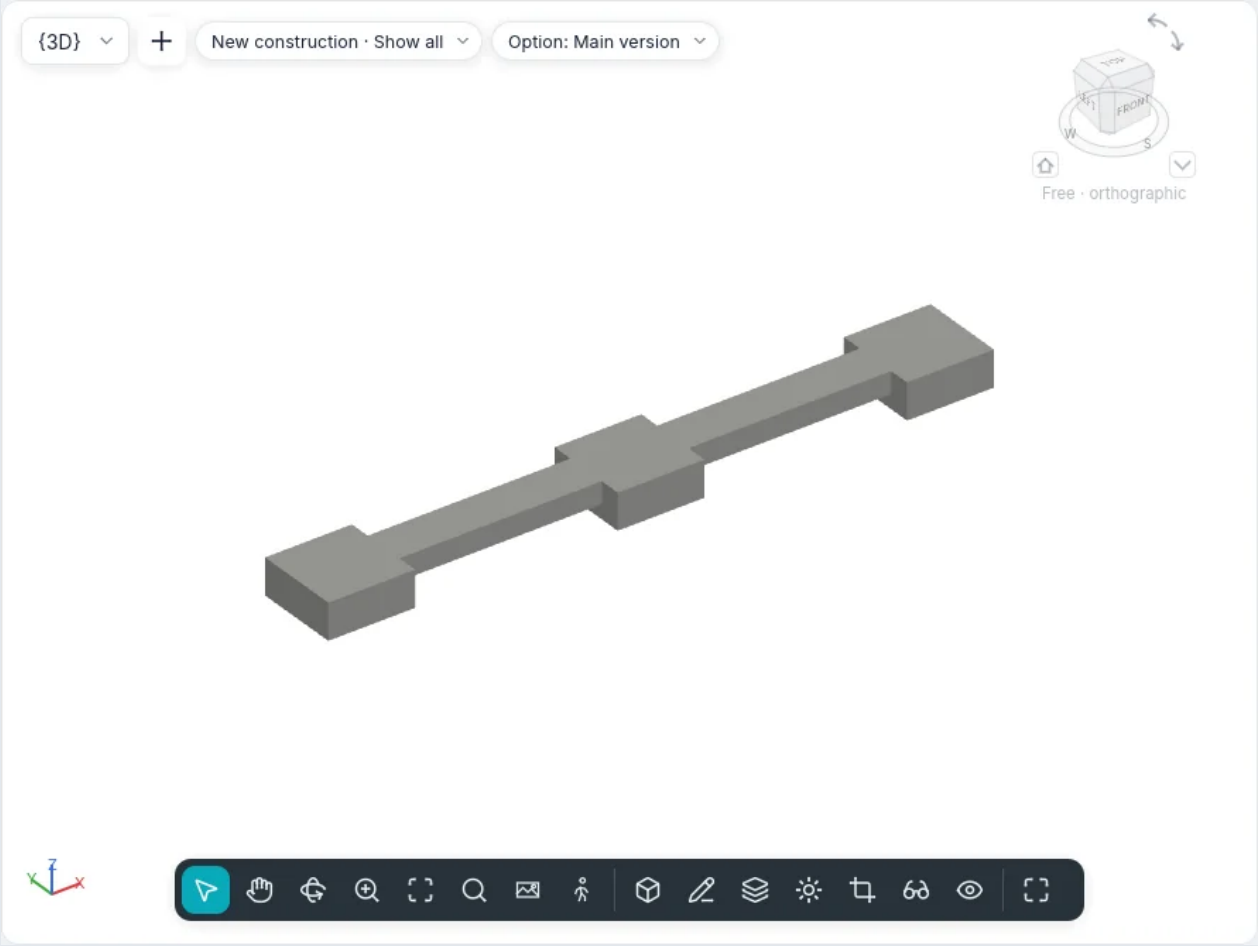
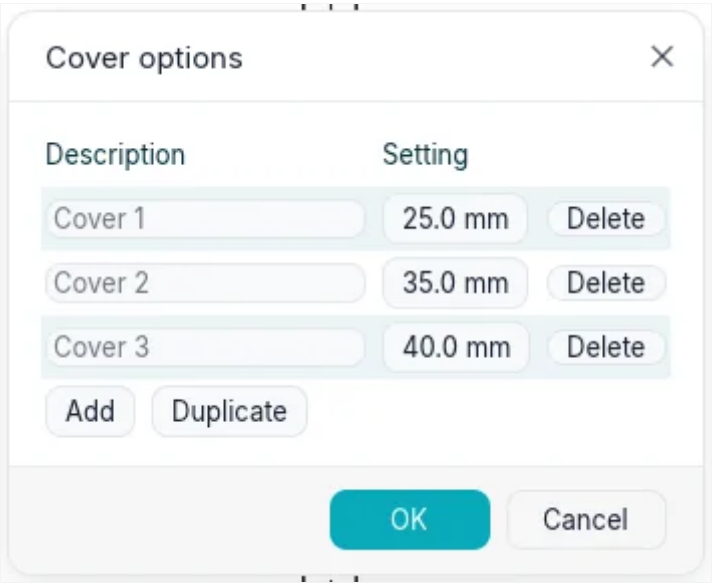


Rebar cover

Reinforced concrete ▸ Cover options lists the project's covers, each with a name and a distance: **Add**, **Duplicate**, **Delete**. Elements that used a deleted cover take the first one. Structural walls, pillars, concrete beams and braces, structural floors, slab foundations, isolated and wall foundations choose their cover in **Properties**: **Rebar cover - Top face**, **Rebar cover - Bottom face** and **Rebar cover - Other faces** (on walls **Rebar cover - Exterior face**, **Rebar cover - Interior face** and **Rebar cover - Other faces**).



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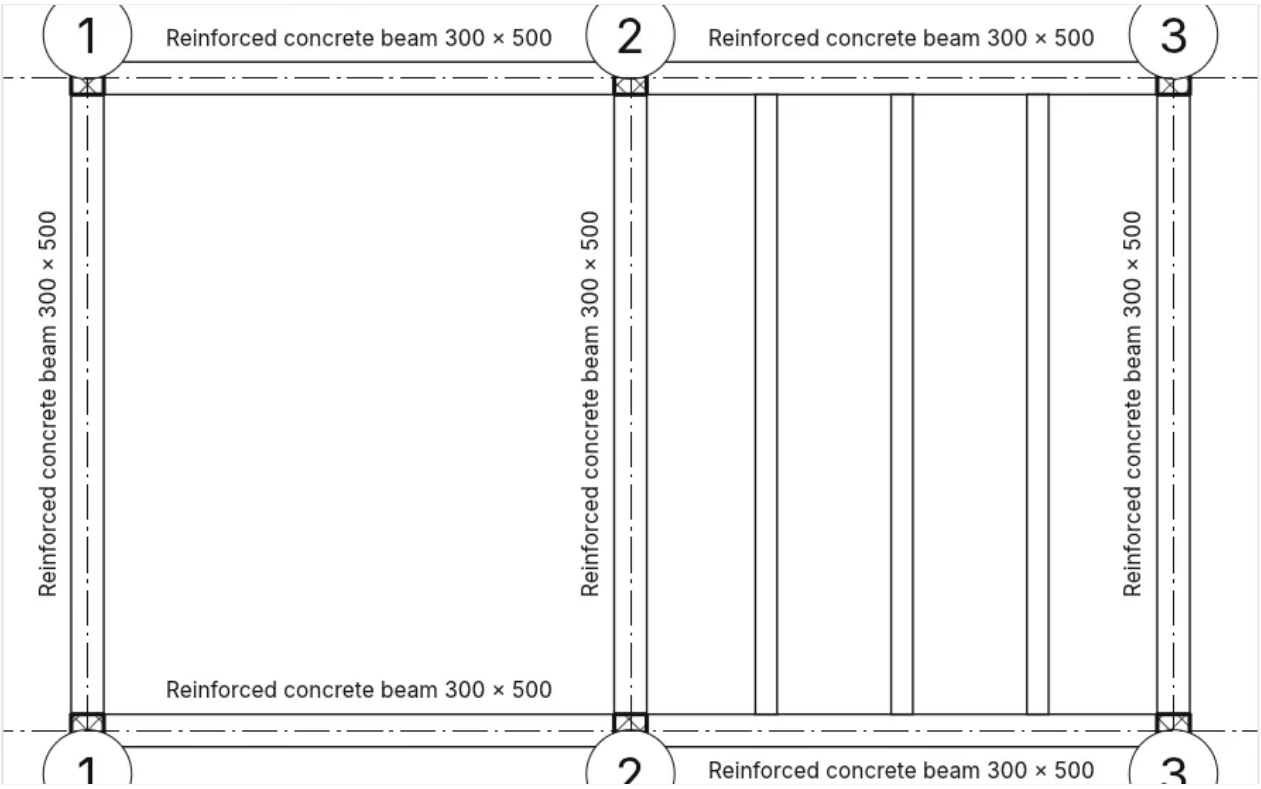
Views, tags, schedules and IFC

- A view's **Discipline**: **Architectural** or **Structural**. The second hides non-bearing walls and dashes what floors cover.
- **Annotations ▸ Beam tags**: framing tags and elevations on all the beams of the plan (or on the selected ones) in six positions, above and below, at the start, the middle and the end; **Display elevation** picks top, bottom, both or actual.
- A beam's tag reads along the beam and stands beside it, on the side where it covers no other beam (above, when both are free); it is dragged like any tag. In **Beam tags ▸ Settings**, **Horizontal end offset** moves the start and end positions towards the middle and **Vertical offset** keeps the annotations away from the beam's side.
- **Schedule/Quantities** has the categories **Structural framing**, **Beam layouts**, **Trusses** and **Structural foundations**, with every parameter as a field.
- In **IFC** girders, joists and purlins are exported as `IfcBeam`, braces and members as `IfcMember`, trusses as `IfcElementAssembly`, pad and strip footings as `IfcFooting`, the raft as a base `IfcSlab`; structural walls and floors carry `LoadBearing`.



From the command line

Command	What it does
`beam`	Beams from two points, in a chain, as an arc or on grids
`brace`	A brace between two points
`beamsystem`, `beamsystemedit`	An automatic beam layout or one from a boundary; reset it or remove it
`truss`, `trussedit`	A truss; attach it, detach it, reset it or remove it
`pad`, `strip`	Pad footings (at a point, on grids, under columns), strip footings
`spansymbol`, `spanalign`	The span direction symbol; align it
`beamjoin`, `structuralsettings`, `beamannotations`	Joins, structural settings, beam tags
`floorshape`	A floor's shape editing: points, split lines, supports, reset
`covers`	Rebar cover settings

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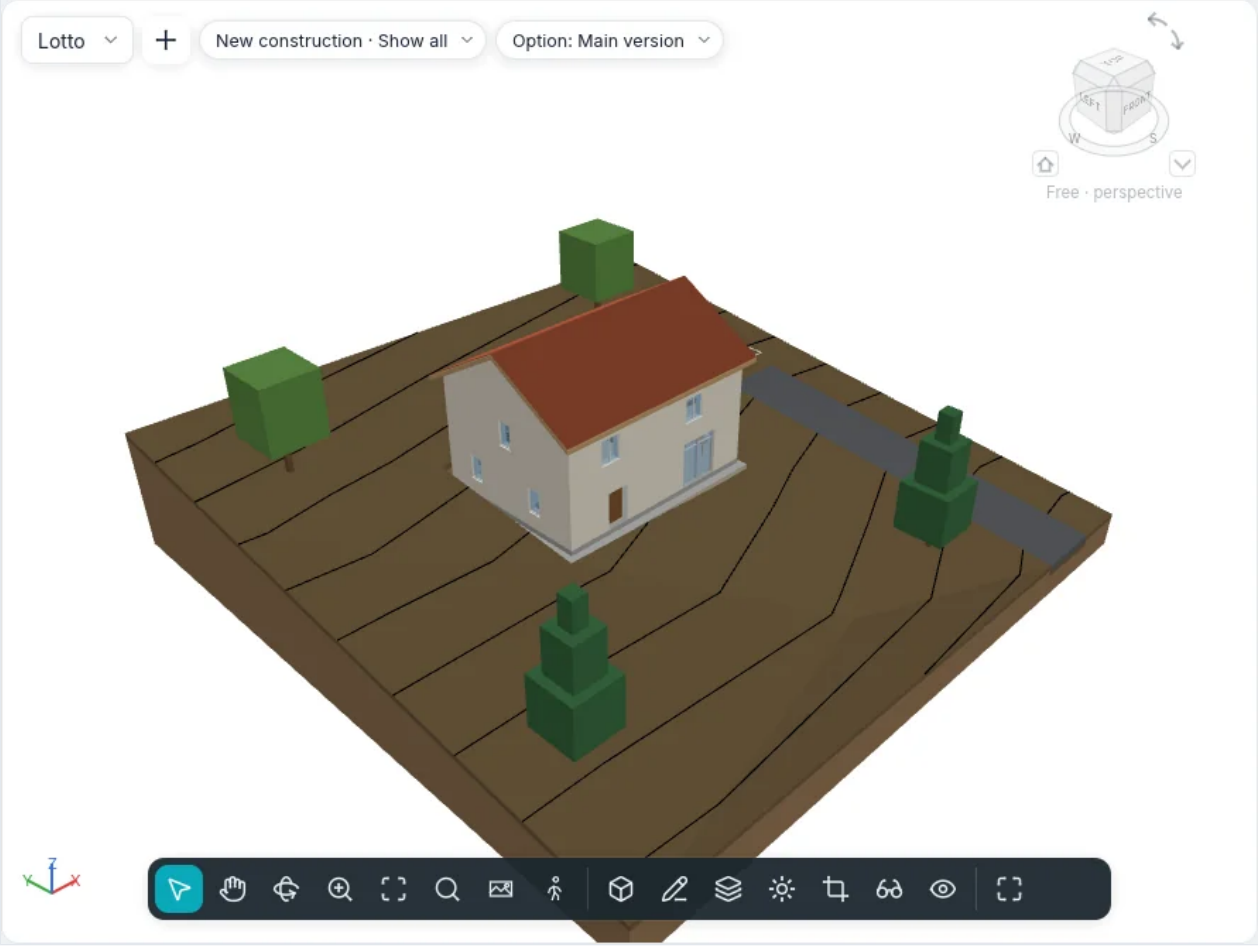
Site plan and terrain

The ground the building stands on and the drawings of the lot around it are made with **Massing & site ▸ Site** and **Edit site**. They work in floor plans and in the **Site plan**; a terrain and a building pad are also sketched in 3D views, on the plane of their level.



Terrain

A terrain is a solid: a sketched outline (with holes) on a level, shaped by points, with the layers and thickness of its type.

Terrain ▾ has two ways:

Way	How it works
Create from sketch	Sketch mode **Create Terrain Boundary**, with the shared **Draw** group (lines, rectangle, polygons, circle, arcs, **Pick Lines**). Draw a closed loop; a loop inside it is a hole. **Finish Edit Mode** makes a flat terrain, its top at **Height Offset From Level**
Create from import	The dialog **Add points from selected layers**: the layers of a DWG or DXF imported with heights (3D lines and points) whose points shape the terrain, or **Create from points file…** (CSV or TXT, its unit chosen in **Format**)

A file of points gives on each row x, y and z first; names and other columns are skipped, and two points at the same x, y keep the higher z. Beyond the limit of **Site settings** (20,000 points by default, 10,000–50,000) the points are sampled down to it and a message says how many were kept.

Shape editing

A selected terrain shows the group **Shape editing**:

Tool	How it works
Add point	Click inside the outline. In the options strip: **Absolute** (the **Elevation** typed, from the **Elevation base**: current level, project base point, survey point, internal origin) or **Relative to surface** (an offset above or below the surface where you click); **XYZ snaps** take the elevation of a floor's corner or a wall's end, plus **Offset from snaps**

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Tool	How it works
Modify sub elements	Click a point: drag it in plan, or type its height above the terrain's original top
Add split line	Two clicks: a fold of the surface, straight between their elevations
Reset shape	Removes every point and split line
Simplify terrain	Keeps a share of the points (1 000 points at 70 % become 700)

The surface is triangulated through the outline and the points; the contours follow at once.

Type, layers and contours

The terrain's type has its **Structure** in the same **Type layers** dialog as walls and floors: the layer marked **Variable** takes up the shape, so the bottom stays flat. **Contour display** (**Contour display** in the terrain's panel) lists groups of contours: **Start**, **End**, **Increment**, **Single value** or **Multiple values**, the line style (primary, secondary, hidden lines, triangulation edges) and a check box to turn a group off. A new project draws primary contours every 5 m and secondary ones every metre; the types **Asphalt road**, **Gravel path** and **Sidewalk** draw none.

The contours are drawn on the terrain in 3D views too, in their line styles, hidden where the building stands in front of them.

In a section the terrain is cut with its layers and their materials' patterns. A **Spot Elevation** placed on the terrain reads its elevation there.

Visibility and Graphics ▸ Terrain has the subcategories **Primary contours**, **Secondary contours**, **Hidden lines**, **Fold lines**, **Boundary points** and **Interior points**; the last three are off unless turned on.

Site options ▾ ▸ Terrain smooth shading makes every terrain smooth in the shaded styles.

Site options ▾ ▸ Cut void stability: a cut that does not come out whole is tried again with its outline moved a few millimetres along x or y; **Excavate** says how many cuts were moved.

Editing the terrain

Tool	How it works
Sub-area	Sketch a loop on the selected terrain (or click a terrain first). Its **Type** gives its material; **Sub-area offset** raises it (a kerb) or sinks it into the host (with its **Excavation volume**)
Divide terrain	Click the terrain, then sketch a closed loop inside it or open lines from edge to edge. A sub-area the division cannot keep asks for **Delete instances**
Merge terrains	Click the main terrain, then the second: they must share an edge or overlap. **Remove points along shared edges** is on by default
Excavate / **Remove excavation**	With the terrain selected, click a floor, a roof, a building pad or another terrain that crosses it. **Individual excavation volume** lists each cut
Grading	**Edit graded region** asks for an exact copy or only the boundary's points; click a terrain of an earlier phase. It is demolished in the current phase and its copy, new, opens in shape editing. The copy reports **Cut**, **Fill** and **Net cut/fill**

A schedule of **Terrains** lists the cut and fill, with a grand total.

Building pad

Building pad sketches a closed loop over a terrain (**Create Pad Boundary**, with **Pick Walls** and **Slope Arrow**). Its top is at **Height Offset From Level**; the terrain under it is cut or filled to its underside. Its type has a **Structure** of layers and the low detail fill pattern.

Site elements and parking

Site element places trees, shrubs, benches, street lamps and hydrants, one at a click (space bar to rotate); **Parking** places parking spaces (90°, 60°, 45°, parallel, accessible). Both stand on the terrain or sub-area under the click and follow it when it changes. **Pick New Host** moves them onto another terrain or sub-area; **Repeat in Series** makes rows of them.

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Property lines

Property line asks how to create it:

- **Create by sketching**: the lines of sketch mode; an open loop warns that its area cannot be computed (finish again to keep it open);
- **Create by entering distances and bearings**: the table **Property lines**, a row per run with **Distance**, **Bearing** (as **N 45°00'00" E**), **Type** (line or arc), **Radius** and **L/R**; **Insert**, **Delete**, **Up**, **Down**, and **Close**, which adds the run back to the start. The runs follow true north.

A selected property line shows **Edit table**. **Site options ▾ ▸ Property data** sets the **Angle display** (bearing or degrees) and the units. Property lines show in site plans and 3D views by default; a schedule of **Property lines** lists their name and area.

Contour labels

Contour labels: click the start and the end of a line across the contours; each contour it crosses shows its elevation, turned to read from project north. The label's type sets the text and whether only primary contours are labelled.

The site plan

Every new project has a **Site plan**; **Views ▸ Plan ▾ ▸ Site plan** makes another, for the active level: 1:200, cut above the building down to 10 m below the level, oriented to **True north**. **Orientation** in its properties turns it back to **Project north**.

Site settings

Site options ▾ ▸ Site settings holds the project's contours (for the terrain types that do not set their own), the **Section cut material** and **Poché base elevation** of terrain types without layers, the property data and the most points an import keeps.

Not yet

Terrain from face, **Link topography** and **Surrounding context** come in a later phase.

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Design options

Design options keep several solutions for one part of the project in the same file: two entrances, three roofs, different room layouts. Everything common to all solutions is the **main version**; each problem gets an **option set** with its **options**. One option of each set is the **primary**: the views show the main version with the primaries, until a view is dedicated to another option or an option is being edited. When the client has chosen, **Accept primary** puts the chosen option into the main version and removes the others.

Option sets and options

Project ▸ Options ▸ Design options (BIMopen style: **Project ▸ Design options...**) opens the **Design options** dialog. On the left, every set with its options; **(primary)** follows the primary's name; under the list, **Editing:** says what is being edited.

Group	Button	What it does
Modify	**Edit option**	the selected option becomes the active one
	Finish editing	back to the main version
Option	**New**	a secondary option of the selected set, «Option N»
	Duplicate	a secondary option «Copy of ...» with copies of the option's elements
	Make primary	the option becomes the primary, the old primary a secondary
	Delete	the option with its elements (not the primary, not while editing)
	Rename	a new name for the option
Option set	**New**	«Option set N» with «Option 1 (primary)»
	Delete	after a confirmation, the set with all its options and their elements
	Rename	a new name for the set
	Accept primary	the primary joins the main version, the set goes

Names are unique within their list and cannot contain «:».

Editing an option

To add elements to an option, make it the **active option** in one of these ways:

- **Edit option** in the dialog (or a double click on the option);
- the drop-down **Main version** of **Project ▸ Options**, of the status bar and, in the BIMopen style, the chip **Option: ...** in the top bar beside the phase chip;
- **Select to edit**, then a click on an element of the option.

While an option is active, every model element you make goes into it, and everything outside it is drawn in halftone in plans, sections, elevations and 3D views and cannot be selected while **Active only** is on in the status bar. Levels, grids, views, dimensions, tags, text and detail lines always belong to the main version or to their view; a section, elevation or callout made while editing is visible only in that option (see Dedicated views). **Finish editing**, or **Main version** in a drop-down, ends the editing. Switching the active option is not an undo step, and a project always opens in the main version.

Doors, windows and wall openings stay in the option of their wall, a stair's railings in its stair's, a strip footing in its wall's: a door cannot be placed in a main-version wall while an option is edited. Move the wall into the option first.

In the status bar, with the main version edited, **Exclude options** keeps the elements of the options from being selected; untick it to select them (to dimension to them, say). Hovering an element of an option shows «(Set : Option) : Category : Type», and its properties show **Design option** under **Identification**.

The status bar's design option controls are always there, greyed in a project with no options; **Views ▸ Windows ▸ Panels ▸ Status bar - Design options** hides or shows them; in the BIMopen style they are also in the menu of the chip **Option: ...**.

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Moving elements into options

Select main-version elements and click **Project ▸ Options ▸ Add to set**. In **Add to option set** choose the set under **Add selection to** and tick the options that get the elements: the first ticked option takes the elements, the others get copies, and they leave the main version. What a selected wall hosts goes with it; a door alone cannot go without its wall.

Elements of one option move to another with **Cut** (Ctrl+X) in the first, then, with the other option active, **Paste ▸ Paste in the same place**.

Groups and arrays are made of elements of the active option only (or of the main version only).

Dedicated views

Visibility and graphics has a **Design options** tab in projects with options: one row per set, with the option the view shows. **Automatic** shows the primary, or the active option when it is of that set; choosing an option **dedicates** the view to it. Duplicate a view (with details too), dedicate it, and put it on a sheet beside the other to compare the solutions.

A schedule's **Visibility and graphics** (VV in the schedule) has only this tab: its rows are the main version and the options it shows.

The view property **Visible in option** (**Graphics**) binds a view to one option: its mark (the section line, the elevation symbol, the callout) shows only in views showing that option, and the view goes when the option is deleted. **All** shows it everywhere.

Rooms, joins and outputs

- A main-version room is bounded by the main version and every primary; a room of an option by its option, the main version and the primaries of the other sets. Volumes take floors, roofs and ceilings by the same rule. Areas and area schemes are never in an option.
- Main-version walls join the primaries' walls as they join each other, but never a secondary option's walls: move such a wall into the option to join it.
- Plans, sections, elevations, 3D views, sheets, PDF and DWG/DXF show each view's options; IFC holds the main version and the primaries.

Make primary, delete, accept

Make primary carries over what refers to the old primary from the main version: the walls attached to its roofs or floors, and the dimensions and tags of the views showing it by **Automatic**, move to the matching elements of the new primary (copies of the same element, else the nearest of the same kind). When some find no match, a window lists them: **Delete and make primary** deletes those dimensions and tags and detaches those walls.

Delete an option, when views are dedicated to it, lists them in **Delete dedicated views**, ticked: the unticked ones stay and show **Automatic** again; those **Visible in option** of it are ticked and cannot be unticked. The primary cannot be deleted: make another option primary, or delete the set.

Accept primary says which option joins the main version and which go, with how many elements, and lists the views dedicated to them. **Accept** does it in one step: **Undo** (Ctrl+Z) brings the set back.

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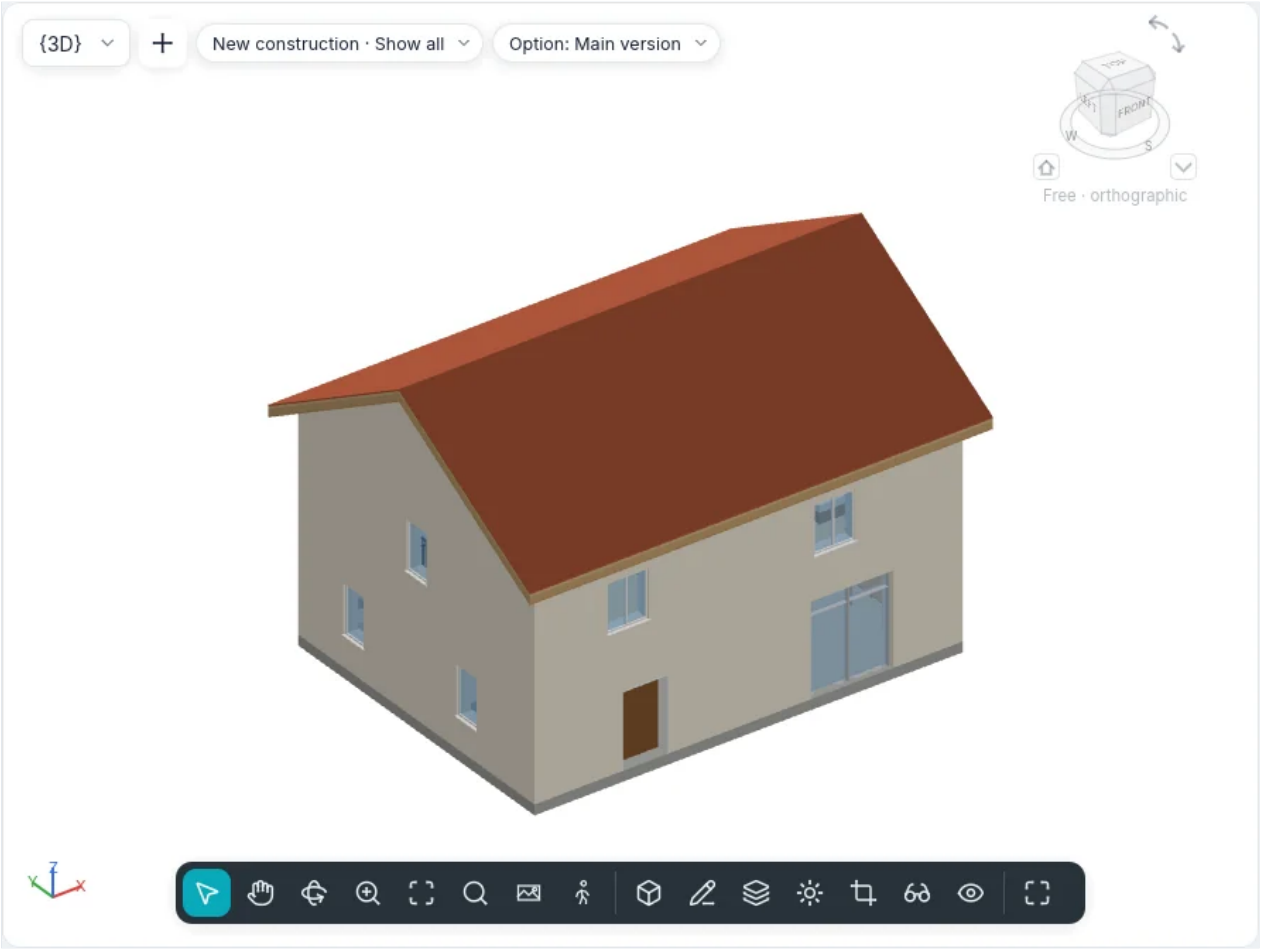
Views and graphics

This chapter describes the views of the model, how each one is drawn, and how to change what a view shows and how: graphic styles, scale and detail, the cut heights of a plan, crop regions, hiding and isolating, **Visibility and Graphics**, filters and view templates. Line weights, line patterns and styles by category are described in the same terms by the program's dialogs of **Settings** on the rail.

3D views

Views ▸ Create ▸ 3D View ▸ Default 3D View opens the default 3D view, **{3D}**, framing the whole model; opened again, it shows the model as it was left. Drag to orbit (or **Shift** and the middle button), drag with the middle button to pan, and turn the wheel to zoom at the cursor; see The interface.



The default 3D view, {3D}

To keep a framing, **Views ▸ Create ▸ 3D View ▸ Save View**: type a name and press Enter. The new view is listed under **3D Views** in the **Project Index** and opens at once; it keeps the framing it is left with. Renaming **{3D}** in the **Project Index** (right-click ▸ **Rename...**) does the same: **{3D}** becomes a named view, and the next default 3D view is a new **{3D}**.

A selected element is washed in blue and half see-through, its edges in blue: what is behind it stays visible.

Floors cut the walls they pass through: where a slab crosses a wall the wall stops under it and goes on above it, so the slab's edge and the wall's face are one surface.

Section box, projection and orientation

{3D} starts orthographic, looking from the south-east corner. Right-click the orientation cube: **Prospettica** or **Ortogonale**, **Vai all'inizio**, **Salva vista...**, **Blocca selezione** (the orbit turns about the chosen elements), **Ripristina Fronte**, **Orienta su direzione** (the compass points and **Alto**), **Orienta su vista** (a plan, section or elevation: the view takes its viewpoint and a section box reproduces its extents) and **Orienta su piano...** (a level, a grid or a reference plane).

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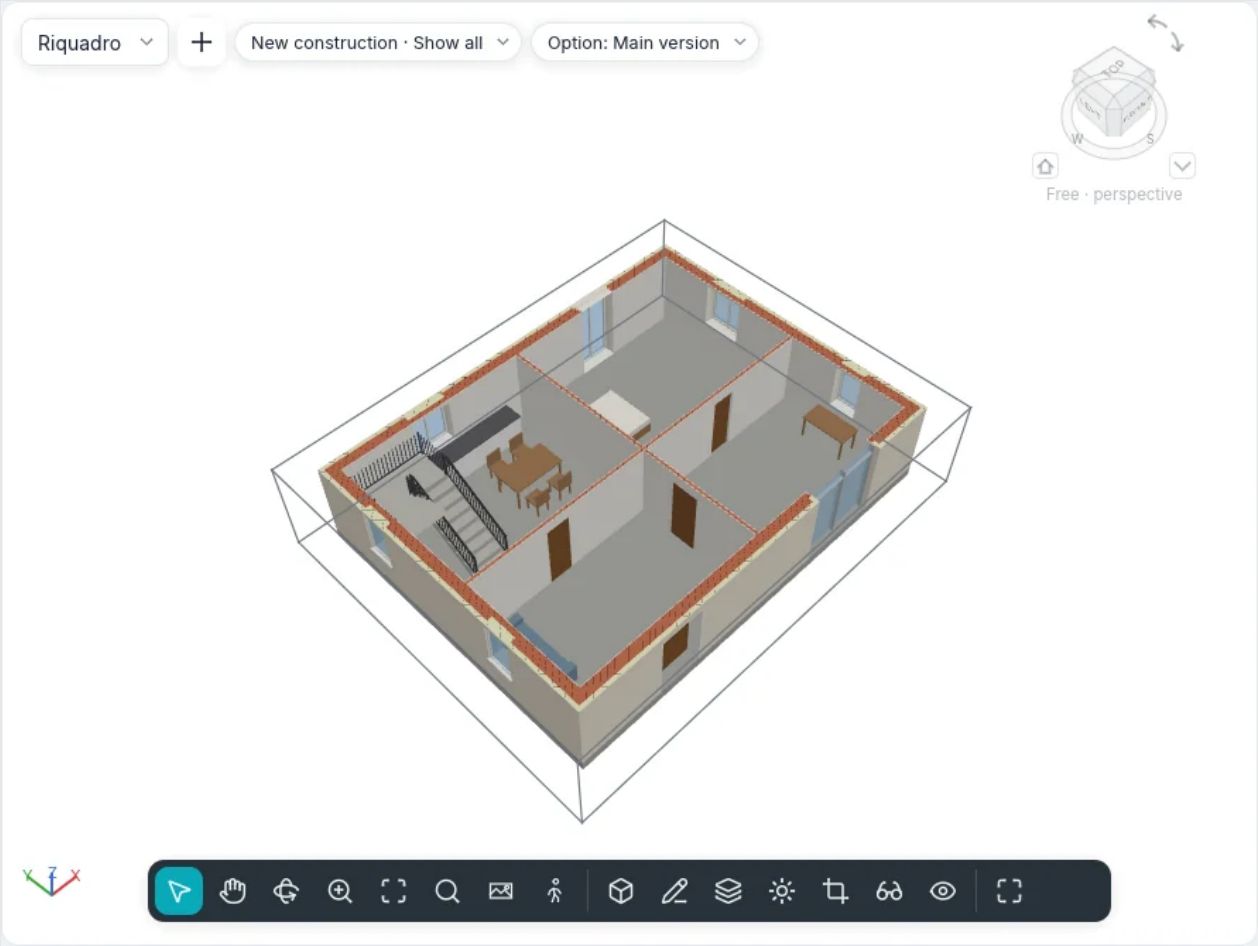
In the 3D view's **Properties**, **Estensioni ▸ Riquadro di sezione** switches the section box on: the first time it wraps the visible model. Outside the box the model is not shown and the parts it cuts are closed on its faces, layer by layer. A click on an edge selects the box: the six arrows move a face, the circle above turns it. Each change is one undo step; switched off, the box keeps its size. In the **Project Browser**, right-click a 3D view ▸ **Mostra riquadro di sezione** shows it in plans and sections too, with grips. **Visibilità e grafica ▸ Riquadri di sezione** hides its drawing, not the cut.



The section box cuts the model at the ground floor

With **Sezione** the options bar offers **Tipo** (**Sezione edificio**, **Sezione di muro**, **Dettaglio**) and **Scala vista**; the Browser lists the view under its type's branch. **Prospetto** is placed with one click beside a wall: the symbol looks at it (**Tab** turns it). Right-click **Viste** in the Browser ▸ **Organizzazione del browser...** chooses how the views are listed: by type, discipline, phase, level, only those not on sheets, or an organisation of your own.

The cut faces show the material's **cut pattern**; at **Basso** detail they show the **Materiale poché a dettaglio basso** chosen in Properties (: each part's own material). With the box selected, **EH** hides its drawing in the view; **BX** puts the box around the selected elements. **{3D}** and the saved 3D views have **Ritaglia vista** and **Regione di taglio visibile** too: the image is limited to the rectangle, dragged by its sides. **Orienta su piano...** also takes a click on a face (**Seleziona un piano**) or on a line (its work plane).

Sezione is drawn in a section, an elevation or a detail too: it is made across that view. **Rif. ad altra vista** makes the section open an existing view (or a new free 2D view) instead of its own; **Offset** moves the line sideways. With an elevation placed by one click selected, its symbol shows four boxes: a click switches the elevation towards that side on or off (interior elevations), the outer grip turns the symbol. In the Browser, **Ctrl** and **Shift** pick several views: **Applica modello vista...**, **Duplica** and **Elimina** act on them all.

Perspective views

Views ▸ Create ▸ 3D View ▸ Perspective view puts a camera in a plan, a section or an elevation. The options bar (in the BIMopen style, the tool panel) has the **Level** and the **Offset**, the eyes' height above the level (1.60 m, as in walk mode); **Perspective** is always on. The first click is where the camera stands; the triangle of what it sees follows the cursor, and the second click is the point it looks at, at the same height. **3D View 1** (then 2, 3...) is made under **3D Views** in the **Project Index** and opens beside the plan, cropped. In a section or an elevation the two clicks give the heights too.

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The view is drawn in its **crop region**, in its proportions (240 × 160 mm on paper); what lies beyond the **far clip plane** (at first, as far as the point looked at) is not drawn. Its properties, in the **Extents** and **Camera** groups: **Crop View**, **Crop Region Visible**, **Far clip plane** and its distance, **Reference level**, **Eye elevation** and **Target elevation** (from the level), **Field of view**.

Right-click the view in the **Project Index** ▸ **Show camera** selects the camera in the plans, sections and elevations (so does a click on the crop region's border in the 3D view). Drag its body to move it (with **Shift** the target stays), the red point to turn it, the triangle's corners for the field of view and the square on its base for the far clip plane; in an elevation or a section you change the heights. The 3D view follows while you drag, and each drag is undone with **Ctrl+Z**. On the **Modify | Cameras** tab: **Reset target** puts the target on the model seen at the middle of the crop region, **Size Crop** changes the width and height on paper by **Field of view** (more is seen) or **Scale (locked proportions)**, **Reset crop** (240 × 160 mm, 50°), **Open the view**. Dragging the crop region's sides in the 3D view changes the field of view too.

The live view of a camera view follows it, walk mode starts at its eye and, when you leave, the camera stays where you walked to; a **New path** starts at its pose. From the command line: `camera {"from":[14,-6],"to":[5,3]}` makes the view, `camera {"eye":[15,-6,1.6],"fov":60,"far":20}` changes it, `camera {"reset_target":true}` resets the target.

Graphic styles

The **Graphic Style** button of the view control bar, at the bottom of every view window, offers:

Style	What is drawn
Wireframe	Every edge, also those behind other faces; no faces
Lines	Faces as paper and the edges in front of them in ink; on the dark background paper and lines are inverted together
Shaded	Faces lit, in the colour of their material
Shaded with Edges	As **Shaded**, with the edges in front
Flat Colors	Faces in the colour of their material, the same whatever the light
Realistic	Faces as in a photograph: the light of the project's sun and sky, reflections, ambient shadows, see-through glass, haze
Pastel	A watercolour drawing: faces washed in their material's colour, close to the paper in the sun, deeper in shade; the lines drawn by hand

Wireframe, Lines and Shaded with Edges

The style belongs to the view and can be undone. The keyboard shortcuts `WF`, `HL` and `SD` choose **Wireframe**, **Lines** and **Shaded**.

The edges are the real ones: where two faces meet at an angle, and the outline of each face and each opening. A wall with a window shows its own outline and the window's, and no line where the pieces of the wall meet.

Plans, sections and elevations have a graphic style too. A plan is drawn in **Lines** until another style is chosen. In **Lines** the surfaces, cut or seen, are not coloured: what is cut shows its material's cut pattern; at **Coarse** detail the type's coarse fill pattern, or else only its outline. In **Shaded**, **Shaded with Edges** and **Flat Colors** every element seen or cut is filled with the shading colour of its material (a part with no material takes its category's, from **Styles by Category**, else the category's default colour; the 3D view follows the same rule): walls (each layer where it is cut), floors, roofs, doors, windows, furniture and components, stairs and railings. Plans are flat; in sections and elevations **Shaded** darkens a little the faces turned away from the viewer, such as the slopes of a roof, while **Flat Colors** keeps them all alike. At **Medium** and **Fine** detail the cut patterns stay on top of the colour. Sheets, prints and exported drawings show the views as the screen does.

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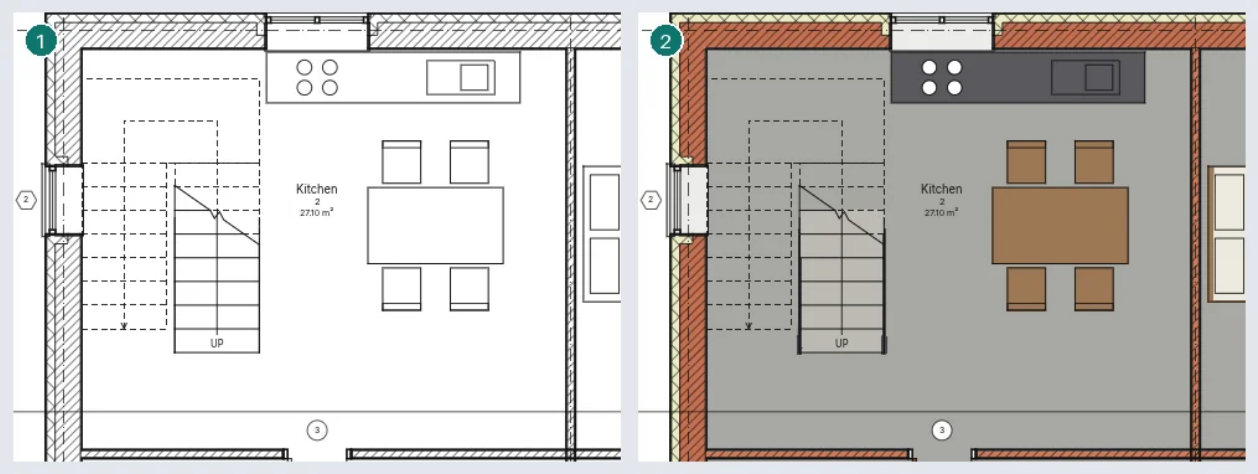
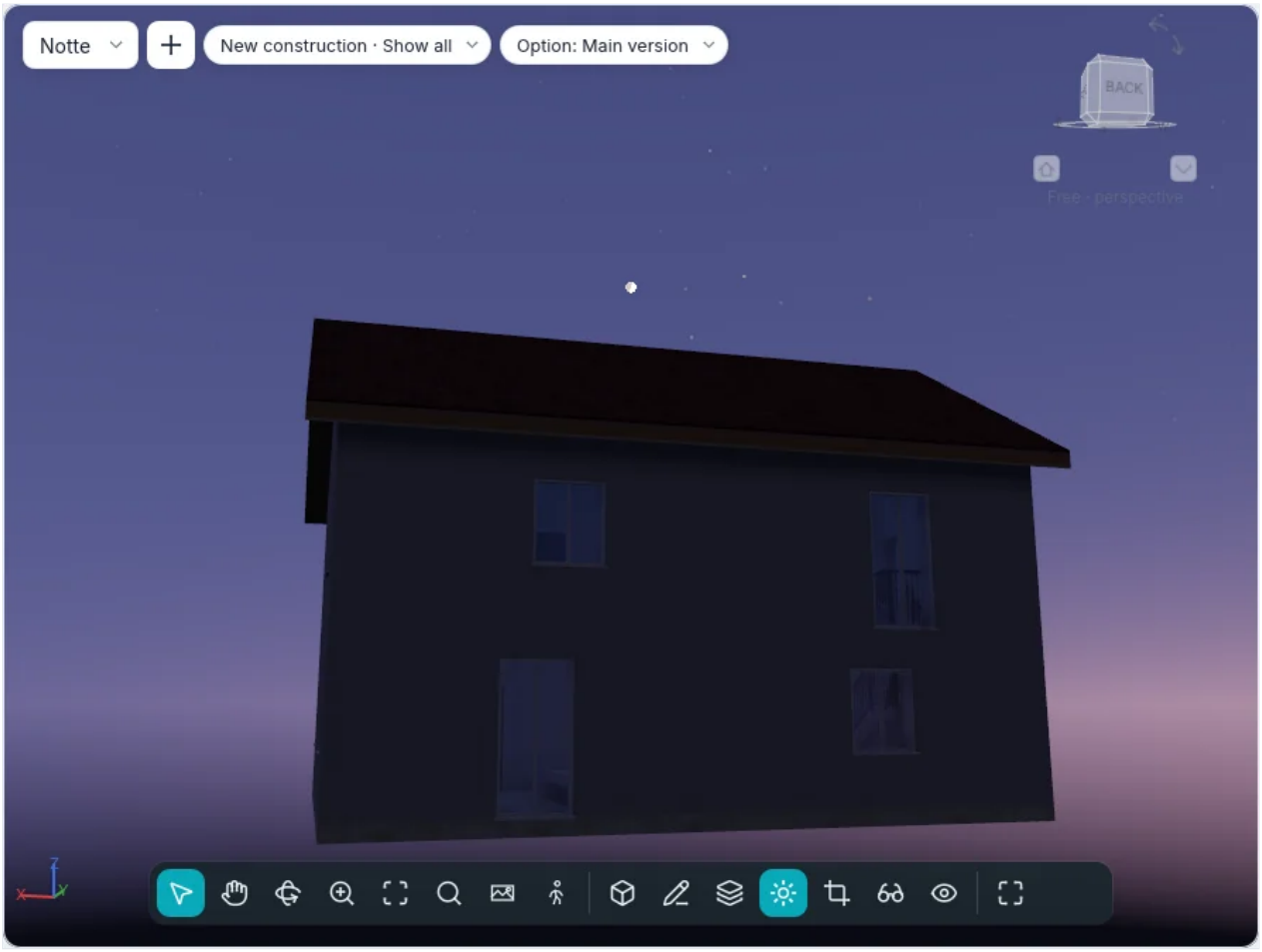


A plan in Lines and in Shaded

The rooms' colour scheme shows in every style. With **Color scheme location** at **Background** it lies under the model's lines: in a plan over the floors and under the furniture, in a section over the surfaces seen through each room it cuts. At **Foreground** it covers the model, under the annotations.

Realistic style

Realistic draws the 3D view like a photograph, in real time, and the view stays editable as in the other styles.



Realistic at night: moon and stars

- **Light of the sun and the sky.** The light is that of the project's sun (place, date and time in **Sun settings**) and of the sky that sun gives: blue by day, warmer and darker at sunset. The sun lights the model even with shadows off; with **Cast shadows** its shadows are those of the other styles. What is in shadow stays lit by the sky and the ground. The sun's disc appears when it comes into the frame, with a glow, and the model hides it. Towards sunset the sun's light turns warm and weaker and the

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shadows lengthen; after sunset only the sky is left, fading into a blue-violet twilight with a pink band on the horizon on the sun's side; at night the sky is dark blue and the scene almost dark (self-illuminated surfaces stay visible). Dragging the sun or following a sun study changes the light continuously.

- **Moon and stars.** The moon is where it is at the project's place, date and time, in its phase: lit on the side facing the sun, the rest faintly grey (earthshine), redder near the horizon. At night it is the light: with **Cast shadows** the model casts moonlight shadows, and a full moon lights the scene softly where a new moon leaves it almost dark. Once the sun is down the brighter stars come out in their places, more of them the darker the sky.
- **Materials.** Surfaces take the look of their material (the **Appearance** tab of **Materials**): the texture image at its real size, the roughness and its map, the relief, metal, glass and water, self-illumination (seen, not lighting others). A material without a look of its own derives it from its colour and transparency; a part without a material takes its category's colour. On the web the images arrive later: until then the colour shows. The preview of the **Appearance** tab shows the material on a sphere, a cube or a 2 × 2 m wall.
- **Glass.** Glass shows what is behind it and reflects the sky and the sun, more when seen at a grazing angle. Panes are drawn from the farthest to the nearest.
- **Ambient shadows.** With **Show Ambient Shadows** corners and recesses, which get less light from the sky, are darker.
- **Haze.** In the distance the air veils the model: **Distance haze** sets how much, and how clear the sky is.
- **Exposure. Automatic** follows the light of the scene and adapts in about half a second when it changes (turning the camera from the sky to the shade, or changing the time). **Manual** uses the **Value (EV)**: higher is darker, about 15 under the midday sun.

In the **Realistic** group of the Graphic display options: **Exposure** and **Value (EV)**, **White point** (the light shown as white: 6500 K is daylight as it is, a lower value cools the image, a higher one warms it), **Saturation**, **Highlights** and **Shadows** (lower or raise the lightest and darkest tones) and **Distance haze**. **Show Edges** draws the edges over the faces. The lighting **Scheme** is for now always **Exterior: sun**. **Sun**, **Ambient light** and **Shadow intensity** set the sun's light, the sky's and how dark the sun's shadows are.

The **Physical sky (time and place) Background** shows behind the model the same sky that lights it, with the sun's disc; with **None**, **Sky** or **Gradient** the background stays as chosen and the sky still lights the model. Switching to **Realistic**, a view whose background is **None** gets **Physical sky (time and place)**, and goes back to **None** on changing to another style. Selected elements and the sun path look as in the other styles.

On the web without WebGPU (with WebGL2) **Realistic** works without ambient shadows, and the automatic exposure is set from the sky instead of from the image. On the web and on Android, while the camera moves, the view may be drawn smaller to stay smooth, and turns sharp again when it stops. Without a graphics card **Realistic** is shown as **Shaded** and the status bar says so. Plans, sections and elevations in **Realistic** are for now drawn as in **Shaded**; sheets, prints and exported images do not show **Realistic** yet.

Pastel style

The **Pastel** style draws the view as a watercolour or marker drawing: each face is a uniform wash of its material's colour, softly graded, never striped. Faces in the sun stay close to the paper; faces turned from the sun take the same colour, deeper and cooler, never grey; where a cast shadow falls the wash goes deeper still, with a clear edge. Glass is a blue-grey wash with a warm glint. The wash fades towards the edges of the drawing, so the paper shows round the model; the paper has a barely visible grain. Choosing **Pastel** turns the Sketchy lines on, in ink; they can be turned off, and they go off when the view changes to another style.

It applies to every view. In 3D views the light is that of **Realistic**: the project's sun and its shadows. In plans, sections and elevations the paper is cream, the faces and floors are washed in their materials' colours and the cut parts are a solid, dark poché in their material's tone. The cast shadows are the view's own: they are turned on, off, deeper or fainter with the same settings as in the other styles (**Shadow Intensity**), in a clear, dark warm tone. Printing and export come out washed as on screen, 3D views too.

Pastel is the same whatever the window's background: always its light paper, with the same washes, poché and shadows.

The **Pastel** group of the Graphic Display Options is set per view and view templates carry it: **Wash Strength** (how deep the colour is laid), **Wash Variation** (how unevenly it lies on a face), **Paper Grain** and **Saturation** (how much of the materials' colours the wash keeps), from 0 to 100, and **Paper Colour** (with **Chosen** a colour of its own; without, the warm white paper). From the command line: `graphicoptions {"style":"Pastello","pastel_density":70,"paper_color":"auto"}`.

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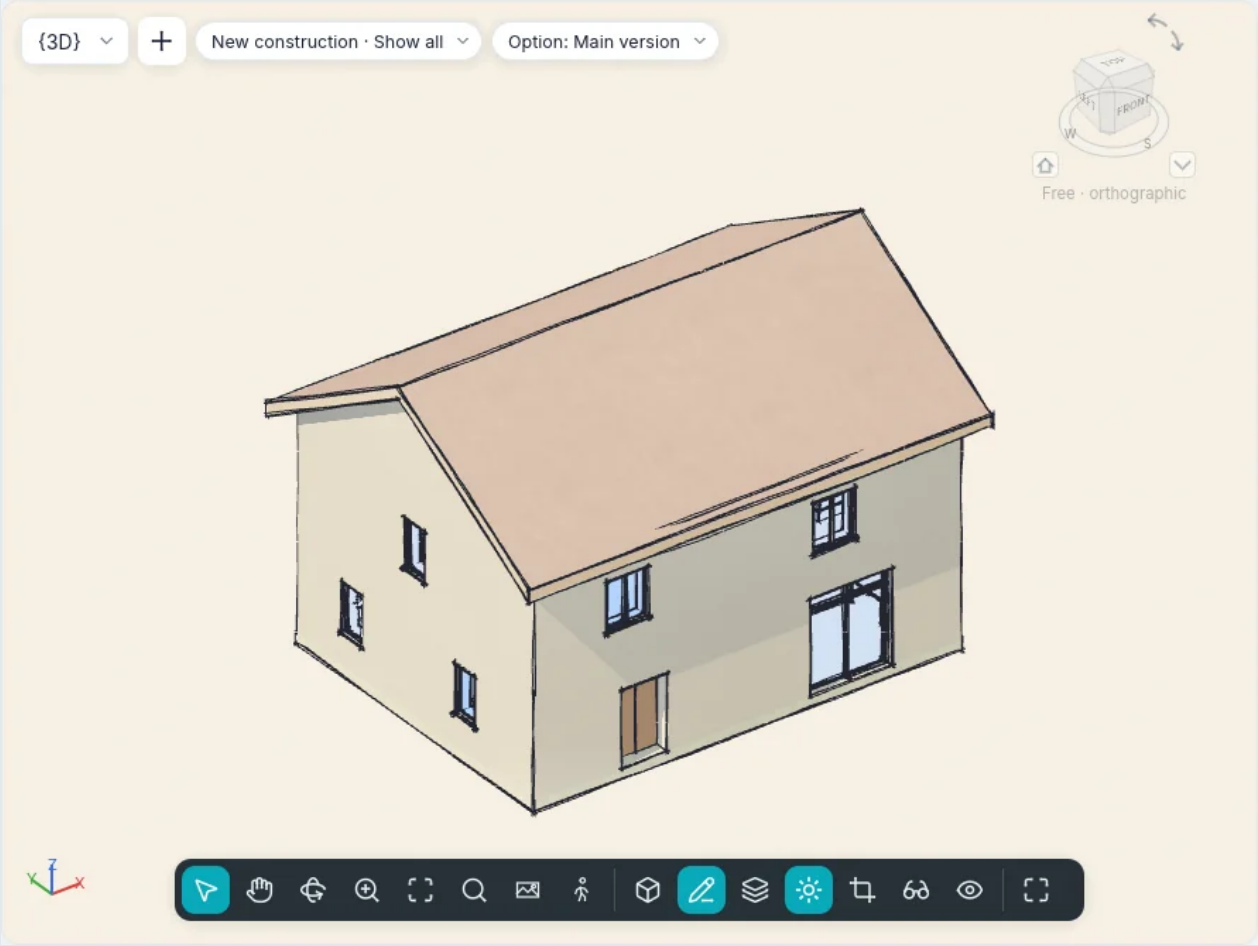

Pastel: wash 70, variation 55, grain 40, saturation 75 (the defaults)

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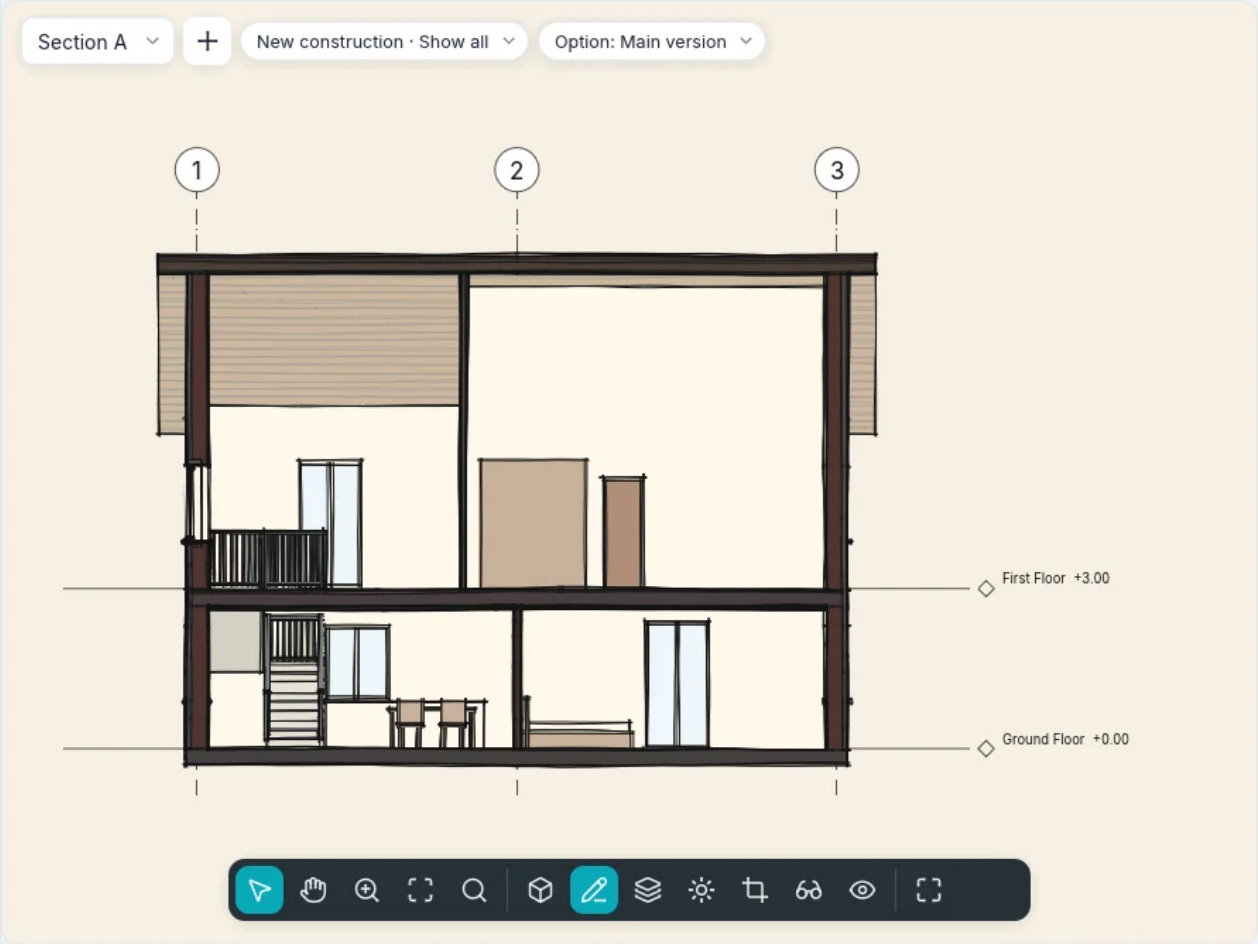

Section A in Pastel, with the default values

Live view and walk mode

Live View (View tab ▸ Presentation, or the button of the 3D view's floating bar) opens a window beside the 3D view that draws it in the **Realistic** style, with the sun's shadows and the physical sky, and follows its camera: orbit or zoom the 3D view and the live view looks from the same place; every change to the model shows at once. While the camera moves the live view draws a draft, then refines. At the bottom of the live view, **Time** sets the hour of the view's own sun (sunrise and sunset are marked on the slider); **Project Sun** goes back to the project's shared sun. The view's properties have the group **Live View**: **Follows the 3D View** and **Draft While Moving**. From the command line: `liveview 3d {"follows":true,"draft":true}`.

Walk (View tab ▸ Presentation, the 3D view's floating bar, the Classic style's navigation bar or the command search) is a command like the others: the **Modify | Walk** contextual tab opens, the Options Bar shows its choices and the status bar says what to do. The active 3D view becomes the eyes of someone walking through the model: you stand on the floor under the middle of the view, facing where the view looked. The pointer is captured at the first click in the view (until then the tab and the Options Bar can still be used):

Key	
W A S D or the arrows	walk
the mouse	look around
Shift	run
comma and full stop, or the wheel	slower and faster
F	**Walk** or **Fly**
E and Q, Page Up and Page Down	rise and sink (flying)
K	a keyframe
R	start and stop recording
Esc	leave walk mode

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The other shortcuts wait until you leave. Walking, gravity keeps you on floors, landings and slabs; a step a little higher than the project's highest riser is climbed without noticing, so stairs and ramps are climbed by walking; walls, columns, railings, furniture and window glass stop you, and you slide along them; doors are walked through. **Fly** drops gravity and collisions. You leave with Esc, **Modify**, **Leave** in the tab or by choosing another command; the view stays where you got to. A plan of the level you are on shows where you are and what you see.

The **Modify | Walk** tab has **Mode** (**Walk**, **Fly**), **Walkthrough** (**Keyframe**, **Record**) and **Walk** (**Options…**, **Leave**); in the BIMopen style they are in the tool panel's tab. The Options Bar has **Walk** or **Fly**, the eye height, the speed, **Walkthrough** (where keyframes and the recording go: **New** or one of the project's walkthroughs), **Move parallel to the ground**, **Turn sensitivity** and **Inertia** (the gaze starts and stops smoothly; 0 % as before). While walking, the Properties palette shows the walking speed, running, the flying speed, the eye height, gravity and collisions, the step climbed, the sensitivity and the inertia, **Invert vertical axis**, **Record into**, **Keyframe every** and the smoothness of the recorded walkthrough. On a touch screen a joystick walks, a drag on the view looks, and buttons run, fly, rise and sink, set keyframes, record and leave.

Options… opens the **Walk Mode** settings. **Movement** holds the personal choices, the same in every project: eye height (1.60 m), walking speed, running, flying speed, **Turn sensitivity**, **Turn inertia**, **Keyframe every**, **Invert vertical axis**, field of view and **Fly parallel to the ground**. **Collisions by category** holds the project's: for each group of categories whether **Disable Collision** is on (it is for doors and curtain wall door panels), and the **Step height**, taken from the stairs unless you type it. From the command line: `walksettings {"step":0.22}`, `collision {"group":"furniture","off":true}`.

Disable Collision

Every element with a body has **Disable Collision** in the **Walk Mode** group of its properties: on, whoever walks passes through it. Its value comes from the type, and the type's from its category; **Collision From Type** (on a type, **Collision From Category**) says so, and setting **Disable Collision** makes the value the element's own. Several elements selected show **‹varies›** where they differ. Schedules can list it as a field and filters can use it as a rule. It changes nothing but walk mode. A floor, stair or ramp with collision disabled carries no one either. From the command line: `collision 42 {"off":true}`, or `{"off":null}` to take the type's value again.

Walkthrough

Walkthrough (View tab ▸ 3D View ▾, or the command search) is drawn in a plan, a section or an elevation like a wall: the Options Bar has **Perspective**, the **Level** and the **Offset** (1.60 m, the walk's eyes); each click sets a keyframe and the next stretch follows the cursor, with the camera's triangle at the start. In a section or an elevation the clicks give the height too. A double click, Esc or **Finish walkthrough** in the tab end the path (**Cancel walkthrough** drops it): **Walkthrough 1** is made, in the **Walkthroughs** branch of the Project Browser, and opens beside, at frame 1. It is a camera view that moves: it has its crop region, its far clip plane and the camera's properties, plus **Walkthrough frames** and **Frame**.

Selected in a plan (a click on the path, or **Show Camera** in the Project Browser), the walkthrough is edited in the **Modify Walkthrough** tab. **Controls** in the Options Bar chooses what the grips do:

- **Active camera**: the camera is dragged along the path, snapping to the keyframes; on a keyframe the red point says where it looks and is dragged, and the triangle's corners change the field of view;
- **Path**: the keyframes become grips; in a section or an elevation their height changes;
- **Add keyframe**: a click on the path adds one;
- **Delete keyframe**: a click on one removes it (at least two stay).

Frame shows the view's frame and can be typed; beside it the previous keyframe, previous frame, next frame and next keyframe buttons and **Play**, which plays the walkthrough in its view from the frame shown to the last; the status bar says "frame 87 of 300", with **Cancel**; Esc stops. **Reset cameras** makes every keyframe look along the path again.

Frames… opens **Walkthrough frames**: **Total frames** (300), **Frames per second** (15), **Total time** (20 s), **Uniform speed** (on: the camera always goes at the same speed; off, each keyframe has its **Acceleration**, from 0.1 to 10, for the stretch starting there), and for each keyframe its frame, speed and elapsed time. **Turn smoothness** says how the gaze turns where the path changes direction: 0 % at once on the keyframe, 100 % along half the stretch before and half after, starting and stopping smoothly; the path has one and each keyframe may have its **own**. The graph below shows the turn of the gaze over time. **Indicators** with **Frame increment** marks the frames along the path in plan.

Export… (also in File ▸ Export ▸ Images and Animations ▸ **Walkthrough…**) opens **Length/Format**: **All frames** or a **Frame range** from–to, **Frames per second** (the total time follows), the **Graphic style** (the view's or another), **Size** and **Zoom**, then the file type: an AVI film (Motion-JPEG), with the images too if wanted, or a sequence of JPEG, PNG, TIFF or BMP images. A bar shows the progress, with **Cancel**.

While walking, **Keyframe** (K) adds a keyframe where you stand, looking where you look, to the walkthrough chosen in the Options Bar; with **New** the second makes **Walkthrough N**. **Record** (R) makes a walkthrough of the walk: a keyframe every second (**Keyframe every**), as long as the walk lasted and as smooth as the **Inertia**. The paths of projects saved before become

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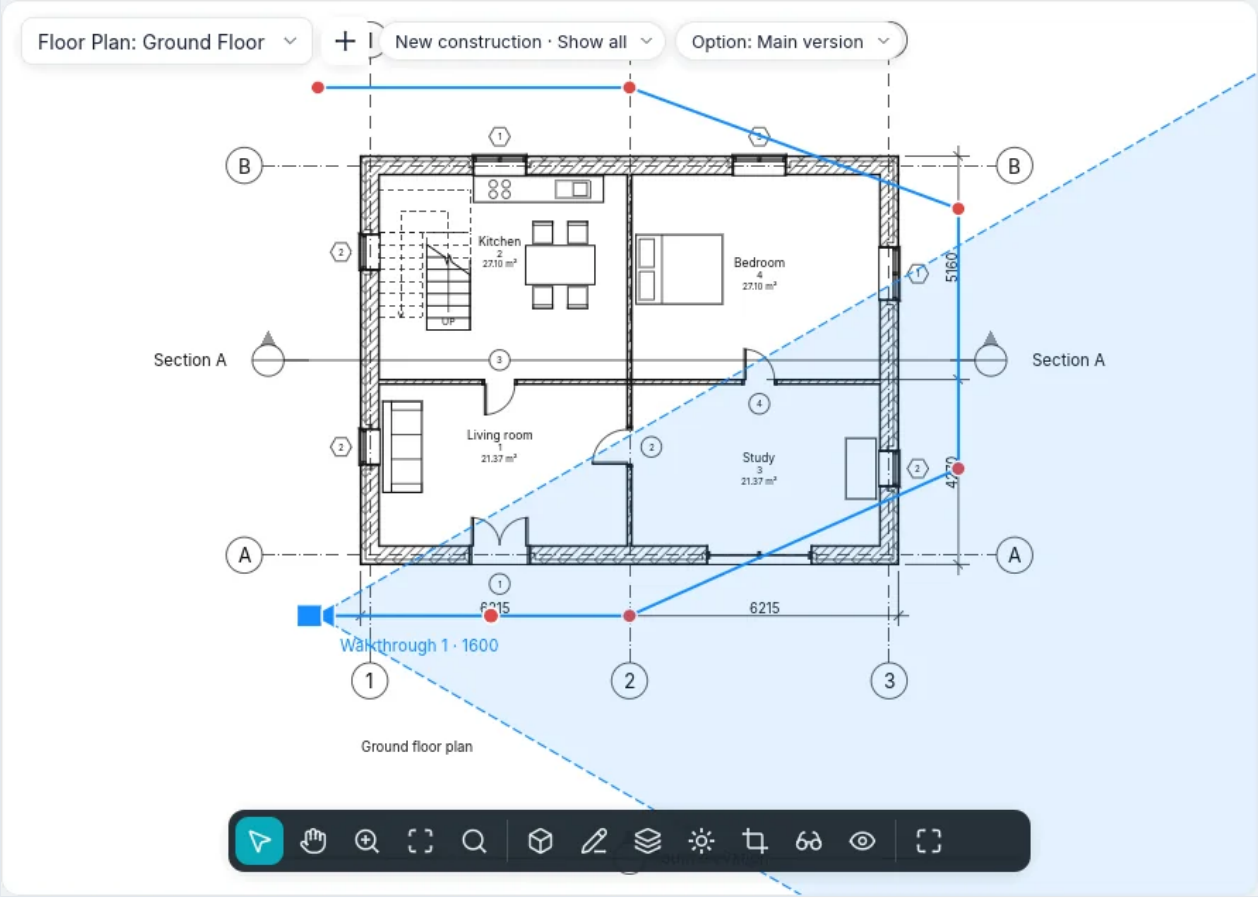
Walkthrough views when opened. From the command line: `walkthrough {"points":[[0,-6],[5,-6],[5,2]],"offset":1.6}`, `walkthrough 42 {"frame":"next_key"}`, `walkthrough 42 {"frames":150,"uniform":false,"key":{"index":0,"accel":2}}`; `walk {"from":[4,2.5],"to":[8,2.5]}` shows where a walk ends.



Sun and shadows

Plans, sections, elevations and 3D views show the sun (drafting views, legends, sheets and schedules do not). The view control bar has two buttons, each with a small arrow for its menu (the right button opens it too):

- **Sun Path**: its menu has **Show Sun Path** / **Hide Sun Path**, **Sun Settings...** and **Fit to Model** (frames the whole sun path);
- **Shadows**: its menu has **Shadows On** / **Shadows Off**, **Graphic Display Options...** and **Preview Sun Study**.

While either is on, the bar also shows the sun's hour and date, each with **−** and **+** (a quarter of an hour, a day). The same buttons and menus are in the **Sun and shadows** menu of the BIMopen style's view bar, in the touch mode's view sheet and in the view's context menu. **Sun Path** is also a check box of the view's **Properties** (**Graphics**), with **Sun Path Size** (100–500 %, 150 % by default) and **Shared Sun**.

Shadows are computed from the model and the sun, the same in every view: in plans they fall on what the plan shows and, where nothing stands, on the plane of a level (**Ground Plane at Level** in the sun settings, otherwise on the ground, the lowest level); in sections and elevations they fall on the façades and on what is behind them, from eaves, balconies, recesses and other buildings, and so in 3D views. In **Lines** (hidden line) a shadow is a grey of the paper, in the shaded styles it darkens the materials' colours. Faces cut by the view stay in the light.

The sun path is a ring round the model with the points of the compass at true north, the day's arc with its hours, the analemma (where the sun stands at that hour through the year: one smooth figure of eight, every day at the clock of the chosen date), the part a study goes through, and the sun with its hour and date; in a 3D view it is a dome over the ring. In plans, sections, elevations and 3D views the sun can be dragged: along the day it snaps to quarters of an hour, along the year it follows the analemma day by day (across a change of summer time the clock moves by the change, so the sun does not jump); `Shift` while dragging takes the other, and **Along the Day** / **Along the Year** in the sun path's menu choose which comes first. The shadows follow the drag: in plans, sections and elevations, while the sun moves, they are a coarser draft, each view showing its last shadows until the new ones are ready, and they are drawn in full when the sun is released. With the view under the cursor (and nothing selected, or after a click on the sun), `Left` and `Right` move the hour, `Up` and `Down` the date. With **Lighting** the sun stands still and has no path.

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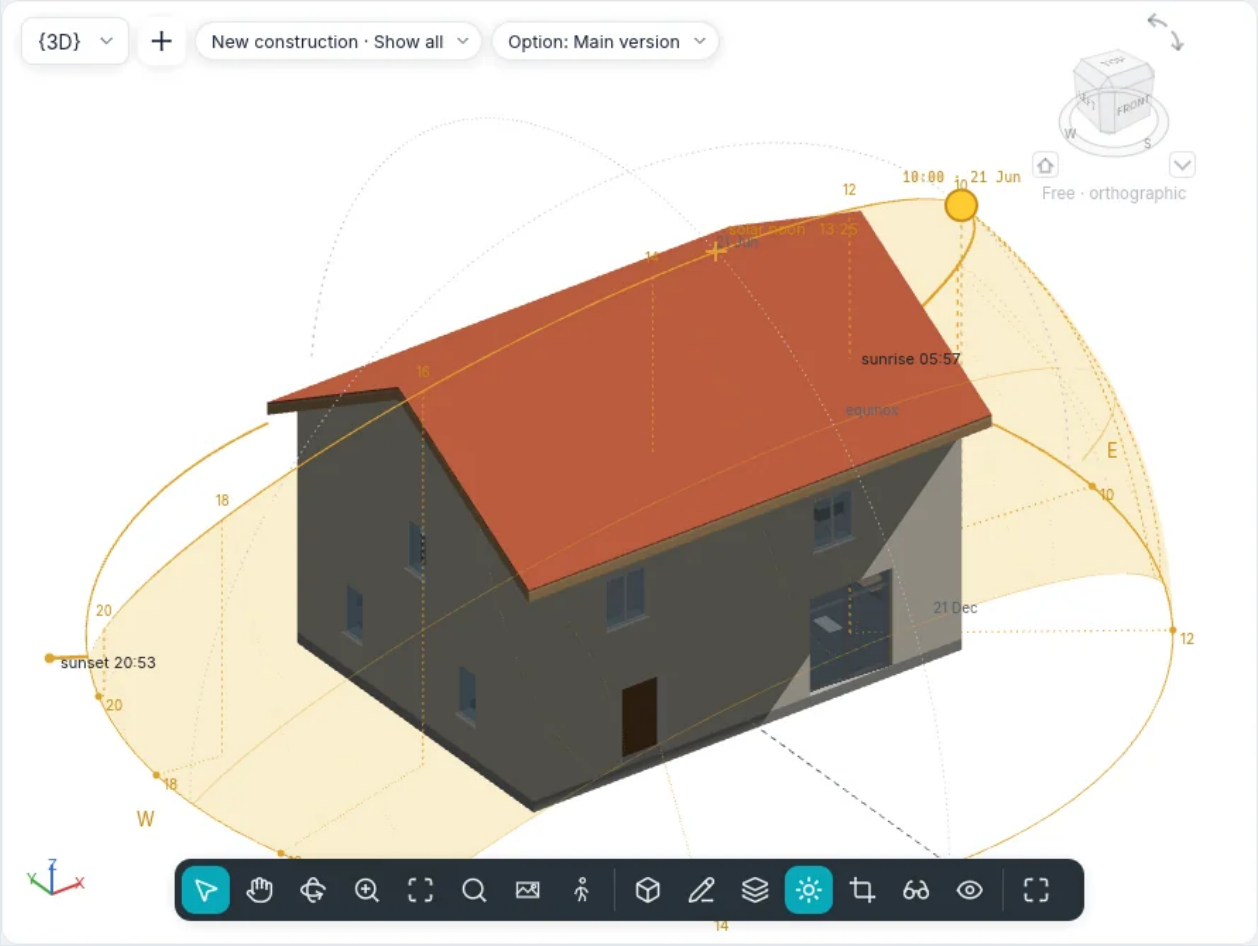

The sun path and shadows in the 3D view

The sun path's menu (in the view bar, in the view's menu and in touch mode) also has **Visual Aids** that help read it: **Vertical Hour Lines** (from each even hour a dashed line down to the ground and out to the ring, with the hour there; in plans a dot on the ring), **Arc Projected on the Ground**, **Solar Noon** (the arc's highest point with its clock time, and the north–south line), **Band of the Year** (the sky the sun crosses in the year, in a light yellow, the arcs of the solstices and of the equinox, the analemmas of the even hours), **Sunrise and Sunset** (marks on the ring with their times; the view bar shows them with the length of the day) and **Analemma**; all but the projection are on at first. In a 3D view what stands behind the model is hidden by it. **Sun Settings** also show the year's sunrises and sunsets as a chart: a click or a drag on it picks the day.

Shared Sun: the views share the project's sun unless they have one of their own (the **Shared Sun** property). Moving the shared sun in one view moves it in every view that shares it; a view with its own sun is not touched. **Shared Sun in All Views** in the sun settings puts every view back on it.

Each change is one step of **Undo**; the `sun` command does the same at the command line and over MCP (see Command line and MCP).

Sun settings

Sun Settings... opens a dialog with the sun's azimuth, altitude, time and date in its header:

- on the left the **presets**: one unsaved row for each mode, the built-in ones (**Summer Solstice**, **Winter Solstice**, **Spring Equinox**, **Autumn Equinox**, **Summer, from 9 to 17**, **Light from top right**) and the project's own; **Save...**, **Duplicate...**, **Rename...** and **Delete** (only the project's presets are renamed or deleted);
- in the middle the mode and its fields:

Mode	What it takes
Still	the **Date** and the **Time**; **Shared Sun in All Views**
One Day	the **Date**, **From Sunrise to Sunset** or a **Start Time** and an **End Time**, the **Frame Interval** (15, 30 or 45 minutes, 1 hour); the number of **Frames** follows
Multi-Day	**From** and **To** (day/month), the **Time**, the **Interval** (1 hour, 1 day, 1 week, 1 month)

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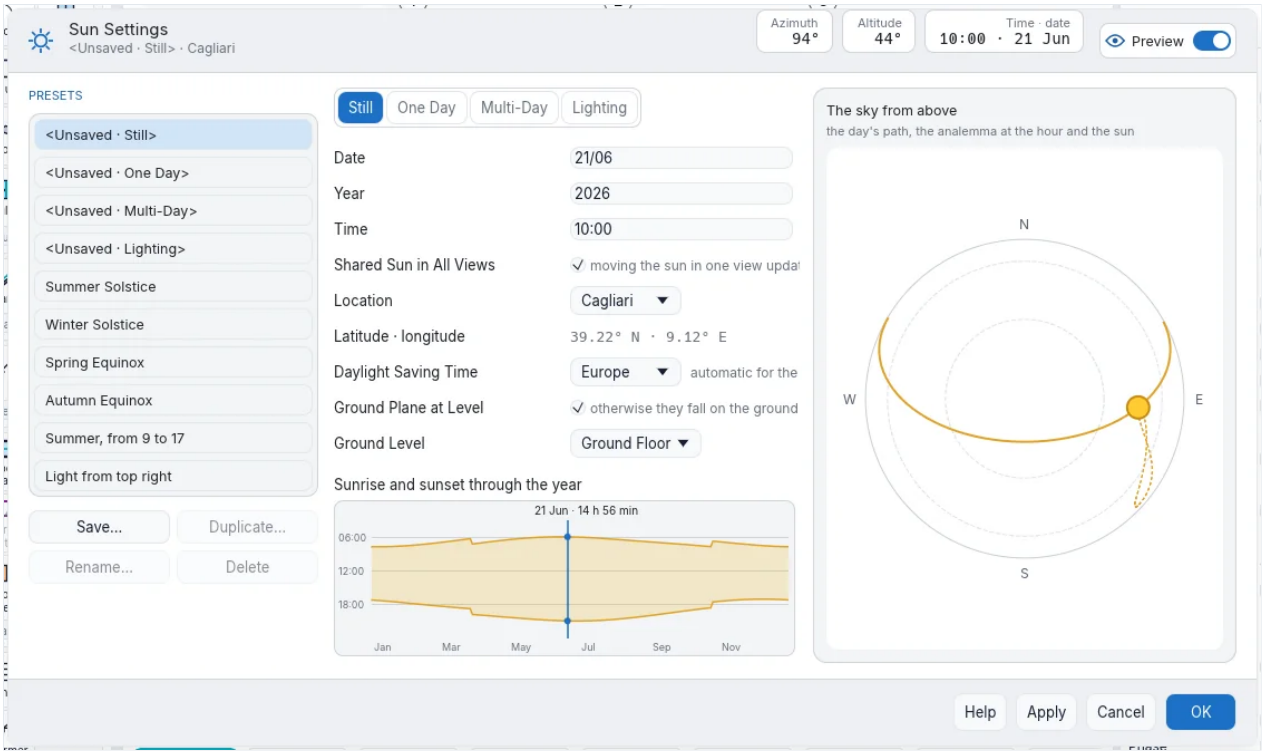
Mode	What it takes
Lighting	**Azimuth (°)** and **Altitude (°)**; **Relative to View**: the light follows the view's direction

Then the project's **Location** (a city, or **Custom** with its latitude and longitude: one for the whole project; a custom place takes the time zone of the nearest city; a new project starts in **Rome**, the sample house stands in **Cagliari**) with **Daylight Saving Time**, which is automatic: the place's rule (Europe, North America, Chile, South-eastern Australia, New Zealand, Egypt, or none) and the exact Sunday of the sun's **Year**, so the times on the sun path and in the studies are those of the clock; the rule can be changed. Then **Ground Plane at Level** with its **Ground Level**; - on the right the preview: the sky seen from above, with the day's path, the analemma and the sun. **Preview** in the dialog's header, right of the figures, is a switch that hides and shows it; the dialog keeps its size, and without the preview the settings take its room.

Apply changes the view's sun and keeps the dialog open, **OK** changes it and closes it, **Cancel** changes nothing. Projects made before the project had a location keep each view's place and date: the first view whose sun was changed gives the project its location, and the views whose sun was changed keep theirs as their own.



Sun Settings

Sun study

Preview Sun Study (in the **Shadows** menu) shows the study bar over the view: ⏮ **−10**, ◀, ▶ **Play** / ⏸ **Pause**, ▶, **+10** ⏭ and **frame n of N**. A sun of one moment is studied through its day. **Save as Image** saves the frame shown as a PNG named after the view, the date and the hour; **Export Study...** saves every frame as a PNG or JPEG image sequence (a film needs a video encoder BIMopen does not have: the choice is greyed, with the reason). × closes the bar; another view closes it too.

Graphic display options

Graphic Display Options... sets the graphic options of the active view, in groups that open and close with a click on their title (the window remembers which are open, and keeps its size and place):

- **Model Display**: the **Graphic Style**; **Show Edges** (the edges over the faces in the **Shaded**, **Consistent Colors** and **Realistic** styles; ticked and grey in the styles that always draw them); **Smooth Lines with Anti-Aliasing** (the 3D view's lines and edges smoothed; without, sharper and faster to draw; plans, sections and elevations always have smooth lines); **Transparency** (0–100: every face of the view turns see-through and what lies behind shows, in the **Hidden Line** style the hidden edges too; it applies to 3D views in the styles with faces, to **Realistic** in a next phase); **Silhouette Edges** (**None**, **Medium**, **Thick**: a thicker line round the model's outline).
- **Shadows**: **Cast Shadows** and **Show Ambient Shadows** (corners and recesses darkened).
- **Sketchy Lines**: see Sketchy lines.
- **Depth Cueing** (sections and elevations): with **Show Depth** what is farther from the view's line fades into the background. **Fade Start** and **Fade End** (as a percentage of the view's depth) set where it starts and ends; **Fade Limit** how much of what

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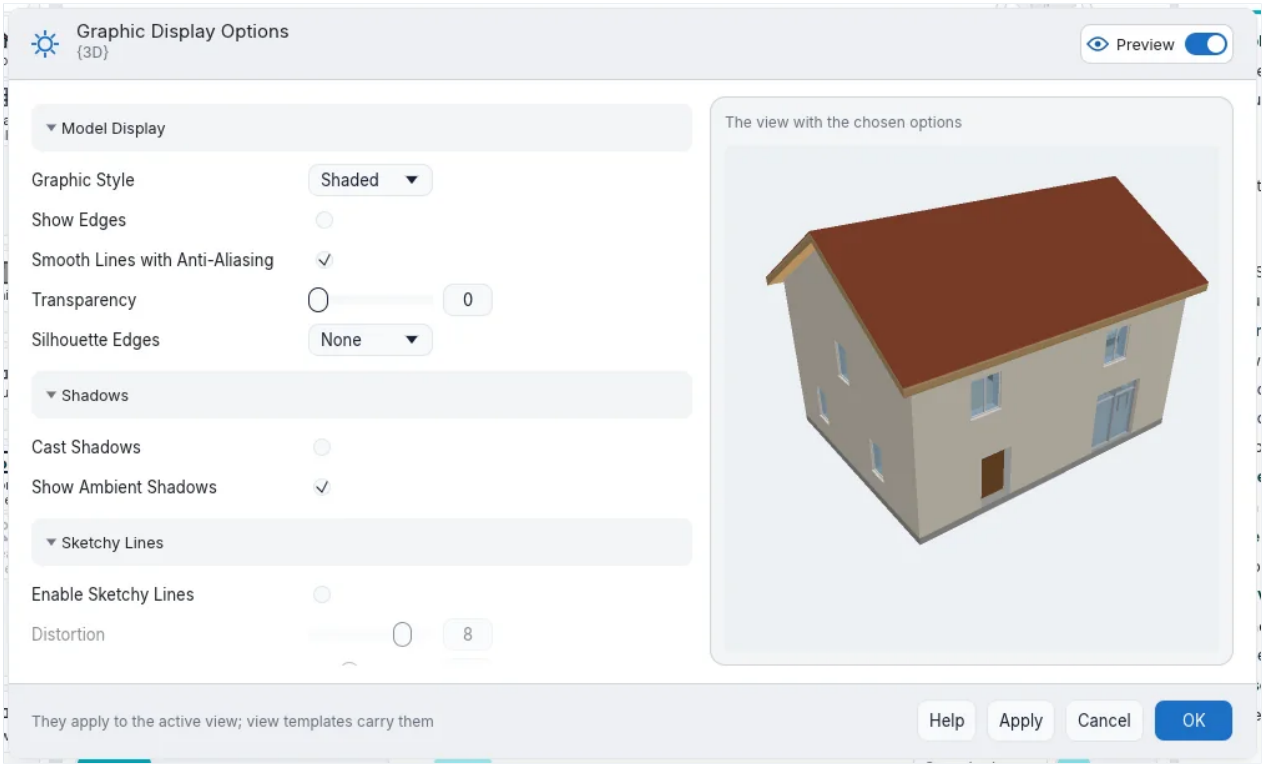
is farthest stays visible (0: it vanishes, 100: no fading). Lines, faces, patterns and sketchy lines fade, on screen as in print and export, on the light and on the dark background; annotations and cut parts stay as they are.

- **Lighting**: the **Scheme** (for now always **Exterior: sun**), the **Sun Settings** (the button opens them), **Artificial Lights...** (grey: they come in a next phase), **Sun**, **Ambient Light** and **Shadow Intensity** (0–100).
- **Realistic**: the settings of the **Realistic** style (see Realistic style).
- **Pastel**: the settings of the **Pastel** style (see Pastel style).
- **Background** (3D views): **None**, **Sky**, **Gradient**, **Physical sky (time and place)**; with **Gradient**, **Sky Color**, **Horizon Color** and **Ground Color**, from the top down. **Image** is grey: it comes in a next phase.

At the bottom, **Save as View Template...** applies the options to the view and asks for the name of a new view template that carries them. Every numeric setting has its slider and a field to type the value. The preview draws the view with the options chosen; the **Preview** switch in the header hides and shows it, as in the sun settings. In the **Realistic** style the preview looks as the view does. View templates carry these options with the graphic style, and the sun and the shadows as **Sun and shadows**. On the command line: `graphicoptions {"edges":true,"transparency":30,"depth":true,"depth_limit":20,"background":"sfumatura","sky_color":"189-209-208"}`.



Graphic Display Options, with the default values

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Transparency at 60%: the faces are see-through; the default is 0

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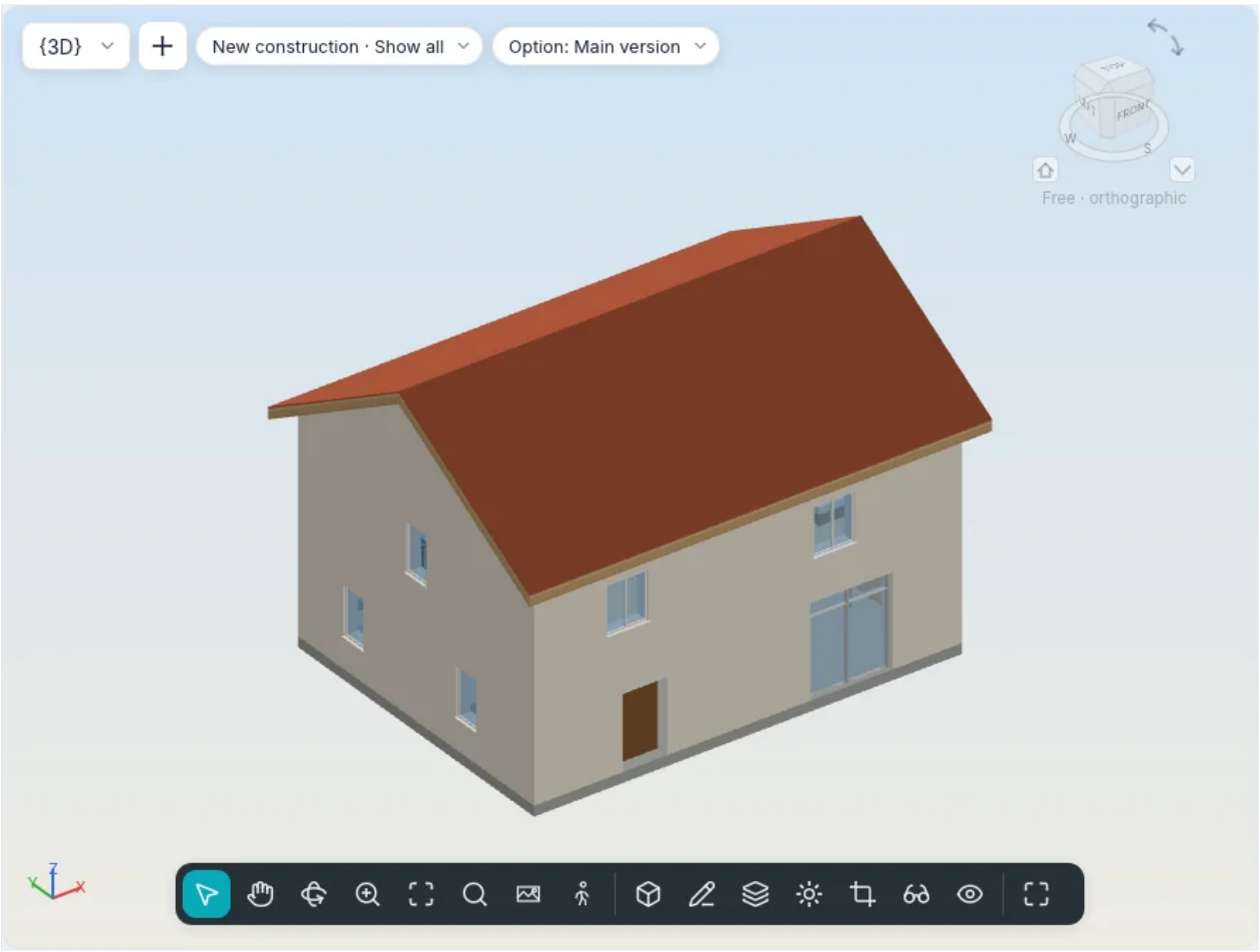

Gradient background, with the default colours

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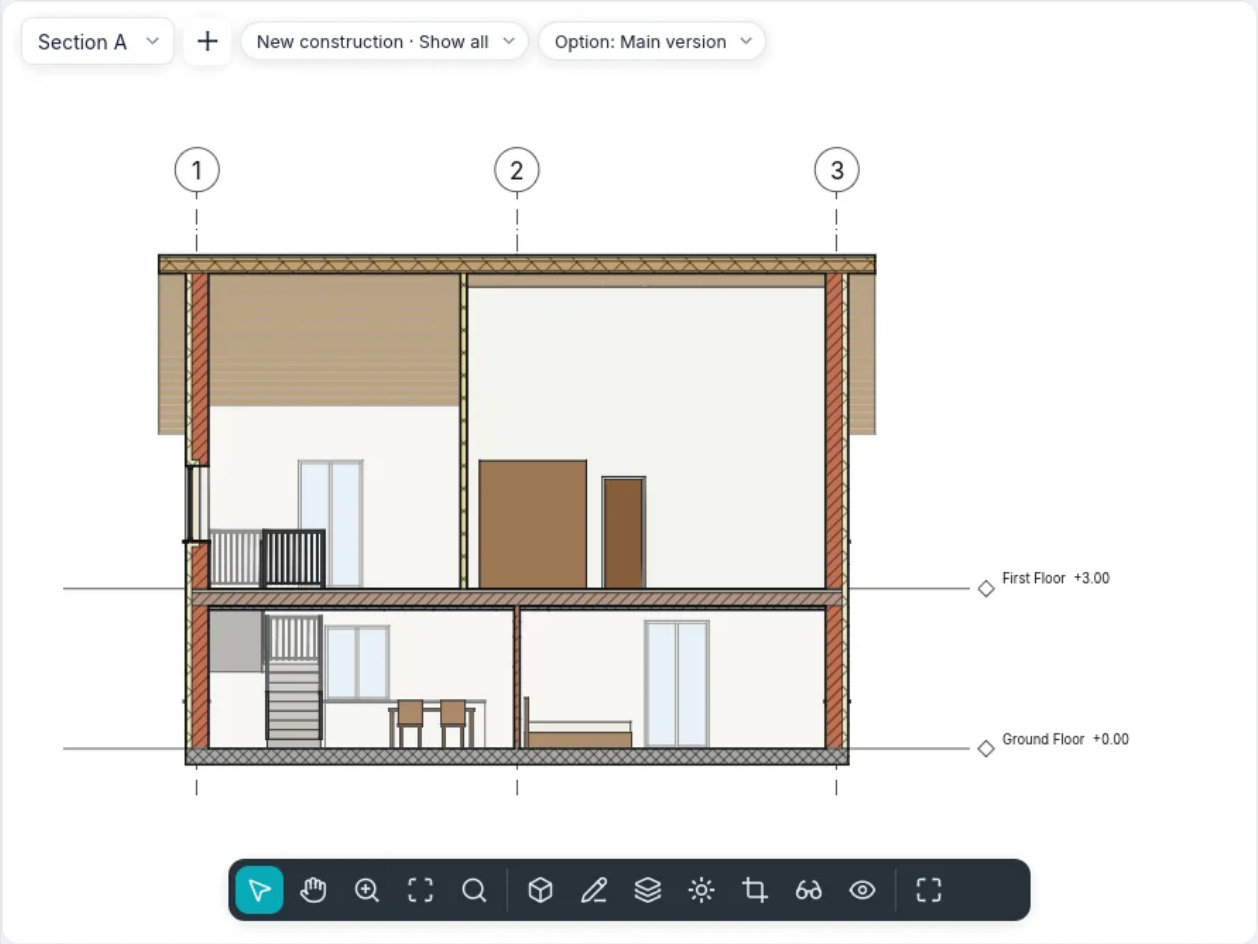

Depth Cueing: start 0%, end 25%, limit 10%

Sketchy lines

The **Sketchy Lines** group of **Graphic Display Options...** draws the view's lines as by hand. **Enable Sketchy Lines** turns it on for the active view; **Distortion** (0–10) sets how much each line wavers and how many times it is traced over (0 straight, twice from 4, three times from 8); long lines waver more than short ones. **Extension** (0–10) sets how far the lines run past their ends, past the corners where they meet (0 stops them there). They start at 5 and 1 in plans, sections, elevations and 2D views, at 8 and 3 in 3D views. The ✎ button of the view bar (in every interface style; in touch mode, in the view's sheet) turns them on and off for the active view: **Enable Sketchy Lines** / **Disable Sketchy Lines**.

They apply in every view (plans, sections, elevations, drafting views, 3D views, views on sheets) and in every visual style: to the lines of the model, of detail items and of fill patterns. Annotations stay straight: dimensions, text, levels, grids, reference planes, tags, symbols and view marks. A view draws the same lines at every redraw and after orbiting back, and prints and exports exactly as the window shows it: PDF and SVG keep the lines as vectors, a printed section or elevation leaves out only the parts nearer surfaces hide. Selecting and snapping still use the straight lines. View templates carry the setting with the other graphic display options. On the command line: `graphicoptions {"sketchy":true,"distortion":5,"extension":1}`.

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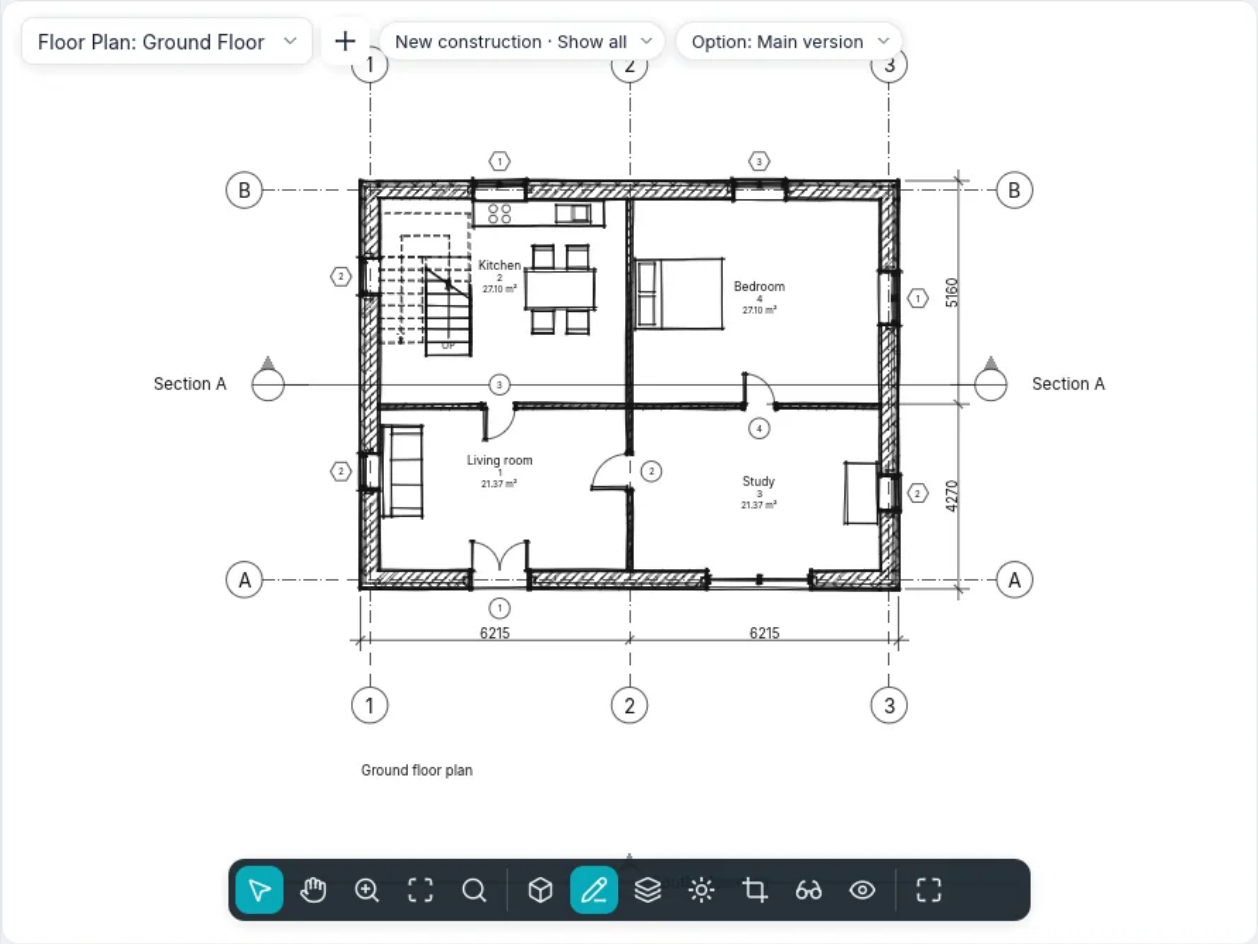

Sketchy Lines: distortion 5, extension 1 (the defaults)

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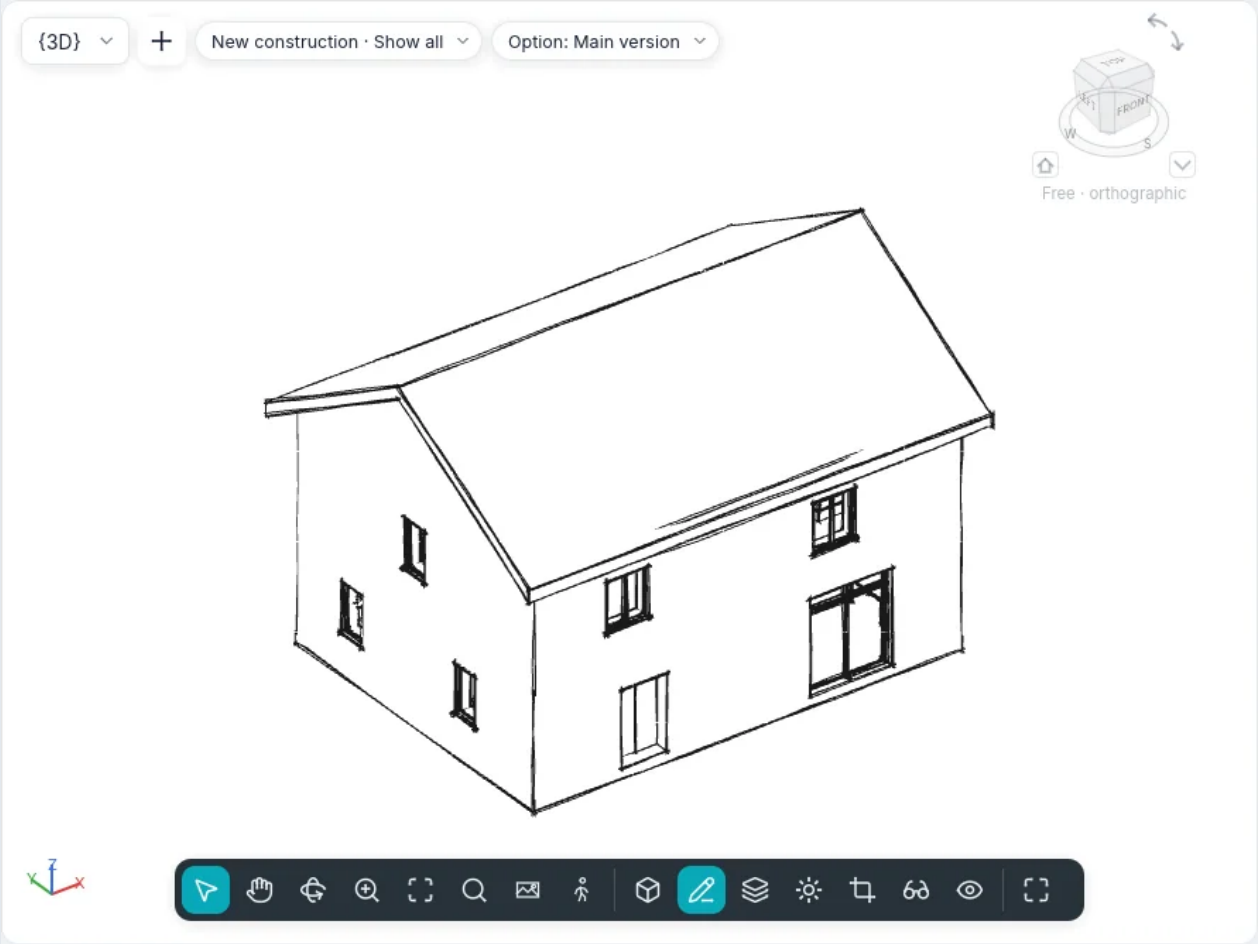

Sketchy lines in a 3D view: distortion 8, extension 3 (the defaults in 3D)

Plans and their cut heights

Views ▸ Create ▸ Plan ▸ Floor Plan opens **New Plan**: pick the levels and the scale, then **OK**. With **Do not duplicate existing views** on, the levels that already have a plan are left out; turn it off to make a second plan of a level.

A plan is a horizontal cut of the model at the **Cut Plane** of its cut heights. Walls crossing the cut plane are drawn cut, with the cut patterns of their layers' materials at **Medium** and **Fine** detail; lower walls and the floors of the level are seen from above.

To change the cut heights, select nothing (so that **Properties** shows the view), then **Cut Heights ▸ Edit...**. Each plane is measured from a level, with an offset:

Plane	Level
Top	The plan's own level (**Associated Level**), the **Level Above**, a level by name, or **Unlimited**
Cut Plane	Always the plan's own level
Bottom	The plan's own level, the **Level Below** or a level by name
View Depth	As **Bottom**, or **Unlimited**

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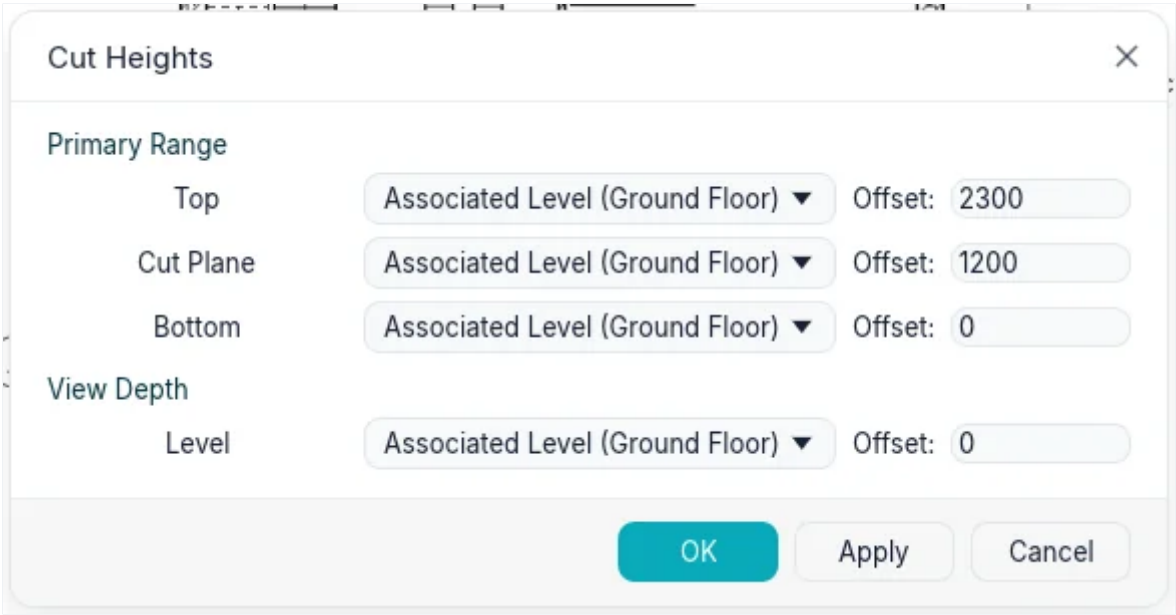

Cut Heights

OK sets the four planes together, as one change. They must stay in order, from the top down (top, cut plane, bottom, view depth): otherwise the window explains what is wrong and stays open. Floors below the bottom, down to the view depth, are drawn *beyond* in thin lines; so are floors a little under the bottom, such as a sunken slab, even with the view depth at the bottom.

Underlay

A plan can show another range of the model under it, lighter: the floor below to trace over, or the floor above to align with. In **Properties** (nothing selected), group **Underlay**:

- **Range: Base Level** turns it on: pick a level (**None** turns it off). **Range: Top Level** fills in with the level above the base; pick another level above it, or **Unlimited**, to show more of the model.
- **Underlay Orientation**: **Look down** shows it from above, as a plan; **Look up** from below, as a ceiling plan.

Only the underlay's lines are drawn, under the plan, and what the view hides stays hidden. While **Select underlay elements** is on (in the status bar) a click or a window selects them too, and they can be edited from here; off, a click passes through them. **Project ▸ Additional settings ▸ Halftone/Underlay** sets, for the whole project, the underlay's **Thickness** and **Pattern**, **Apply halftone**, and the **Brightness** of every halftone (also of **Display Model ▸ Halftone**).

Other properties of the views:

- **Orientation**: **True North** turns a plan on screen so that the project's true north is up; drawing and editing work as usual.
- **View Scale ▸ Custom** lets **Scale Value 1:** be typed.
- **Depth Clipping** (plans) and **Far Clipping** (sections and elevations): **No clip**, **Clip without line**, **Clip with line**; **Far Clip Offset** types a section's depth.
- **View template**: a template assigned to the view; while it is, the properties it includes follow it and are set in the template.
- **Show In** (sections): **Intersecting Views** shows the section in every plan, **Primary View Only** only in the plans of the level it was drawn in.
- **Associated Level** and **Dependency** are read-only.

Light and dark background

Options ▸ Theme ▸ Dark (or **Invert background color**) turns the drawing area dark. The background changes one thing only: black lines (edges, text, patterns) are drawn light, so they stay legible (text over the rooms' colours stays dark, as on the light background); in the line styles alone (**Wireframe** and **Lines**) the paper and the black lines are inverted together, in plans, sections, elevations and 3D. In the styles that show the model's colours (**Shaded**, **Flat Colors**, **Realistic**, **Pastel**) faces, materials, fills, washes, shadows, poché, transparency and depth cueing are identical on any background, and **Pastel** always draws on its light paper. **Options ▸ Interface size** makes the whole interface bigger, from 100 % to 250 %, on top of the screen's own scale: useful on screens with many pixels. It is kept for the next start.

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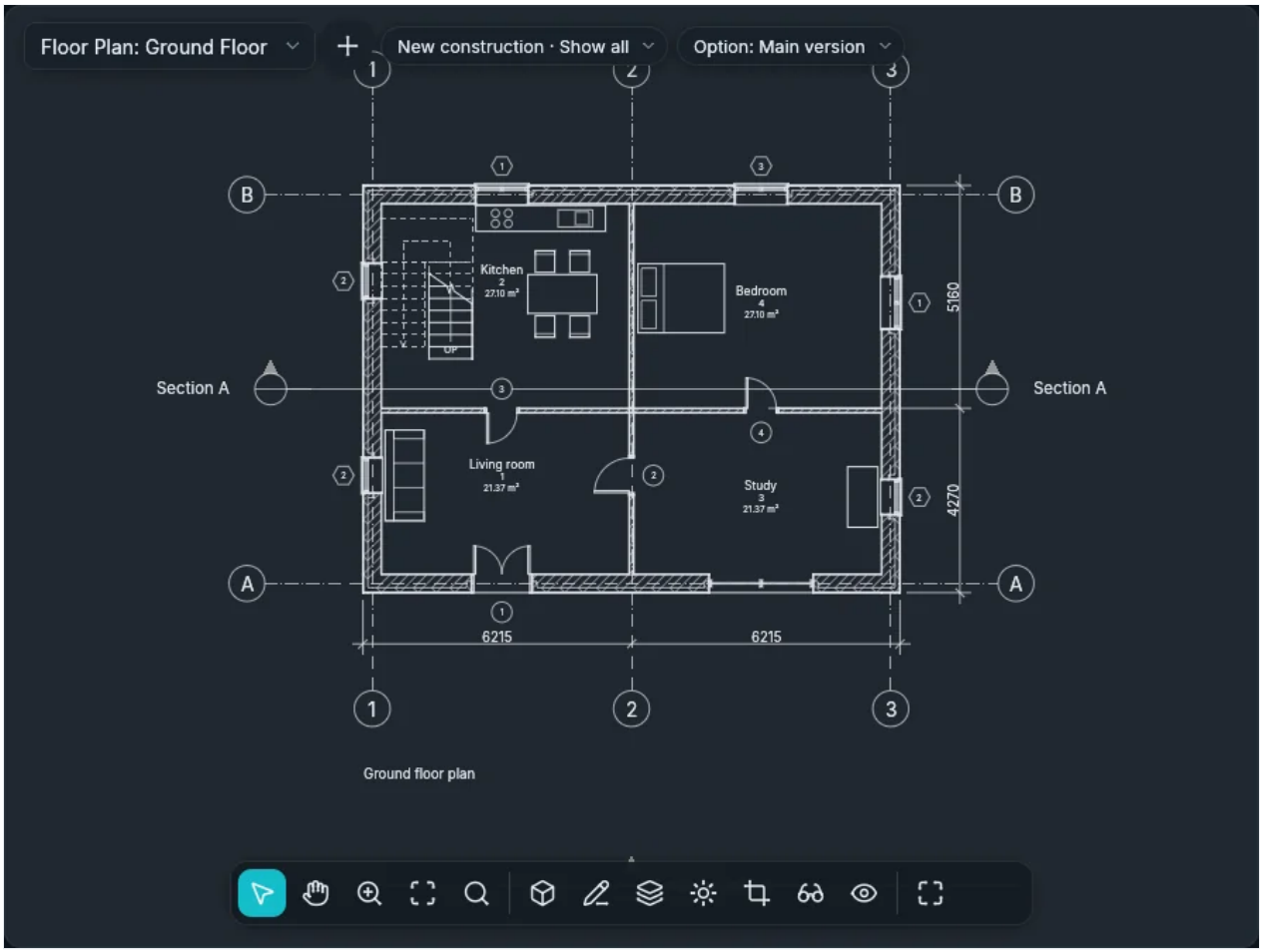

A plan on the dark background

Opening a view from its symbol

Double-click a view's symbol to open its view: a section line or its head, an elevation symbol, a callout's boundary (a reference callout opens the view it points at), a level's line in a section or elevation (its floor plan opens), or a view placed on a sheet. The view's window becomes the active one.

Views ▸ Create ▸ Elevation ▾ ▸ Framing Elevation makes an elevation of a grid: click beside the grid, on the side to look from (before the click, a line and an arrow show the view). The elevation looks at the nearest grid, as wide as the bay between the grids that cross it and as deep as the next grid parallel to it.

The heads of sections, elevation symbols and callouts show the view's detail number over the number of the sheet it is placed on; their circles stay empty while the view is on no sheet. A new project comes with four elevations, **North**, **South**, **East** and **West**, looking at the area round the origin: move them where the building is.

Scale, detail and crop region

The view control bar shows the view's **View Scale** and **Detail** (in 3D views too).

- At a smaller scale the model is drawn smaller on paper while text, tags and symbols keep their size on paper: at 1:200 a text covers twice as much of the model as at 1:100.
- At **Coarse** a cut wall is one fill; at **Medium** and **Fine** its layers show, each with its material.

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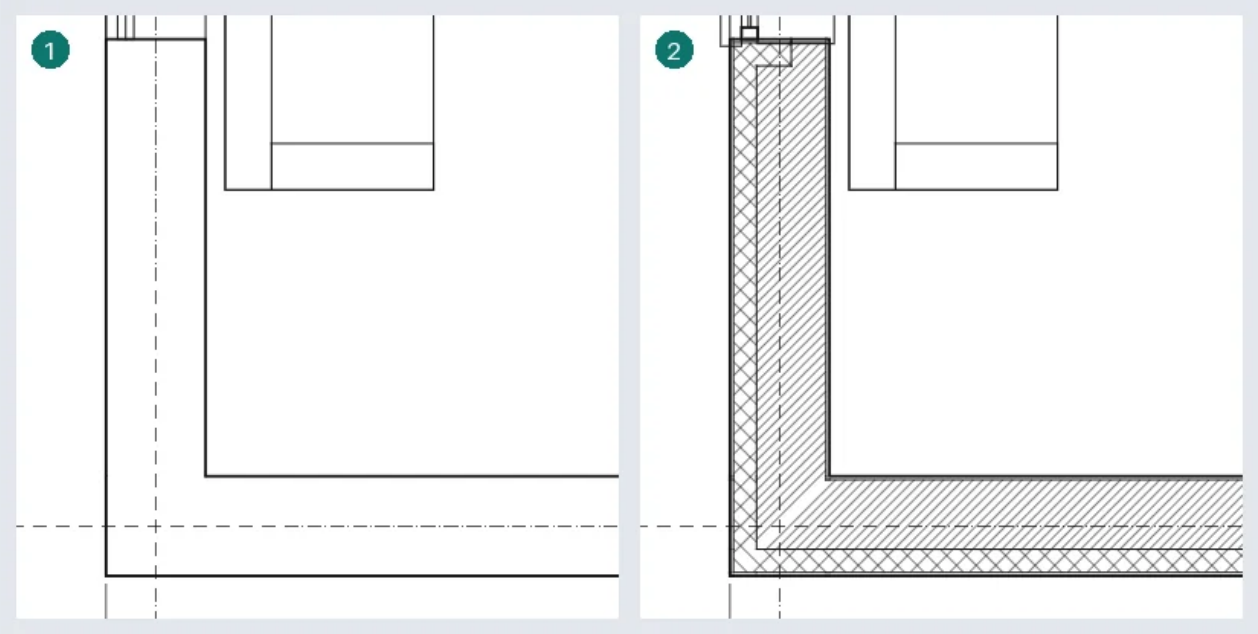

The same corner at Coarse and at Fine detail

Sections cut the model the same way. Walls, floors and roofs that meet in the cut are drawn as one: a thick outline round them, a thin line only where two materials meet, none between two parts of the same material; walls attached to a roof stop under it. Each cut layer has its material's cut pattern (at **Coarse**, the type's **Coarse Scale Fill Pattern** in its **Coarse Scale Fill Color**, set on wall, floor and roof types, else only its outline). Furniture and plumbing fixtures are never cut: they are seen whole. A window is cut with its frame in the thick cut line and its glass in thin lines; what lies beyond the cut is drawn in thinner projection lines. Beyond the cut, and in elevations, a face in front hides what is behind it, and a window's glass is opaque: each window shows its frame and glass, not the rooms behind it.

A section: what is cut, and what is seen beyond

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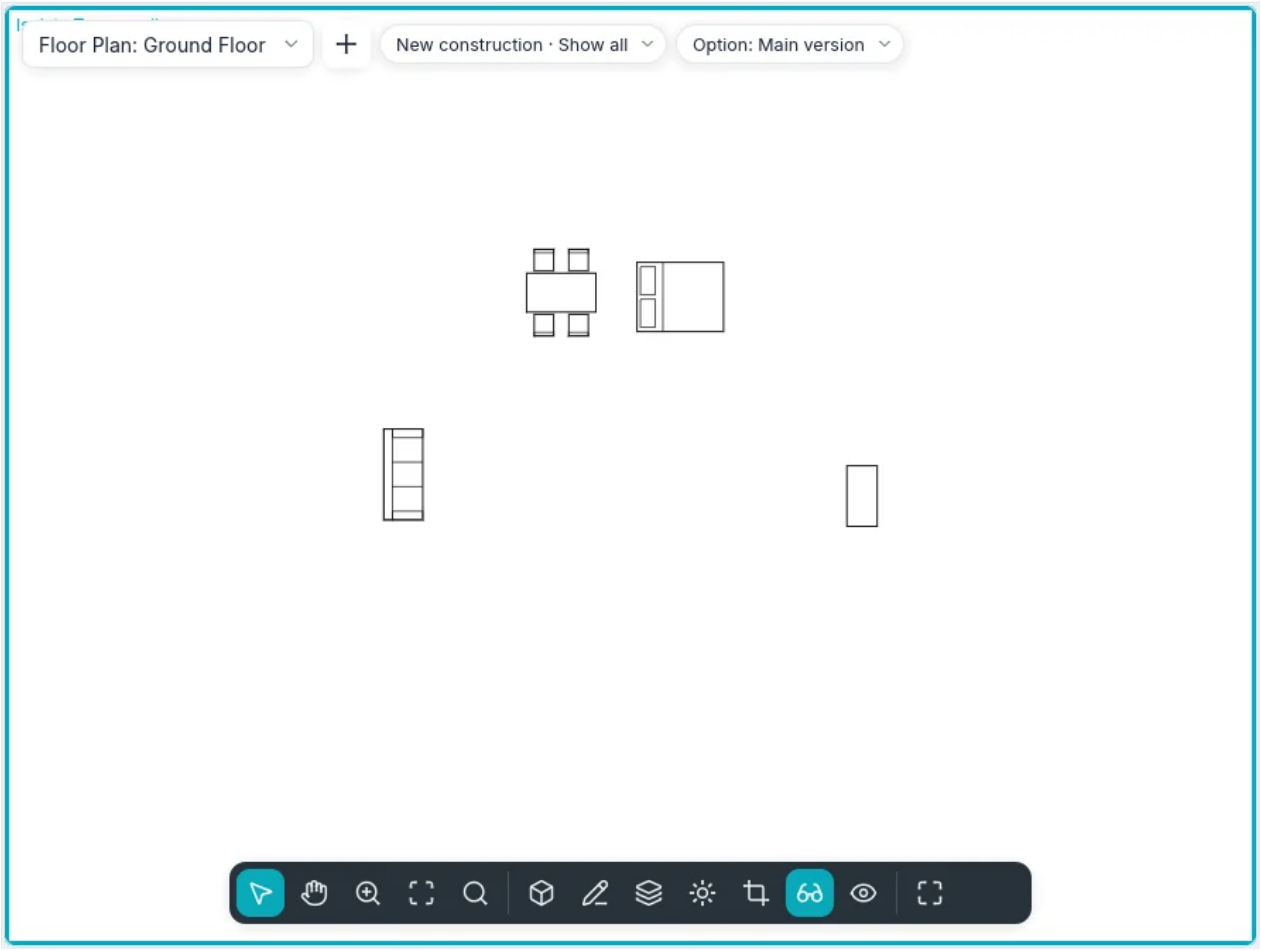
Crop View clips the view to its crop region, **Show Crop Region** draws the region; drag its edges to resize it. To type its size: **Properties** ▸ **Size Crop** ▸ **Edit…**, then **Width** and **Height**; the region keeps its centre.

Hiding and isolating

Isolate Temporarily (the glasses on the view control bar) acts on the selected elements, in the active view only, and does not change the project:

Command	Shortcut	What stays visible
Isolate Element	`HI`	Only the selected elements
Hide Element	`HH`	Everything but the selected elements
Isolate Category	`IC`	Only the categories of the selected elements
Hide Category	`HC`	Everything but those categories



Isolate Category: only the furniture, a coloured border

The view gets a coloured border. Each use narrows what is shown: hiding one element and then another hides both. **Reset Isolate Temporarily** (`HR`) shows everything again; **Apply Hide/Isolate to View** hides for good what was hidden for now.

To hide for good in the view, select the elements, then right-click ▸ **Hide in view** ▸ **Elements** or **Category**, or the contextual card's **View** ▸ **Hide in View**, or type `EH` (elements) or `VH` (category).

Reveal Hidden (the light bulb on the view control bar, or `RH`) frames the view in magenta and shows what is hidden in magenta, the rest in halftone. A hidden element can then be selected and shown again: right-click ▸ **Unhide in View** ▸ **Elements** (or **Category**). Click the light bulb again, or type `RH`, to leave.

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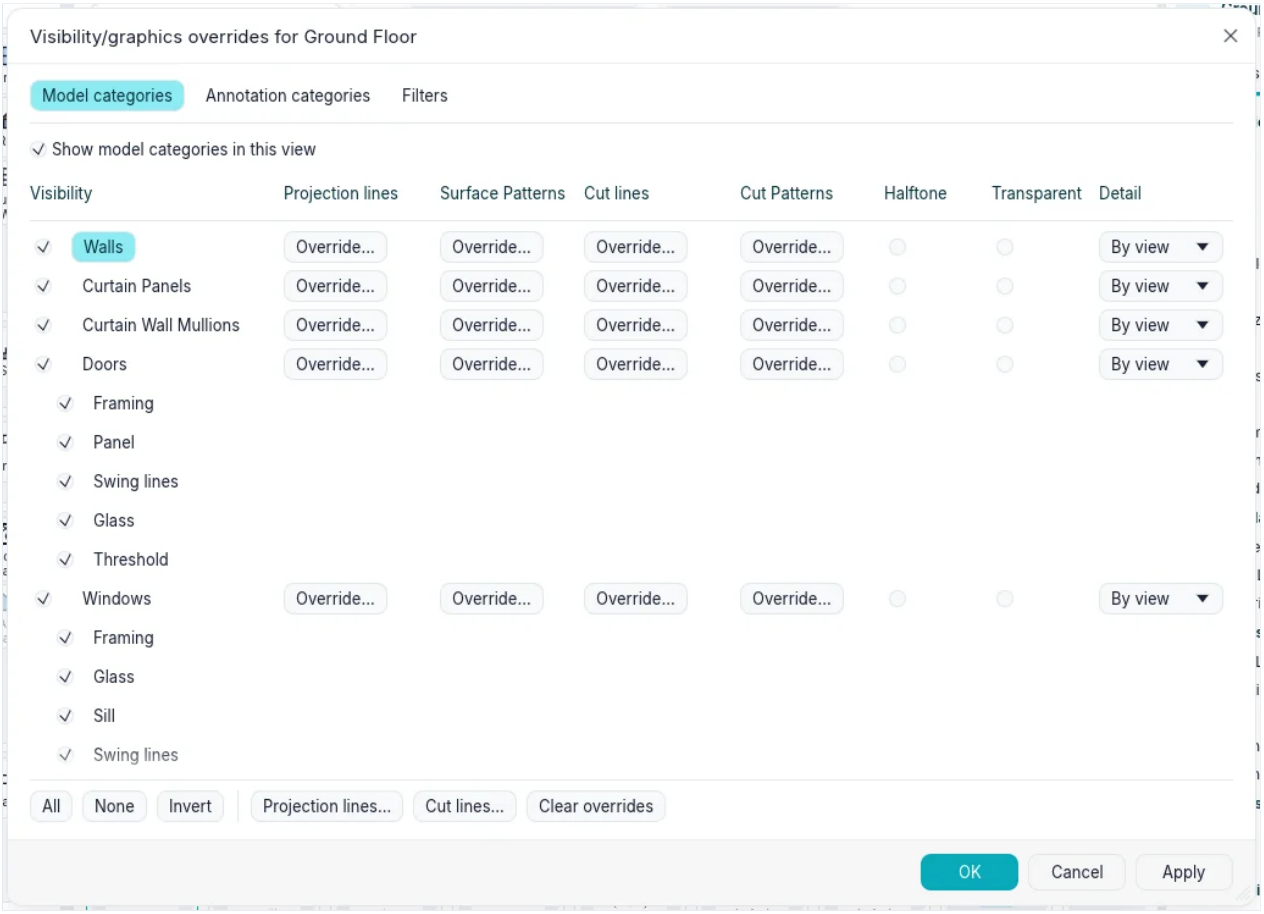
Visibility and Graphics

Views ▸ Graphics ▸ Visibility and Graphics (or `VG`, or **Properties ▸ Visibility/graphics overrides ▸ Edit...**) shows, for the active view, one row per category:

- the check box shows or hides the category; **Show model categories in this view** shows or hides them all;
- **Projection lines** and **Cut lines**: **Override...** sets the weight, colour and pattern;
- **Surface Patterns** and **Cut Patterns**: the material's patterns, hidden, in another colour or replaced by another fill pattern;
- **Halftone** blends the lines with the background;
- **Transparent** makes the surfaces see-through in the 3D view;
- **Detail** draws the category at another level than the view.



Visibility and Graphics

Clear overrides takes the selected rows back to the styles by category. Changes show at once: **Apply** keeps them and the window, **Cancel** (or the window's ×) takes back what was not applied, **OK** keeps them and closes.

One element can have its own graphics in a view: select it, then right-click ▸ **Custom Graphics ▸ By Element...** (or the contextual card's **View** group). **View-Specific Element Graphics** has **Visible**, **Halftone**, **Transparent**, the surface and cut patterns and the projection and cut lines. They apply only in this view, over the category's and the filters'.

Filters

Views ▸ Graphics ▸ Filters lists the project's filters: **New**, **Duplicate**, **Rename**, **Delete**. A filter has categories and up to four rules on their parameters, all of which an element must meet.

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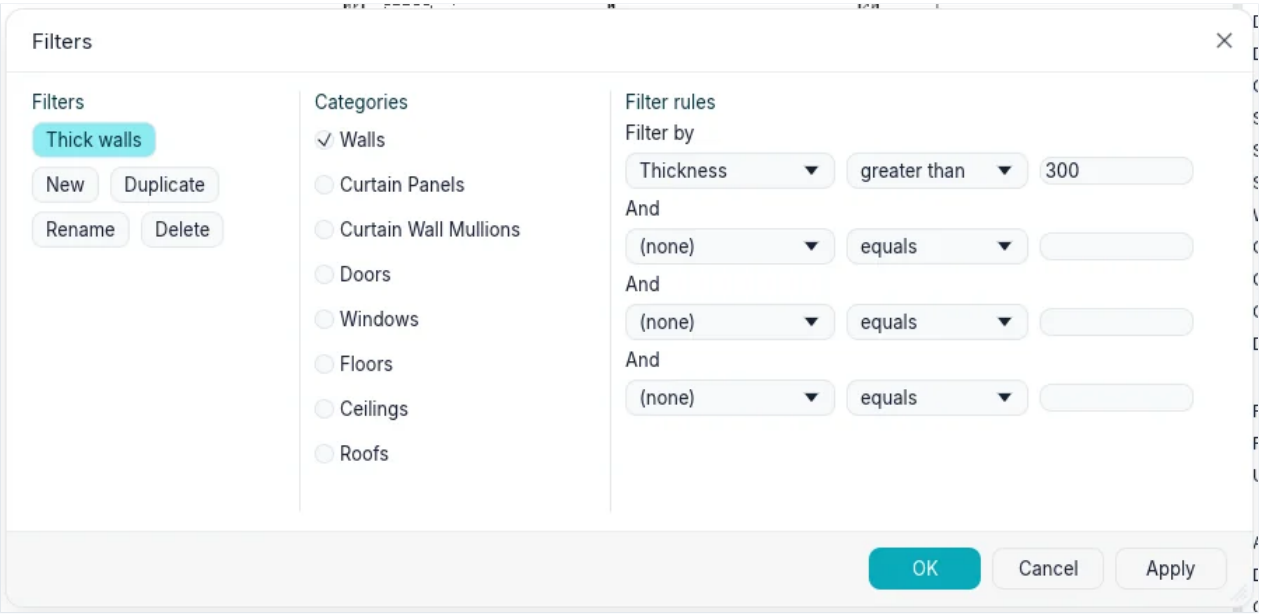

A filter: categories and rules

In **Visibility and Graphics ▸ Filters**, **Add** applies a filter to the view; its row then sets visibility, lines, **Halftone** and **Transparent** for the elements that pass it. Where filters disagree, the one higher in the list wins: **Up** and **Down** move the selected filter.

View templates

Views ▸ Graphics ▸ View templates ▸ Create template from current view keeps the view's scale, detail, graphic style, cut heights and graphics as a template, by name. **View templates ▸ Apply template to current view** gives them to another view of the same kind: a plan (a second plan of a level too), a section or elevation, or a 3D view. A view does not follow later changes of its template until it is applied again, and the graphics of single elements are never part of a template.

A new project starts with two templates, **Architectural - Plan** and **Architectural - Section and Elevation**, applied to every new view of their kind: plans, ceiling plans, sections and elevations are drawn with lines only, the materials' cut patterns on what is cut and their surface patterns on what is seen, with no colour fills and no shadows. They carry the graphic style and the sun, so a new view keeps the scale, detail and view range it is made with; change them to change how new views start. Area plans keep their colour scheme.

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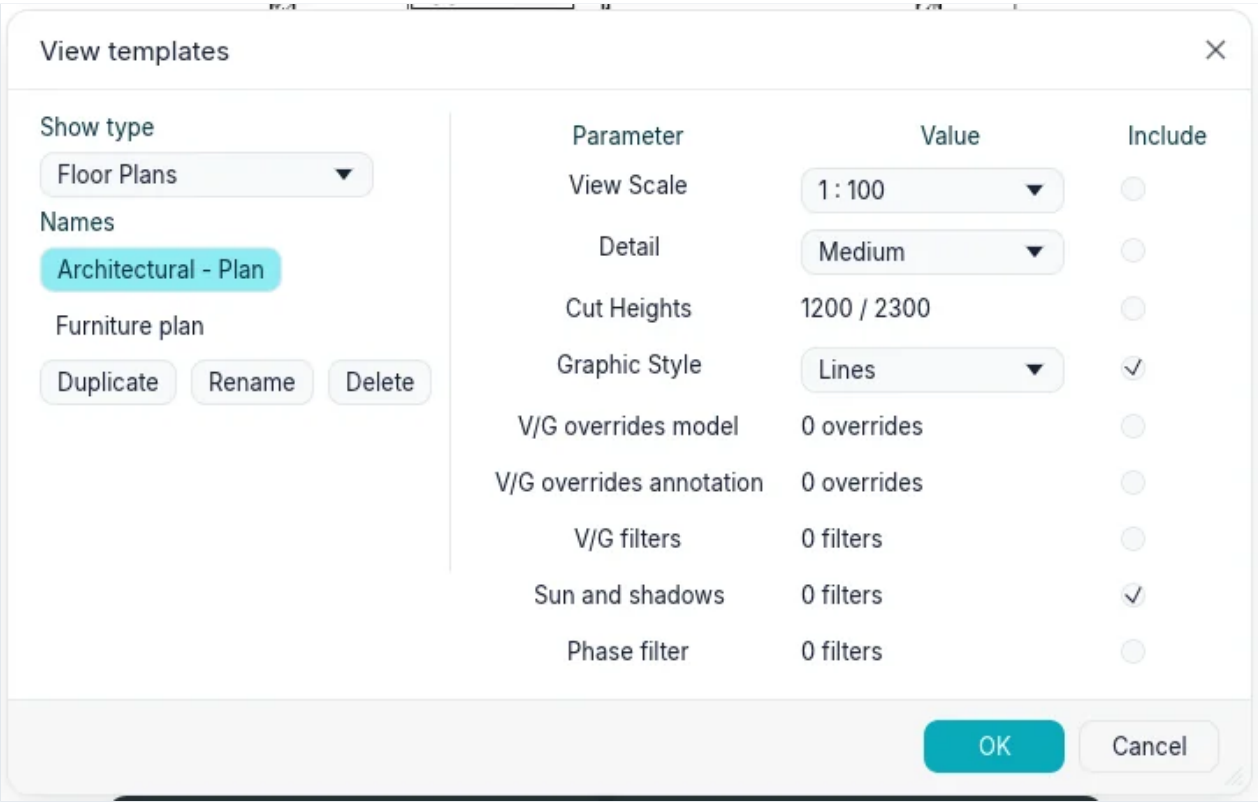

View templates

Managing views

Right-click a view in the **Project Index**:

- **Duplicate View ▸ Duplicate** copies the view with its properties; **Duplicate with Detailing** copies its dimensions, text, tags and 2D details too. The copy opens;
- **Rename...** asks for the new name (**Rename View**); names stay unique among views.

Views ▸ Windows ▸ Tile shows all the open views side by side, filling the drawing area: with three windows, the active one on the left at full height and two stacked on the right.

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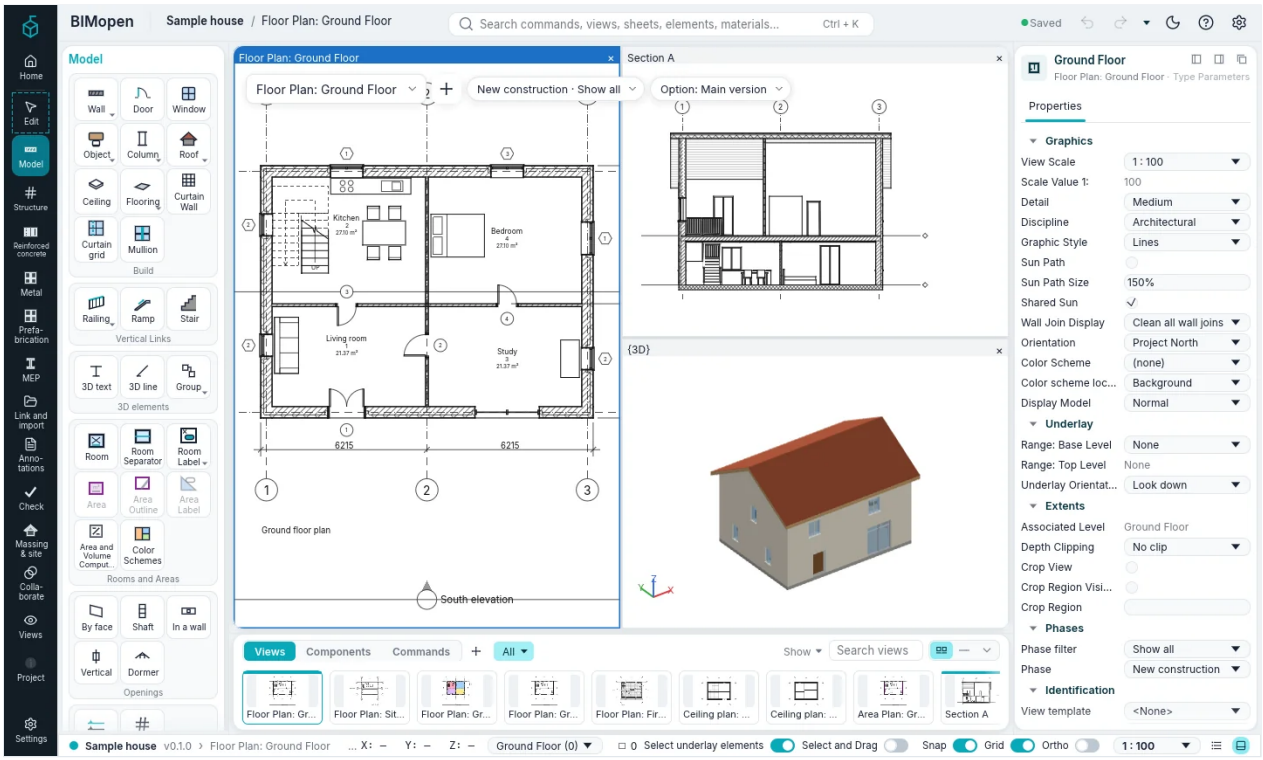

Tile: three open views

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Annotation and 2D details

This chapter describes dimensions and constraints, text notes, tags, rooms and areas with their colour schemes, legend views, and the 2D drawing tools: detail lines, regions, detail components, repeating details, insulation and free 2D views. Annotations belong to the view they are made in: a dimension placed in a plan does not appear in another plan.

Before each click, every tool shows what it would make where the cursor is: the dimension with its value, the note's box, the tag on the element under the cursor, the region a room or area would fill, the line, region, component or insulation being drawn.

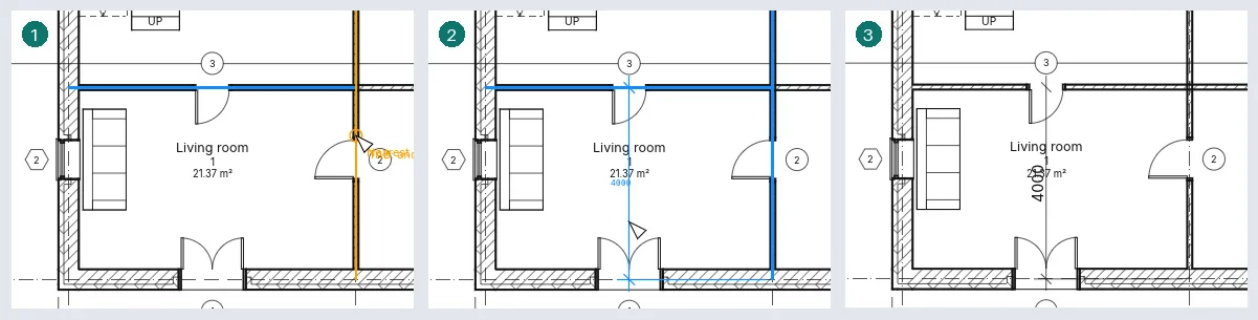
Dimensions

Annotations ▸ Dimension holds the dimension tools: **Aligned**, **Linear**, **Angular**, **Radial**, **Diameter**, **Arc length**, and the spot dimensions **Spot Elevation**, **Spot Coordinate** and **Spot Slope**. They work in plans, sections, elevations, free 2D views and legends. Pick the dimension type in the type selector while the tool is on; **Type Parameters** changes the type, and with it every dimension of that type.

Aligned and linear dimensions

With **Aligned**, the options bar offers which part of a wall is picked (**Wall centerlines**, **Wall faces**, **Center of Core**, **Faces of Core**) and what a click picks:

- **References**: click references one after the other (wall faces or centre lines, grids, door and window edges, detail lines, the edges of legend symbols), then click an empty spot to place the dimension line there. While you choose, the string follows the cursor. Press **TAB** over a wall to step through its references.
- **Entire Walls**: one click dimensions a whole wall. **Options** opens the automatic dimensioning options: **Openings** with their **Centers** or both **Widths**, **Intersecting Walls** and **Intersecting Grids**; openings by their widths and intersecting walls are ticked to begin with. Hover a wall: it lights up with every reference the click will add. Click it, click more walls to add them to the string, then click away to place the line. At each end the string reaches the wall joined there (its centre line or outer face, as the options bar's preference says). A curtain wall embedded in the wall counts as an opening. A grid that falls on a wall end or on another reference adds no witness line. It works the same in plans, sections and elevations.



Aligned: pick two references, then place the line

Linear measures along the horizontal or the vertical of the view, between points such as wall ends: where you place the line decides which, and the **space bar** switches between the two.

Angular measures the angle between two walls, grids or lines that meet; more lines through the same point make a string of angles, each with its value; the last click picks the sector.

Radial, diameter and arc length dimensions

Radial, **Diameter** and **Arc length** dimension a circle or an arc: a detail line's, or a curved wall's. Click it, then where the dimension goes. Over a curved wall, the options bar's **Preference** decides what comes first (its centre line, its faces, its core) and **TAB** steps to the others. A diameter dimension runs through the centre with an arrow on the curve at each end and **Ø** before the value; placed outside the curve, its line runs on to hold the value. On an arc shorter than half a circle it still spans the whole diameter. The dimensions follow the arc when its radius changes and go when it is deleted. In a section or an elevation only the arcs drawn there are offered.

A selected radial or diameter dimension has a grip at its end on the curve: drag it to turn the line round the centre; dropped on another arc or circle of the same centre, the dimension measures that one. Its value opens **Dimension Text**, and the grip under it moves the text, as for aligned dimensions. The values of angular and arc length dimensions open it and move in the same way.

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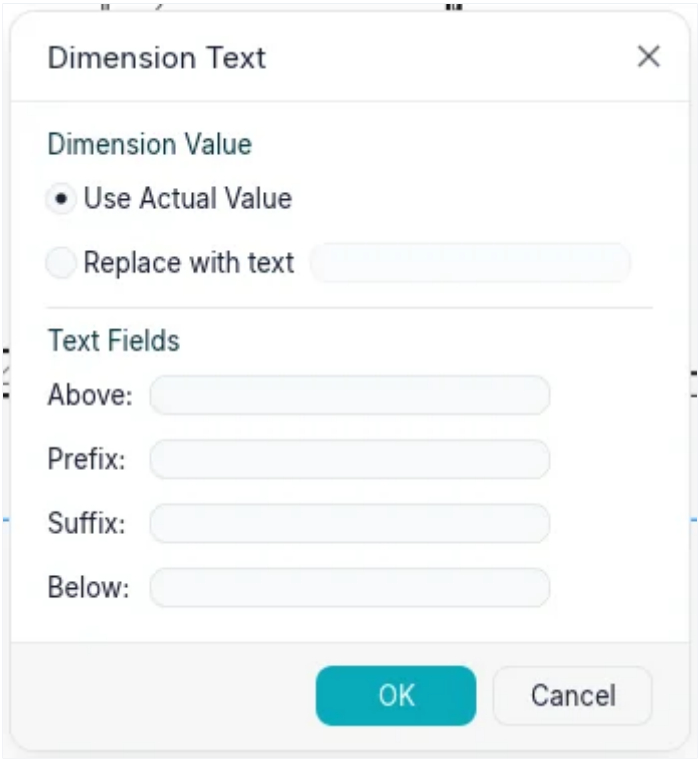
An ordinate type has **Ordinate Dimension Settings** in its type parameters: **Edit...** opens a dialog with the **Text Orientation** (**Aligned** with the witness lines or **Horizontal**), the **Text Location** (**In line with the witness line** or **Above the witness line**), the **Origin Visibility**, the **Origin Marker** (**Zero value**, **Circle** or the type's **Tick Mark**) and the **Dimension Line Style** (**Continuous** or **None**, which leaves only the witness lines and ticks).

Changing a dimension

- Select a dimension and click its value: **Dimension Text** opens. **Use Actual Value** keeps the measure and adds text above, below, before or after it; **Replace with text** shows words instead (a number is refused).
- Drag the blue control under the text to move it away from the line; dragged back near the line, it snaps into place. Text moved far away gets a leader to its segment (see the type's **Leader Type** below); **Leader** on the options bar removes it.
- **Selection: Dimensions ▸ Witness Lines ▸ Edit Witness Lines**: click elements to add witness lines, click an existing witness line to remove it, then click an empty spot to finish. Each witness line of a selected dimension has a control in its middle: right-click it ▸ **Move Witness Line**, then click the new reference; or ▸ **Delete Witness Lines**. The control at the element's end of a witness line drags its gap from the element; near another witness line's gap it takes the same.
- Select an element the dimension measures and click a value next to it: type a new distance and press Enter, and the element moves: a wall, a grid, a door or window (along its wall) or a detail line.



Dimension Text

The type sets the tick (diagonal, filled arrow or dot), the extensions of the dimension and witness lines (**Witness line control**: a gap to the element, or **Fixed to Dimension Line** with its **Witness Line Length**), the **Dimension Line Snap Distance** (while placing or dragging, the line snaps at multiples of it from what it measures, and one step from a parallel string so that strings stack evenly; a dashed line and a tip show the snap; 0 turns it off), the **Line Weight** and **Tick Mark Line Weight**, the **Centerline Symbol**, **Centerline Pattern** and **Centerline Tick Mark** (**Default** is the type's tick) of witness lines to centre lines, the colour, the text size, offset, background, bold, italic and underline, the units (**Units Format** opens **Format**: **Use project settings**, or a unit with its rounding, which may be **Custom** with a **Increment factor**, its symbol and options), and for linear dimensions the **Dimension String Type**: **Continuous**, **Baseline** (stacked values from the first reference) or **Ordinate** (values from the first reference along one line). **Show Opening Height** writes the height of a door or window under the width measured jamb to jamb. Radial and diameter types have their centre mark (**Center Marks**, with its size) and their symbol: **Radius Symbol Text** (**R**) or **Diameter Symbol Text** (**Ø**), and where it goes, **Before the value**, **After the value** or **Hidden** (**Radius Symbol Location**, **Diameter Symbol Location**). The leader of moved text is set by **Leader Type**: **Arc**, or **Line** with its **Shoulder Length** and **Text Location** (**Above**: the shoulder runs under the text; **In line**: it ends at the text's side); **Leader Tick Mark** marks its end on the dimension line, and **Show Leader When Text Moves** draws it as soon as the text moves (**Away from Origin**) or only past the witness lines (**Beyond Witness Lines**). With arrows, a segment too short for them has them outside, and the line runs on past them by **Flipped Dimension Line Extension**. **Annotations ▸ Dimension ▸ Dimension Types** opens the type parameters of each family's current type, to set before dimensioning.

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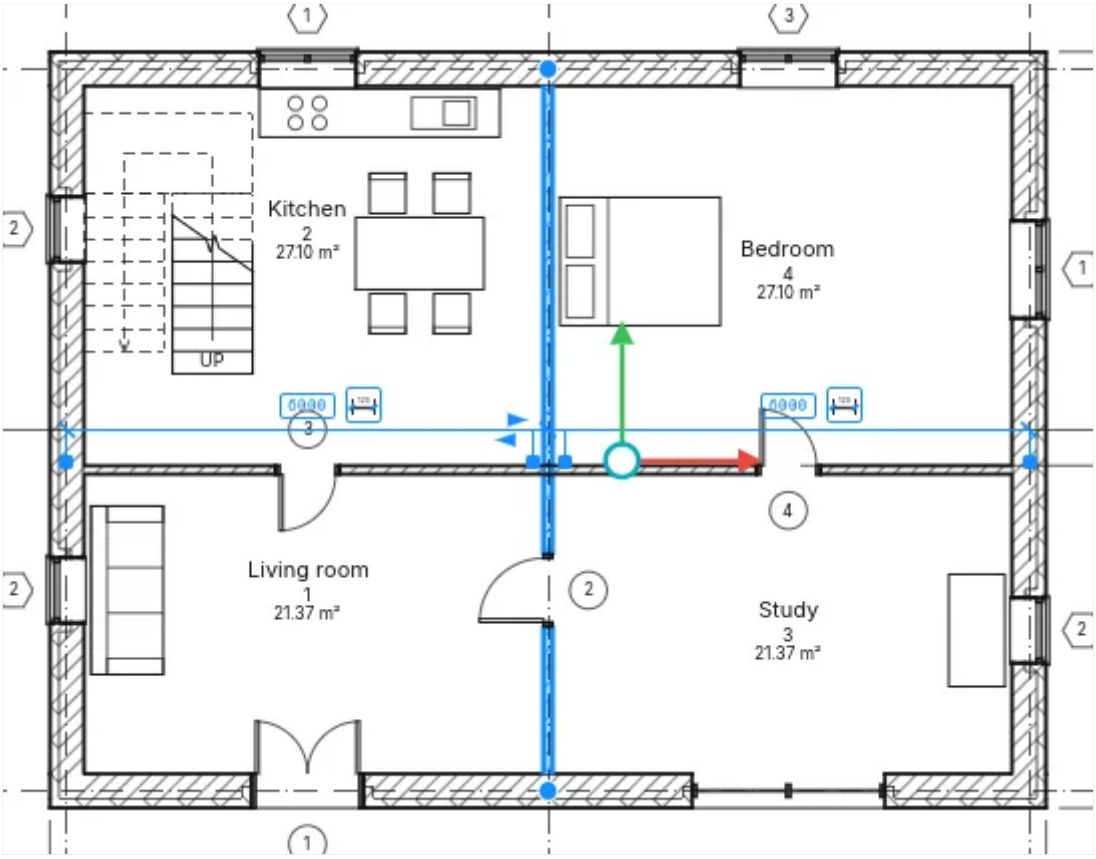
A selected baseline string shows a flip control at its origin: click it, or press the **space bar**, to measure from the other end. **Baseline Offset** in its properties sets the distance between its lines on paper.

Temporary dimensions

Select an element and temporary dimensions show its distances to the nearest references parallel to it: walls, grids, levels, reference planes, floor edges, room separation lines and detail lines. They show for every category in every view the element is drawn in: plans, sections, elevations and 3D views, where they lie on the element's level. A selected level shows its distances to the levels above and below it, in a section or an elevation.



Temporary dimensions of a selected wall

Click a value, type a new one and press Enter to move the element: a length takes a unit of its own (`2.5 m`), and a formula starts with `=` (`=8000/2`). Click the dimension symbol beside it to make it permanent.

Each witness line has a blue control. Drag the one on the reference onto any other reference to measure from there; or right-click it ▸ **Move Witness Line**, then click the new reference. A click on a control steps through the references of the same object: a wall's centerline, its faces, its core faces and its core centerline. The element remembers the references chosen, for each view, and they follow when the reference moves; right-click ▸ **Reset References** goes back to the nearest ones.

With several elements selected none are shown, and the options bar offers **Activate Dimensions** to show them for all the selected elements.

While a component, a door or a window is placed, temporary dimensions show its distances to the nearest references; **Tab** steps to the next ones. Rest the pointer for a moment: the element waits while the pointer goes to one of its values, and a click on the value types the distance it goes at. While drawing, the length and the angle are shown; **Rotate** shows the angle as an angular dimension, and an angle typed turns the selection.

Project ▸ Additional settings ▸ Temporary Dimensions sets where walls measure from (**Centerline**, **Faces**, **Center of Core**, **Core Faces**) and doors and windows (**Centerline** or **Openings**), for the whole project. The same dialog sets the text of every temporary dimension, for the application: from 8 to 20 pt (8 at first), on a transparent or an opaque background. Beside the settings a preview shows three walls measured as chosen; the **Preview** switch in the dialog's header hides it, the dialog keeping its size, and the choice is kept.

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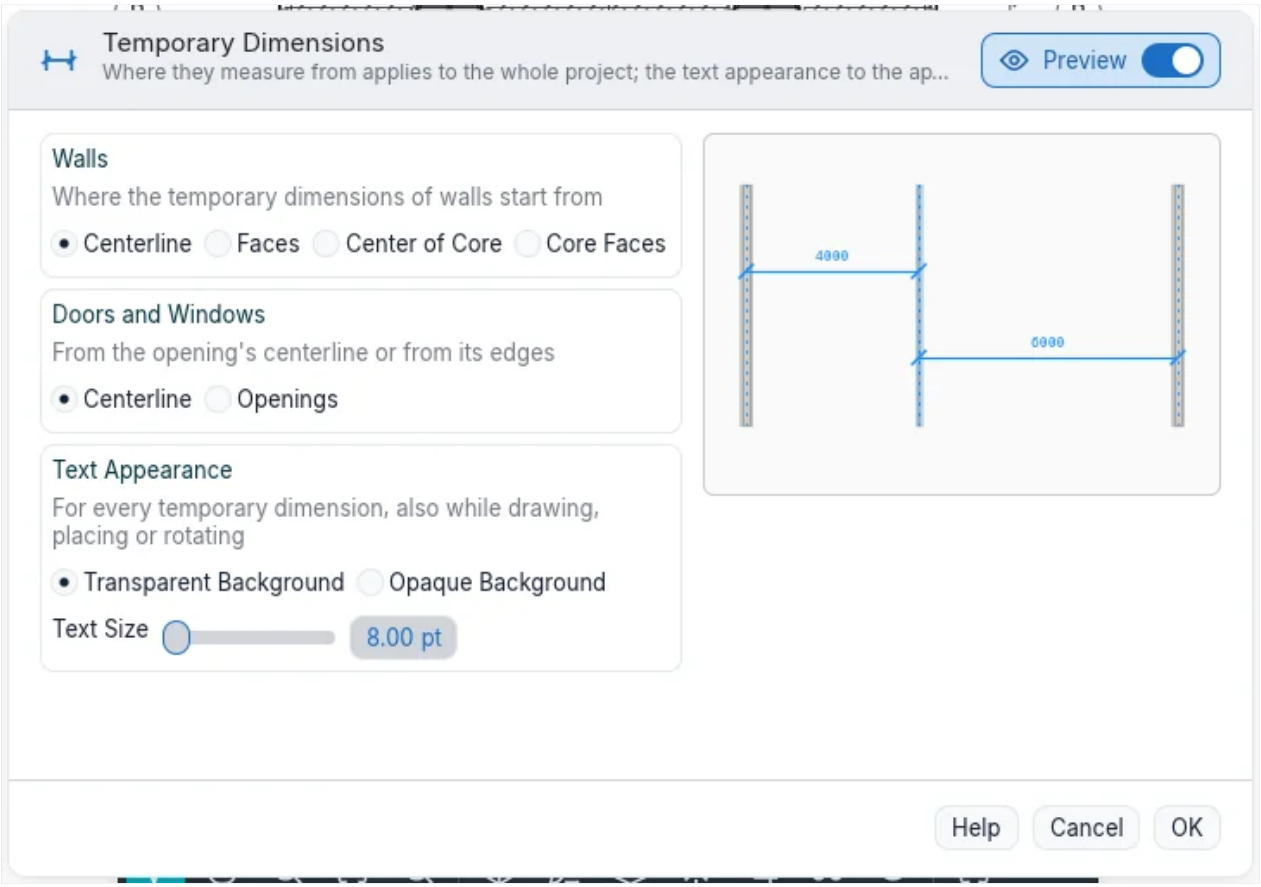

Temporary Dimensions settings

Constraints

Select a dimension: a padlock is shown on each segment. Click it to lock the segment: the distance then holds when either element moves, and a move that would break it is refused. Click it again, or right-click it ▸ **Toggle Lock**, to unlock; right-click ▸ **Show Related** selects the two elements the lock ties. A string of two segments or more shows **EQ**: click it and the references become equally spaced, and stay so when one of them moves. **Equality display** in the properties shows the values instead of **EQ**.

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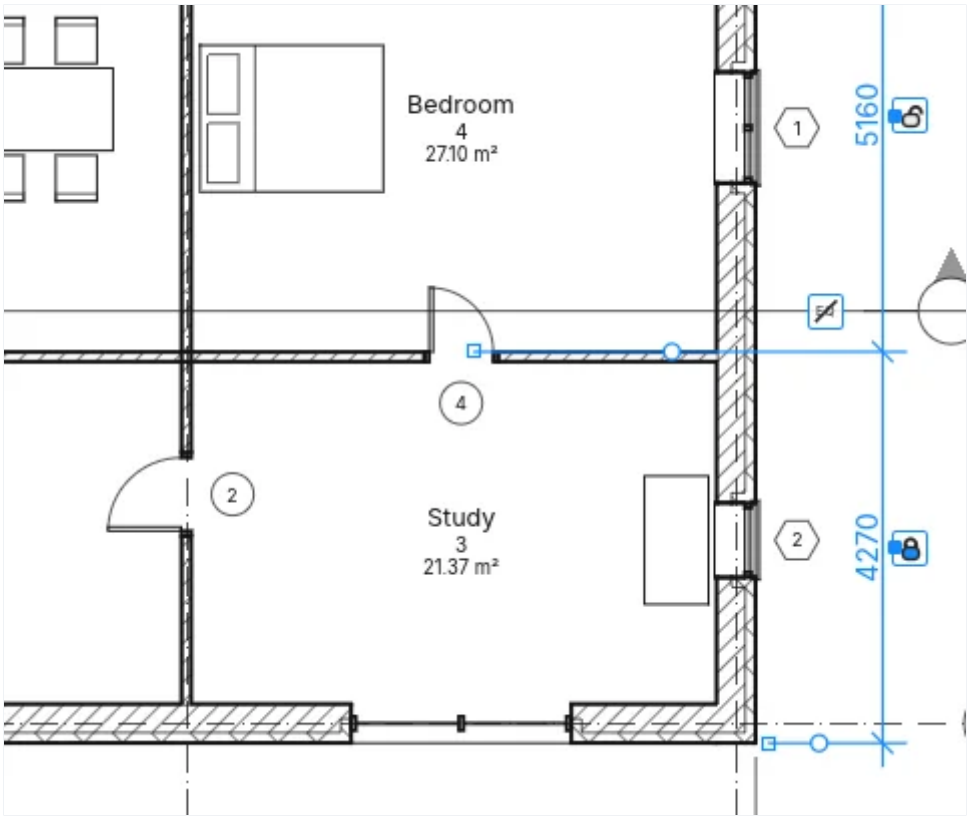

Padlocks and EQ on a selected dimension

Selecting an element a constraint holds shows the constraint as blue dashed lines, unless the category **Constraints** is hidden in **Visibility and Graphics**. Deleting a dimension that holds a constraint asks what to do: **OK** deletes the dimension and keeps the constraint; **Delete Elements** deletes both. **Modify ▸ Align** with its padlock closed is a constraint too.

Spot dimensions

- **Spot Elevation**: in a plan, click a floor, a roof or the level; in a section, any point. The value is in metres with its sign.
- **Spot Coordinate**: the north and east coordinates of a point in a plan.
- **Spot Slope**: the slope of a roof pitch in a plan, or of a sloped line in a section, with an arrow pointing down.

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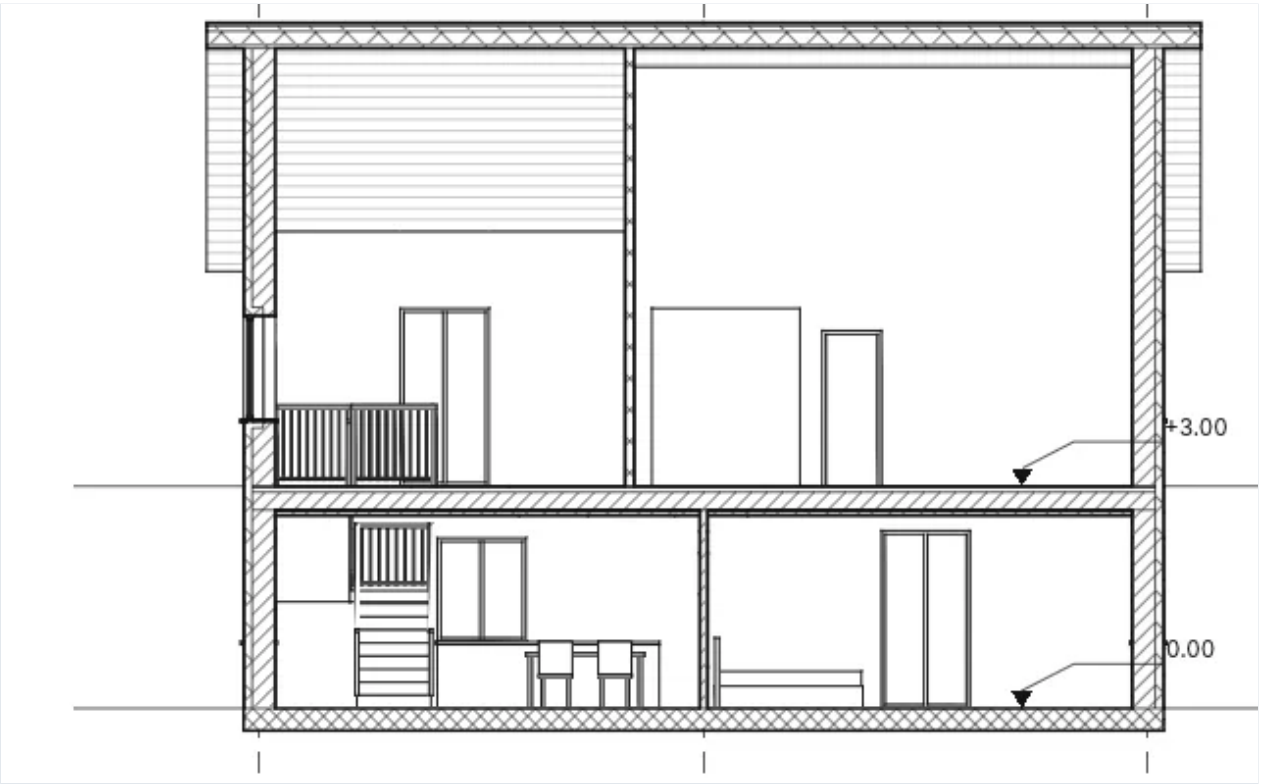

Spot elevations in a section

The options bar offers **Leader** and, with a leader, **Shoulder**: with no leader one click places the spot; with a leader, click the point and then the text; with a shoulder, the point, the bend and the text. While you hover, the value shows at the cursor. For spot elevations it also offers **Relative Base:** (for a type whose **Elevation Origin** is **Relative**) and, in plans, **Display Elevations:**: the actual elevation, the top, the bottom, or both. For spot slopes, **Slope Representation:** draws an **Arrow** or, in sections, a **Triangle**.

A selected spot has grips at its point, its shoulder and its text: drag the point and the spot measures what is there. A spot on a floor or a roof is deleted with it. While placing a spot, **Esc** lets go of the one half placed; a second **Esc** ends the tool.

The spot type sets **Elevation Indicator**, **Top Indicator** and **Bottom Indicator** (the top or bottom indicator, then the elevation's, then the value; each one's **... as Prefix/Suffix** puts it after the value instead), the **Symbol**, **Include Elevation** and the **North/South Indicator** and **East/West Indicator** of coordinates (with **Indicator as Prefix/Suffix**), and the **Leader Line Length** and **Slope Direction** (**Down** or **Up**) of slopes.

A spot elevation's **Properties** also give **Single/Upper Value Prefix** and **Single/Upper Value Suffix**, and, for the bottom value, **Lower Value Prefix** and **Lower Value Suffix**: words written before and after the value.

Arrows, fonts and alternate units

Project ▸ Additional settings ▸ Arrows opens **Type Parameters** of the project's arrow types. Each has an **Arrow Style** (Arrow, Diagonal, Dot, Heavy end tick, Elevation mark), **Filled**, **Opening angle**, **Mark size** on paper and, for the heavy tick, its own pen; what the style does not use is greyed. **Duplicate...** and **Rename...** work as for other types; the preview shows a note's leader and a dimension drawn with the arrow. Every leader arrow (text, tags, symbols, spot dimensions) and every tick of a dimension type lists **None** and these types, so changing a type changes every drawing that uses it. **Copy Settings from Project** copies them, **Purge unused** removes the ones nothing uses.

In a dimension type, **Text Font**, **Width Factor** and **Reading convention** (Aligned with the line, or Horizontal: always level, also on vertical dimensions) shape the value; **Alternate units** shows a second value To the right or Below, with its own **Format...**, prefix and suffix. On the desktop **Text Font** also lists the fonts of the computer; a project opened on a computer without that font draws it with BIMopen Sans and says so when it opens. In the web version only our two fonts are offered. A spot elevation type also chooses its **Symbol** (None, Triangle, Filled triangle, Crosshair) and its **Leader Arrowhead** with the weights of the leader and the arrow.

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Text

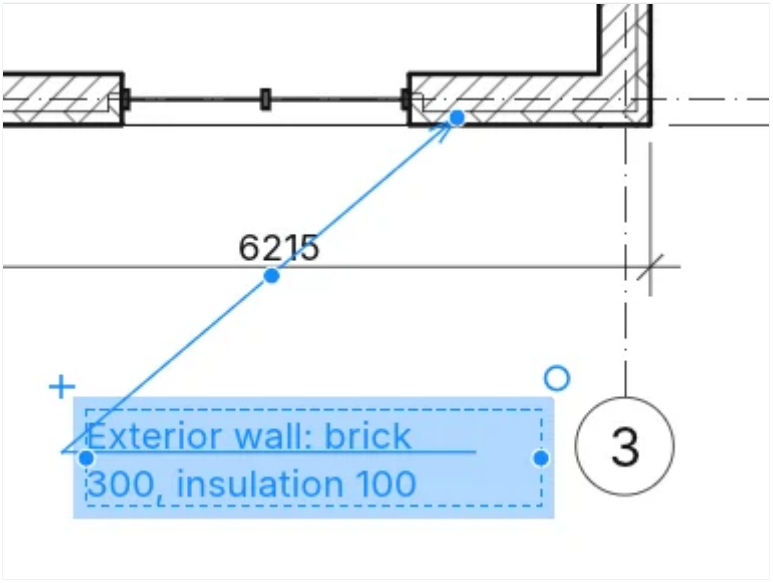
Annotations ▸ Text ▸ Text (`TX`): click where the note goes and type; Enter starts a new line. A click gives a note that does not wrap; dragging gives it a width, and the words wrap in it. Clicking elsewhere finishes the note, and **Esc** ends the tool. `Tab` while typing indents the paragraph. The text size is on paper: the note keeps its printed size at any view scale.

The **Format** panel sets, before placing or on a selected note:

- the leader: **No Leader**, **One Segment** (click the tip, then the note), **Two Segments** (tip, elbow, note) or **Curved**;
- the alignment, left, centre or right, and **Bold**, **Italic** and **Underline** (Ctrl+B, Ctrl+I, Ctrl+U on the selected characters while typing, or on the whole selected note);
- on a selected note, **Add Left Straight Leader** and the other three leader buttons, and **Remove Last Leader**;
- **Left Attachment** and **Right Attachment** (**Top**, **Middle**, **Bottom**): where the leaders meet the note; chosen while placing, they stay for the next notes.
- **Paragraph Format**: the paragraphs as a list, with **Bullets**, **Numbers**, **Lowercase letters** or **Uppercase letters**; the text of each item starts at the first tab stop, and wrapped lines are indented by one.

Near the line of another note's corner, a new note snaps to it, and a dashed line shows it.

On a selected note, drag the cross to move the box (the leader tips stay), the round side controls to change its width, and the rotation control to turn it. A grip in the middle of each leader makes an elbow when dragged, or bends a curved leader. Double-click inside it to edit the words. Its properties include **Arc Leaders**, **Left Attachment** and **Right Attachment**, and **Keep Readable**, which never shows it upside down. The text type sets the colour (of text, border and leaders), the background (**Opaque** masks what is behind), **Show Border** with its offset, the leader arrowhead, the **Line Weight** of the border and leaders, the size, the **Tab Size** (12.7 mm), bold, italic and underline. The small **Text Types** button of the **Text** panel opens **Type Parameters**, where **Duplicate...** makes a new text type.



A selected note with its leader and controls

Annotations ▸ Text ▸ Find/Replace finds words in the notes of the current selection, the current view or the whole project (**Scope:**), with options for case and whole words. **Find All** lists the matches with the view they are in; **Find Next**, or a click on a row, selects that note and shows its whole text under **Context**. **Replace** changes the match highlighted (the first when none is), **Replace All** every one, and **Close** closes the dialog. On selected notes it is also under **Tools** of their tab.

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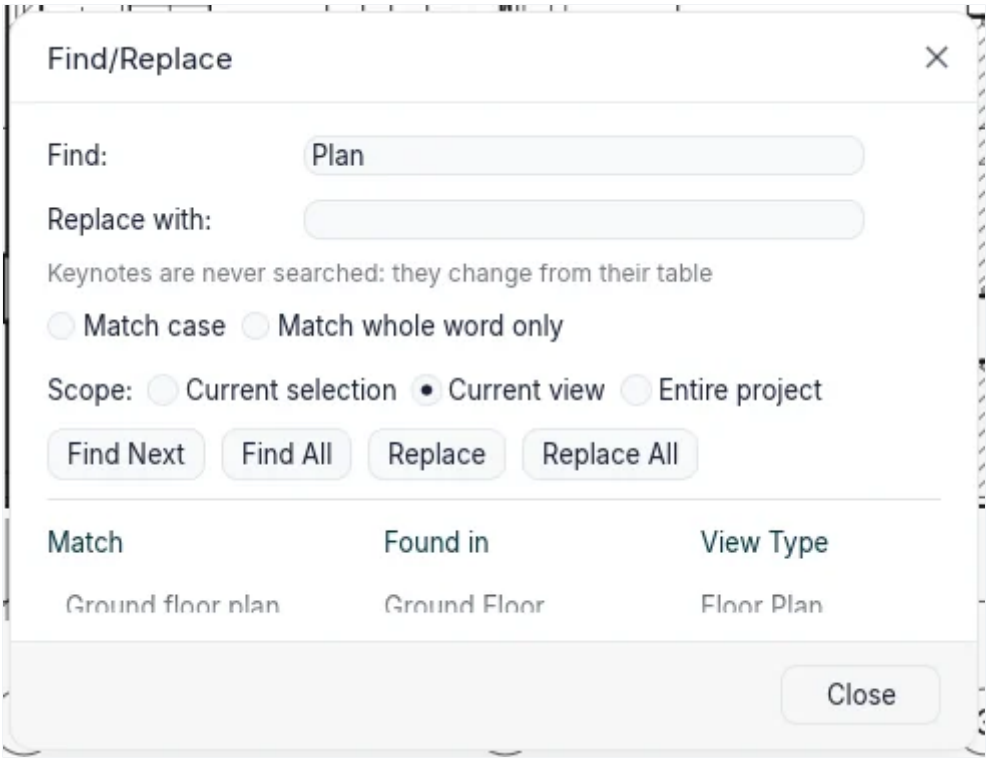

Find/Replace

Inside a group, **Find All** shows one row per group type, with a lock and the number of copies; those notes change from **Edit group**, where the scope becomes **Group being edited** and **Finish** carries the change to every copy.

Annotations ▸ Text ▸ Spell checker (`F7`) checks the selected text notes, otherwise those of the view or sheet (with the views placed on it); tags, dimensions, parameters and schedules stay out, and in 3D the command is grey. For each word not in the dictionary the dialog shows the note with the word highlighted, **Change to** and up to six **Suggestions**: **Ignore**, **Ignore all** (until the project closes), **Add to dictionary** (the **Personal** dictionary), **Change** and **Change all** (also in the notes still to check); a repeated word goes with **Delete**. The whole check is one undo step. Notes in groups are skipped and the final message lists the groups; in **Edit group** only the group's notes are checked. **Options ▸ Spelling** holds the five boxes (words in capitals, with numbers, in mixed case, web addresses and paths, repeated words), **Restore defaults**, the **Main dictionary** (Italian is built in, the other languages come with the language packs) and the **Personal** dictionary, a text file with one word per line, with **Edit...**.

The text type's **Text Font** is **BIMopen Sans** or the fixed-width **BIMopen Mono**, and its **Width Factor** draws the letters narrower (below 1) or wider, on screen, in prints and in exports.

Text format and leaders

While you type, **Bold**, **Italic** and **Underline** (or Ctrl+B, Ctrl+I, Ctrl+U) apply to the selected characters; with nothing selected they apply to what you type next. With the note selected but not being typed, they apply to the whole note. When the selection is partly formatted, the button shows it with a dashed border.

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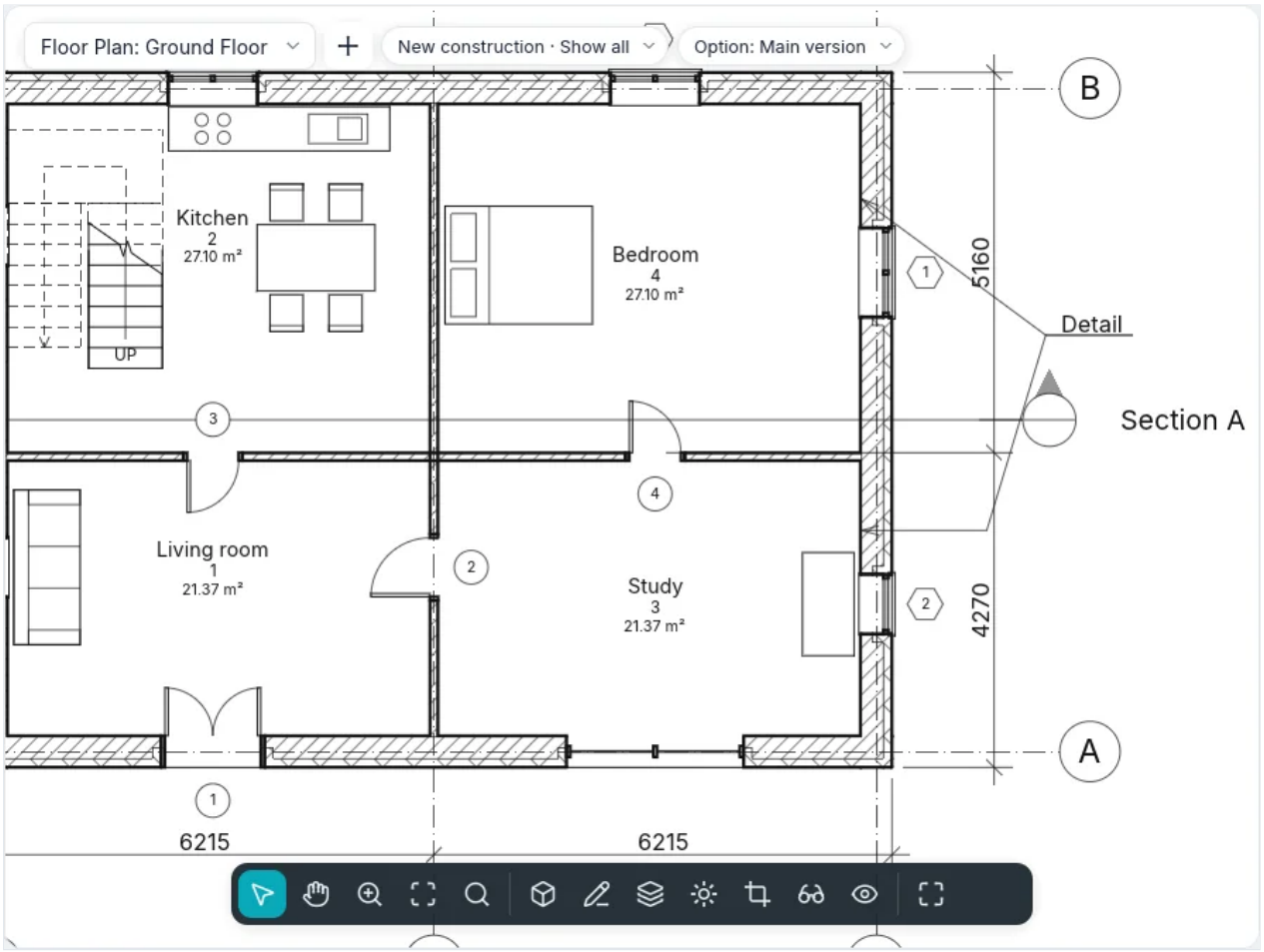

A text note with two leaders

Paragraph format applies to the paragraphs the selection touches (or the one with the cursor), and numbering starts again after a paragraph without a list. `Tab` indents the paragraph's text by one tab stop (the marker stays), `Shift+Tab` removes the indent. Pasted text takes the paragraph's format. While you type, **Paste**, **Cut** and **Copy** in the **Clipboard** group work on the text, at the cursor.

With a leader, near another note the leader's end snaps to its six attachment points (left and right, top, middle and bottom), with the mark and the hint **Note attachment**; `Tab` moves to the next point. Without a leader the note lines up with the origins of the other notes (left, centre or right by alignment) and with the texts of tags. With **Top** or **Bottom** attachment the leader goes on under the first or last line; with **Middle**, or with **Show border**, it stops at the border.

Clicking away, the note just typed stays selected with its controls and the tool stays on: the first **Esc** clears the selection, the second ends the tool. **Superscript** and **Subscript** shrink and raise or lower the chosen characters; **All caps** shows them in capitals without changing the text typed.

Symbols

Annotations ▸ Symbol ▸ Symbol (`SY`) places a 2D annotation symbol in a plan, a section, a free 2D view or a legend: a **North Arrow**, a **Centerline Symbol**, a **Flow Arrow**, a **Target** or a **Revision Delta**, the last two with a text of their own. The symbol follows the cursor before the click; the **space bar** turns it a quarter, or **Rotation** on the options bar sets its angle. With **Number of leaders:** above zero, the click places the symbol and the next clicks put the ends of its leaders. The type sets the symbol's size on paper and its leaders' arrowhead; **Duplicate...** in **Type Parameters** makes a new one.

Tags

Annotations ▸ Tag ▸ Tag by Category (`TG`): hover an element (a door, window, wall, room or area) and click to tag it. The options bar sets **Horizontal** or **Vertical**, the **Leader** and its length; the **space bar** switches the orientation, while placing or on a selected tag. Door and window tools have **Tag on Placement**, so each new door or window comes with its tag.

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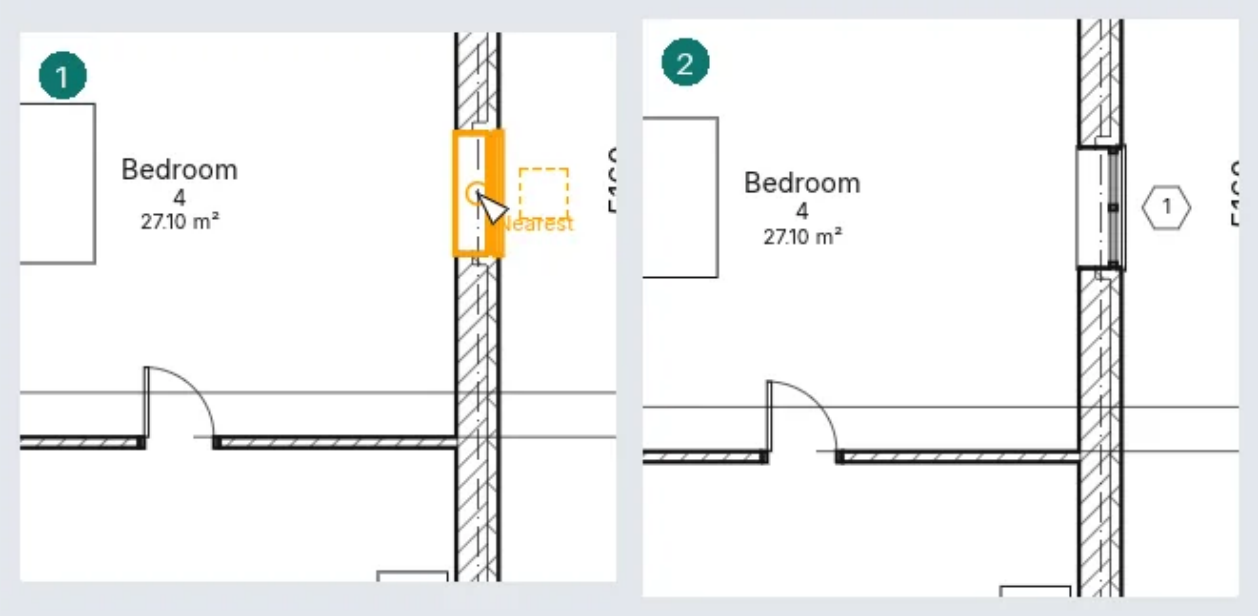

Tag by Category: the preview, then the tag

Tag All opens **Tag All Not Tagged**: tick the categories, the tag type of each, the leader and the orientation, for all the elements of the view or only the selected ones; only elements with no tag get one.

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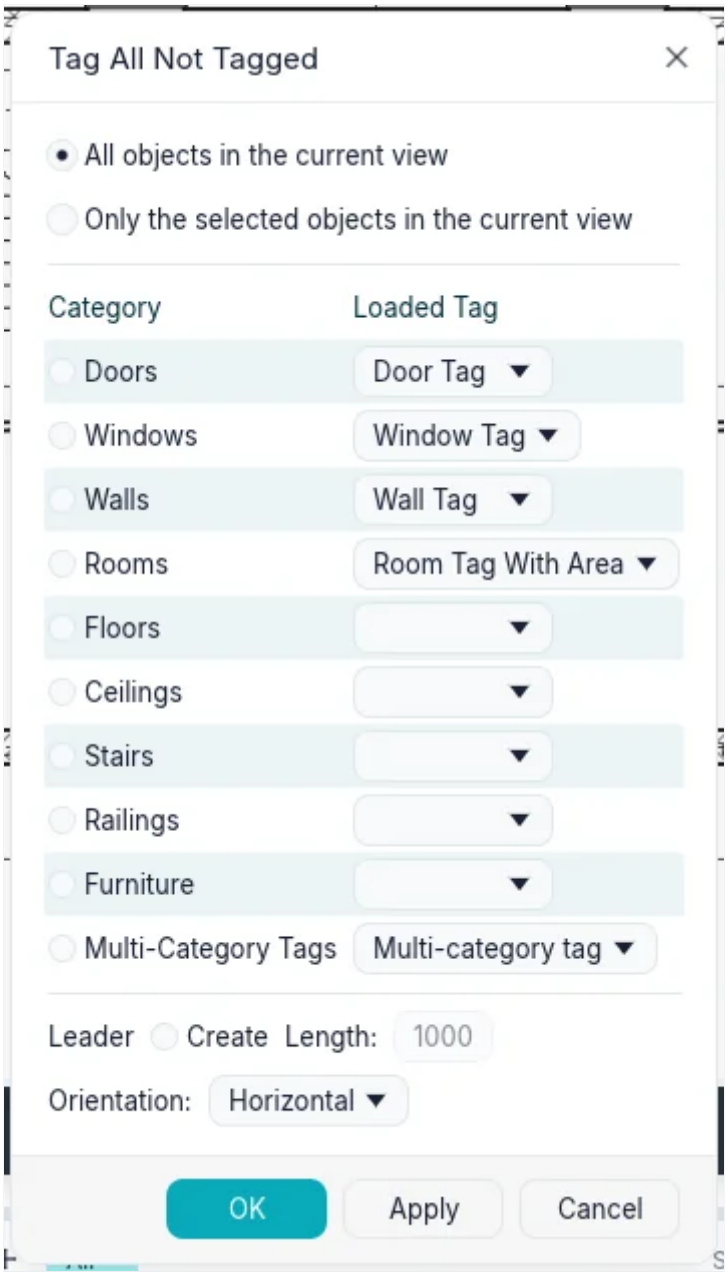

Tag All

With a leader, **Attached End** ends it on the element; **Free End** ends it where you clicked, and a selected tag shows a grip to drag that end anywhere (**Leader End** in the properties switches between the two). A tag lines up with the other tags of its kind while you place it or drag it: near the line of another tag's centre it snaps to it, and a dashed line shows it. Tags hidden in the view (**Visibility and Graphics**) are turned on when you tag, and the message says so.

Double-click a tag to edit what it shows: a door's mark or a wall's type mark in place; a room's or area's number and name together in **Edit Label Values**. A tag takes another type of its own category from the type selector, and **Pick New Host** (**Host** panel) moves it to another element of the same category. Room label types choose whether they show the number, the area and the volume. A room or area label set to **Pattern** keeps an angle: it turns with **Rotate**, and the **space bar** lines it up with each wall near it in turn (**Rotation** in its properties). Deleting a room's label leaves the room, and the message says so.

Loaded tags and numbering

Annotate ▸ Tag ▸ Loaded Tags (in the BIMopen style the command of the same name; in the options bar of **Door** and **Window** the **Loaded Tags…** button) lists every taggable category with its default tag. A filter narrows the list; in the tag column you pick a type of the category or, for model categories, a multi-category tag; **Edit Type…** opens the type of the picked row. **Load…** stays greyed for now. The choice is saved in the project and applies to **Tag**, **Tag All** and **Tag on Placement**, which is on from the start for doors and windows.

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Every model category has its own tag (floors, ceilings, roofs, columns, stairs, ramps, railings, panels and mullions, furniture, plumbing, generic models), showing the **Type Mark**. The tag just placed stays selected until the next one is placed or ESC is pressed, and the tool stays on.

Doors, windows, rooms and the other elements are numbered from the last mark typed: its last number goes up keeping its zeros (P-09 → P-10), or its last letter (1A → 1B, Z → AA); a value already used is skipped.

A selected tag shows its controls: the **cross** above its text, dragged, moves the text and the leader follows; the **elbow** halfway along the leader bends it (double click: straight again); the **attached end** slides along the element. The options bar of the selected tag has Orientation, **Leader** and Attached/Free End; a click on the text of the tag already selected types its value. Detail components and revision clouds have their tags too (the cloud shows the revision's number).

Tag All tags only what the view shows (it skips hidden elements, categories turned off, elements a filter hides or outside the crop, and says how many); with a selection it starts from **Only selected objects** and each row says how many it has; when tags are hidden in the view, it asks first whether to show them. In **Object Styles** and **Visibility/Graphics**, under **Tags**, each kind of tag (doors, windows, walls, rooms, furniture, multi-category, material...) has its row with weight, colour and pattern.

Multi-category tags

Annotations ▸ Tag ▸ Multi-category tag tags any model element: walls, doors, windows, floors, ceilings, roofs, columns, stairs, railings, furniture, plumbing, curtain panels and generic models (rooms and areas keep their own tags). It has the options of **Tag** and the same properties once placed; the space bar turns it. What it shows is set in the type (**Type Parameters**): **Label**, **Edit...** opens **Edit Label**, the fields one per line, each with a prefix and a suffix, chosen among Category, Type Name, Mark, Type Mark, Comments, Description, Host Count and the project's and shared parameters. A value that is empty, or a parameter the element does not have, reads **?**.

With a **Filter Parameter** (a shared parameter) in the type, only the elements that have that parameter light up and can be tagged. Double-click a line of a selected tag to type its value: a parameter of the element changes at once; a parameter of its type asks first, since every element of the type changes. **Tag All** lists the multi-category tags too: ticked, every element of the view that has the filter parameter and no tag of that type gets one. In a section or elevation the tag goes where you click, its leader to the point of the element. It follows the element: when the element moves or changes, the tag and its leader's end move with it, as in plan; the space bar turns it and a double click types its value there too. Material tags follow their element the same way, in plans and sections.

Material tags

Annotations ▸ Tag ▸ Material tags the one material under the cursor, in a plan, section or elevation at **Medium** or **Fine** detail: a layer of a wall, floor, roof or ceiling, cut or seen, or the material of a door, window, component or column. At **Coarse** detail the layers are not drawn, so they do not light up and the status bar says so. The tag shows the material's **Description**; when it is empty it reads **?**: double-click it and type, and the material's description is filled (every tag of that material shows it). Drag the grip at the end of the leader onto another layer and the tag reads that material; the space bar turns it. If the element later loses that material (its type changes), the tag reads **?** and the change warns. The type's **Label** chooses what it shows: Description, Name, Comments, Mark, Material class, Keywords, Manufacturer, Model, Cost, URL, Keynote or Host Count.

Tags with several leaders

Turn on **Add/Remove Host** on the options bar of **Tag**, **Multi-category tag** or **Material**, click the elements one after the other (a second click takes one out again), then click an empty place: one tag goes there, with a leader to each element (**Enter** puts it by the first). A category tag takes elements of its own category, a material tag places of the same material. On a selected tag, **Host ▸ Add/Remove Host** adds or removes leaders until **End**; **Leaders ▸ Merge Leaders** joins them at one elbow before the tag, and **Select Leaders to Show** hides or shows the leader of each host clicked. **Host Count** in the properties tells how many elements the tag points at. Deleting an element takes its leader away; the tag stays while it points at something.

Several values

When the hosts of a tag differ (two doors with different marks, a field of a multi-category tag), that line reads **‹varies›**; a line they share reads its value, and **Host Count** never varies. Double-click it: the field opens empty, or with the shared value, and what you type goes to every host in one step (a parameter of their types asks first). A room or area tag with several rooms opens **Edit Label Values** with what they share; only what you type is written. **Project ▸ Additional settings ▸ Multiple Values Indication** chooses what such lines show: **‹varies›**, or **Show custom text** with a text of your own. The **Properties** palette of several selected elements shows the same text where their values differ.

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Keynotes

Annotations ▸ Tag ▸ Coded note has four entries (the BIMopen style shows them as tiles):

- **Element note**: click the element (a wall, door, component, detail component… whose type has **Keynote**), then the leader's bend, then where the note goes. It reads the type's **Keynote**; a type without one opens **Keynotes**.
- **Material note** works as **Material**: click a material (a layer at **Medium** or **Fine** detail) and the note reads its **Keynote**; its leader always ends on the material.
- **Free note**: the same three clicks; **Keynotes** always opens and the value stays in the note, the element does not change.
- **Coded note options**: see below.

Each kind has four types: **Number**, **Number in large box**, **Number in small box** and **Text** (the note's text from the table). The options bar has the orientation, **Leader** and **Attached end** or **Free end** (a material note's end is always free).

Keynotes shows the table as a tree: **Key Value** and **Keynote Text**, each row under its parent value; + and − open and close the branches, **Search** finds words in values and texts. **OK** gives the row picked, **Cancel** leaves the box empty. **Edit table…** edits the project's table (value, text, parent value). Select a note and click its value, or its empty box, to pick another: an element note writes its type's keynote after asking (every element of the type changes), a free note only itself, a material note its material. **…** by **Keynote** in **Type Parameters** and in a material's **Identity** opens the same tree.

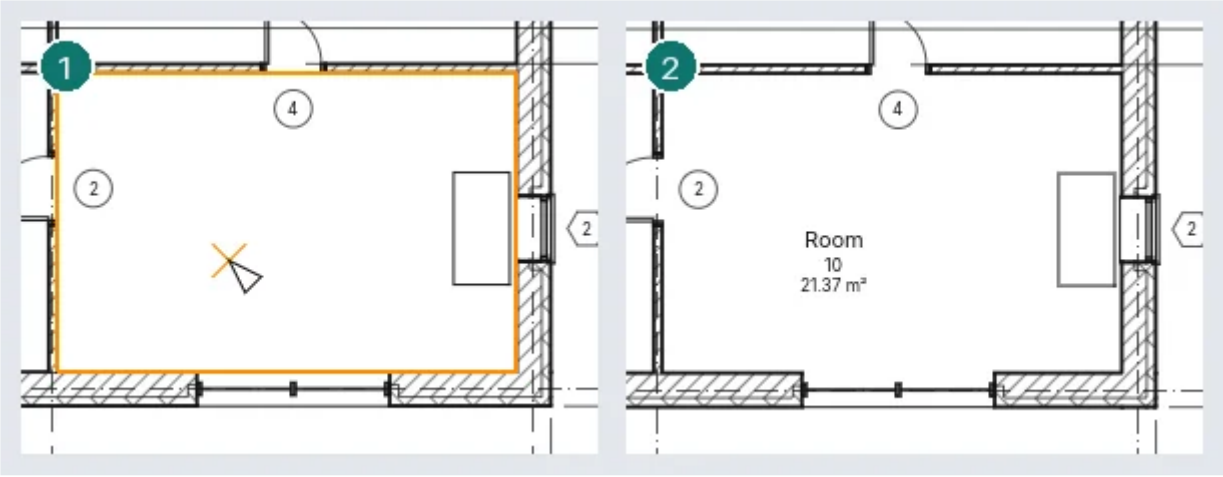
Coded note options chooses where the table comes from: **Table in the project** (edited here) or **Linked file**, a text file with one row per line — value, TAB, text, TAB, parent value (a spreadsheet saved as tab-separated text works). **Full path** shows where the file is, **Saved path** how it is written, with **Browse…**; **Path type** is **Absolute**, **Relative** (to the project's folder) or **In the library**. **View…** shows the table read only. **Sample table** links the table we ship: short, in Italian, with building works, insulation, windows and doors and a section of user notes. **Reload** reads it now, and the dialog warns when the file changed since it was read. The file is read again every time the project opens; the project keeps a copy, used when the file is missing (it says so), and values no longer in the file are listed. **Numbering method**: **By keynote** shows the value; **By sheet** numbers the notes 1, 2, 3… on each sheet in the order they were made, and a view not on a sheet shows an empty box.

View ▸ Legends ▸ Coded note legend makes a schedule, **Keynote Legend**, with **Keynote Value** and **Keynote Text**, one row per value, with the usual schedule tabs (a filter such as «begins with» on the value, sorting, formatting, appearance). **Filter by sheet** (in **Filter**) lists on each sheet only the notes of its views; numbered by sheet it shows the same numbers. **Find/Replace** never changes keynotes: they change from the table.

Rooms

Home ▸ Rooms and Areas ▸ Room (`RM`), then click inside a region enclosed by walls or room separators: the room fills it, around free-standing walls too. In the options bar:

- **Upper Limit** and **Offset**: the room reaches that level plus the offset (4000 mm by default); a higher level gives a room over several storeys, such as an atrium;
- **Tag on Placement**, on by default, with the tag's orientation and leader;
- **Room**: **New**, or a room that is not placed, to put it back;
- **Highlight Boundaries**: walls and separators that bound rooms turn orange, and the warning lists them.



Room: the region it fills, then the room and its tag

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A room placed in open space is kept with a warning, and takes its shape once the space is closed; a second room in a region already taken is marked **Redundant Room** and not counted twice. A number another room already has, a room lower than the computation height (1200 mm) and a base above it are warned about too, and so is a room separator drawn over another.

Walls bound the rooms of a level when they stand at its computation height: 1200 mm while **Automatic Room Computation Height** is ticked in the level's properties, otherwise its **Computation Height**.

A section shows the rooms it cuts when they are selected or under the cursor (**TAB** reaches a room behind other elements). A room selected there has an arrow at its top and one at its base: drag them to change its **Limit Offset** and **Offset at Base**.

Room Separator draws lines that split a space into rooms without walls. A wall bounds rooms while its **Room Bounding** property is on. Floors and roofs above cap the room's volume.

Deleting a room removes it from the plan but keeps it in the project, **Not Placed**, with its data; its tags go with it. Deleting a room label leaves the room. A room label dragged out of its room asks whether to **Move to Room**.

Rooms show through their tags. In **Visibility and Graphics**, **Rooms** has **Interior Fill** and **Datum** (the cross lines). **Area and Volume Computations** opens on **Computations**: the room boundary at the wall finish, centre, core layer or core centre, and **Areas and Volumes** to compute volumes too (by default only areas; the volume is always measured to the wall finish). Its second tab is **Area Schemes**.

Areas

Areas belong to an area scheme: **Gross Building** (which cannot be deleted) and **Rentable** come with every project, and **Area Schemes ▸ New** adds more.

1. **Rooms and Areas ▸ Area ▸ Area Plan**: pick the scheme and the levels. Answer **Yes** to draw area outlines along the exterior walls, **No** to draw them yourself.
2. **Area ▸ Area Outline**: draw the outlines, or pick walls; with the area rules on, an exterior wall of a gross plan gives its outer face.
3. **Area ▸ Area**: click inside the outlines. The area is tagged when placed; **Tag ▸ Area Label** tags it later. Placed where no outlines close a region, the area is kept with a warning and takes its shape once they do; a second area in a region is **Redundant Area**.

Deleting an area leaves it in the project, not placed, without its tags; **Area:** in the options bar of the **Area** tool lists the areas not placed, to put one back with a click.

Colour schemes

A colour scheme fills the rooms or areas of a plan (and the rooms a section cuts) by a value (name, number, department, area type) or by ranges of area. Set it in the plan's **Color Scheme** property; **(none)** turns it off. **Color Schemes** (**Edit Color Scheme**) creates and duplicates schemes and edits the title, the parameter, the values with their colour and visibility, their order and the ranges. **Apply** keeps the changes and sets the scheme of the view with the dialog open. **Color scheme location**, a view property, draws the fills in the **Background** (under the model) or in the **Foreground** (over the model, under the annotations).

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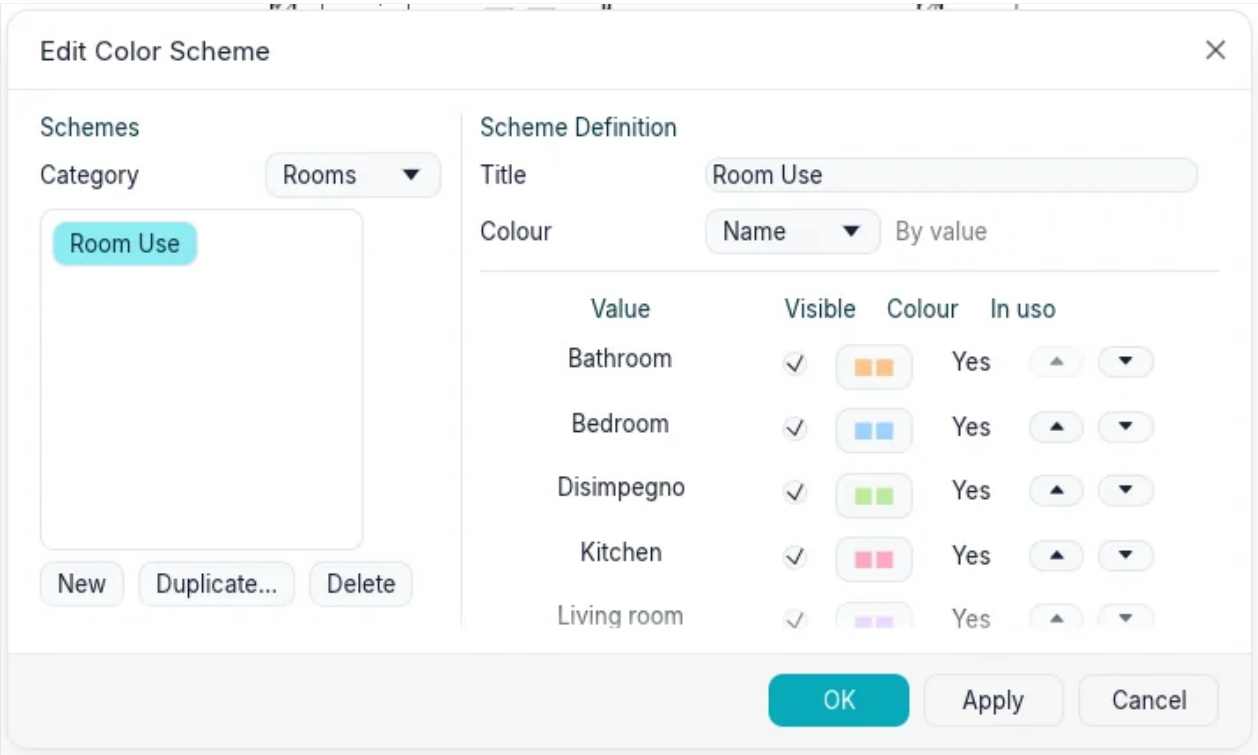
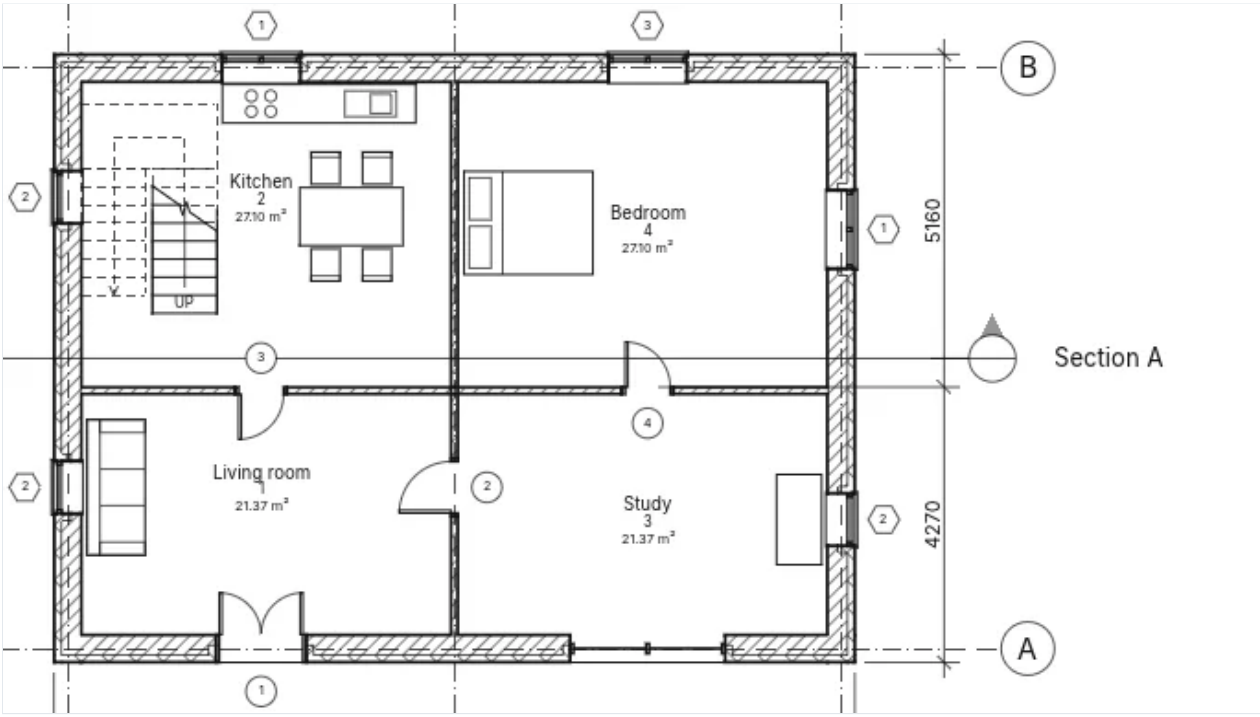


Edit Color Scheme

Rooms and Areas ▸ Legend places the legend of the plan's scheme: a swatch and a label for each value. Its **Values displayed** are **By view** (those in the plan) or **All**, and **Columns** flows the swatches into several columns. A plan can have several legends; deleting one leaves the scheme.



Rooms by colour, and the legend

Legend views

Views ▸ Legends ▸ Legend opens **New Legend View**: a name and a scale. The legend opens and is listed under **Legends** in the Project Index. **Annotations ▸ Component ▸ Legend Component** places the symbol of a type: in the options bar, the type, the **View** it is seen from (a wall in plan or section; a floor or roof in section or plan; a door, window or component in plan, front or back) and the **Host Length**, the piece of wall a door or window sits in. Legend components draw at the legend's detail, can

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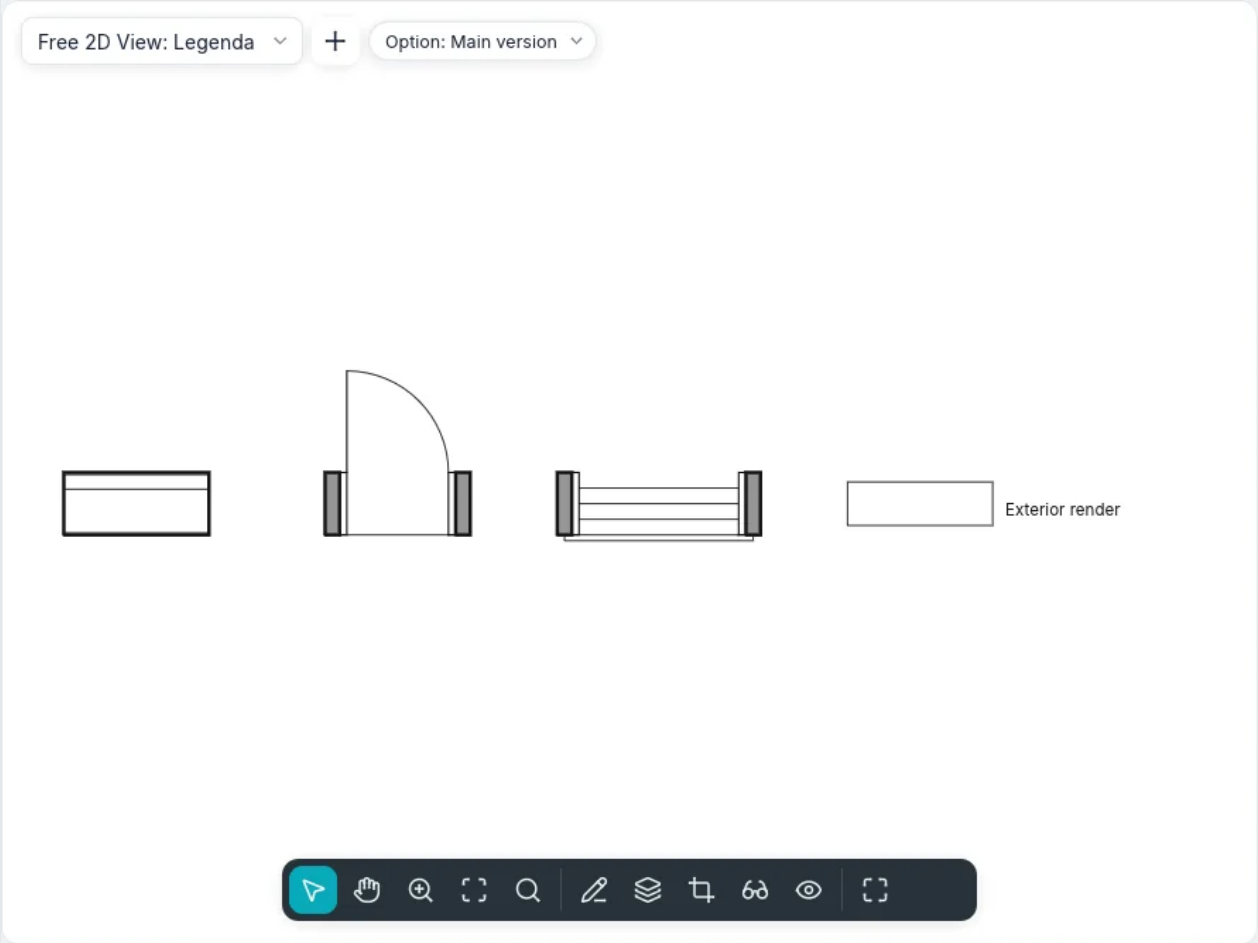
be moved and dimensioned (the edges of doors, windows and components; walls, floors and roofs are not), and are not counted by schedules. Text, detail lines and regions work as in a free 2D view. A legend can be placed on several sheets.



A legend: wall, door, window and a material

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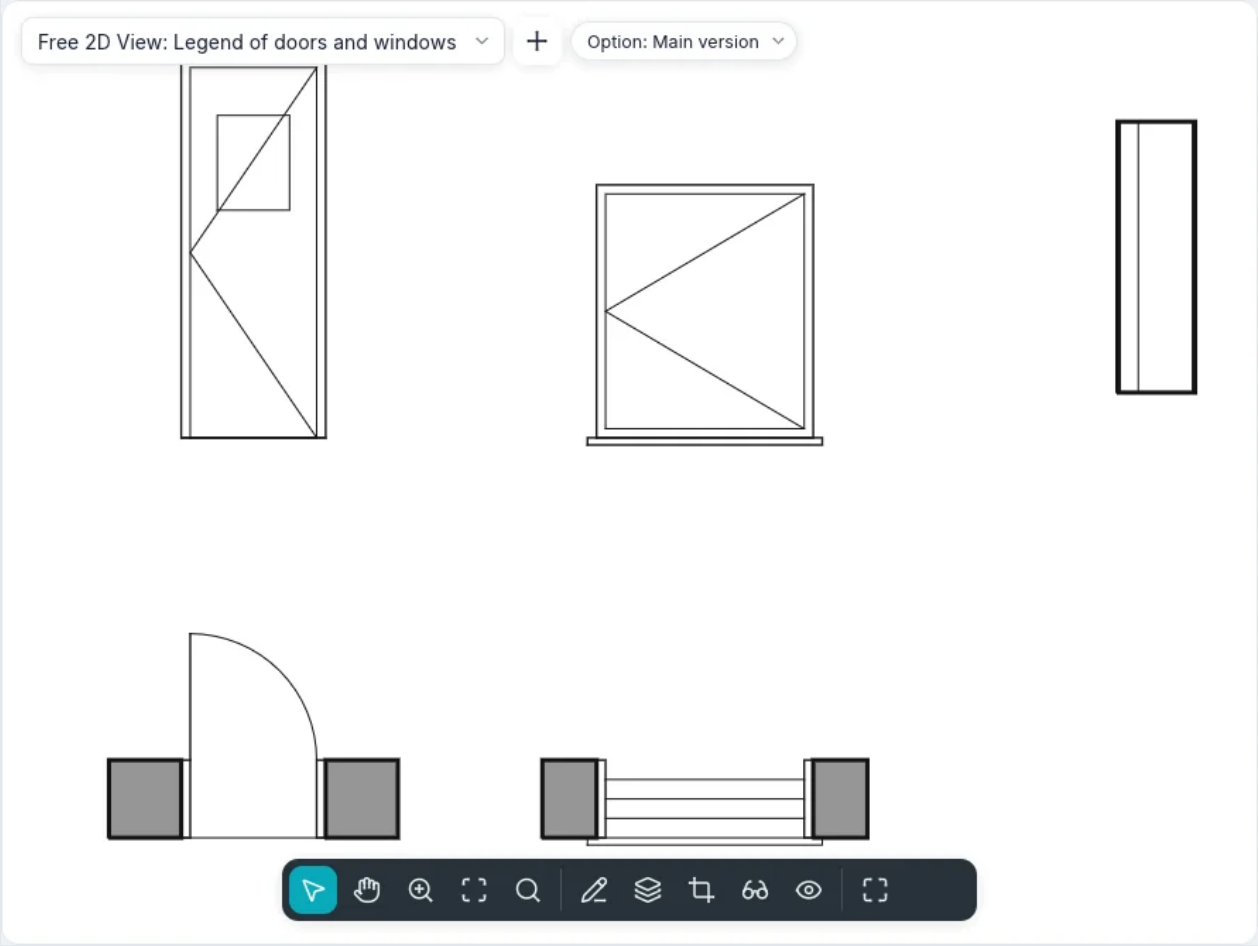

A legend view

Component also lists ceilings, stairs, railings and columns, and two kinds of linked samples: **Materials: …** draws a sample of the material's **Cut** or **Surface** pattern, a **Swatch** wide × high, and **Line Styles: …** draws a line as long as **Length**; **With name** writes the name beside. A sample follows what it samples: change the material's pattern or name, or the line style's weight, colour or pattern, and the legend follows. In the legend's **Visibility/Graphics**, doors (Frame, Panel, Swing lines, Glass, Threshold), windows (Frame, Glass, Sill, Swing lines), stairs (Treads, Risers, Walk line) and railings (Handrail, Balusters) have their subcategories under them, switched one by one: a legend of frames only is a door legend with the other rows off. The colour fill legend (**Rooms and Areas ▸ Legend**) also goes in a section or an elevation with a colour scheme.

A type dragged from **Components** in the Project Index into a legend becomes its legend component where it is dropped; a type of **Components ▸ Symbols** places that annotation symbol. **Create Same**, in the context menu of a selected element, starts the tool that places it with its type: for a legend component, with its view and host length too.

Detail lines and regions

Annotations ▸ Detail ▸ Detail line (`DL`) draws lines in the open view only, in the line style of the **Line style** panel. The **Draw** panel offers **Line**, **Rectangle**, **Inscribed Polygon**, **Circumscribed Polygon**, **Circle**, **Start-End-Radius Arc**, **Center-ends Arc**, **Tangent End Arc**, **Spline**, **Ellipse**, **Partial Ellipse** and **Fillet Arc**. The options bar shows what the shape takes:

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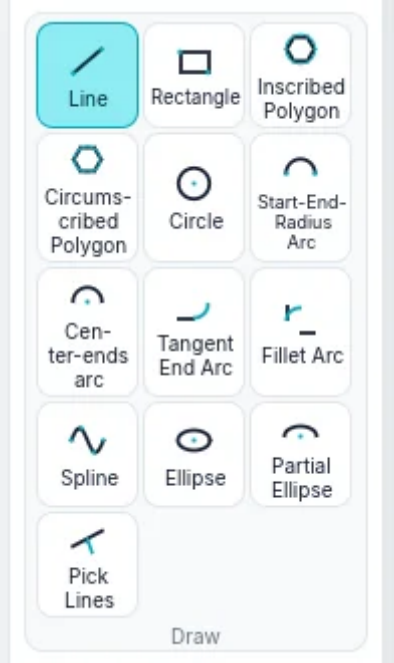

The Draw group

Option	What it does
Continue	Each line (or arc, or spline) starts where the last ended; off, each one stands alone
Offset	What is drawn moves that far sideways; chained lines join at their corners
Radius	A circle, polygon or centre-ends arc of that radius (a circle takes one click); chained lines and rectangles get round corners. **Offset** is then ignored
Sides	The number of sides of a polygon

After the first click, type a length (a line, a chord) or a radius (circles, polygons, arcs, ellipse axes) and press `Enter`; the value shows at the cursor. A centre-ends arc never exceeds 180°: past it, it turns the other way, and a typed angle sets its end. A tangent arc starts at the end of a line or arc you click. `Shift` keeps lines and polygon radii horizontal or vertical and turns chords, arc radii and ellipse axes in steps of 45°, and keeps the sweep of an arc at 90°, 180° or 270°. `Esc` ends a chain or a spline. The same shapes and options are in every sketch (floors, roofs, filled regions, revision clouds, wall boundaries).

Pick Lines makes a detail line on a wall's face, axis or layer, a grid or another line under the cursor (`Tab` for the next one), moved by **Offset**; with **Lock** the line stays on it and follows when it moves. **Fillet Arc** picks two lines and rounds their corner, with the arc under the cursor or of the **Radius** set. A selected detail line has grips at its ends and a temporary dimension of its length: type a value to change it. The end of a line selected on its own drags alone and leaves the next line where it is; with a chain selected (`Tab`), the ends where its lines meet drag together, in one undo step. The cursor snaps to the ends, middles, centres and crossings of the view's detail lines.

A spline cannot close on itself: a click on its first point is refused, and a second spline closes the boundary. A selected spline has a grip at each control point: drag one in the middle to reshape the curve; drag an end to move the whole spline, or press `Tab` on the end first to move only that end. **Edit Spline ▸ Add Control** puts a control point in the middle of the chord you click (the status bar says **Reference** over it) and **Delete Control** removes the one you click; `Esc` ends them. A selected ellipse shows its two radii as temporary dimensions (type a value to change one) and a grip at each quadrant: dragging one resizes that axis and turns the ellipse towards the cursor. With **Editing Retains Proportions** on the options bar both radii change together. A partial ellipse also has a grip at each end: **Drag End** changes how far it goes round, `Tab` switches to **Drag Axis End**, which moves that end of the axis while the other stays. **Pick Lines** on an arc, circle, ellipse or spline copies it; with **Offset** an ellipse or a spline gives splines.

The third point of a **Start-End-Radius Arc** snaps when the arc becomes tangent to the line or arc that ends at its start or its end: the tangency mark shows and the status bar says **Tangent**. A line drawn from the end of an arc snaps tangent to it in the same way. A selected arc has a grip at each end, which changes its sweep, and one in its middle: with **Remain Concentric** on the options bar the radius changes about the same centre, otherwise the ends stay where they are. The same happens in plans, sections, elevations and 3D.

Filled region ▸ Filled region opens sketch mode: the boundary is sketched with every shape of the **Draw** panel and with **Pick Lines**, and loops inside another are holes. A spline or an ellipse stays one curve of the sketch, with its grips; the region's outline

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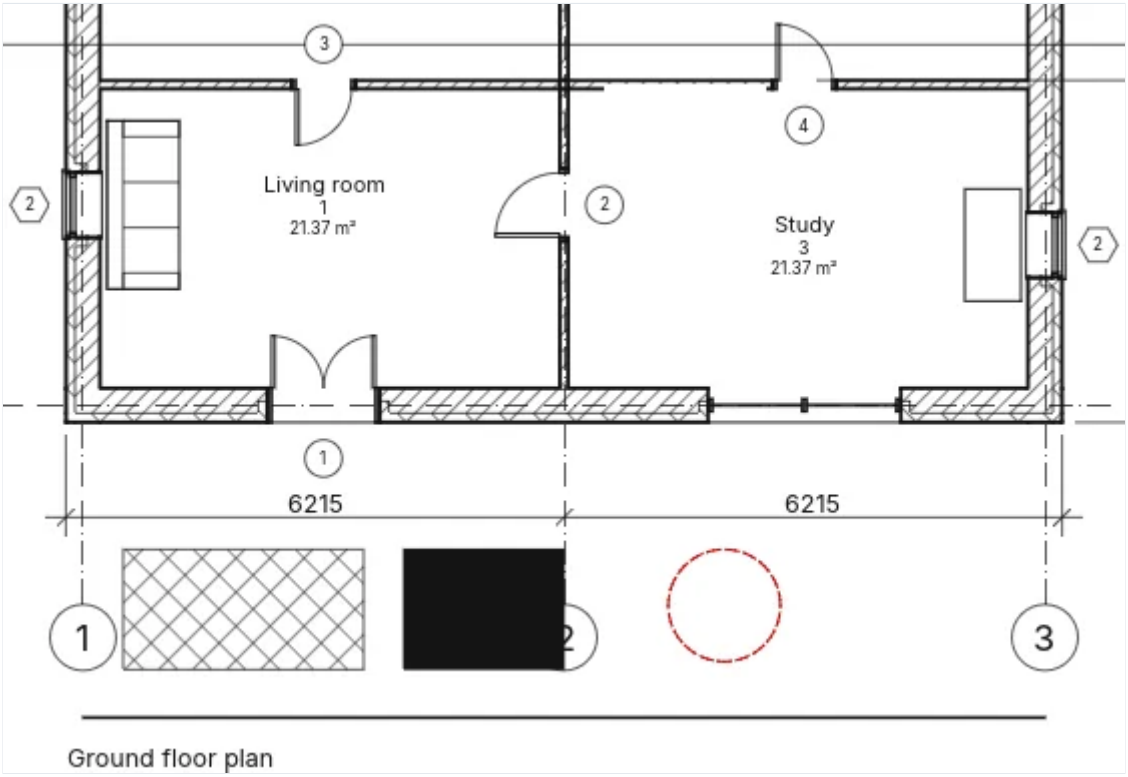
is made of its short pieces. Esc ends a chain and keeps the sketch; **Modify** and the Modify tools work on its lines. **Finish Edit Mode** makes the region in the current region type, or says why it cannot (an open or crossing boundary). A selected region's **Edit Boundary** opens its sketch again. The type sets its fill pattern, colour and background (**Opaque** hides what is behind). **Filled region ▸ Masking Region** makes a region in the paper colour that hides what is beneath it. A region shows its **Area** and has **Comments**. A selected region has a grip at each vertex: drag one to change its shape.



Filled regions, detail lines and a masking region

The tangency padlock

Select a line (a sketch line, a detail line or a 3D line) that meets another one end to end with the same direction: an open padlock shows beside the connection, outside the bend (none between two straight lines). Hover it to highlight both lines; click it to lock the tangency, click again to unlock. Locked, the lines stay tangent while either is edited: a line that turns makes an arc refit through its far end; an arc between two lines keeps its radius and slides on them; an ellipse turns about its far end; a spline's second control follows the direction. An edit that cannot keep it asks **Constraint not kept**: **Remove constraint** unlocks and does it, **Cancel** cancels. Right-click the padlock for **Toggle Lock** and **Show Related**. In the BIMopen style the command search also has **Lock Tangencies** and **Unlock Tangencies** for the selected lines.

Detail components, repeating details and insulation

- **Component ▸ Detail component**: a 2D symbol (brick, block, timber, steel section, insulation board, break line) placed in a drafting or detail view; the **space bar** lines it up with each wall or detail line near the cursor in turn (a quarter turn when there is none).
- **Component ▸ Repeating detail**: click a start and an end; the type sets the repeated component, the **Layout** (fill the available space, fixed distance, fixed number, maximum spacing), the **Spacing**, **Inside** and the rotation within the pattern. The instance has the **Number** when the layout is a fixed number.
- **Insulation**: click two points; **Width** and **Offset** (the insulation runs that far to the left of the clicks) in the options bar; with **Pick Lines** ticked, a click on a detail line or a wall draws it along that line. The insulation width and the bulge ratio in its properties. Selected insulation, and a selected repeating detail, have a grip at each end.

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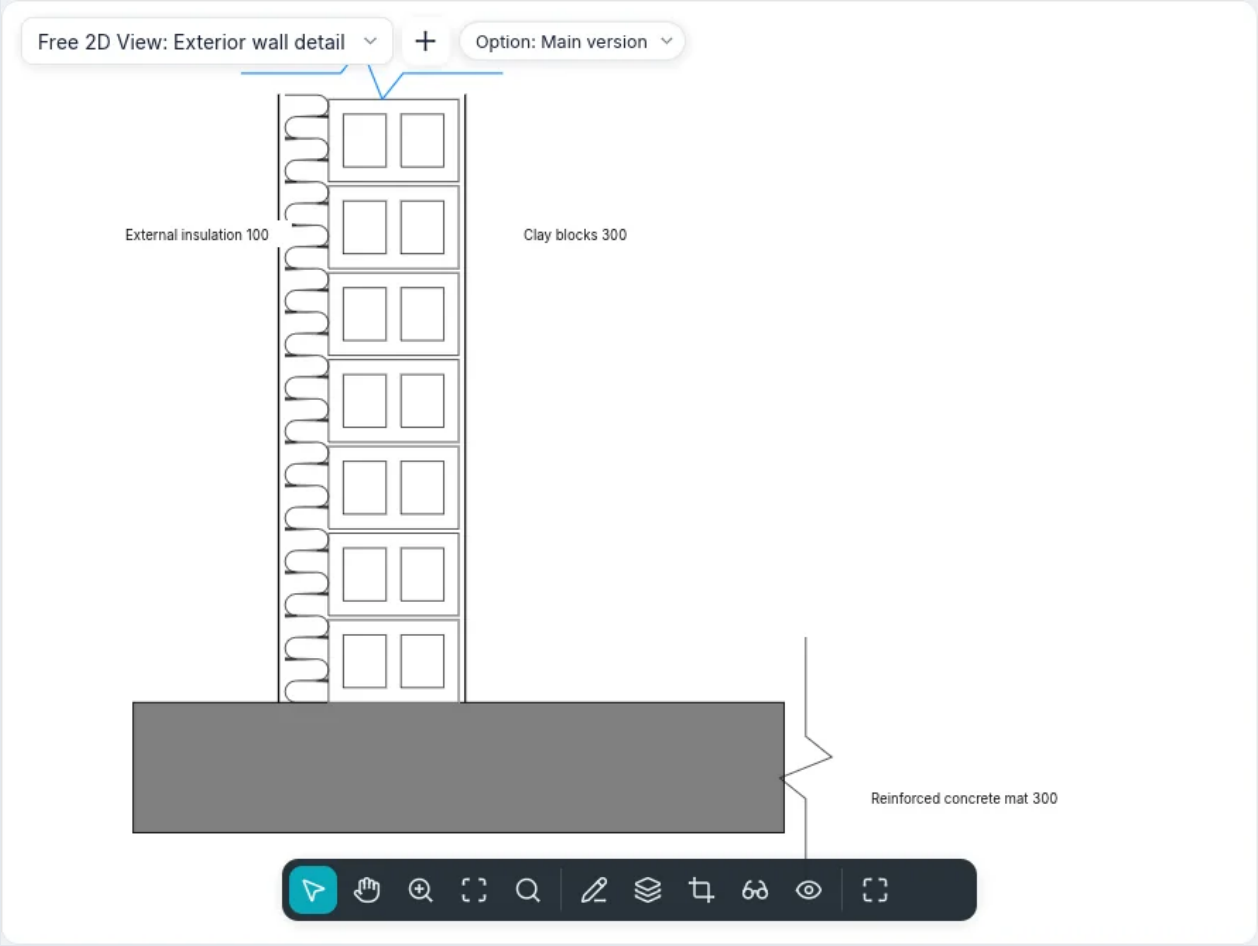

A wall detail in a free 2D view

In **Fill patterns**, a pattern's **Line weight** draws its lines with one of the project's line weights; left at **(thin)** they stay hairlines.

Draw order

Select detail items only: **Arrange** offers **Bring to Front**, **Bring Forward**, **Send to Back** and **Send Backward**. Detail items always draw over the model, and a new one goes on top. **Undo** restores the previous order.

Free 2D views

Views ▸ Create ▸ Free 2D View asks for a name and a scale (or a custom one). The view opens and is listed under **Free 2D Views**; everything drawn in it is 2D, with no model behind it. It can go on one sheet. **Duplicate View ▸ Duplicate with Detailing** copies a view with its detail items, text, dimensions and legend components.

Save to New File... in the menu of a free 2D view in the Index writes a project with the view, its 2D elements and the types they use. **Insert ▸ Insert Views from File...** lists the file's free 2D views and schedules to tick; **Insert 2D Elements from File...** copies the 2D elements of one of its views into the active view (free 2D, plan or section): they follow the cursor and a click drops them, selected. Types with the same name stay this project's and are listed.

Type in the dialog and in the type selector picks one of the two shipped types, **Detail** and **Scheme**; others are made with **Duplicate...** in **Type Parameters**, which sets **Section tag**, **Callout tag**, **Reference label** (written beside the head of a callout pointing at the view, e.g. «Sim.») and **Template for new views**. **Graphic Style** is **Hidden lines** (patterned regions hide what lies beneath) or **Wireframe** (everything shows). A **Reference Plane** drawn here belongs to this view only and is not a work plane.

In a callout with **Reference other view**, the first item makes, once placed, an empty free 2D view at the callout's scale, «Detail of »; views already on a sheet show «(detail / sheet)» after their name.

Display Model in the properties of a plan, a section or a detail view draws the model **Normal**, **Halftone** (greyed under the details) or not at all (**Do not display**), leaving the details and annotations.

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Views ▸ Create ▸ Detail with **Reference other view** ticked on the options bar draws a callout that points at an existing detail or free 2D view instead of making a new one: its head names that view and a double-click opens it; **Referenced view** in its properties changes it. In a free 2D view the callout is always of this kind.

Hide at scales coarser than in the properties of a section or a callout hides its symbol in the views whose scale is coarser, for example a 1 : 50 detail left out of the 1 : 100 plan.

At the command line

Every tool has its command: `dim`, `lineardim`, `angulardim`, `radialdim`, `diamdim`, `arcdim`, `spotelevation`, `text`, `findreplace`, `tag`, `room`, `roomsep`, `areaplan`, `areaboundary`, `area`, `colorscheme`, `legend`, `legendview`, `legendcomponent`, `detailline`, `region`, `mask`, `draworder`, `lock`, `eq`. `help` followed by a command shows its usage; Keyboard shortcuts lists the typed shortcuts.

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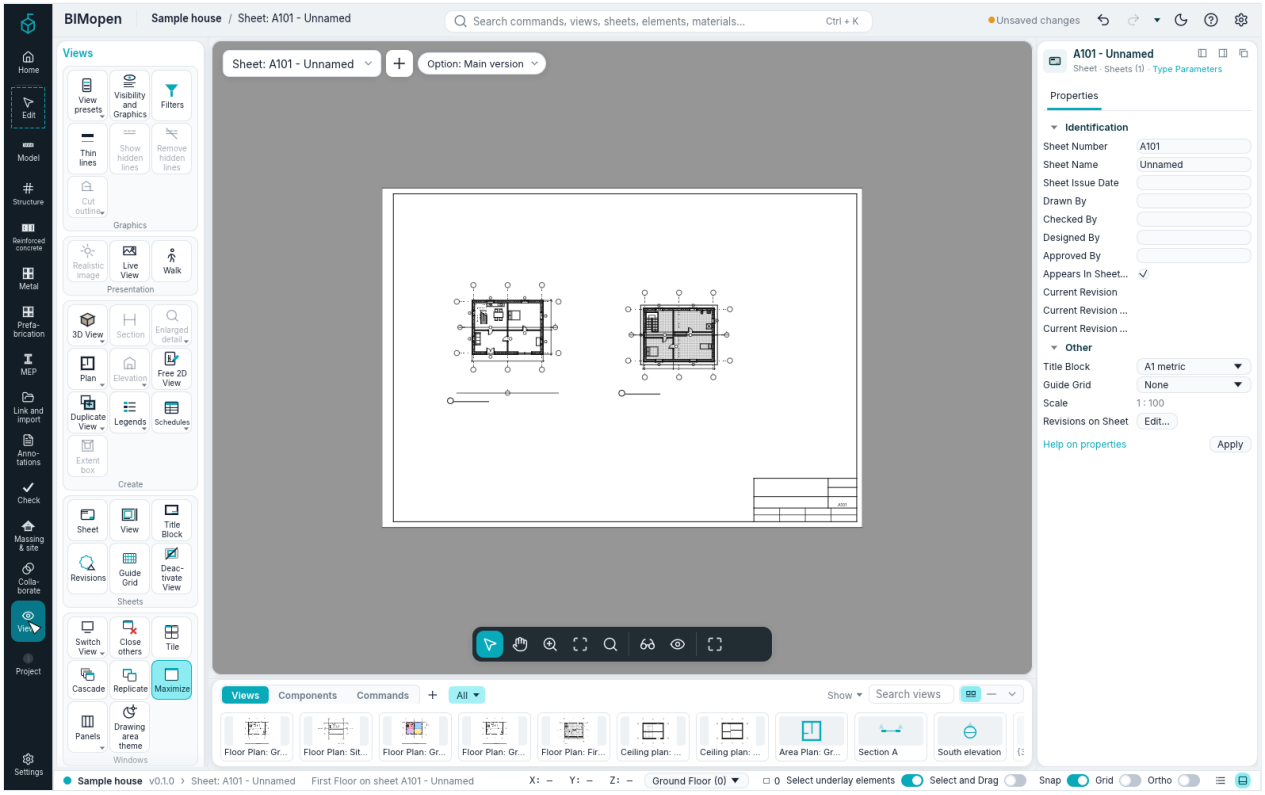
Sheets, schedules and printing

This chapter describes how the project is documented: sheets and their title blocks, the views placed on them, revisions, schedules, printed drawings and exported images.

Sheets

A sheet is a drawing to print: a title block and the views placed on it. Sheets are listed under **Sheets (all)** in the Project Index, by number.



A sheet with both floor plans, and the Document tools on the rail

Creating a sheet

Document ➤ Sheets ➤ Sheet opens **New Sheet**: choose a title block (its paper size is shown), **None**, or one of the placeholder sheets of the sheet list. **OK** creates the sheet with the next free number (**A101**, **A102**...) and opens it.

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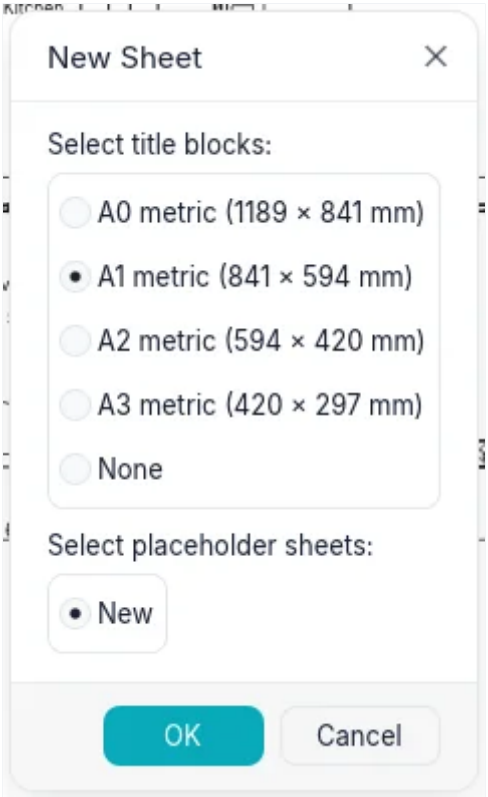

New Sheet

Right-click a sheet in the Project Index ➤ **Rename...** to change its number and name in **Sheet Title**. The other parameters of the sheet are in the element panel: **Sheet Number**, **Sheet Name**, **Sheet Issue Date**, **Drawn By**, **Checked By**, **Designed By**, **Approved By**, **Appears In Sheet List**, its guide grid and its revisions. Several sheets picked together in the Project Index (with `Ctrl` or `Shift`) are changed together: the panel shows the values they share.

Settings ➤ Settings ➤ Project information sets what every sheet shows: **Project Issue Date**, **Project Status**, **Client Name**, **Project Address**, **Project Name** and **Project Number**.

Views ➤ Create ➤ Schedules ➤ Note Block makes a schedule of the symbols placed with **Symbol**, of one family only (north arrows, targets, revision deltas...): in **New Note Block** choose the **Family** and type the **Note Block Name**, then **OK**. **Note Block Properties** opens first, with the same tabs as any schedule; by default each row shows the symbol's type, its text and the count.

Title blocks

The title block shows the project's information, the sheet's number and name, the people and the dates, the current revision and the scale (**As indicated** when the views have different scales); above it, a table lists the sheet's revisions with their number, description and date. A double click on one of its fields types the value in place.

- Click a line of the title block to pick it: the type selector changes it to another paper size, `Del` removes it (the sheet and its views stay).
- **Document ➤ Sheets ➤ Title Block** inserts one on the open sheet: choose it in the type selector, its frame follows the cursor, click to place it.
- **Type Parameters** shows its **Sheet width** and **Sheet height**; **Duplicate...** makes a new paper size.

Views on sheets

Drag a view from the Project Index onto the sheet, or use **Document ➤ Sheets ➤ View** (**Views**, then **Add View to Sheet**): the view follows the cursor at its scale, and a click places it as a viewport. A view goes on one sheet only; a schedule on several.

A picked viewport:

- moves with the mouse or the arrow keys, unless it is pinned (**Fix**); near another view, its title lines up with the other's and a dashed line shows it;
- has its **View Scale**, its **Title on Sheet** (also typed by double-clicking the title) and its **Rotation on Sheet** (90° clockwise or counter-clockwise; the title stays horizontal);
- shows the end of its title line, which can be dragged;

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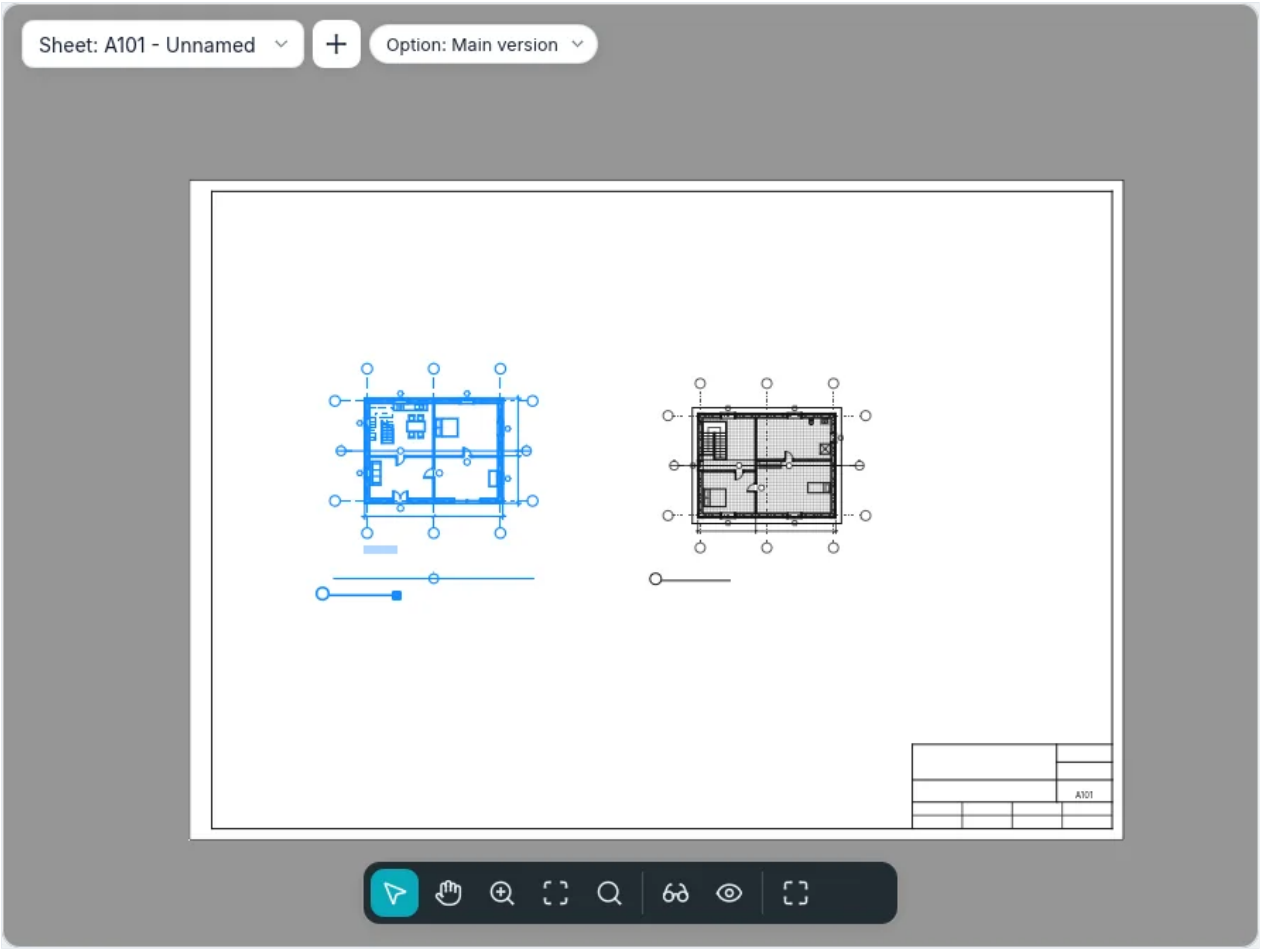
- takes a viewport type from the type selector: **No Title**, **Title w/ Line** or **Title w/o Line**, or a type of your own (**Show Title**, **Show Extension Line**, line weight, colour and pattern of the line);
- leaves the sheet with `Del`; the view stays in the project.



A viewport picked on a sheet

Activate View (on the **Selection: Viewports** card or the right button) edits the view where it is on the sheet, the rest of the sheet in halftone; **Deactivate View** goes back to the sheet. A view turned on its sheet cannot be activated: take its rotation off first. A double click on a view opens it in its own window.

Each view on a sheet has a **Detail Number**: the first number free on that sheet (1, 2, 3...), shown in a circle left of its title. Change it in the viewport's **Properties**; a number already used on the same sheet is refused. The view's own **Properties** show where it is placed, read only: **Referencing Sheet** and **Referencing Detail**.

Text, lines and dimensions on a sheet

With a sheet open, **Text**, **Detail line** and the dimensions work on the sheet itself: what is drawn there belongs to the sheet, keeps its printed size and is printed with it. Dimensions measure the lines drawn on the sheet. The other tools still open **Go to View**.

Guide grids

Document ➤ Sheets ➤ Guide Grid asks for a name and covers the sheet with a grid. Pick it to drag its corners or to set its **Guide spacing**; the sheet's **Guide Grid** parameter shows the same grid on other sheets, or none. Views dragged near the grid's lines snap onto them. The grid is never printed.

Sheet list

Views ➤ Create ➤ Schedules ➤ Sheet List makes a schedule of the sheets: number, name and any other field. Sheets with **Appears In Sheet List** unticked are left out. **Rows ➤ New** adds a placeholder sheet: a row that is not a sheet yet and is not in the Project Index. **Filter placeholder sheets** shows every row, hides the placeholders or shows only them. A placeholder chosen in **New Sheet** becomes a real sheet.

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Revisions

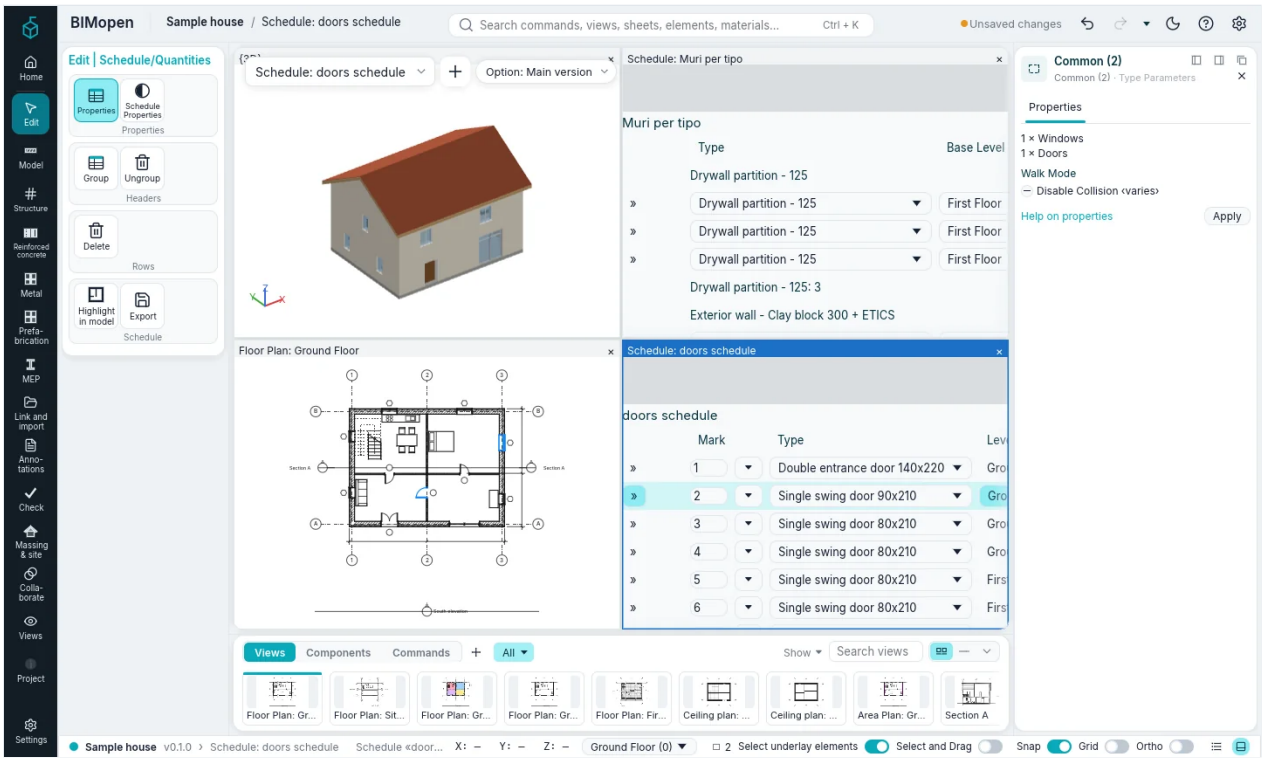
Document ➤ Sheets ➤ Sheet Issues/Revisions lists the revisions of the project, one per row: **Sequence**, **Numbering** (**Numeric**, **Alphabetic** or none), **Date**, **Description**, **Issued**, **Issued to**, **Issued by** and **Show**. **Add** adds one; a row picked by its sequence number is merged into the one above or below (its clouds go with it) or moved up and down. **Numbering** is **Per Project** or **Per Sheet** (the place of the revision among the sheet's revisions); **Alphabetical order ➤ Options...** sets the letters used. An issued revision cannot be changed, and neither can its clouds, until **Issued** is unticked.

Document ➤ Detail ➤ Revision Cloud draws a cloud in a plan, a section, a free 2D view or on a sheet: sketch closed loops with the shapes of the **Draw** group, then **Finish Edit Mode**; a selected cloud's **Edit Sketch** opens them again, unless its revision is issued. The cloud takes the latest revision not issued; picked, it shows its **Revision** (to change), its number, date and people, its **Mark** and **Comments**. **Show** of a revision draws its clouds and tags, keeps them but does not draw them, or hides them.

A sheet lists the revisions of the clouds of its views and of its own clouds. Its **Revisions on Sheet** (in the element panel, **Edit...**) adds other revisions to the list.

Schedules

A schedule is a table of the elements of the model, and a view like the others: it opens in a window of the drawing area, is tiled, cascaded and switched with plans, sections and 3D views, and is listed under **Schedules/Quantities** in the Project Index. It follows every change of the model at once, and every change made in it changes the model.



Schedules beside the plan, selected together

Creating a schedule

Views ➤ Create ➤ Schedules ➤ Schedule/Quantities opens **New Schedule**: choose a category, a name (the default follows the category until one is typed) and **Schedule building components** or **Schedule keys**. **OK** opens **Schedule Properties**; its second **OK** creates the schedule and opens its view. **Material Takeoff** in the same menu lists the materials of the layers of walls, floors and roofs.

Schedule Properties

The dialog opens again from the element panel of the schedule (each tab has its **Edit...** button) or from the right button on the schedule in the Project Index. Its tabs:

Tab	What it sets
Fields	The columns, chosen from **Available fields:** and put in order; the fields of the rooms the elements are in; **Add Parameter...** for a project or shared parameter; **Calculated Value...** for a formula (with `+ - * / ^`, `sqrt`, `round`, `IF`, `AND`, `OR`, `NOT` and units) or a percentage

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Tab	What it sets
Filter	Up to four criteria, all of which an element must meet
Sorting/ Grouping	Up to four sort levels, each with **Header**, footer and **Blank line**; the grand totals; **Itemize every instance** (off: one row per type)
Formatting	Per field: heading, orientation, alignment, hidden field, **Format…** (units, rounding, symbol), totals and **Conditional Format…** (a background colour where a value meets a test)
Appearance	Grid lines and outline with their line styles, the title and the column headers, a blank row, **Header text** and **Body text** (size, **Bold**, **Italic**); shown when the schedule is on a sheet

Working in a schedule view

- A cell edits the element, or its type for a type parameter; every edit is one step of undo.
- A type parameter (a door's width, a component's depth) changes every instance of the type: **Edit Type** says how many and asks first.
- The predefined schedules of the Project Index (doors, floors, rooms, stairs, curtain panels…) work the same way: the type and the level are picked from a list, the values the model computes (areas, volumes, the host wall) are read-only, and in the curtain panel and mullion schedules the type of each panel or inner mullion is picked.
- A row selects its elements in every view. **Highlight in model** on the **Edit | Schedule/Quantities** card opens **Show Elements in View**, which brings a plan forward framed on them; **Show** goes on to the next view.
- **Rows ➤ Delete** deletes the elements of the selected rows, after asking.
- Headings picked together are joined under a common heading with **Group**, and parted with **Ungroup**; **Hide column** and **Unhide all columns** work on the column picked.

Schedules on sheets

Drag a schedule from the Project Index onto a sheet. Selected on the sheet, it shows its controls: the Z on the right edge of a segment splits it in two, the arrow in the middle of a segment moves it (dropped on the segment beside it, the two join), the dot at the bottom of a segment moves its last row, and the triangles at the top of the columns set their widths. A double click, or **Edit Schedule** on the right button, opens the schedule view. The same schedule can go on several sheets.

A schedule on a sheet, with its controls

Graphic schedule by type

In **View › Create › Schedules › Schedule/Quantities** choose **Graphic schedule by type** and the category: *Doors*, *Windows*, *Curtain walls* or *Openings* (all three). The schedule has one entry per type, with the type's elevation drawn and dimensioned from the model and its data below; curtain walls have one entry per type and size. With **Number missing type marks** types without a mark get F1, F2, P1, FC1…, written into the type.

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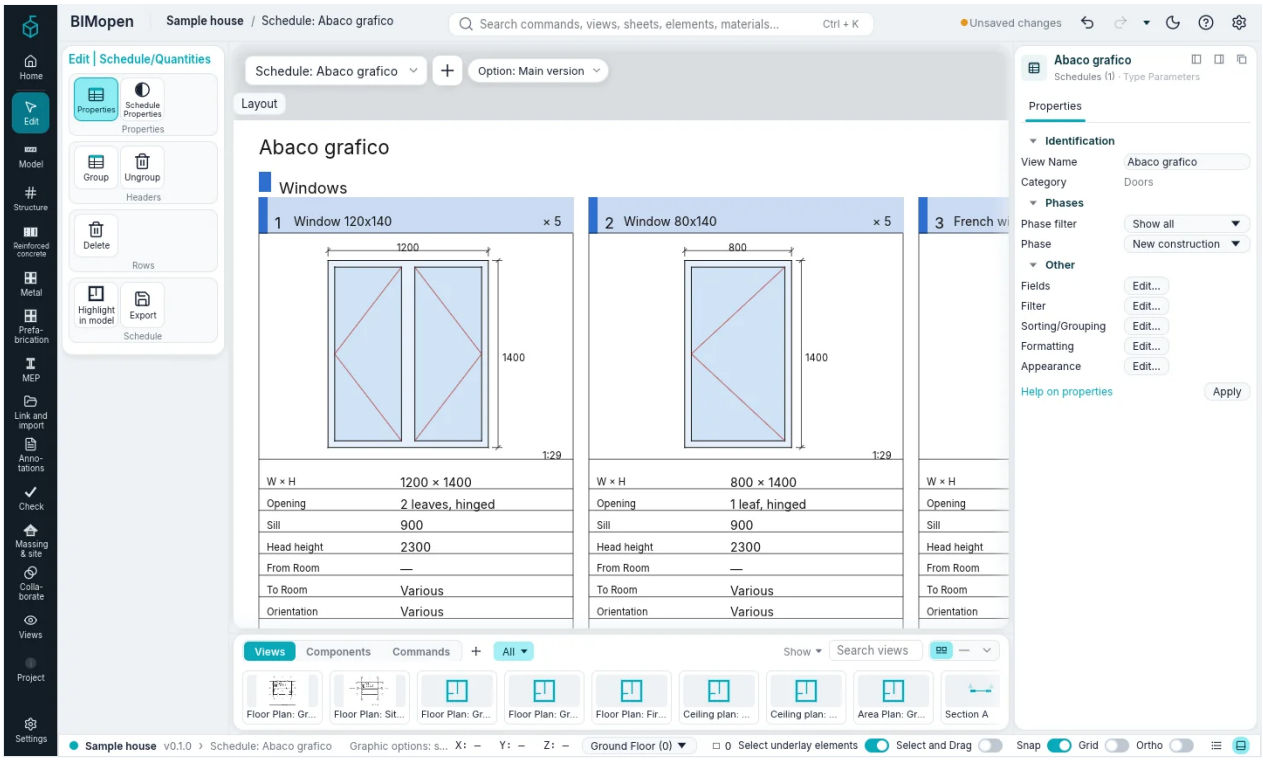

Graphic schedule by type

In **Schedule Properties** the **Layout** tab chooses between *Table*, *In columns* (one column per type, quantity per level and total), *As cards* (cards side by side with the area colour, quantity and a Notes row) and *Elevations by level* (the types of each level on the ground line, with a label and dimensions between types). **Type drawing** sets front or back view, scale, detail level, width, height and sill dimensions, swing symbol and ground line. **Areas and colours** splits the schedule into areas with a condition and a colour; **Coloured opening tag in plans** uses the same colours in plans. *Thermal transmittance U* and *Acoustic class* are type fields you fill in. A double click on an entry opens the type's properties. On a sheet the schedule splits like the others.

Graphic schedule by element has one column per door or window (or one row, with a *Normal* table): a computed classification code (*Window - 003*), the rooms on both sides, sizes, orientation, sill and head height. In the **Fields** tab the *2D symbol*, *Back view* and *Type elevation (dimensioned)* fields draw the model in the cell; for the selected field you set scale, detail level and row height. In **Layout** you choose the level and whether curtain wall panels are included.

Key schedules and material takeoffs

A key schedule (**Schedule keys** in **New Schedule**) defines named sets of values, one row per key, added and deleted with **Rows**. Its **Key Name** becomes a parameter of the category: choosing a key for an element gives it all the values of that row.

A material takeoff lists, for each element, the materials of its layers with their area and volume; its fields can be sorted, grouped and totalled like any other.

Project and shared parameters

Settings ➤ Settings ➤ Project Parameters adds parameters to the elements of some categories, or to their types, with a name, a type of value and the group they show in the element panel.

Settings ➤ Settings ➤ Shared Parameters opens **Edit Shared Parameters**: the shared parameter file in use (**Browse...**, **Create...**), its groups and the parameters of each group. A shared parameter keeps its name and type, and has an identifier of its own, so that several projects add the same parameter. In **Parameter Properties**, choose **Shared parameter** and **Select...** to add one to the project; **Export...** writes a shared parameter of the project back to the file.

Exporting and reusing schedules

- **File ➤ Export ➤ Schedule...** writes the schedule as delimited text: **Export Schedule** sets the field delimiter, the text qualifier and which headings and group rows go in the file.
- **Create view template from view...** on the right button of a schedule makes a view template with its fields, filter, sorting, formatting and appearance; **Apply view template...** gives them to another schedule.

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- **Save to New File...** on the right button writes the schedule's setup, without the elements, to a file of its own. **Create ➤ Load from library ➤ Insert from File ➤ Insert Views from File...** brings the schedules of another project into this one, with their project parameters.

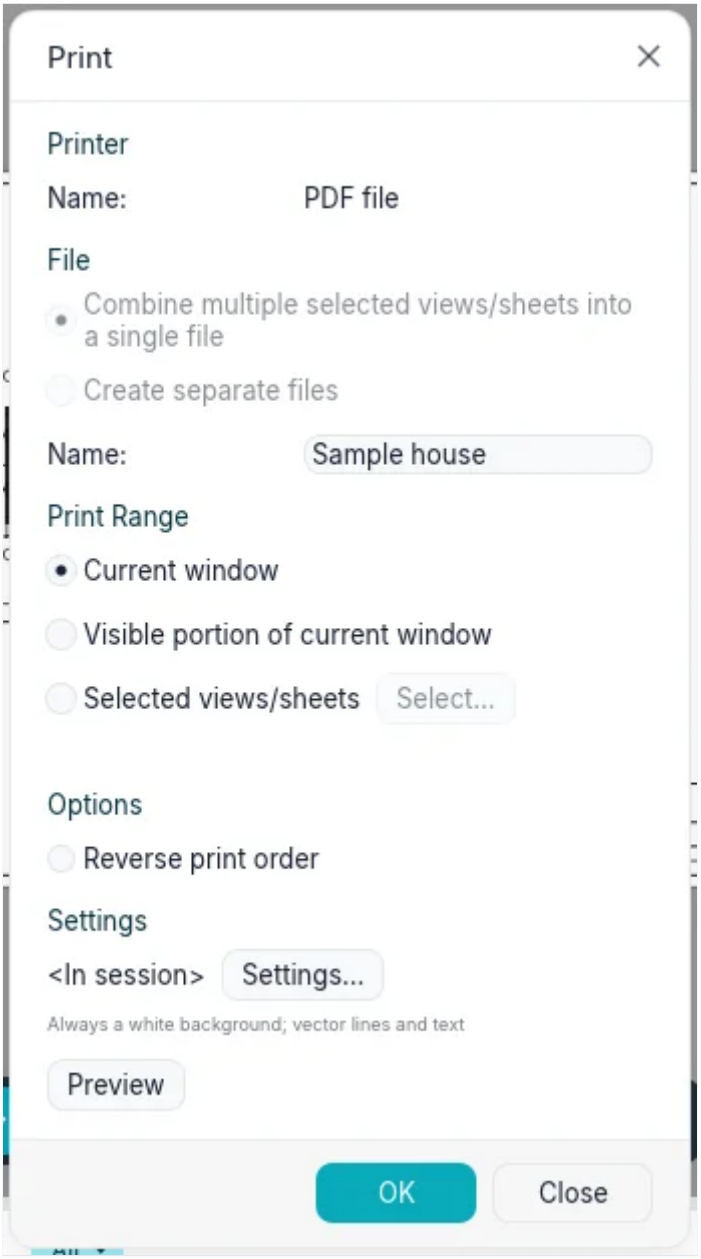
Printing

Printing writes PDF files: on the desktop they are saved where you choose, in the browser they are downloaded.

The Print dialog

`Ctrl+P`, or **File ➤ Print ➤ Print...**, opens **Print**:

- **File**: the name of the file and, when several views or sheets are printed, **Combine multiple selected views/sheets into a single file** (one page each) or **Create separate files** (one file each, named after the sheet or view).
- **Print Range**: **Current window**, **Visible portion of current window** (only what the window shows), or **Selected views/sheets** with **Select...**.
- **Reverse print order** prints the last page first.
- **Settings**: the print settings in use, changed with **Settings...**; **Preview** shows the pages.



Print

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Plans, sections, elevations, free 2D views and sheets print as drawings; the 3D window prints as an image of the view; a schedule prints its table on a page. The page is always white, elements hidden temporarily in a window are printed, and each printed sheet carries the date and time of the print.

View/sheet sets

Select... opens **View/Sheet Set**: tick the sheets and views to print (**Show** sheets, views or both; **Check all**, **Check none**). A set is saved by name with **Save As...** and kept in the project; **Save**, **Reset**, **Rename...** and **Delete** manage the saved sets, and the name list picks one again.

Print Setup

File ➤ Print ➤ Print Setup..., or **Settings...** in **Print**, sets:

Setting	What it does
Size and **Orientation**	The paper (A0 to A4, or the sheet's own size) and its orientation
Paper Placement	**Center**, or an offset from the lower-left corner
Hidden line views	**Vector Processing** (sections and elevations print only the lines that are seen: the lines behind nearer surfaces are removed, not just covered, and lines lying on one another print once, so the PDF opens clean in other programs) or **Raster Processing** (the pages as images, at the raster quality chosen)
Zoom	**Fit to page**, or a percentage of the real size
Colors	**Black Lines**, **Grayscale** or **Colour**
Options	**Show view links in blue**, **Hide unreferenced view tags**, **Hide crop boundaries**, **Replace halftone with thin lines**, **Keep coincident lines** (off: two lines lying on one another in printed and exported sections and elevations become one; on: both stay, each with its own element)

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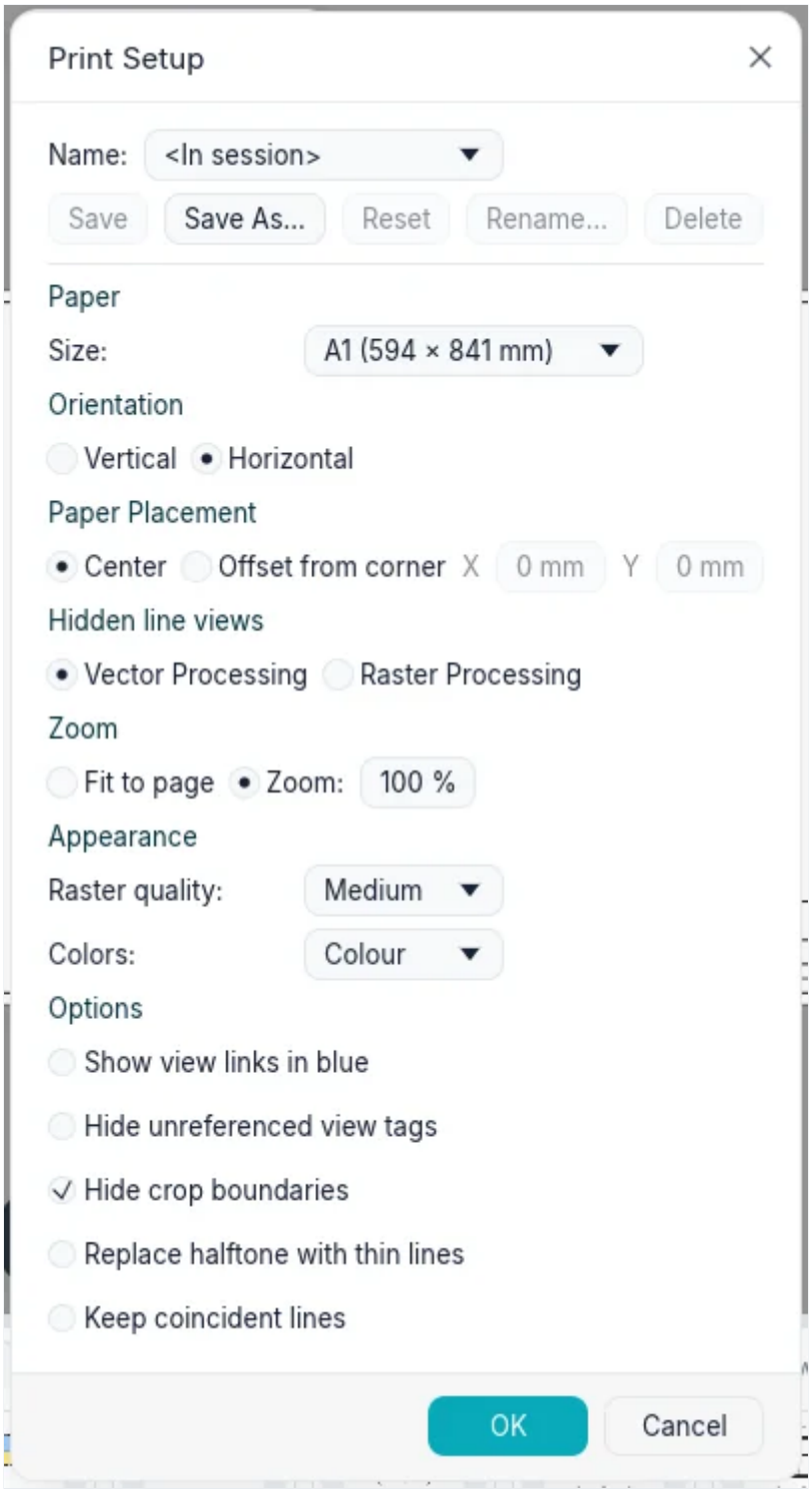

Print Setup

Settings are saved by name with **Save As...**, and kept in the project. Closing the dialog after changing saved settings asks whether to save the changes; **No** keeps them for this session only.

Print Preview

File ➤ Print ➤ Print Preview, or **Preview** in **Print**, shows the pages as they will print, in their colours: **Previous Page** and **Next Page** go through them, the wheel zooms and dragging pans; **Print...** prints, **Close** goes back.

Exporting images

File ➤ Export ➤ Image... opens **Export Image**:

- the file name and the **Export Range**: the current window, its visible portion, or the views and sheets chosen (one file each);

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- the **Image Size**: **Fit view(s)/sheet(s) to** a number of pixels across or down, or a zoom percentage of the real size at a resolution;
- the options, as for printing: view links in blue, crop boundaries and unreferenced view tags hidden;
- the **Format** of views without shading and of shaded views: PNG, JPEG, BMP or TIFF.

The 3D window is exported as it is shown, with its graphic style.

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Command line and MCP

Everything you do in BIMopen is a command. A click on **Wall** and two points in the plan run the same `wall` command you can type yourself. This chapter describes the command line of the window, the `bimopen-cli` program, which runs the same commands without a window, and the MCP server, through which an AI agent can model, check and render a project.

The command line in the window

The command line sits under the drawing area. Its upper part is the log; its last row is where you type.

- You do not need to click it first: whatever you type while no other field is being edited goes there.
- `Enter` runs the line. The command is echoed in the log after a `>`, followed by its result; an error is shown in red and changes nothing.
- `Up` and `Down` bring back the lines typed before.
- On the left of the row, the prompt says what is expected: `Command:` when no tool is active, or the request of the active tool, such as `WALL next point [Esc to finish]:`.
- `help` lists every command with a line of description; `help wall` shows the usage of one. **? ▸ Commands** in the top bar does the same as `help`.

Typed shortcuts start the tools

A command typed **without arguments** starts the interactive tool, as if you clicked its button: `WA` (or `wall`) starts the wall tool, `DR` the door tool, `VG` opens Visibility and Graphics. The same command **with arguments** does the work at once, without the tool: `wall 0,0 5,0` draws a wall from the origin to 5 m along x.

Command names, their short forms and their arguments are the same in every language of the interface; most also accept an Italian name, such as `muro` for `wall`. Parameter names, type names and view names follow the language of the interface: in English you type `set Free Height 2.7m`, in Italian `set Altezza libera 2.7m`. The keyboard shortcuts chapter lists every short form.

Points and lengths

You type	It means
`3,4`	The point x = 3 m, y = 4 m
`@2,0`	2 m along x and 0 along y from the previous point
`@4<90`	4 m from the previous point, at 90° (angles counter-clockwise from the x axis)
`2.7m`, `270cm`, `2700mm`	A length with its unit
`2.7` or `2,7`	A length in metres: at the command line a number without unit is in metres

For example, `wall 0,0 @6,0 @4<90 0,4 c` draws a 6 × 4 m rectangle; the final `c` closes it.

> Warning
>
> At the command line `set Free Height 2700` means 2700 **metres**. Write the unit: `2700mm`. In the element panel, instead, a number without unit is in the project's units, millimetres in a new project.

Arguments in JSON

Every command also takes its arguments as a JSON object, which gives access to all its options. The usage line of `help` lists them:

```
wall 0,0 10,0 {"height":3.2,"location":"Linea d'asse del muro"}
roof walls {"slope":30,"overhang":0.5}
sheet A102 Plans {"title_block":"A3"}
export doors.csv {"schedule":"doors"}
```

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Commands you will use often

Command	What it does
`new`, `sample`	A new project, the sample project
`open house.bpr`, `save house.bpr`	Opens and saves a project
`undo`, `redo`	Undoes and redoes the last change
`view 3d`, `view Ground Floor`	Opens the default 3D view, or the plan of a level
`wall`, `door`, `window`, `floor`, `roof`	Model elements: see the usage with `help`
`select walls`, `select all`, `deselect`	Selects by category, everything, nothing
`set <parameter> <value>`	Sets a parameter of the selection, as in the element panel
`type <type name>`	Changes the type of the selection, or of the next element
`move 1,0`, `copy 0,3`, `rotate 90`	Modifies the selection
`delete`	Deletes the selection
`list walls`, `params 102`, `info`	Lists elements, the parameters of one, a summary of the project
`schedule doors`	Shows a schedule at the command line
`export house.ifc`	Exports IFC, SVG, PNG, CSV or USD (`.usdz`, `.usda`), by the file's extension
`walk`, `collision`, `walksettings`, `liveview`	Walk mode: a walk computed, Disable Collision, the step climbed, the live view (see Live view and walk mode)
`walkthrough`	Walkthroughs: makes one from the points clicked, changes its frame, total frames and frames per second, speed, acceleration and turn smoothness, adds and removes keyframes (see Walkthrough)
`camera`	Perspective views: makes a camera view from two points, moves its camera, its field of view, far clip plane and crop region, resets its target (see Perspective views)
`bundle house.zip`	Project bundle: the project and the files it uses in one .zip; `open house.zip` unpacks and opens it

Every element has a number, its id, shown in brackets after it is created: `Door «Single swing door 80x210» in wall 102 (106)`. Commands accept it where they need an element: `params 106`, `delete 106`.

Clipboard, types and groups

The commands of Copy, cut and paste, Transfer type and Groups. Without ids they work on the selection.

Command	Example	What it does
`copyclip`	`copyclip 102 103`	Copies the elements to the clipboard
`cut`	`cut 102`	Cuts the elements to the clipboard
`paste`	`paste 4,0`	Pastes at the point; without a point, in the same place; `{"mode":"view"}` in the active view, `{"mode":"levels","levels":[…]}` on levels, `{"mode":"views","views":[…]}` in views, by id or by name
`matchtype`	`matchtype 102 105 106`	The elements after the first take its type, if they are of the same category
`group (GP)`	`group 102 103 104 Bathroom`	A new group of the elements, with its name (`detail_name`: the name of the attached 2D group); with `"empty":true` (and `"detail":true` for a 2D group) a group with no members, to fill in the editor
`placegroup`	`placegroup 10,0 Bathroom {"level":"First Floor"}`	A new copy of the group at the point, on the level; `rotation` in degrees, `view` for a 2D group
`ungroup`	`ungroup 120`	The members of the copy become ordinary elements

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Command	Example	What it does
`groupedit`	`groupedit {"group":120,"add":[130],"remove":[121]}`	Adds and removes members of a copy and updates the other copies
`groupsync`	`groupsync 120`	The other copies of the group are made again from this one
`groupexclude`	`groupexclude 121 keep`	The members leave their copy only; with `keep` (Move to project) they stay in the project as ordinary elements
`grouprestore`	`grouprestore 120`	The members excluded from the copy come back
`grouptype`	`grouptype {"name":"Bathroom","rename":"Bathroom A"}`	Renames (`rename`), duplicates (`duplicate`) or deletes (`"delete":true`) a group without copies; `"duplicate":true` takes the next free number
`groupattach`	`groupattach {"group":120,"detail":"Bathroom dimensions"}`	Shows in the active view (or `view`) the attached 2D group with the copy; `"show":false` hides it
`groupdetach`	`groupdetach 130`	A copy of an attached 2D group, with no dimensions or tags, becomes a 2D group of its own
`groups`	`groups`	The groups of the project, with their copies and attached 2D groups

bimopen-cli

`bimopen-cli` runs BIMopen without a window: to convert files, to build a project from a script, to check a model, or to serve agents over MCP. Build it with `cargo build --release -p bimopen-cli`.

```
bimopen-cli info <file>
bimopen-cli convert <in> <out> [--level NAME] [--view 3d] [--schedule walls|doors|windows|floors|rooms|materials]
bimopen-cli run [<file> | --sample] [--cmd LINE]… [--script FILE] [--save FILE] [--export FILE]…
bimopen-cli commands
bimopen-cli mcp [<file> | --sample]
```

Subcommand	What it does
`info <file>`	A summary of the project and, in JSON, its levels, element counts and active view
`convert <in> <out>`	Opens `<in>` (`.bpr` or `.ifc`) and writes `<out>`: `.ifc`, `.svg` or `.png` of a plan (the lowest level, or `--level`), `.png` of the 3D view (`--view 3d`), `.csv` of a schedule (`--schedule`), or `.bpr`
`run`	Opens a file, the sample, or a new project, then runs the options in order: `--cmd` one line, `--script` a file of lines (empty lines and lines starting with `#` are skipped), `--save` and `--export` a file
`commands`	Every command, with its description and usage
`mcp`	The MCP server on standard input and output
`--version`	The version

The messages are in the language of the environment variable `BIMOPEN_LANG` (`it`, `en`, `de`, `fr`, `es`, `ar`), or of the system (`LANG`), or Italian. `bimopen-cli` ends with code 0 when everything worked, 1 when a command failed (the message says which), and 2 when the arguments are wrong (it prints the usage).

```
export BIMOPEN_LANG=en
bimopen-cli run --script house.txt --save house.bpr --export house.ifc
bimopen-cli convert house.bpr plan.svg
bimopen-cli convert house.bpr doors.csv --schedule doors
bimopen-cli convert house.bpr first.png --level "First Floor"
bimopen-cli convert house.bpr house-3d.png --view 3d
bimopen-cli convert model.ifc model.bpr
```

A script is a text file with one command per line, exactly as typed at the command line:

```
# a small house
wall 0,0 10,0 10,6 0,6 c
door 5,0
window 10,3
```

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```
floor walls
roof walls
```

The schedules are written as CSV with `;` between the columns, the headings in the language of the messages and a total row at the end.

MCP: BIMopen for AI agents

The Model Context Protocol (MCP) lets an AI agent use programs as tools. `bimopen-cli mcp` is an MCP server: it speaks JSON-RPC 2.0 on standard input and output, one message per line, and keeps one project open for the whole conversation. Without a file it starts from a new project; `--sample` starts from the sample house.

To add it to an agent, give the agent the command to start. For example, in Claude Code:

```
claude mcp add bimopen -- bimopen-cli mcp --sample
```

or, in a client configured with JSON:

```
{ "mcpServers": { "bimopen": { "command": "bimopen-cli", "args": ["mcp", "--sample"] } } }
```

The server offers seven tools:

Tool	Arguments	What it does
`command_line`	`line`	Types a line at the command line, exactly as a person would
`execute`	`command`, `args`	Runs a command with its arguments in JSON
`inspect_model`	–	A summary: levels, element counts, active view, selection
`query_elements`	`category` (optional)	The elements, with id, category and name: `walls`, `doors`, `windows`, `floors`, `rooms`, `roomseps`, `grids`, `levels`, `types`
`get_params`	`id`	Every parameter of an element
`schedule`	`kind`	A schedule: `walls`, `doors`, `windows`, `floors`, `rooms` or `materials`
`render`	`view`, `width`, `height`, `yaw`, `pitch`	A PNG image of a view: `3d`, the name of a level for its plan, of a section, elevation, callout or free 2D view, or the number or name of a sheet; `yaw` and `pitch` turn the 3D camera, in degrees

The answers carry the message of the command and, when there is one, its data in JSON, such as the ids of the elements created. An agent can check its own work by asking for a render.

> **Tip**
>
> Ask the agent to render the plan and the 3D view after each step: it sees what it built and corrects itself.

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Keyboard shortcuts

BIMopen has two kinds of shortcuts: **keys**, such as `Ctrl+Z`, which act at once, and **typed shortcuts**, such as `WA`, which you type letter by letter. At the command line a typed shortcut waits for `Enter`, like any command; typed over the drawing area, it acts as soon as it is complete when nothing longer begins with its letters (see below). Typed shortcuts are the same in every language.

On macOS, `Cmd` takes the place of `Ctrl`.

Keys

Key	What it does
`Ctrl+Z`	Undoes the last change
`Ctrl+Y` or `Ctrl+Shift+Z`	Redoes the change just undone
`Ctrl+S`	Saves the project; a project never saved asks for a file name
`Ctrl+P`	Opens **Print**
`Ctrl+C`	Copies the selection to the clipboard
`Ctrl+X`	Cuts the selection to the clipboard
`Ctrl+V`	Pastes the clipboard: in a plan it follows the cursor and the click drops it; elsewhere, in the same place
`Ctrl+A`	Selects the elements of the open plan's level, or of the whole model in a 3D view
`Del`	Deletes the selected elements, and the elements they host
`Home`	Fits the whole view in its window
`F1`	Opens the page of this manual about the open dialog, the tile under the mouse or the active tool
`F9`	In the BIMopen style, puts the views strip away or shows it again
`Esc`	Cancels the step in progress; pressed again, ends the tool; with no tool, clears the selection
`Enter`	Runs the line typed at the command line; confirms a value in the element panel
`Up` / `Down`	At the command line, the lines typed before
`Space`	While drawing walls, swaps the exterior and interior side; while placing a component or a detail component, rotates it
`Shift`	While drawing, keeps segments horizontal or vertical (or frees them, when **Ortho** is on); with a click or a window, takes elements out of the selection; while dragging, frees a joined wall or keeps the move horizontal or vertical
`Ctrl` + click	Adds an element to the selection; held when a drag starts, places a copy
`Tab` / `Shift+Tab`	Highlights the next (previous) element under the cursor, or a chain of joined walls and lines
`Ctrl+Left`	Selects again what was selected before
`Left` `Right` `Up` `Down`	Move the selection a small step; with `Shift`, ten steps

`Ctrl+Z`, `Ctrl+Y`, `Ctrl+A`, `Del` and `Home` act on the model only while you are not typing in a field: in a field they act on the text.

`Ctrl+C`, `Ctrl+X` and `Ctrl+V` act on the model while no tool is active (see Copy, cut and paste).

In a floor, roof, stair or railing sketch, `Esc` (or a double-click) also finishes the sketch when it has enough points.

In the **Text Note** dialog, `Ctrl+Enter` inserts the note and `Esc` cancels it.

`Alt` pressed and released, or `F10`, shows the key tips of the rail and the tool panel. With no tool active and nothing typed, `Enter` repeats the last tool. After a click in the Project Index, the arrow keys and `Enter` move in its tree.

Mouse

	In plans and sections	In 3D views
Wheel	Zoom at the cursor	Zoom

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	In plans and sections	In 3D views
Middle button, drag	Pan	Pan; with `Shift`, orbit
Left button, drag	Selection window (left to right: inside; right to left: touching), or moves the selected elements	The same
Right button	Context menu; with a tool active, the tool's menu (**Cancel**, **Snap Overrides ▸**...)	The same; dragging, orbit
Double-click	Finishes a sketch	–

Typed shortcuts

Type the letters while no field is being edited, then press `Enter`. Without arguments, a shortcut starts the tool or opens the dialog; with arguments, it runs the command at once (see Command line and MCP).

With the cursor in the drawing area, the letters go to the command line (in the BIMopen style, to the search), and a shortcut acts as soon as it is typed, with no `Enter`, when no longer shortcut or command name begins with its letters: `VR` (cut heights), `WT` (tile), `ZE`, `ZF`, `ZA`, `ZP`, `ZR`, `ZO`, `ZS`, `HR`, `HH`, `HC`, `IC`, `DL` (detail line), `RT` (room label), `TG` (tag by category), `TX`, `CC` (copy), `MV`, `RM`, `LL` (level), `RC` (revision cloud), `PP` (shows or hides **Properties**), `PT`, `RP`, and the shortcuts you added. The others begin longer names and wait for the next letter, a space (arguments follow) or `Enter`: `VG` (`vgsub`...), `HI` (`hide`), `DI` (`dim`), `AL` (`align`), `WA` (`wall`), `DR` (`draworder`), `UN` (`undo`), `MA` (`mask`); so `dim`, `wall 0,0 5,0` or `WA 0,0 5,0` typed over the view run as typed. `MA` starts **Transfer type**; `PT` says in the status bar that it is not available yet; `RP` starts the reference plane. Typed in the command line itself, a line waits for `Enter`.

While a tool waits for a point, in every view (plans, free 2D views, sections, elevations, the 3D view on its work plane, sketches), two letters override the snaps for that point only: `SE` (endpoint), `SM` (midpoint), `SN` (nearest), `SW` (work plane grid), `SQ` (quadrant), `SI` (intersection), `SC` (center), `SP` (perpendicular), `ST` (tangent), `SR` (remote objects), `SZ` (close), `SO` (no snap); `SS` ends the override in force. They are also chosen in the right button's **Snap Overrides ▸**; those a longer name begins with need `Enter`. The status bar says so, and `Esc` or the point taken ends it. With several snaps under the cursor, `Tab` goes to the next and `Shift+Tab` to the previous. Shortcuts added in **Keyboard Shortcuts** under a snap's name work as its two letters.

Modelling

Shortcut	Command	Starts
`WA` or `W`	`wall`	Wall
`DR`	`door`	Door
`WN`	`window`	Window
`SB`	`floor`	Floor (the **Flooring** button)
`RF`	`roof`	Roof
`CM`	`component`	Object
`ST`	`stair`	Stairs
`RA`	`railing`	Railing
`CG`	`curtaingrid`	Curtain grid
`MU`	`mullion`	Mullion
`CP`	`curtainpanel`	Curtain wall panel
`GR`	`grid`	Grid
`RM`	`room`	Room
`RS`	`roomsep`	Room separator
`PA`	`areaplan`	The **Area Plan** dialog
`CA`	`areaboundary`	Area outline
–	`area`	Area
`LG`	`legend`	Color fill legend

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Shortcut	Command	Starts
SC	colorscheme	The **Color Schemes** dialog

Modifying

Shortcut	Command	Starts
MV or M	move	Move
CO	copy	Copy
RO	rotate	Rotate
MM	mirror	Mirror on Axis
DM	mirror	Mirror on Line
AR	array	Repeat in Series
OF	offset	Offset
AL	align	Align
TR	trim	Shorten/Lengthen
SL	split	Split Element
SG	split	Split with Joint
RE	resize	Scale
PN	pin	Fix: pins the selection
UP	unpin	Unpins the selection
MA	matchtype	Transfer type
GP	group	Group: a new group of the selection
SA	selectinstances	Selects all instances of the selection's types visible in the view
DEL	delete	Deletes the selection
U	undo	Undoes the last change

Annotating and detailing

Shortcut	Command	Starts
DI	dim	Aligned dimension
DL	lineardim	Linear dimension
DA	angulardim	Angular dimension
TX	text	Text
TG	tag	Tags the untagged doors and windows of the level
–	detailline	Detail line
RG	region	Filled region
–	mask	Masking region
CD	detailcomponent	Detail component
RD	repeatingdetail	Repeating detail
IS	insulation	Insulation

> Note
>
> Typed alone, DL starts the detail line; followed by points at the command line, it draws a linear dimension (lineardim).

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Views and settings

Shortcut	Command	Starts or opens
`SE`	`section`	Section
`EL`	`elevation`	Elevation
`DT`	`callout`	Callout (the **Detail** button)
`DS`	`splitsegment`	Split Segment of a section
`VD`	`draftingview`	The **Free 2D View** dialog
`PP`	`floorplan`	The **Floor Plan** dialog
`DV`	`duplicateview`	Duplicates the active view
`VG` or `VV`	`vg`	**Visibility and Graphics**
`WF`, `HL`, `SD`	–	**Wireframe**, **Lines**, **Shaded**: the graphic style of the active view
`HH`, `HI`	–	**Hide Element**, **Isolate Element** for now
`HC`, `IC`	–	**Hide Category**, **Isolate Category** for now
`HR`	–	**Reset Isolate Temporarily**
`EH`, `VH`	–	**Hide in view**: the selected elements, their category
`RH`	–	**Reveal Hidden**, on and off
`TL`	`thinlines`	**Thin lines**, on and off
`SO`	`objectstyle`	**Styles by Category**
–	`material`	**Materials**
`UN`	`units`	**Project units**
`PU`	`purge`	**Purge unused**
–	`filter`	**Filters**
–	`viewtemplate`	**View templates**
–	`applytemplate`	Apply a view template to the active view
–	`linestyle`	**Line styles**
–	`fillpattern`	**Fill patterns**
–	`saveview3d`	Saves the {3D} view with a name
–	`renameview`	Renames the active view

The commands with **–** have no short form, but their whole name works the same way. `help` lists every command and its short forms.

The zoom commands have typed shortcuts too: `ZR` zoom in region, `ZO` zoom out (2x), `ZF` zoom to fit, `ZA` zoom all to fit, `ZS` zoom sheet size, `ZP` previous pan/zoom. **Views ▸ Windows ▸ Panels ▸ Keyboard Shortcuts** gives any command shortcuts of your own (The interface).

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Troubleshooting

This chapter collects situations that can surprise you, and what to do about them. It grows with the manual: each area of the realignment adds its own.

Starting BIMopen

The web app shows «BIMopen non si è avviato» and an error

The browser could not give BIMopen a graphics surface: the message ends with `webgl2 not available` or with a WebGPU error. BIMopen in the browser needs WebGPU or WebGL2.

- Turn on hardware acceleration in the settings of the browser, then reload the page.
- Try another browser, or a newer version of the same one.
- Add `?webgl` to the address to use WebGL2 even where WebGPU exists but does not work.

The desktop app does not build on Linux

The interface needs the development files of xkbcommon, Wayland and GTK 3. On Debian and Ubuntu:

```
sudo apt-get install -y libxkbcommon-dev libwayland-dev libgtk-3-dev
```

The interface

The element panel or the Project Index has disappeared

Views ▸ Windows ▸ Panels shows or hides them: tick them again there. The **Properties** tile of the contextual card also shows the element panel again. A click on the rail entry already chosen folds the tool panel: click it again to open it.

What I type ends up at the command line

Whatever you type while no field is being edited goes to the command line, the search box of the top bar: this is how typed shortcuts work. To type in a field of the element panel or of a dialog, click the field first.

A tile is greyed

Some tools work only in the right view, and their tip says why. **Area Outline** and **Area** need an open area plan: the tip says **Open an area plan first**. **Save View** needs an open 3D view; **Document ▸ Sheets ▸ View** needs an open sheet; **Undo** and **Redo** are greyed when there is nothing to undo or redo.

`DL` starts a dimension instead of a detail line

`DL` is the short form of the linear dimension. To draw detail lines, use **Document ▸ Detail ▸ Detail line**, or type `detailline`.

Modelling

«no wall at that point»

A door or a window must be placed in a wall. Click on the wall itself, or give a point on its line at the command line.

A height set at the command line is enormous

At the command line a number without unit is in metres: `set Free Height 2700` makes the wall 2700 m high. Undo with `Ctrl+Z` and write the unit: `2700mm` or `2.7m`. In the element panel a number without unit is in the project's units.

Files

Unsaved work is gone after New, Open or Sample project

These commands replace the current project without asking. Save before using them: the asterisk **Unsaved changes** in the top bar shows that there are changes to save.

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Open… in the browser does not show a file dialog

In the browser, drag the `.bpr` file into the page: the command line says **Drag a .bpr file into the window to open it**.

bimopen-cli prints its usage and stops

The arguments are not the ones it expects: it ends with code 2 and shows the usage. A command that fails ends with code 1 and a message saying which line failed. See Command line and MCP.

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